


The University of Western Australia

School of Physics, Mathematics and Computing

# Physics-Informed Quantum Hamiltonians for Solving ODEs, PDEs and Navier–Stokes

Master of Physics (Computational Physics)

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# Declaration

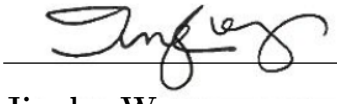
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# Summary of Student Achievement

My research was an individual project, outside the main focus of the QUISA group, although a subgroup working on quantum linear algebra (QLA) has since formed.

The year began with an idea my supervisor put to me: that the quantum singular value transformation (QSVT) is a "grand unification" of quantum algorithms, and that its reach had not yet been fully mapped. I set out to find new applications for it and for the newer QLA methods, GQSP and GQSVT. I was first introduced to differential equations in my Bachelor's degree and studied their numerical solution in my first postgraduate semester, so a quantum solver was where I most wanted to focus.

I found the preprint of Wu et al., which told a compelling story. A single physics-informed Hamiltonian whose ground state is the solution, for linear and nonlinear equations alike. It was close to what I had hoped to find, one unified quantum method for differential equations. I first rebuilt quantum signal processing, QSVT and GQSP for myself. GQSP is enough for a Hermitian Hamiltonian, so I set GQSVT aside.

The preprint had not been peer reviewed, so I reproduced it before building on it, and the reproduction changed the course of the project. Its error bounds were not stated precisely, it didn't discuss post-selection probability, and its treatment of nonlinear equations turned out to produce degenerate kernels. The paper was less substantial than I had first thought, so I rebuilt the method with its printed defects removed as the standard form, and wrote an exact structured encoding of its differentiation operator, for which the paper gives only an approximate form.

The dense differentiation operator was expensive to encode. Building the exact encoding showed me that the real cost sits in the Hamiltonian itself, whose norm grew so quickly that the filter degree grows as a power of the resolution, however good the circuit. That realisation led me to the descriptor form.

I wrote a python package, `pihm`, which reproduces my method generally for (almost) any differential equation we claim to be able to solve. Generality was the principle that had drawn me to the paper, so I kept it through the model and code design.

I then extended the method's generality to coupled systems, and explored non-degenerate treatments for nonlinearity. This solved two things at once, since we can carry complex-valued equations as coupled real systems. For nonlinear equations I replaced the doubled space with a Carleman lift. On my supervisor's advice I then pushed the method onto harder problems, ending with two-dimensional Navier–Stokes, and ran the production campaigns on Setonix at the Pawsey Supercomputing Research Centre.

# Abstract

The cost of solving a differential equation numerically grows with the resolution it demands. A register of $n$ qubits holds $N = 2^n$ amplitudes, so a quantum algorithm can store a discretised solution in exponentially few qubits. What such an algorithm costs is the number of times it must apply a circuit encoding the problem, and that number is set by how well conditioned the encoded problem is. Wu et al. [1] recently proposed writing a differential equation as the ground state of a physics-informed effective Hamiltonian and preparing that state with a spectral filter in place of a variational search. Every term of their Hamiltonian has an efficient circuit, but how many applications a preparation needs, a number that grows with the square root of the Hamiltonian's norm over its spectral gap, was left open.

This thesis locates the cost of that approach and reduces it. Rebuilding the method and reproducing its examples for ordinary differential equations, we separate the cost of encoding the Hamiltonian from that of the Hamiltonian itself. For a critically damped oscillator on eight qubits, an exact encoding of the differentiation matrix lowers the estimated number of applications from $9.1 \times 10^{22}$ with the published encoding to $1.2 \times 10^{10}$, and no encoding of the same Hamiltonian can bring it below $1.5 \times 10^{9}$. What remains is intrinsic to the Hamiltonian of this standard form, whose state holds the solution alone: each derivative is eliminated through the dense Chebyshev differentiation matrix, whose norm grows as $N^2$, so for a second-order equation with constant coefficients the Hamiltonian's norm grows as $N^8$ and the number of applications as $N^4$, however it is encoded.

We then propose a descriptor form of the Hamiltonian, which has the same ground state and carries the solution's derivatives as additional unknowns linked by sparse, banded relations. The idea adapts two classical methods, the ultraspherical spectral method and first-order-system least squares. Its circuit needs only $\mathcal{O}(\log n)$ auxiliary qubits. For linear problems on one axis with polynomial coefficients, analytic data, a leading coefficient that never vanishes and a solution unique up to scale, its norm grows only as $N^2$ and we prove that the spectral gap stays bounded away from zero as the resolution grows, so the number of applications grows about linearly in $N$; for the oscillator the estimate falls to 9128, or 8736 under an ideal encoding, although on constant-coefficient problems each application can cost more gates. The carried derivatives erode the gap wherever the equation fails to control them. On one axis, scaling each derivative block by a single weight restores it without extra qubits, and the number of applications then grows about as $N^2$ when the leading coefficient has a simple zero. On two axes the gap still closes, and the standard form can need fewer applications on Legendre's equation under an ideal encoding and on the heat equation at small sizes.

Both formulations extend to coupled systems and, through Carleman linearisation, to nonlinear equations. We show that the doubled-space construction proposed for nonlinear terms has a degenerate ground space, onto which a filter can only project, whereas the Carleman system has a unique ground state, which we prepare in both formulations for Burgers' equation by exact emulation and for two-dimensional incompressible Navier–Stokes flow on up to 48 Fourier modes by exact emulation and resource estimate. A well-isolated ground state does not certify the nonlinear solution, however: at strong nonlinearity the linearisation diverges while its ground state stays unique.

All results are classical computations: the field is decoded from the prepared state's amplitudes, and nothing was run on quantum hardware. Circuits of up to 24 qubits are verified against the operators they encode, and simulations of up to 20 qubits with the minimax filter agree with exact emulation to within $10^{-10}$. Neither formulation offers an exponential speed-up, and no advantage over classical solvers is claimed. The broader lesson is that once the encoding is tight, a ground-state solver's cost is set by the operator being prepared, so reformulating the problem can matter more than optimising its circuit. The descriptor form moves the conditioning of differentiation from the norm into the gap, which can be secured on one axis but not yet on two.

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# Notation and Conventions

A quick reference for symbols used across the thesis.

**Typeface.** Matrices are italic. Calligraphic letters are maps that change the register, the lifts $\mathcal{M}$ and the folds $\mathcal{N}$, and also $\mathcal{T}$ (Gram map), $\mathcal{H}$ (Hilbert space) and $\mathcal{O}$ (order symbol). The constraint rows $\mathbb{B}$, $\mathbb{D}^{(0)}$, $\mathbb{S}$ and $\mathbb{F}$ are blackboard bold; otherwise blackboard bold is kept for $\mathbb{R}$, $\mathbb{C}$ and the expectation $\mathbb{E}$. Wu et al. print every non-unitary matrix in blackboard bold (§ 3.2).

**Discretisation**

| Symbol | Meaning | Introduced |
|---|---|---|
| $n$ | Qubits per Chebyshev axis (the resolution). | § 3.1 |
| $N = 2^n$ | Truncation dimension of one Chebyshev axis. | § 3.1 |
| $k$ | Order of an ODE. | § 3.2 |
| $p$ | Degree of a polynomial nonlinearity. | § 6.5 |
| $n_f$ | Qubits in the field (block) register of the descriptor form; for a chain, $\lceil \log_2(1 + \sum_a k_a) \rceil$ with order $k_a$ along axis $a$ ($\lceil \log_2(k+1) \rceil$ on one axis). | § 5.2 |
| $\dim w$ | Padded descriptor state dimension, $2^{n_f} \prod_a N_a$ ($2^{n_f} N$ on one axis). | Appendix D.2 |

**The Chebyshev model**

| Symbol | Meaning | Introduced |
|---|---|---|
| $T_\ell$ | Chebyshev polynomial of the first kind, degree $\ell$. | § 3.1 |
| $\lvert\tau(x)\rangle$ | The Chebyshev feature state. | § 3.1 |
| $\varphi_k$ | Normalised Chebyshev basis functions, $\varphi_0 = T_0/\sqrt{N}$, $\varphi_k = T_k/\sqrt{N/2}$; $\lvert\tau(x)\rangle$ holds their values at $x$. | § 4.1 |
| $\psi$, $\eta$, $\eta_e$ | Normalised coefficient state; its scale; the ideal $\eta_e = \lVert b \rVert^2$. | § 3.1 |
| $\psi_e$ | The ideal state $b/\lVert b \rVert$ of a truncated solution $f_N$ with coefficients $b$; the model represents $f_N$ exactly at $(\psi_e, \eta_e)$. | § 4.1 |
| $\hat{f}^{(j)}$ | Coefficients of $f^{(j)}$, the $j^{\text{th}}$ derivative of $f$: one block of the descriptor state. | § 5.2 |
| $w$, $E$ | The descriptor state $(\hat{f}^{(0)}; \ldots; \hat{f}^{(k)})$; the map from a field's coefficients to the descriptor state that represents it. | § 5.2 |

**Operators and constraints**

| Symbol | Meaning | Introduced |
|---|---|---|
| $G$ | Differentiation matrix, $f \mapsto f'$, in coefficient space: dense, upper triangular, $\lVert G \rVert_2 = \Theta(N^2)$ (Wu et al. print it as $\mathbb{G}_n^\top$). | § 3.2 |
| $\tilde{B}$, $\tilde{D}$ | Banded factors of the differentiation matrix, $G = \tilde{B}^{-1}\tilde{D}$. | § 5.1 |
| $\mathcal{M}_{x^p}$, $\mathcal{M}_a$ | Multiplication by $x^p$ (the exact lift, $2N \times N$) and by a smooth $a(x)$. | §§ 3.2 and 4.2 |
| $\mathcal{N}_1$, $\mathcal{N}_x$ | Product-to-sum folds. | § 3.4 |
| $\mathbb{B}(x)$, $\mathbb{D}^{(0)}(x_s)$ | Rank-one zero (invariant) and collapse (regular) constraint rows. | § 3.2 |
| $x_z$, $x_m$, $x_s$ | Constraint points: a zero, an extremum, the datum point ($f(x_s) \neq 0$). | § 3.2 |
| $R_0$, $U_{\text{odd}}$, $\Lambda$ | Factors of the structured exact encoding of $G$. | § 4.5 |
| $\mathbb{S}_a(x^*)$, $\hat{g}$ | Datum-slice row on axis $a$ at $x^*$; the coefficients of the datum function. | § 4.3 |
| $U_{\text{CG}}$ | The Chebyshev–Gauss transform used by the nodal fold. | § 6.6 |
| $\lVert G \rVert_S$ | Wu et al.'s norm $\max\big(\lVert GG^\top \rVert_1, \lVert G^\top G \rVert_1\big)$. | § 3.3 |

**Residuals and Hamiltonians**

| Symbol | Meaning | Introduced |
|---|---|---|
| $A_{\text{std}}$, $A_{\text{desc}}$ | Equation block of each form's stacked residual; the descriptor's includes its recurrence rows. | § 4.2 and definition 3 |
| $H_{\text{std}}$, $H_{\text{desc}}$ | Effective Hamiltonian of each form; bare $H$ only in regime-independent statements. | § 4.4 |
| $R = [R_1; \ldots; R_L]$ | Stacked residual: the equation block over the constraint rows, with $H = R^\dagger R$; the thesis works with $R$. | § 4.4 |
| $H_\otimes$ | The doubled-space Hamiltonian. | § 6.4 |

**Block encoding, filtering and state preparation**

| Symbol | Meaning | Introduced |
|---|---|---|
| $\alpha$, $\alpha_R$, $\alpha_H = \alpha_R^2$ | Subnormalisation: generic, of $R$, of $H$. | § 2.4 |
| $V$, $W$ | Stacked-residual reflection; qubitised walk. | § 7.3 |
| $H' = 2H/\alpha_R^2 - I$ | The normalised Hamiltonian that $V$ block-encodes. | § 7.3 |
| $\sigma_0$, $\sigma_1$, $\sigma_{\max}$ | Smallest, second-smallest and largest singular values of $R$; $\lambda_j = \sigma_j^2$. | § 7.2 |
| $\Delta$ | Spectral gap $\lambda_1 - \lambda_0 = \sigma_1^2 - \sigma_0^2$. | § 7.2 |
| $x_0$, $x_1$; $h_j$ | Ground and first excited values of $H'$; their heights $1 + x_j = 2\sigma_j^2/\alpha_R^2$. | § 7.4 |
| $w(\cdot)$, $\theta_0$ | The minimax filter's edge map of $[x_1, 1]$ onto $[-1, 1]$; its rate $\operatorname{arccosh}(-w(x_0))$. | § 7.4 |
| $\beta$ | Imaginary time. | § 7.4 |
| $d$, $q_k$, $p_G$ | GQSP degree; filter coefficients; success probability. | § 7.4 |
| $\varepsilon$ | Target: the excited spectrum's suppression relative to the ground state, and the infidelity bar ($10^{-6}$). | § 7.4 |
| $\psi_{\text{init}}$, $\gamma$ | Initial state; its overlap $\lvert \langle \psi_{\text{ref}} \vert \psi_{\text{init}} \rangle \rvert$ with the reference. | § 7.4 |
| $\zeta$, $S$ | Block-rescaling constant; the diagonal of block weights, so that the rescaled residual is $R_{\text{desc}}S$. | § 5.4 |
| $w_f$ | Weight of the field block in the descriptor's ground state; $\gamma^2 \leq w_f$. | § 5.3 |
| $\alpha_A$, $Q_m$ | Subnormalisation of the descriptor's system rows; the reflection $I - 2\lvert m \rangle \langle m \rvert$. | § 5.5 |
| $P_t$, $D_t$ | A descriptor term's block word on the block register; its diagonal (the identity, a $Z_j$ or a $Q_m$). | § 5.5 |
| $\epsilon_G$, $\mu$ | Entry error and per-bit coefficient of Wu et al.'s encoding of $G$. | Appendix I.6.2 |

**Nonlinearity**

| Symbol | Meaning | Introduced |
|---|---|---|
| $K$ | Carleman truncation order. | § 6.5 |
| $M$ | Dimension of the base state $u$. | § 6.5 |
| $\rho$ | Nonlinearity-to-dissipation ratio; $\rho < 1$ is sufficient, not necessary, for the Carleman truncation error to fall as $\rho^K$. | § 6.5 |
| $F_1$, $F_2$, $\Gamma_K$ | Linear and quadratic terms; the lifted Carleman generator. | § 6.5 |

**Systems and complex coefficients**

| Symbol | Meaning | Introduced |
|---|---|---|
| $C$ | Number of components (fields) of a system. | § 6.1 |
| $z$ | A system's state: one block per component. | § 6.1 |
| $\Gamma$ | A system's coupling matrix, $u_i' = \sum_j \Gamma_{ij} u_j$. | § 6.1 |
| $\mathbb{F}$ | Ratio row: fixes the direction of a vector datum and leaves its scale free. | § 6.2 |
| $y$, $y_a$, $y_\perp$ | The vector datum; its anchor component; its off-anchor part. | § 6.2 |
| $w_0$ | A system's stated initial data; it weights the initial state. | § 6.2 |
| $c = \operatorname{Re} c + i \operatorname{Im} c$ | A complex coefficient, in the realification example. | § 6.3 |

# Acronyms

**DCT** discrete cosine transform.

**DNS** direct numerical simulation.

**FOSLS** first-order system least squares.

**GQSP** generalised quantum signal processing.

**HHL** Harrow–Hassidim–Lloyd.

**IR** intermediate representation.

**IVP** initial-value problem.

**LCU** linear combination of unitaries.

**MCX** multi-controlled $X$.

**NISQ** noisy intermediate-scale quantum.

**NLFT** nonlinear Fourier transform.

**ODE** ordinary differential equation.

**PDE** partial differential equation.

**PIHM** physics-informed effective Hamiltonian method.

**PINN** physics-informed neural network.

**QFT** quantum Fourier transform.

**QITE** quantum imaginary-time evolution.

**QSVT** quantum singular value transformation.

**SVD** singular value decomposition.

# 1 Introduction

Differential equations govern how dynamic systems evolve in broad fields of physics and engineering. For a classical numerical solver, the cost grows with the resolution: $N$ unknowns along $D$ axes makes $N^D$ unknowns in all. A register of $n$ qubits holds a vector of $N = 2^n$ amplitudes, and this exponential compression is the reason quantum algorithms for differential equations are proposed [2–4]. Such algorithms return a quantum state, from which we measure only a few properties of the solution [5]. Cost is primarily determined by asymptotic scaling in circuit depth, gate complexity, qubit count, and/or shots.

Each established route has its own limitation. For quantum linear-system algorithms applied to a discretised equation [2, 3], the cost depends on the condition number of the discretisation, which grows with the resolution. Hamiltonian simulation and its extensions to dissipative dynamics [6, 7] solve only linear systems, and the generator's norm times the evolution time also grows with the resolution. Linearisation such as Carleman's has a truncation that is controlled only where dissipation dominates [4]. Variational circuits trained on the equation's residual [8, 9] can solve nonlinear equations, but their training can stall on a barren plateau, where the cost landscape is flat and resolving a descent direction takes exponentially many measurements [10]. The remaining route prepares the solution as the ground state of a Hamiltonian built from the equation, with a ground-state filter and nothing to train [11]. This thesis takes up that route, following the physics-informed effective Hamiltonian method (PIHM) of Wu, Paine, Philip, Gentile and Kyriienko [1].

## 1.1 Contributions and Scope

The principal contribution is a reformulation of the residual which the Hamiltonian is built from, which we call the descriptor form. Chebyshev differentiation is dense, but it factorises into two banded matrices which are (up to a factor of two) the conversion and differentiation operators of Olver and Townsend's ultraspherical spectral method [12]. We were motivated by the query count, which is set by the Hamiltonian's norm against its gap, and by the observation that forming the dense product inflates the norm. The descriptor form therefore uses the two factors, and carries intermediate derivatives as unknowns, the idea of first-order system least squares (FOSLS) [13]. The reformulation does not change the kernel of the Hamiltonian, so the state prepared is the same solution, but its Hamiltonian is banded and its norm is smaller.

We measure the reformulation against a reconstruction of PIHM. We first restate their construction as the standard form, and then reconstruct their differentiation operator $G$, the Gram structure, and application to nonlinear problems. The paper provides an approximate encoding of $G$, but we present an exact structured encoding whose cost is constant for constant coefficients. The encoding serves the comparison and is not offered as a competing solver. One classically verified preparation pipeline serves both forms, so that a difference measured between them is a difference between their residuals. The pipeline replaces the imaginary-time filter of Wu et al. with a minimax edge filter, chosen by a direct comparison of the two, which needs fewer filter degrees at equal suppression of the excited spectrum.

Both forms are extended to coupled first-order systems, whose fields share one state, and to complex-valued linear equations, by realification into coupled real ones. Nonlinear equations were the harder test, since Wu et al.'s doubled space encodes them exactly but only projects onto a degenerate solution space. We use Carleman linearisation instead, which gives a linear problem with a unique ground state, and carry it to two-dimensional incompressible Navier–Stokes. A unique ground state certifies the lifted linear problem and not the lift's convergence to the nonlinear solution, so we measure the two separately.

Neither formulation has a cost that is polylogarithmic in the resolution, and we claim no advantage over classical solvers. Every result in the thesis is a classical computation, and nothing ran on

quantum hardware. The evidence comes from three methods, named wherever a number is quoted: circuit simulation, exact emulation, which is provably equal to the circuit's post-selected output, and resource estimates beyond them. Readout is a research field of its own and lies outside the scope of this work. Therefore we decode the field from the prepared state classically.

The work departs from its research proposal, which intended an extension of the PIHM to complex-valued nonlinear equations, including the nonlinear Schrödinger and complex Ginzburg–Landau equations, through a generalised quantum singular value transformation (QSVT). Reproducing Wu et al.'s examples showed that the growth of the query count, which their Discussion leaves open, is set by the dense differentiation matrix. Controlling it matters more than extending the method, since any extension would inherit the cost. The thesis therefore addresses the growth of the query count first. The generalised QSVT also proved unnecessary, since it offers no advantage over generalised quantum signal processing (GQSP) when the operator to be filtered is Hermitian, which our Hamiltonian is. GQSP was therefore enough, and by removing the parity constraint it replaces the mixed-parity linear combination of unitaries (LCU) that Wu et al. use for their filter.

### 1.2 Outline

In § 2, we first examine the current landscape of quantum algorithms for differential equations, and then detail classical spectral methods we rely on further in. We present PIHM in § 3, assess it as prior art and articulate three specific questions later chapters answer. § 4 states our construction of the standard form, which we reformulate in § 5. Both are extended into coupled systems, complex-valued fields and nonlinear equations in § 6. The design of the numerical experiment and its results are given in §§ 7 and 8. We tie these together in the discussion, § 9 and present conclusions and future work in § 10.

The main body is accompanied by a rather extensive appendix, which holds the identities, proofs and more detailed constructions behind each chapter, as well as a comprehensive set of results from 3155 computation runs.

## 2 Background and Literature Review

Every quantum algorithm for a differential equation pays a ratio that its discretisation fixes before any circuit is built: the condition number of a linear system, the norm of a generator times the evolution time, the ratio of nonlinearity to dissipation in a lift, or the norm of an encoded Hamiltonian against its gap. An accurate discretisation can still make that ratio large, and it typically grows with the resolution. The review follows the ratio through three bodies of work: the quantum routes to differential equations, judged by what each assumes, which ratio it pays, what it returns and where it fails; the classical methods the thesis adapts, the spectral discretisations on which the physics-informed construction rests, for which numerical analysis has already shown how the choice of basis and of unknowns sets the ratio and how it can be lowered, and Carleman's linearisation of a nonlinear equation; and the quantum primitives that turn a ratio into a count of queries, the filter degree of Equation (2.2). The PIHM itself is presented and judged separately, in § 3.

### 2.1 Quantum Computation Preliminaries

We use the circuit model of Nielsen and Chuang [14]. The state of $n$ qubits is a unit vector in $\mathbb{C}^N$, $N = 2^n$, and a vector $v \in \mathbb{C}^N$ is *amplitude-encoded* as $|v\rangle = v/\|v\|$. The encoding stores only the direction of $v$, so its length must be carried separately (for a differential equation, the scale $\eta$ of § 3.1), and it gives access to $v$ only through measurement, which returns the index $j$ with probability $|v_j|^2/\|v\|^2$; recovering all $N$ entries takes at least of order $N$ repetitions (§ 2.4). Circuits of unitary gates act on these states, and a non-unitary operation is applied by acting on the system together

**Table 2.1: Each quantum route's cost is set by a ratio its discretisation fixes.** Rows: the routes of § 2.2; columns: assumptions, cost driver, output, failure mode. $A$, the generator of a linear system; $\kappa$, its condition number; $T$, evolution time; $x(t)$, the solution at time $t$; $\alpha_H$, $\Delta$, subnormalisation and spectral gap of a block-encoded Hamiltonian $H$.

| Route | Assumes | Cost depends on | Returns | Fails where |
|---|---|---|---|---|
| Linear systems [3, 16–19] | linear equation in first-order form; efficient access to $A$ and loading of the data | $\kappa$ of the discretised system, which grows with $T$ | the solution, or its history, as a state | $\kappa$ grows with the resolution; readout; at most polynomial gain at fixed dimension |
| Hamiltonian simulation; linear combination of Hamiltonian simulations; Schrödingerisation [6, 7, 20] | linear dynamics; for the linear combination, a positive semi-definite dissipative part | $\|A\|T$; the decay $\|x(0)\|/\|x(T)\|$ | the state at time $T$ | nonlinear dynamics; $\|A\|$ grows with the resolution; readout |
| Variational [8, 9, 21, 22] | a trainable parametrised circuit | optimisation steps times samples, unbounded | a trained circuit | no guarantee of convergence; barren plateaus |
| Ground-state preparation [11, 18, 23, 24] | a positive semi-definite $H$ whose null vector is the solution; a known gap; an initial state of non-vanishing overlap | $\alpha_H/\Delta$, or its square root for a filter at the spectral edge; for a linear system, $\Delta$ of order $1/\kappa$ | the solution as a ground state | the gap closes as the problem's conditioning grows; readout |

with *ancilla* qubits prepared in $|0\rangle$ and keeping only the outcomes in which they are measured in $|0\rangle$ again. This *post-selection* succeeds with a probability that enters the cost; every non-unitary matrix of the thesis, from a residual to a filter, acts on a state this way (§ 2.4).

An algorithm that accesses a matrix only through a circuit encoding it is costed in *queries*, the uses of that circuit, and its gate count is the number of queries times the gates per query, on a register of system and ancilla qubits. The asymptotic results reviewed below, and the filter degree of Equation (2.2), count queries; the gates per query belong to the encoding circuit. § 4 separates the two, and § 7.7 fixes how gates are counted. A tilde on $\mathcal{O}$, $\Omega$ or $\Theta$ suppresses factors logarithmic in $N$ and in $1/\varepsilon$, and a cost polynomial in $n = \log_2 N$ is called poly-logarithmic.

Every cost in the thesis is that of a fault-tolerant machine, whose logical qubits and gates are protected by error correction. Current noisy intermediate-scale quantum (NISQ) processors have no error correction, and noise limits the depth of a reliable circuit [15]; the variational algorithms of § 2.2 target them. A filter applies its block encoding as many times in sequence as its degree, which grows with the resolution, so its feasibility is judged against fault-tolerant resource estimates (§ 9.4) and NISQ hardware is out of scope.

## 2.2 Quantum Algorithms for Differential Equations

Aaronson's caveats on the linear-systems algorithm apply to every route below: the input must load efficiently, the operator must be well conditioned and efficiently accessible, and the output is a quantum state, useful only where a few properties of the solution suffice [5]. Table 2.1 compares the routes under these headings.

**Linear systems.** The first route discretises the equation, in time and where needed in space, into one linear system whose solution holds the trajectory, and solves that system with a quantum linear-systems algorithm. The Harrow–Hassidim–Lloyd (HHL) algorithm prepares a state proportional to $A^{-1}b$ for an $s$-sparse $N \times N$ matrix of condition number $\kappa$ in time $\tilde{\mathcal{O}}(s^2\kappa^2 \log N/\varepsilon)$, and shows that

time $\kappa^{1-\delta}$ for any $\delta > 0$ would imply BQP = PSPACE [16]. Discretising a differential equation into such a system gave the first solvers: Berry with a linear multistep method, at a cost polynomial in $1/\varepsilon$ [2]; Berry, Childs, Ostrander and Wang with a truncated Taylor series, at a cost polynomial in $\log(1/\varepsilon)$ [17]; and Childs and Liu with Chebyshev collocation in time, whose collocation matrix is dense and whose condition number they bound by a polynomial of degree 3.5 in the number of nodes [3]. Childs, Liu and Ostrander extended finite-difference and spectral discretisations to partial differential equations (PDEs) at the same dependence on precision [25], Costa et al. brought the linear-systems solver to the optimal $\mathcal{O}(\kappa \log(1/\varepsilon))$ with a discrete adiabatic theorem [18], and Jennings et al. gave the first non-asymptotic costs, constant factors included, preparing the history state of any stable linear system with complexity $\mathcal{O}(T^{1/2})$ in the evolution time, a fast-forwarding that classical stability permits [19]. The ordinary-equation solvers take the equation as a first-order system $x' = Ax + b$, reducing any higher order before discretising, so the conditioning they pay is that of a first-order operator. The route's strength is its generality at a proven cost: any linear equation whose discretised operator can be accessed efficiently is solved at a cost logarithmic in $N$ and, from Berry et al. on, in $1/\varepsilon$. Its weakness is that the cost runs through $\kappa$, which the discretisation fixes and which grows with the resolution and with the evolution time, so the discretisation is part of the cost (§ 2.3). Learning all $N$ amplitudes of the output takes at least of order $N$ repetitions, and for equations in a fixed number of dimensions with smooth solutions the advantage over the best classical methods is at most polynomial [26].

**Hamiltonian simulation and Schrödingerisation.** A linear equation $x' = -\mathrm{i}Hx$ with Hermitian $H$ is a Schrödinger equation, and integrating it to time $T$ is Hamiltonian simulation, which qubitisation performs with $\mathcal{O}(\alpha T + \log(1/\varepsilon))$ queries to a block encoding of $H$ at subnormalisation $\alpha$, an optimal dependence on both [20, 27]. Two constructions carry non-unitary linear dynamics $x' = -Ax$ to this primitive. The linear combination of Hamiltonian simulations writes the propagator of a dissipative system as a weighted integral of unitary evolutions and implements it as a LCU, without dilating the problem into a larger linear system and at optimal state-preparation cost [6]. Schrödingerisation maps a linear ordinary or partial differential equation to a Schrödinger equation in one more dimension by a warped phase transformation, and recovers the solution from the auxiliary variable [7]. Neither solves a linear system, so neither pays a condition number: the cost runs through $\|A\|T$, the norm of the generator times the evolution time. The route is linear, and its ratio also grows with the resolution, since the norm of a discretised spatial operator grows with the number of unknowns, as the square of the resolution for a second derivative. It returns the state at the final time only, and recovering it from a dissipative evolution needs a number of repetitions that grows with the decay $\|x(0)\|/\|x(T)\|$ of the solution's norm. The linear combination, as stated, also requires the dissipative part of $A$, its Hermitian part, to be positive semi-definite, which restricts it to stable dynamics.

**Variational algorithms.** A variational algorithm represents the solution by a parametrised circuit and minimises a cost built from the equation with a classical optimiser. Lubasch et al. evaluate a nonlinear cost on several copies of the state, as demonstrated on the nonlinear Schrödinger equation [8]. The differentiable quantum circuits of Kyriienko, Paine and Elfving represent the solution as the expectation value of a circuit with input $x$, differentiate it with respect to $x$ by parameter shifts on the feature map, and train it on the residual of the equation and its data [9], the loss of a physics-informed neural network (PINN) [28]. The variational linear solver finds a ground state the same way, minimising $\langle x|A^\dagger(I - |b\rangle\langle b|)A|x\rangle$, whose zero-energy ground state solves $Ax \propto b$ [21]; encoding time in a register, as the Feynman–Kitaev clock does, puts a whole trajectory in one ground state [29], and a space–time encoding of this kind has given the evolution of Burgers' equation from a single ground-state computation [22]. The route's strengths are its circuits, shallow enough for NISQ devices, and its reach: a nonlinear term enters the cost directly, with no linearisation. Its weakness is that nothing bounds the number of optimisation steps or guarantees the optimum. The gradients of

a sufficiently random parametrised circuit vanish exponentially in the number of qubits, the barren plateau [30], and each evaluation of the cost is a sampled expectation value, so the cost of a solution is unknown before it is found.

**Ground-state preparation.** The last route builds a positive semi-definite Hamiltonian whose null vector is the solution and prepares that vector without optimisation. Subaşı, Somma and Orsucci do so for a linear system by randomised adiabatic evolution at cost $\mathcal{O}(\kappa \log \kappa/\varepsilon)$ [23], An and Lin by a time-optimal adiabatic schedule [24], and Costa et al. by a discrete adiabatic walk [18]. In Lin and Tong's form [11], for positive-definite $A$ with $\|A\| = 1$ and $Q_b = I - |b\rangle \langle b|$, the Hamiltonian with off-diagonal blocks $AQ_b$ and $Q_bA$ has $|0\rangle |x\rangle$ in its null space beside $|1\rangle |b\rangle$, which a measurement of the first qubit removes; the rest of its spectrum lies at a distance of order $1/\kappa$ from zero, and an optimal Chebyshev filter projects onto the null space. The route's strength is a proven cost with nothing to train, in a form that separates the circuit from the spectrum: a filter's degree depends on the Hamiltonian only through the subnormalisation of its block encoding and its gap, and its error falls exponentially with the degree (§ 2.4). Its weakness is that the gap carries the problem's conditioning, of order $1/\kappa$ for a linear system, so the route pays the ratio of the linear-systems route in another form. A filter also needs the gap, or a bound on it, in advance, and an initial state whose overlap with the solution does not vanish. The physics-informed effective Hamiltonian of Wu et al. belongs to this route: it builds the Hamiltonian from a Chebyshev discretisation of the equation and its data and prepares the ground state with a filter (§ 3).

**Critical assessment.** Each route in Table 2.1 pays a ratio set by how the equation is discretised and encoded, which an accurate discretisation can still make large: the condition number of a linear system, the norm of a generator times the evolution time, or the norm of an encoded Hamiltonian against its gap. Every route returns a quantum state, so its advantage holds only where a few properties of the solution suffice [5, 31], and is at most polynomial in a fixed, low number of dimensions [26]. The thesis therefore compares quantum methods with one another, at equal accuracy, in queries and gates, and claims no advantage over classical solvers. The guarantees differ: the linear-systems, Hamiltonian-simulation and ground-state routes have proven costs, and the variational route has none. Only the variational route takes a nonlinear equation as it stands; the others are linear and reach a nonlinear equation through a linearisation such as Carleman's, which adds a ratio of its own (§ 2.3).

## 2.3 Classical Discretisation and Linearisation

The ratios of Table 2.1 are set when the equation is discretised. For the Chebyshev discretisation on which the physics-informed construction is built, classical numerical analysis locates the source of the ratio, the density of differentiation in coefficient space, and two ways of lowering it: a basis in which differentiation is banded, and derivatives carried as unknowns. A nonlinear equation reaches the linear routes only after a second classical step, Carleman's linearisation, which brings a ratio of its own.

**Chebyshev discretisation and differentiation.** A function analytic on $[-1, 1]$ has a Chebyshev series $f = \sum_k b_k T_k$ with geometrically decaying coefficients, so the $N$-term truncation converges faster than any power of $N$ [32]; the physics-informed construction, like the spectral methods it draws on, works with these coefficients for that reason. A differential equation becomes a finite system by Galerkin projection onto the basis, by the tau method, which projects but replaces the highest-degree rows by the boundary conditions, or by collocation at a set of nodes [33, 34]. Wu et al.'s construction is closest to the tau method, with the equation acting on the coefficients and the data imposed as rows at points, but its system is overdetermined and solved by least squares (§ 3.2).

Differentiation in coefficient space is exact on polynomials of degree below $N$ and dense, because the derivative of $T_p$ contains every $T_k$ of the opposite parity below it: the matrix is upper triangular,

about a quarter of its entries are non-zero, and its norm grows as $N^2$ (§ 4.2). An operator of order $k$ built from its powers grows as $N^{2k}$, and so, typically, does the condition number of a tau or collocation discretisation [12, 33].

**The ultraspherical method.** Olver and Townsend make differentiation banded by letting the derivative live in another basis [12]. Since $T_k' = k\,C_{k-1}^{(1)}$, with $C_k^{(1)} = U_k$ the Chebyshev polynomials of the second kind, differentiation from Chebyshev to $C^{(1)}$ coefficients is a single superdiagonal, and conversion of a series' own coefficients to $C^{(1)}$ has two non-zero diagonals. Raising the basis one ultraspherical (Gegenbauer) level $C^{(\lambda)}$ per derivative gives, for an equation of order $k$ with polynomial coefficients, a leading coefficient that does not vanish and $k$ boundary conditions, an operator banded apart from its boundary rows, whose condition number grows linearly with the truncation and is bounded after a diagonal preconditioner, in the 2-norm for Dirichlet conditions [12]. The dense differentiation matrix we use factorises as $G = \tilde{B}^{-1}\tilde{D}$, with $\tilde{B}$ and $\tilde{D}$ the method's first conversion and differentiation operators up to a factor of two (Appendix A.5); the descriptor form of § 5 keeps the two factors, and its block rescaling takes the place of the preconditioner.

Olver and Townsend eliminate the lower derivatives and solve one banded system for the solution's coefficients, whereas the descriptor keeps every derivative as an unknown in the Chebyshev basis, tied to the next by the same pair $(\tilde{B}, \tilde{D})$ (§ 2.3). The full reduction, banded on $N$ unknowns without a block register, is not compared here, although its conditioning suggests, untested, that it would match the descriptor's degree on a regular problem (§ 9.5).

**First-order system least squares.** The normal equations of a least-squares method square the residual operator, so high order compounds the conditioning: least squares applied directly to a second-order elliptic equation needs continuously differentiable finite elements and gives condition numbers of order $h^{-4}$ in the mesh width $h$, against $h^{-2}$ for a Galerkin method [35]. FOSLS introduces derivatives as unknowns, such as a flux $v = \nabla\phi$, and minimises the residuals of the resulting first-order system [13]. Where the functional is equivalent to a product of $H^1$ norms, the condition numbers return to order $h^{-2}$ and the unknowns nearly decouple, so block-diagonal preconditioners are effective [35, 36].

The physics-informed Hamiltonian is such a functional. Stacking the operators of the equation and of its data into one residual operator $R$ gives $H = R^\dagger R$, whose energy $\|R\psi\|^2$ is the squared residual of the equation and its data (§§ 3.2 and 4.4). Eliminating every derivative through $G$, as Wu et al. do, is the high-order least-squares method; carrying the derivatives as unknowns is FOSLS in a spectral basis. The quantum ordinary-equation solvers of § 2.2 already work with first-order systems, and the physics-informed construction departs from them in eliminating the derivatives into one dense operator. A filter applied through a block encoding of $R$ pays the conditioning of $R$ (§ 2.4); the squaring falls only on an eigensolver acting on $H$. A first-order formulation still lowers what the filter pays, because each block of $R$ then carries one derivative instead of $k$ (§ 4.6).

**What we add.** Both ideas are classical. The increment is their transfer to a block-encoded physics-informed Hamiltonian and its consequences for preparing the ground state: a proof that the reformulated residual has the standard form's kernel, so that the prepared state is the same solution (§ 5.2); an encoding of the banded residual as one flat linear combination of unitaries with $\mathcal{O}(\log n)$ ancilla qubits (§ 5.5); the finding that the carried derivatives erode the Hamiltonian's gap, and a block rescaling that restores it on one axis without ancilla and not on two (§§ 5.3 and 5.4); and a measurement of what the reformulation does to the filter's degree, as built and at an ideal encoding (§§ 8.2 and 8.3).

**Fourier–Galerkin truncation.** On a periodic domain the Fourier basis diagonalises differentiation, $\partial_x e^{\mathrm{i}kx} = \mathrm{i}k\,e^{\mathrm{i}kx}$, so linear terms act mode by mode and a quadratic term becomes a convolution of coefficients [33]. A Galerkin truncation keeps exactly the convolution terms whose output lies in the retained set of modes; a pseudo-spectral evaluation on a grid aliases the discarded terms

onto retained modes unless it is dealiased [33]. The fluids of § 6.7 use Galerkin truncations on real Fourier modes, so their quadratic coupling is a fixed matrix of exact coefficients, as the Carleman lift requires; a pseudo-spectral simulation serves only as the reference for the truncation's error.

**Carleman linearisation.** Carleman's embedding replaces a polynomial system of ordinary differential equations by an infinite linear system for the monomials of its unknowns [37, 38]. For $u' = F_1 u + F_2 u^{\otimes 2}$ the vector $(u, u^{\otimes 2}, u^{\otimes 3}, \dots)$ obeys a linear equation in which each level couples only to itself and to the next, and truncation at order $K$ leaves a finite linear system whose first level approximates $u$. Its use in quantum algorithms followed an earlier approach of Leyton and Osborne, which transforms the amplitudes of many copies of the state nonlinearly, at resources exponential in the integration time [39]. Liu et al. solve the truncated Carleman system with a quantum linear-systems algorithm for dissipative quadratic equations in $M$ variables, at cost $T^2 q\,\mathrm{poly}(\log T, \log M, \log 1/\varepsilon)/\varepsilon$, with $q$ the decay of the solution, when the ratio $R$ of nonlinearity and forcing to dissipation satisfies $R < 1$, and prove the problem intractable in the worst case for $R \geq \sqrt{2}$ [4]; Krovi makes the dependence on precision logarithmic and relaxes the normality of the linear part to a negative log-norm [40].

The lift is linear and exact before truncation, so a nonlinear equation inherits the guarantees of whichever linear route solves the truncated system, and the dimension $M^K$ of the lift fits in about $K \log_2 M$ qubits. Its weakness is the truncation. The condition $R < 1$ bounds the truncation error uniformly over states, so it is sufficient and not necessary: a lift may converge for $R > 1$, which § 6.5 (writing the ratio $\rho$) measures and § 8.4 reports. Dissipation must still dominate, since the truncation is controlled only where the nonlinearity is weak against the damping, and the lift's dimension, of order $M^K$, enters the cost of every route that solves it.

## 2.4 Block Encodings, Signal Processing and Filtering

A block encoding applies a non-unitary matrix on a quantum computer, a polynomial transformation of the block encoding builds a filter, and the filter's degree counts the queries that a prepared ground state costs.

**Block encodings and linear combinations of unitaries.**

**Definition 1 (Block encoding [41]):** A unitary $U$ on $m + n$ qubits is an $(\alpha, m, \epsilon)$ *block encoding* of an $N \times N$ matrix $A$ if

$$\left\| A - \alpha \left( \langle 0^m | \otimes I \right) U \left( |0^m\rangle \otimes I \right) \right\| \leq \epsilon; \tag{2.1}$$

$\alpha$ is its *subnormalisation*, and the encoding is *exact* when $\epsilon = 0$.

Applied to $|0^m\rangle\,|\psi\rangle$ and post-selected on the ancillas' zero, an exact block encoding applies $A/\alpha$ with success probability $\|A\psi\|^2/\alpha^2$. The block of a unitary has norm at most one, so $\alpha \geq \|A\|$ for every exact block encoding, and the excess $\alpha/\|A\|$ multiplies the cost of every algorithm that uses it. The standard construction is the LCU [42]: for $A = \sum_{t<L} c_t U_t$ with unitaries $U_t$, PREPARE loads $\sum_t \sqrt{|c_t|/\alpha}\;|t\rangle$ on $\lceil \log_2 L \rceil$ ancilla qubits, SELECT applies $U_t$ controlled on $|t\rangle$, with the phases of the $c_t$ absorbed into the $U_t$, and PREPARE$^\dagger$ returns the ancillas, giving an exact block encoding at $\alpha = \sum_t |c_t|$. An exact combination whose SELECT indexes $L$ distinct operations needs $\lceil \log_2 L \rceil$ selector qubits [43]. Explicit circuits exist for many structured and sparse matrices [44, 45]; each carries its own $\alpha$, and the two forms of the residual compared in § 8 differ in both $\|A\|$ and $\alpha/\|A\|$.

If $U$ is Hermitian and block-encodes $H/\alpha$, the walk $W = (2\Pi - I)\,U$, with $\Pi = |0^m\rangle\langle 0^m| \otimes I$, satisfies $\Pi W^k \Pi = T_k(H/\alpha)$: its $k$-th power block-encodes the $k$-th Chebyshev polynomial of the encoded matrix at one query per power [20], so a polynomial of degree $d$ in $H/\alpha$ costs $d$ queries once its combination is formed.

**Quantum signal processing.** Quantum signal processing interleaves a single-qubit signal rotation with fixed phase rotations to realise a polynomial of the signal [27]. The QSVT lifts the construction to block encodings, applying a degree-$d$ polynomial $P$ to the singular values of the encoded matrix with $d$ queries and one further ancilla [41], and places search, Hamiltonian simulation and linear systems in one framework [46]. Its polynomials are bounded by one on $[-1, 1]$ and of definite parity, so a function of indefinite parity, such as the imaginary-time filter $e^{-\beta x}$, is split into even and odd parts, each with its own phase sequence, and the two are summed by a further linear combination at a cost in subnormalisation; Wu et al. use this mixed-parity construction (§ 3.3). GQSP drops the parity constraint [47]. SU(2) rotations on one signal qubit, with the encoded unitary applied controlled on it, realise any $P(z) = \sum_{k\le d} q_k z^k$ with $|P| \le 1$ on the unit circle at $d$ queries, and applied to the walk $W$ they give $\sum_k q_k T_k$ of the encoded matrix (§ 7.4). The phases follow from a complementary polynomial, computed by a fast-Fourier-transform formula with explicit error guarantees [48], through the inverse of the nonlinear Fourier transform (NLFT) to which GQSP is equivalent [49, 50].

**Ground-state preparation by filtering.** A filter is a function of $H$ that keeps the ground eigencomponent of an initial state $\psi_{\text{init}}$ and suppresses the rest. Imaginary-time evolution $e^{-\beta H}$ does so once $\beta$ exceeds the inverse gap [51], and inverse iteration $H^{-k}$ by powers [52]; expanded in polynomials, both are implementable by the transformations above. For an eigenvalue at zero with the rest of the spectrum in $[-1, -\Delta]\cup[\Delta, 1]$, Lin and Tong's eigenstate filter, a Chebyshev polynomial in $x^2$, has the least maximum on that set among polynomials of its degree equal to one at zero, and needs degree $\mathcal{O}(\Delta^{-1}\log(1/\varepsilon))$ [11]. Their ground-state algorithm approximates a shifted sign function, given a bound that separates the ground energy from the rest of the spectrum, with $\mathcal{O}\big((\alpha/(\gamma\Delta))\log(1/\varepsilon)\big)$ queries for subnormalisation $\alpha$, gap $\Delta$ and overlap $\gamma = |\langle\psi_0|\psi_{\text{init}}\rangle|$ with the ground state; they argue, by reduction from unstructured search, that its dependence on $\gamma$ and $\Delta$ is essentially optimal [53]. Both assume the gap, or bounds that imply it, known in advance, a standard assumption that we share (§ 7.4).

Both place the ground eigenvalue inside the encoded spectrum; at its edge the degree falls quadratically. Let $H \succeq 0$ be block-encoded with its spectrum mapped into $[-1, 1]$ and the ground value at the lower edge, as by $2H/\alpha_H - I$, with gap $\Delta = \lambda_1 - \lambda_0$ and a ground value small against $\Delta$. The excited spectrum lies in $[x_1, 1]$, at a distance of order $\Delta/\alpha_H$ from the ground value. Among polynomials of degree $d$ bounded by one on $[-1, 1]$, the Chebyshev polynomial of $[x_1, 1]$ has the least maximum there relative to its value at the ground state, by Chebyshev's classical inequality [54]; it is the edge counterpart of Lin and Tong's filter. Outside its interval it grows as the hyperbolic cosine of the degree times the square root of the distance, so suppressing the excited spectrum to $\varepsilon$ relative to the ground state needs the degree

$$d = \Theta\left(\sqrt{\frac{\alpha_H}{\Delta}}\,\log\frac{1}{\varepsilon}\right), \qquad \alpha_H = \alpha_R^2. \tag{2.2}$$

The square root comes from the ground value's place at the edge of the encoded spectrum, where a Chebyshev polynomial grows as the square root of the distance; for frustration-free Hamiltonians, sums of positive terms that all vanish on the ground state, it is Somma and Boixo's spectral gap amplification [55], and $H = R^\dagger R$ is one exactly when its kernel is exact. A small non-zero ground value leaves the degree unchanged, since the degree is set by the gap, and costs only the square of the filter's value at the ground state, a factor close to one, in the probability of success (§ 7.4, Appendix F.4). Here $\alpha_R$ is the subnormalisation of a block encoding of $R$, from which one reflection block-encodes $H$ at $\alpha_H = \alpha_R^2$ (§ 7.3); the walk's eigenphases are set by the singular values of $R/\alpha_R$, and the gap the filter resolves is of order $\sigma_1/\alpha_R = (\Delta/\alpha_H)^{1/2}$ (§ 7.3).

Each degree is one query, so $d$ counts the queries of one filtered preparation before amplification. The circuit enters only through $\alpha_H \ge \|H\|$ (§ 2.4), so with this filter no encoding brings the degree below order $\sqrt{\|H\|/\Delta}\,\log(1/\varepsilon)$. With $\alpha_H = \alpha_R^2$ the degree is linear in $\alpha_R$; for an exact kernel,

whose gap is the square of the smallest non-zero singular value of $R$, $\sqrt{\alpha_H/\Delta}$ is at least the ratio of the largest singular value of $R$ to that one, the residual's effective condition number. The cost of a filtered ground state is therefore set by the norm of the operator it is built from, measured against its gap. The bound holds for polynomial filters given the edge and the gap, among which Chebyshev's inequality makes the degree least; over all algorithms the thesis claims no more than Lin and Tong's optimality supports (§ 9.2).

Post-selection succeeds with probability of order $\gamma^2$, and amplitude amplification raises it to near one with $\mathcal{O}(1/\gamma)$ repetitions [56], so a preparation costs of order $d/\gamma$ queries; an answer-independent initial state with non-vanishing overlap is part of the method (§ 7.4).

**Readout.** An overlap or expectation value is estimated to additive error $\delta$ by a Hadamard or swap test with $\mathcal{O}(1/\delta^2)$ repetitions, or with $\mathcal{O}(1/\delta)$ by amplitude estimation, including variants without phase estimation [56–58]; a full field needs of order $N$ such estimates, the readout problem of quantum differential-equation algorithms [31, 59]. Wu et al. read the field point by point with an interferometric overlap measurement (§ 3.3). We decode the field classically from the prepared state, which our classical computations make available (§ 7.6), and leave readout unadressed, following the convention of every route of Table 2.1.

## 2.5 Synthesis

The physics-informed effective Hamiltonian sits where the three strands meet. It is a ground-state route, so it inherits the filters' guarantees and their cost law, Equation (2.2): the square root of an encoded Hamiltonian's norm over its gap. It is a least-squares functional on a Chebyshev discretisation, so it inherits spectral accuracy together with the conditioning of dense differentiation, the property that the ultraspherical method and first-order-system least squares each address. And it descends from physics-informed learning but trains nothing. Its authors name the gap's behaviour and the cost of differentiation as questions for future work; in the terms of Equation (2.2) the first is the degree's denominator and, as § 3.5 argues, the second sets its numerator. § 3 presents the method on the same attributed footing as the material above, judges it, and poses those questions in the form we answer.

# 3 The Physics-Informed Effective Hamiltonian Method

This chapter presents the PIHM of Wu, Paine, Philip, Gentile and Kyriienko [1] and evaluates it. The method turns a differential equation into a ground-state problem. The unknown function is expanded in Chebyshev polynomials whose coefficients are the amplitudes of an $n$-qubit state $|\psi\rangle$; the equation and each of its data conditions become matrix operators that annihilate $|\psi\rangle$ when it holds the solution's coefficients; and the Gram matrices of these operators are summed into an effective Hamiltonian $H$ whose ground state is the solution. Preparing that ground state needs a block encoding of $H$, a filter that removes its excited spectrum, and a measurement that reads the field back.

The method is claimed to need no variational training, prepare a continuous analytic function for $x \in [-1, 1]$, and "necessary terms [operator circuits] have $O(\mathrm{poly}(n))$ scaling and can be implemented efficiently" [1]. We first present the method in Wu et al.'s own terms, their model, Hamiltonian, circuits and nonlinear extension, and evaluate it; § 4 then states the construction in the form every later chapter runs.

## 3.1 The Chebyshev Latent-Space Model

On $n$ qubits, with $N = 2^n$, the quantum Chebyshev state or feature map is

$$|\tau(x)\rangle_n = \frac{1}{2^{n/2}} T_0(x) |0\rangle^{\otimes n} + \frac{1}{2^{(n-1)/2}} \sum_{k=1}^{N-1} T_k(x) |k\rangle , \tag{3.1}$$

whose components are the first $N$ Chebyshev polynomials of the first kind. The model is a scaled overlap

$$f_Q(x) = \sqrt{\eta}\, f_q(x), \qquad f_q(x) = \langle \tau(x)|\psi\rangle_n , \qquad \langle\psi|\psi\rangle = 1, \tag{3.2}$$

where the amplitudes of $|\psi\rangle$ are the series' weighted coefficients and $\eta > 0$ is a scaling factor [1, 60]. The $N$ states $|\tau(\bar{x}_j)\rangle_n$ at the Chebyshev nodes $\bar{x}_j = \cos[\pi(j + \frac{1}{2})/N]$ are orthonormal, but elsewhere $|\tau(x)\rangle_n$ is not a unit vector; a circuit prepares its normalised version by post-selection, and Wu et al. discard the norm, "since it can be implicitly incorporated into the scaling factor", so their derivations use the unnormalised state. Every constraint below is linear in $|\psi\rangle$ and fixes only its direction. The scale is recovered after the ground state $|\psi_g\rangle$ has been prepared, from one known value of the solution, the *regular* datum $f(x_s) = y_s$: $\sqrt{\eta} = y_s / \langle \tau(x_s)|\psi_g\rangle_n$.

## 3.2 Hamiltonian Construction

Differentiation acts on the coefficients by a fixed matrix, $f'_q(x) = \langle \tau(x)|G|\psi\rangle_n$, with $G$ the Chebyshev differentiation matrix.[1] $G$ is dense, because the derivative of $T_p$ contains every $T_k$ of the opposite parity below it [12]: one high coefficient of $f$ reaches every lower coefficient of $f'$. Its entries are given in § 4.2. Higher derivatives are its powers, so a homogeneous linear ODE with constant coefficients, $\sum_j a_j f^{(j)} = 0$, is represented by

$$A |\psi\rangle = 0, \qquad A = \sum_j a_j G^j. \tag{3.3}$$

Boundary and initial conditions enter as data constraints built from the rank-one row $\mathbb{B}_n(x) = \sqrt{2^n}\, |0\rangle^{\otimes n} \langle \tau(x)|_n$, which maps the model to the constant function equal to its value at $x$. An *invariant* constraint has a zero target and is unchanged by scaling: $f(x_z) = 0$ gives $\mathbb{B}_n(x_z) |\psi\rangle = 0$, and $f'(x_m) = 0$ gives $\mathbb{B}_n(x_m) G |\psi\rangle = 0$. Wu et al. place these at boundaries where the solution vanishes or at interior zeros and extrema of the known solution, and use one invariant constraint for each second-order ordinary example, with the regular datum fixing the scale.

A variable coefficient enters through the lift of Paine et al. [60]: for each power $p$, a $2N \times N$ matrix $\mathcal{M}_{x^p}$ with $x^p \langle \tau(x)|_n = \langle \tau(x)|_{n+1} \mathcal{M}_{x^p}$, which follows from the Chebyshev product-to-sum identity. It carries $x^p$ onto a register one qubit wider, so that every term of the equation shares the $(n+1)$-qubit basis, and a polynomial coefficient $a_j(x)$ enters as the sum of the lifts of its monomials. A source $r(x)$ is brought into the same form through the regular datum: the collapse row $\mathbb{D}^{(0)}(x_s) = \mathbb{B}_n(x_s)/f(x_s)$ maps the solution's coefficients to the constant $1/\sqrt{\eta}$, so a source expanded in a Maclaurin series truncated at order $\bar{p}$ enters the equation's operator as $-\big(\sum_{p \le \bar{p}} c_p \mathcal{M}_{x^p}\big)\mathbb{D}^{(0)}(x_s)$, with $c_p$ the Maclaurin coefficients of $r$.

A function of several variables is carried on one register per variable, with the product of the feature maps as its basis; Wu et al. treat two. For variables $x$ and $y$ the state is $\langle \tau(x)|_n \otimes \langle \tau(y)|_n$ on two $n$-qubit registers; derivatives act as $G \otimes I$ and $I \otimes G$, and a constraint on a boundary as $\mathbb{B}_n(x_z) \otimes I$. The invariant sets of the three PDE examples are chosen by hand: Laplace's equation uses its three homogeneous sides, so its non-zero side appears in no constraint, and the heat and wave

[1] Wu et al. define the lower-triangular $\mathbb{G}_n$ and apply its transpose, $\mathbb{G}_n^\top$, to the coefficient vector. We write that upper-triangular matrix as $G$ throughout.

examples place interior points at zeros of the known solutions (the sets are listed in Appendix I).

Each operator is turned into a Gram matrix, and the Gram matrices are summed:

$$H = \mathcal{T}(A) + \sum_i \mathcal{T}(\mathbb{B}_i), \qquad \mathcal{T}(X) = X^\top X, \qquad \langle\psi|H|\psi\rangle = \|A\psi\|^2 + \sum_i \|\mathbb{B}_i\psi\|^2, \tag{3.4}$$

where $\mathbb{B}_i$ runs over the invariant constraints. The energy of a state is its total squared violation of the equation and its data. Wu et al. describe $H$ as "a real symmetric positive-definite matrix with distinct eigenvalues" and recast the equation as the eigenvalue problem $H\,|\psi_g\rangle = \lambda_{\min}\,|\psi_g\rangle$ with $\lambda_{\min} \approx 0$. Wu et al.'s worked example at $N = 4$ and the entries of the lifts are given in Appendix B.

## 3.3 Circuits, Filtering and Readout

**Block encodings.** Each term of $H$ is block-encoded separately and combined by an LCU (§ 2.4). The rank-one rows $\mathbb{B}_n(x)$ and $\mathbb{D}^{(0)}(x_s)$ pair the inverse Chebyshev feature map with a reflection about the all-zero state, on $n+2$ qubits at subnormalisation $\sqrt{2^{n+1}}$, and run "with high probability" [1]. $G$ is encoded *approximately* by two conditioned subroutines that load its entries as rotation amplitudes. The circuit's top-left block is $G$ divided by the prefactor

$$\alpha_G^{\mathrm{Wu}} = \left(2^{n-1} + 2^n\right) \|G\|_S, \qquad \|G\|_S = \max\!\left(\|GG^\top\|_1,\ \|G^\top G\|_1\right), \tag{3.5}$$

on $2n+3$ qubits, "with negligible errors in matrix entries that exponentially decrease as $n$ increases", and with $O(n^3)$ gates once the multi-controlled gates are decomposed: "polynomial in $n$ and thus efficient" [1]. The registers, subroutines and angles are given in Appendix I.6.1.

**Ground-state preparation.** The ground state is prepared by quantum imaginary-time evolution (QITE) [51], which multiplies each eigencomponent of an initial state by $e^{-\beta E}$, so that the lowest dominates once $\beta$ exceeds the inverse gap. Wu et al. apply it to $\tilde{H} = H/\|H\|_F$ by an indefinite-parity QSVT sequence [41] whose phases are fitted to $e^{-\beta x}$ (§ 2.4). The time is set by an "approximate lower bound",

$$\beta \;\geq\; \frac{\lambda_{\max}}{\lambda_2\,(n-1)}, \tag{3.6}$$

with $\lambda_{\max}$ and $\lambda_2$ the largest and second-smallest eigenvalues of $\tilde{H}$ (or $H$). Wu et al. expect a preparation time of order $\Delta^{-1}\,\mathrm{polylog}(1/\varepsilon)$ by eigenstate filtering [53], or $\Delta^{-1/2}$ by algorithmic thermalisation [61], and note that the gaps "are problem-dependent and in general are not easy to bound". Their phase sets and the dilation of $\tilde{H}$ are given in Appendix I.7.

**Readout.** Wu et al. read the field out point by point by an interferometric measurement [52, 62], combining the all-zero probabilities of the prepared state and of its superposition with a reference into the signed overlap $\langle\tau(x)|\psi_g\rangle_n$, which $\sqrt{\eta}$ scales to $f_Q(x)$; the identity is given in Equation (F.6).

## 3.4 Doubled-Space Extension to Nonlinearity

A nonlinear term such as $(f')^2$ cannot be expressed as a linear operator on $|\psi\rangle$. Wu et al.'s treatment doubles the space: the unknown becomes $|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle$ on $2n$ qubits, a product of two model functions is the overlap of $|\Psi\rangle$ with a product of two feature states, and the product-to-sum rule folds that product back into one Chebyshev series on one extra qubit,

$$\langle\tau(x)|_n \otimes \langle\tau(x)|_n = \langle\tau(x)|_{n+1}\,\mathcal{N}_1, \qquad \mathcal{N}_1 \in \mathbb{R}^{2N\times N^2}, \tag{3.7}$$

so that $f^{(i)}f^{(j)}$ becomes $\mathcal{N}_1(G^i \otimes G^j)$ on $|\Psi\rangle$. Linear terms are carried into the same space by tensoring them with the collapse row $\mathbb{D}^{(0)}(x_s)$, which supplies the constant of one copy. With the

Hamiltonian built as in Equation (3.4), Wu et al. count at least $2^{2n-1}$ degenerate zero-energy states, so that solving the equation is "converted to that of seeking a degenerate state" that "closely matches the analytical solution" [1]. A term of degree $p$ needs $p$ copies, a register of dimension $N^p$, as Paine also notes [63]; the $p$-copy construction is given in Appendix E.2.

## 3.5 Critical Assessment

**Strengths.** The construction is unified: ODEs, PDEs and their data take one least-squares form, with a route for nonlinear terms. Given exact data, its kernel is exact for any polynomial solution within the truncation. With no grid, the model and its derivatives are available at every $x \in [-1, 1]$ from one prepared state, and $H$ has few terms, each of polynomially many gates.

**Weaknesses in the cost.** A filter's degree grows as $\sqrt{\alpha_H/\Delta}$ with $\alpha_H \geq \|H\|$ for any encoding (Equation (2.2)), which for a sum of Gram matrices is bounded below by the effective condition number of the equation and constraint operators taken together, so the spectrum of $H$ may set the cost whatever its circuits. Each derivative enters $A$ as a power of the dense $G$, whose norm grows as $N^2$ (§ 4.2) [12]. For an equation with constant coefficients, at a gap that does not grow with $N$, the query count therefore grows with each derivative eliminated through $G$. Classical least squares avoids the same growth by carrying the derivatives as unknowns, in first-order-system least squares [13]. Wu et al.'s expected time hides the norm in the normalisation of $\tilde{H}$ (Appendix I.7). Separately, the prefactor of Equation (3.5) exceeds $\|G\|_2$ by a factor growing with $N$; § 4.5.1 and § 4.6 separate this looseness from the operator's norm.

**Weaknesses in the evidence.** The printed examples carry error floors that no increase in $n$ removes: their constraint points are rounded zeros and extrema of the known solution, demanding a zero the solution lacks, and their truncated Maclaurin sources pose a different equation (§§ 4.2 and 4.3, Appendix I). The PDE examples' invariant sets are read off the known solution and omit the initial or boundary function that fixes it, so the kernel degenerates beyond the printed resolution; the heat example's initial function, at $t = 0$ inside the time domain, would pose a backward, ill-posed problem (§ 4.3). Wu et al. describe $H$ as positive definite, yet it is singular in the very case the method aims at, a solution represented exactly (§ 4.4). The time rule (Equation (3.6)) contains no $\varepsilon$ and needs $\lambda_{\max}$ and $\lambda_2$ in advance, so it certifies no accuracy (Appendix I.5).

**Weaknesses in scope.** The nonlinear route projects rather than prepares: any function of $H$ is a scalar on its zero-energy eigenspace, of dimension at least $2^{2n-1}$, so a filter returns the initial state's projection onto it, a direction set by the initial state and not by the equation (§ 6.4). The readout identity assumes the ground state is prepared with certainty, though post-selection succeeds with probability below one, and yields one point of the field per measurement campaign (§ 7.6). The construction carries one real scalar field: complex-valued states are set aside, since with a Fourier basis the energy is "of a complicated form, rendering it difficult to find the ground state" [1] (§ 6).

**Open problems.** Wu et al. leave open "the spectral gap dependence", the choice of basis, to be guided by "the cost of differentiation", and the data constraints, "crucial for regularizing solutions and growing the gap" [1]. In concurrent work, Paine compares Hamiltonian constructions and function encodings (grid values against spectral coefficients) on linear and nonlinear examples, and finds that both choices affect the recovery of the solution, the spectral gap, its degeneracy and the readout [63]. On the few small instances studied the normalised gap depends on both choices without a general ordering, no scaling with resolution or block-encoding circuit is given, and the lift for products is polynomial only, needing more registers as the degree grows; the comparison raises the first problem without giving its scaling in $N$. The first two are the two terms of Equation (2.2). The gap is its denominator and has meaning only against the norm, and the cost of differentiation, which Wu et al. measure in circuit depth, is also paid in its numerator at every query. Three questions follow, the

first two on the numerator and the third on the denominator at its limit, where a degenerate kernel leaves no gap and the degree is undefined:

(i) *How does the query count grow with the resolution, and how much of that growth belongs to the encoder and how much to the Hamiltonian?* (§ 4)

(ii) *Can the residual be reformulated so that its Hamiltonian is banded and well conditioned without changing its solution, and what does the reformulation cost?* (§ 5)

(iii) *Can the solution of a nonlinear equation be prepared as the unique ground state of a Hamiltonian, rather than selected by projection from a degenerate kernel, and does the reformulation's advantage survive there?* (§ 6, after the coupled linear systems on which its answer is built)

# 4 The Standard Form and Its Cost

Every term of Wu et al.'s Hamiltonian has a circuit of polynomially many gates, so one application of it is cheap (§ 3.3). The filter applies the Hamiltonian as many times as its degree, and the degree depends on the circuit only through its subnormalisation $\alpha_H$, which no block encoding brings below $\|H\|$ (Equation (2.2)). We call the ratio $\alpha_H/\|H\| \geq 1$ the *encoder's cost* and $\|H\|$ itself the *Hamiltonian's cost*: the first belongs to the circuit, the second to the operator. The first question of the assessment asks how the query count grows with the resolution and how that growth divides between the two, and answering it needs an encoding tight enough that whatever growth remains cannot be the circuit's.

The construction under examination is that of Wu et al. [1], built on the Chebyshev latent space and lifts of Paine et al. [60]. We call it the *standard form*: the state holds the field alone, and every derivative is eliminated through the dense differentiation matrix $G$, so that a term $f^{(j)}$ acts as $G^j$ on the field's coefficients. Wu et al.'s Hamiltonian, a sum of Gram matrices (Equation (3.4)), is the Gram matrix $H = R^\dagger R$ of one operator, the *stacked residual* $R$, whose rows are the equation block followed by the constraint rows, so that a state's energy is $\|R\psi\|^2$. The thesis assembles, solves, encodes and reformulates $R$; $H$ is reached from it, and its eigenvalues are the squared singular values of $R$ (§ 4.4). Where a printed choice would make a problem inaccurate, answer-dependent or ill posed, the standard form replaces it at the point where the construction meets it, and Table 4.2 collects the departures.

## 4.1 The Model

We use Wu et al.'s model, Equations (3.1) and (3.2), unchanged, and write it in the orthonormal basis that makes its scale explicit. The prefactors of the feature map define the basis functions $\varphi_0 = T_0/\sqrt{N}$ and $\varphi_k = T_k/\sqrt{N/2}$ for $1 \leq k < N$, which are orthonormal under the discrete inner product on the $N$ Chebyshev nodes, and $\langle\tau(x)| = (\varphi_0(x), \ldots, \varphi_{N-1}(x))$ is the row of their values at $x$. A truncated solution $f_N = \sum_{k<N} b_k\varphi_k$ is then $f_N(x) = \langle\tau(x)|b\rangle = \|b\|\,\langle\tau(x)|\psi_e\rangle$ with $|\psi_e\rangle = b/\|b\|$, so the model $f_Q$ of Equation (3.2) represents $f_N$ exactly at $|\psi\rangle = |\psi_e\rangle$ and $\eta = \eta_e$, where, by Parseval's identity under the Chebyshev weight (Appendix A.3),

$$\eta_e = \|b\|^2 = \frac{N}{\pi}\int_{-1}^{1} \frac{f_N(x)^2}{\sqrt{1-x^2}}\,dx. \tag{4.1}$$

The state carries the direction of $b$ and the scale its length, and $\eta_e$ is fixed by the solution and the resolution alone, growing with $N$ for a smooth solution. It describes representability only: the ground state $|\psi_g\rangle$ is set by the residual's kernel and approaches $\pm|\psi_e\rangle$ only as far as $f_N$ satisfies the constraints. The regular datum recovers the scale and its sign, $\sqrt{\eta} = y_s/\langle\tau(x_s)|\psi_g\rangle$, which is

negative when the ground state, defined only up to sign, opposes the datum; Wu et al.'s positive root is therefore a convention, and the decoded field carries the sign.

The feature vector is kept unnormalised, as Wu et al.'s derivation uses it, so every constraint row inherits its squared norm, which oscillates about one and reaches $(2N-1)/N$ at $x=\pm1$; normalising it point by point would reweight the least-squares problem and change its solution (Appendix A.3). A function of several variables is carried on one register per axis, each with its own $n$, with the feature vector $\langle\tau(x_1)|\otimes\langle\tau(x_2)|\otimes\dots$ in Kronecker order. The running example of this chapter is Wu et al.'s Fig. 3b: the critically damped oscillator $f''+4f'+4f=0$ with $f(-1)=0$ and $f(0)=\frac{1}{2}$, whose solution is $f=\frac{1}{2}(1+x)e^{-2x}$. It is R2a in the benchmark problems of Table 4.1, which the rest of the thesis uses. They are ordered so that each adds one capability to those above it, and a failure therefore localises to the capability that introduced it; each label names the problem with the data of § 4.3, not the printed panel.

**Table 4.1: The benchmark problems R1–R10, each adding one capability to those above it.** R2–R6 are the examples of Wu et al. [1] (their Figs 3–7) with our data; R7–R10 go beyond one real field. $\{a_2,a_1,a_0\}$, coefficients of $a_2f''+a_1f'+a_0f=0$; $l$, $m$, Legendre degree and order; $a/\nu$, initial amplitude over viscosity. Equations, data and references in full: Tables H.1 and H.2.

| | Problem | New capability |
|---|---|---|
| R1 | $f'+2f=0$, $f(-1)=1$ | the smallest end-to-end system |
| R2 | three second-order equations, $\{1,4,4\}$, $\{1,-2,-3\}$, $\{1,5,400\}$ | order two; Dirichlet and Neumann data; stiffness (R2c) |
| R3 | Legendre, $m=0$ ($l=4,5$), and associated Legendre, $m=1$ ($l=5,6$) | variable coefficients and lifts; an exact kernel; the substitution of § 4.3 (R3c, R3d) |
| R4 | four inhomogeneous equations | sources by their Chebyshev coefficients |
| R5 | Laplace (two space axes), heat and wave (one space and one time axis) | two axes; initial and boundary functions as datum slices |
| R6 | $4f''+2f'^2+f=0$; $f''-2f^2+x=0$ | the doubled space (a degenerate kernel) |
| R7 | $u'=v$, $v'=-\pi^2u$ | two coupled fields; ratio rows |
| R8 | $y'=-y+y^2$, $y(0)=\frac{1}{2}$ | the Carleman lift on a time axis, both forms |
| R9 | viscous Burgers, periodic, $a/\nu=0.03$–$10$ on 4–16 sine modes | a field-scale lift across its convergence boundary |
| R10 | two-dimensional Navier–Stokes in vorticity form, periodic, 12–80 modes | incompressible flow; the tensor layout to 316 modes |

## 4.2 Differentiation and Multiplication

In the normalised basis the differentiation matrix has the entries

$$[G]_{k,p}=\begin{cases}2p, & p>k\geq1,\ p-k \text{ odd},\\ \sqrt{2}\,p, & k=0,\ p \text{ odd},\\ 0, & \text{otherwise},\end{cases}\tag{4.2}$$

and it is exact on the represented space, since the derivative of a polynomial of degree below $N$ stays in it [12]. $G$ is upper triangular, dense and nilpotent, and its norm grows as $N^2$: $\|G\|_2/N^2$ converges to $1/(2j_{-3/4,1})\approx0.472$, with $j_{-3/4,1}$ the first positive zero of the Bessel function $J_{-3/4}$, and $\|G^k\|_2/N^{2k}$ to a positive constant for every $k$. Because the basis is orthonormal at the Chebyshev nodes, $G$ is orthogonally similar to the matrix that differentiates nodal values, so the growth is not an artefact of the normalisation; its order is the classical one for dense Chebyshev differentiation [12, 33]. The structure and the similarity are proved in Appendix A.4, and the limits in Appendix C.4.

A polynomial coefficient $a(x)$ acts by the lift of Paine et al. [60], $x^p\,\langle\tau(x)|_n=\langle\tau(x)|_{n+1}\,\mathcal{M}_{x^p}$ [1], extended by linearity to $\mathcal{M}_a=\sum_p a_p\mathcal{M}_{x^p}$. It is exact whenever $\deg a\leq N$, a constant enters as

$a_0\mathcal{M}_1 = a_0\sqrt{2}\,[\,I;\,0\,]$, and $\|\mathcal{M}_a\|_2 \le \sqrt{2}\sup|a|$ (Proposition 13). The equation $\sum_j a_j(x)f^{(j)} = r(x)$ then has the equation block

$$A_{\text{std}} = \sum_j \mathcal{M}_{a_j} G^j - \mathcal{M}_r\,\mathbb{D}^{(0)}(x_s), \qquad \mathcal{M}_r\,\mathbb{D}^{(0)}(x_s) = \frac{|\hat{r}\rangle\,\langle\tau(x_s)|}{f(x_s)}, \tag{4.3}$$

in which the source enters through the collapse row of § 4.3, with $\hat{r}$ the coefficients of $r$ on the lifted register. The subscript marks the standard form, defined in Definition 2.

Wu et al.'s truncated Maclaurin source (§ 3.2) poses a different equation, whose solution the ground state then approximates, with an error floor that no increase in $n$ removes (Appendix I.3). The standard form loads the Chebyshev coefficients of the source instead, through the same collapse row, so the source is as accurate as the series that represents the solution, geometrically convergent for a source analytic on the interval [32].

Wu et al. treat at most two variables. On any number of axes a term is the Kronecker product over the axes of $\mathcal{M}_a G^{j_a}$, each axis lifted if any term on it needs the exact product, and every axis if the equation has a source, and a constant coefficient on a lifted axis enters through $\mathcal{M}_1$ (Appendix B.1). The benchmark problems use at most two.

## 4.3 Constraints and Data

The constraints are built from Wu et al.'s rank-one rows [1]. The zero row $\mathbb{B}(x) = \sqrt{N}\,|0\rangle\,\langle\tau(x)|$ maps a series to the constant function $f_N(x)$, so $\mathbb{B}(x)G^j$ annihilates exactly the series whose $j$-th derivative vanishes at $x$. The collapse row $\mathbb{D}^{(0)}(x_s) = \mathbb{B}(x_s)/f(x_s)$ maps the solution to the constant 1 whatever its scale, which is what lets a source enter Equation (4.3).

A zero row is exact only at a point where the solution's derivative of that order vanishes. Wu et al.'s rounded constraint points (§ 3.2) demand a zero the solution lacks, least squares trades it against the equation, and the error floors at every $n$ (Appendix I.3). The standard form takes the exact point, the root of the solution's derivative nearest Wu et al.'s value. The root depends on the answer: we compute it from the reference solution, outside the method, so a new problem must supply its constraint points with its data (Appendix B.4). Only R4 and R6b, the nonlinear $f'' - 2f^2 + x = 0$ of Wu et al.'s Fig. 7c, need such a point, and for both we keep Wu et al.'s invariant row, moved to the exact root, rather than impose the problem's stated data, so that the residual differs from the printed one only in the defect and the comparison with their construction stays one to one; the stated data could enter without the answer, as the point-datum rows of Definition 2, which no run uses in place of an invariant row.

For an ordinary equation of order $k$ whose leading coefficient has no zero on $[-1, 1]$, the invariant constraints fix the solution up to its scale, which the regular datum then fixes, exactly when they have rank $k - 1$ on the equation's $k$-dimensional solution space (Proposition 12). The rank is that on the solutions, not of the functionals, so $k - 1$ constraints chosen without regard to the equation can fail. Where the count does not apply, for an inhomogeneous equation or a vanishing leading coefficient, and for the gap at a given $n$, the computed kernel decides. We call it *trusted* when the residual's near-kernel is one-dimensional, its smallest singular value separated from the next by a ratio of at least 20 (§ 7.2). Trust certifies that the residual singles out one state, not that the field it decodes to has converged, which is measured separately.

A PDE has an infinite-dimensional solution space and no such count. Wu et al.'s invariant sets (§ 3.2) omit the function that fixes the solution: on Laplace's equation (R5a) a whole family of modes satisfies the equation and its three homogeneous sides, only one of which matches the omitted top side, so the near-kernel grows once the truncation resolves the others (Table I.3); the sets of the heat equation (R5b) and the wave equation (R5c) fail in the same way. The standard form imposes the

initial or boundary function $g$ itself, on its slice $x_a = x^*$, by the *datum-slice* rows

$$\mathbb{S}_a(x^*) - \frac{\hat{g}\ \langle \tau(x_s)|}{f(x_s)}, \tag{4.4}$$

with $\mathbb{S}_a(x^*)$ the zero row $\mathbb{B}(x^*)$ on axis $a$ and the identity on the others, $\hat{g}$ the coefficients of $g$ broadcast along axis $a$, and $x_s$ a datum point, where $f(x_s) \neq 0$. They vanish on the solution and its multiples and leave the scale free (Proposition 10), take $g$ from the problem's statement, serve both forms, and on R5a leave a one-dimensional kernel that is trusted at every resolution computed from $n = (3, 3)$ to $(6, 6)$. The heat problem is the one exception: data at the printed $t = 0$ would pose a backward heat problem on $[-1, 0]$, so the standard form poses it forward from $t = -1$, with the known solution evaluated there as its initial function (Appendix B.2), an answer-dependent set-up like the exact constraint points.

Wu et al.'s associated Legendre examples R3c and R3d (their Figs 4c and 4d, $l = 5$ and 6 with $m = 1$) multiply the equation through by $1 - x^2$, which leaves a double zero of the leading coefficient at each end and a solution $P_l^1 = -\sqrt{1 - x^2}\, P_l'$ [64, §14.6] with a branch point there, which a Chebyshev series approximates only algebraically, so that neither problem has a trusted ground state at the printed resolution (Appendix B.3). The standard form solves for $v = f/\sqrt{1 - x^2}$ instead, whose equation has a leading coefficient with simple zeros and the polynomial solution $v = -P_l'$, represented exactly once $N > l - 1$ (Proposition 11): the error on R3c is $4.9 \times 10^{-15}$ at $n = 3$. It is also the form the descriptor can run (§ 5.4).

## 4.4 The Standard Form

The equation block and the constraint rows are stacked into one residual,

$$R_{\text{std}} = \begin{pmatrix} A_{\text{std}} \\ C_1 \\ \vdots \\ C_L \end{pmatrix}, \qquad H_{\text{std}} = R_{\text{std}}^\dagger R_{\text{std}} = A_{\text{std}}^\dagger A_{\text{std}} + \sum_i C_i^\dagger C_i, \tag{4.5}$$

every block at weight one, so that $H_{\text{std}}$ is a sum of Gram matrices of Wu et al.'s form, Equation (3.4), with the datum slices among the $C_i$. The eigenvalues of $H_{\text{std}}$ are the squared singular values of $R_{\text{std}}$, and its ground state is the smallest right singular vector. The classical reference computes that vector by a singular value decomposition (SVD) of $R_{\text{std}}$ or, beyond its reach, by Lanczos iteration, because an eigensolver on the formed $H_{\text{std}}$ would square the condition number that limits its accuracy (§ 7.2) [65]. The circuit block-encodes $R_{\text{std}}$ at $\alpha_R$ and reaches $H_{\text{std}}$ by one reflection, which block-encodes $2H_{\text{std}}/\alpha_R^2 - I$, so $\alpha_H = \alpha_R^2$ and the encoder's cost enters the degree as $\alpha_R/\|R_{\text{std}}\|_2$ (§ 7.3). For a small ground value the degree's $\sqrt{\alpha_H/\Delta}$ is close to $\alpha_R/\sigma_1$, so the filter pays the conditioning of $R_{\text{std}}$ unsquared, through any encoding (Equation (2.2)). The reformulation of § 5 also acts on the rows of $R$, and its kernel theorem is a statement about $\ker R$.

$H_{\text{std}}$ is positive semi-definite, and singular in the very case the method aims at, a solution the truncation represents exactly, contrary to Wu et al.'s description (§ 3.2). When the solution is a polynomial of degree below $N$, any source is represented exactly and every constraint holds exactly at its point, $G$ differentiates the solution and the lifts multiply it exactly, so its coefficients are annihilated by $R_{\text{std}}$ and give $H_{\text{std}}$ an exact zero eigenvalue; Legendre's equation $(1 - x^2)f'' - 2xf' + 20f = 0$ (R3a, Wu et al.'s Fig. 4a), whose solution is $P_4$, has one from $n = 3$. Otherwise the truncated solution leaves a residual, $\lambda_0 > 0$, and the ground state minimises the total squared residual: the construction is least squares. Nor need the bottom of the spectrum be distinct, as Wu et al.'s partial differential sets show (§ 4.3). The thesis relies on neither property; it relies on an isolated ground state, measured as a trusted kernel (§ 7.2).

**Definition 2 (The standard form):** For a linear problem on one or more Chebyshev axes, the *standard form* is the stacked residual $R_{\text{std}}$ of Equation (4.5), whose equation block is Equation (4.3), lifted on every axis where a coefficient or a source needs the exact product, and whose constraint rows, each at weight one, are built from four kinds of row: zero rows $\mathbb{B}(x^*)G^j$, at a point or on a slice, for data that vanish; the collapse row $\mathbb{D}^{(0)}(x_s)$, which carries a non-zero point datum $f^{(j)}(x^*) = y^*$ as the row $\mathbb{B}(x^*)G^j - y^*\mathbb{D}^{(0)}(x_s)$, as it carries a source into the equation block; ratio rows for vector data (§ 6.2); and datum slices (Equation (4.4)) for initial and boundary functions. Its Hamiltonian is $H_{\text{std}} = R_{\text{std}}^\dagger R_{\text{std}}$, and its solution is the smallest right singular vector of $R_{\text{std}}$, scaled by the regular datum $f(x_s) = y_s$.

Table H.3 lists the four kinds as model inputs, and Table 4.2 collects every departure from the printed method, the pipeline's included. The residual's departures change its data, never $G$, the lifts or the Gram structure, so $H_{\text{std}}$ keeps Wu et al.'s norm to leading order, the constraint rows growing more slowly than the equation block (Appendix C.4); R3c and R3d, whose substitution changes the equation, are costed in the form the standard form runs. Only the pipeline's departures change the cost, and they lower it, through the structured encoding of § 4.5. The baseline therefore keeps Wu et al.'s Hamiltonian, with the cheapest exact encoding built and better accuracy; how each comparison with the descriptor accounts for the two forms' different encodings is set out with the experiment's design (§ 7.7).

The baseline is validated against Wu et al.'s printed results. With every printed choice restored, the rounded points, the Maclaurin sources and the hand-picked invariant sets, the same construction reproduces the matrices printed in Wu et al.'s supplementary material at $n = 2$ entry by entry, which checks the operators. Its basis gives, for the exact solution, the printed scale factor of each of Wu et al.'s eleven ordinary examples to within one unit of the last printed digit, which checks the normalisation: for Fig. 3b, as printed at $n = 3$, the exact solution gives $\eta_e = 1.29009$ against the printed 1.29. The printed scale factors of their three partial differential and two nonlinear examples are not reproduced, by the exact solution's $\eta_e$ or by any ground state we computed, and the cause is not settled. The record of every example, the triage of the printed choices and the verdict on each of Wu et al.'s claims are given in Appendix I.

## 4.5 Encoding the Derivative

Every term of the standard form is a product of lifts, rank-one rows and powers of $G$. The rank-one rows have exact circuits whose subnormalisation is fixed by their own norm, so the term whose encoding decides the encoder's cost is $G$.

### 4.5.1 Wu et al.'s Encoding

Wu et al.'s circuit for $G$ (their Fig. 9; § 3.3) loads the column value $2p$ of each entry as an amplitude through a ladder of small controlled rotations, and its claims about gates and entry errors hold. The claim that decides its cost is the prefactor of Equation (3.5), which is valid but not compared with $\|G\|_2$: it grows as $N^5/2$ against $\|G\|_2 \sim N^2/(2j_{-3/4,1})$, so its looseness tends to $j_{-3/4,1}N^3$, and at $n = 7$ it is $2.2 \times 10^6$ (exact classical computation, Appendix C.6). Rebuilt so that it can be costed beside the others, the circuit has a subnormalisation one third of the printed prefactor, so every cost quoted for Wu et al.'s encoding is the kinder reading. Its entries carry an error of order $N^{-5}$ (Proposition 38), so it enters resource estimates only, inside the same stacked residual as the exact encoding below. The rebuild and the verdict on each of Wu et al.'s circuit claims are given in Appendix I.6.1.

**Table 4.2: Every departure from the printed method removes a defect of the printed choice or extends what the residual can carry.** Our standard form and pipeline against the method of Wu et al. [1]; only the pipeline rows change the cost. $G$, Chebyshev differentiation matrix; $R_{\text{std}}$ and $H_{\text{std}} = R_{\text{std}}^{\dagger} R_{\text{std}}$, the standard form's stacked residual and Hamiltonian; $\alpha_R$, the residual's subnormalisation; $\|H\|_F$, Frobenius norm; $\varepsilon$, target suppression; $p_G$, post-selection success probability; Where, the section that argues the row. Evidence for every row: Appendix I.

| Choice | This thesis | Defect of the printed choice | Where |
|---|---|---|---|
| *Data: what the residual carries* | | | |
| Constraint points | the exact point, set up from the reference solution | a rounded zero the solution lacks floors the error | § 4.3 |
| Sources | Chebyshev coefficients on the lifted register | a truncated Maclaurin series poses a different equation | § 4.2 |
| PDE data | datum slices for initial and boundary functions; the heat data at $t = -1$ | hand-picked invariant sets let the near-kernel grow with $n$; data at $t = 0$ pose a backward heat problem | § 4.3 |
| Double zero | R3c and R3d solved for $v = f/\sqrt{1 - x^2}$ | a branch point limits convergence to algebraic | § 4.3 |
| *Reach: what the residual can express* | | | |
| Axes | any number (two run) | a limit of scope: two axes | Appendix B.1 |
| Fields | several fields on one state, ratio rows, realification | a limit of scope: one real scalar field | § 6 |
| Nonlinearity | the Carleman lift, in both forms; the doubled space kept in the standard form | the doubled space projects onto a kernel of dimension at least $2^{2n-1}$ | §§ 6.4 and 6.5 |
| *Pipeline: how the residual is encoded, filtered, solved and read* | | | |
| Hamiltonian | the stacked residual, solved and encoded from $R_{\text{std}}$; $H_{\text{std}}$ never formed | forming $H$ squares the residual's conditioning | §§ 4.4 and 7.2 |
| Encoding of $G$ | an exact factorisation, $\alpha$ within a factor 2.12 of $\|G\|_2$ | an approximate encoding, $\alpha$ looser than $\|G\|_2$ by about $N^3$ | § 4.5 |
| Filter | the minimax edge filter, its degree set a priori | the time rule does not reach the suppression $\varepsilon$ | § 7.4 |
| Phases and scale | GQSP phases from the target alone; scaled by $\alpha_R^2$ | fitted phases carry no guarantee; $\|H\|_F$ is not a subnormalisation | §§ 7.3 and 7.4 |
| Initial state | the geometric product state; its overlap $\gamma$ reported | not specified | § 7.4 |
| Readout | the field decoded classically from the prepared state | the interferometric identity assumes $p_G = 1$ | § 7.6 |

### 4.5.2 An Exact Factorisation

Every non-zero entry of Equation (4.2) is a product of three factors, each depending on one index alone: a row weight, equal to $1/\sqrt{2}$ in row 0 because of the normalisation of $T_0$ and to one elsewhere; a selector depending only on the offset $p - k$, non-zero when $p > k$ and $p - k$ is odd; and the column value $2p$, the weight with which differentiation passes $T_p$ to the lower polynomials. Each factor has an exact block encoding with no amplitude loaded by rotation.

**Proposition 1** (Exact factorisation of the differentiation matrix)**.** *For every $n \geq 1$ the differentiation matrix on the normalised Chebyshev coefficients factorises as*

$$G = R_0 \left(2U_{\text{odd}}\right) \Lambda,$$
$$R_0 = I - \left(1 - \tfrac{1}{\sqrt{2}}\right) |0\rangle \langle 0|, \qquad \left(U_{\text{odd}}\right)_{kp} = \left[\, p > k,\ p - k \ \textit{odd} \,\right], \qquad \Lambda = \operatorname{diag}(p), \tag{4.6}$$

*where $U_{\text{odd}}$ is the sum of the $N/2$ truncated shifts by odd offsets.*

The proof compares the two sides entry by entry with Equation (4.2) and is given in Appendix C.1.

### 4.5.3 Circuits and Cost

Each factor is encoded as a LCU [42] whose terms are unitary already, so no part of the encoding is approximate. The column index $\Lambda$ is a sum of $n+1$ Pauli-$Z$ strings, encoded at $\alpha_\Lambda = N-1 = \|\Lambda\|_2$, and the row weight $R_0$ a positive combination of the identity and the reflection about $|0\rangle$, encoded at $\alpha_{R_0} = 1 = \|R_0\|_2$ (Appendix C.1). The sum of shifts is the one factor that needs arithmetic: a single ripple-carry subtraction [66] serves all $N/2$ terms, and post-selecting its borrow flag removes exactly the terms that would wrap around, so $U_{\text{odd}}$ is encoded exactly at $\alpha = N/2$ with $O(n)$ Toffoli gates (Figure 4.1; Lemma 3).

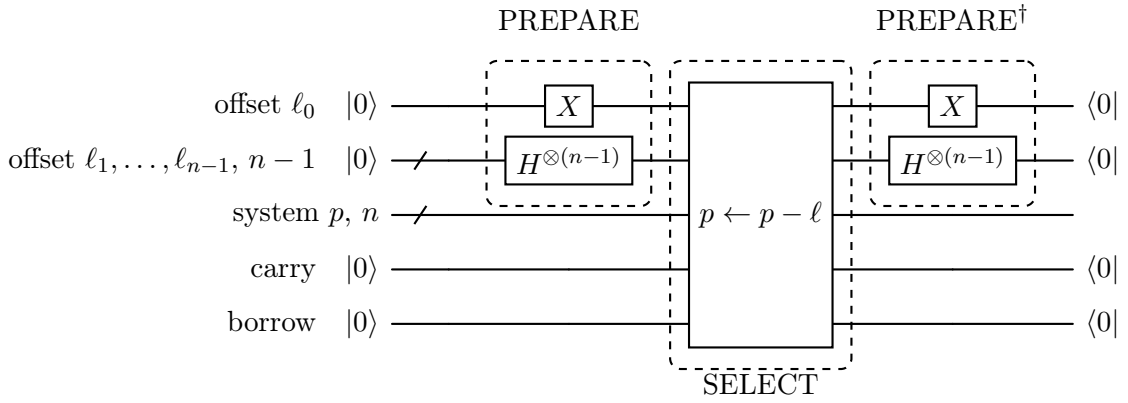



**Figure 4.1: The odd-offset shifts are one subtraction.** PREPARE (left) puts the offset register $\ell$ in a uniform superposition over the $N/2$ odd offsets, by an $X$ on its lowest qubit and Hadamards on the rest. SELECT is a ripple-carry subtractor [66] that replaces the system register $p$ by $p - \ell$ and sets the borrow flag when $p < \ell$. PREPARE$^\dagger$ (right), with post-selection of $\ell$ and the borrow flag on $|0\rangle$, keeps $\frac{2}{N}|k\rangle\langle p|$ for every odd $\ell = p - k \geq 1$, and the wrapped terms vanish exactly. The block is $U_{\text{odd}}/(N/2)$, so $\alpha = N/2$, on $n$ system and $n+2$ ancilla qubits.

The three encodings multiply, and so do their subnormalisations. Their product is an exact block encoding of $G$ at $\alpha_G = 1 \cdot N \cdot (N-1)$.

**Proposition 2** (Cost of the structured encoding)**.** *The factorisation of Proposition 1 compiles to an exact block encoding of $G$ on $n$ system qubits with*

$$\alpha_G = N(N-1), \qquad \frac{\alpha_G}{\|G\|_2} \longrightarrow 2j_{-3/4,1} \approx 2.117, \qquad O(n^2) \text{ gates}, \qquad n+4 \text{ ancilla}, \tag{4.7}$$

*of which $n-1$ ancilla are the selector of the $N/2$-term combination $U_{\text{odd}}$.*

The limit is that of $\|G\|_2/N^2$ (§ 4.2). At the sizes reached the ratio $\alpha_G/\|G\|_2$ rises from 1.62 at $n = 2$ to 2.11 at $n = 10$ (Table C.1), against a ratio for Wu et al.'s encoding that grows as $N^3$. The proofs, the ancilla accounting and the gate count under each synthesis of the multi-controlled gates are given in Appendix C.3. The $n-1$ selector qubits are the price of writing $U_{\text{odd}}$ as one combination of its $N/2$ shifts, which § 5.6 sets against the descriptor's ancilla.

The remaining terms of the residual are built from the same pieces: a polynomial coefficient through its lift, a combination of truncated shifts built by the same post-selected subtraction (Appendix B.5); the rank-one rows by Wu et al.'s exact circuits; a source by one rotation tree over its Chebyshev coefficients; and a power $G^k$ as a product of encodings that share one ancilla register with one flag per product, at subnormalisation $\alpha_G^k$ while $\|G^k\|_2 < \|G\|_2^k$. The stacked residual is assembled into one reflection (§ 7.3) and verified against the independently assembled Hamiltonian (§ 7.5).

## 4.6 The Standard-Form Hamiltonian

For constant coefficients the structured encoding holds the encoder's cost of the whole residual, as well as that of $G$, to a constant. On R2a at $n = 8$ the stacked residual's encoding has $\alpha_R = 4.26 \times 10^9$

against $\|R_{\text{std}}\|_2 = 5.51 \times 10^8$, both computed exactly from the built circuit and the assembled residual, a ratio of 7.74, levelling off towards the constant looseness of the residual's dominant term $G^2$ (Appendix C.4), though no limit is proved for the whole residual. Since the degree is linear in $\alpha_R$, this ratio is the whole of the encoder's cost in the degree; in $\alpha_H = \alpha_R^2$ it is squared, 59.9 at $n = 8$. With Wu et al.'s encoding the same ratio is $9.4 \times 10^{11}$ at $n = 7$ and $6.1 \times 10^{13}$ at $n = 8$: it grows as $N^{3k}$, a factor $N^3$ for each of the $k$ factors of $G$ in the leading term.

For an equation of order $k$ with constant coefficients, $a_k \neq 0$ and constraints of order below $k$, $\|G^k\|_2/N^{2k}$ tends to a positive constant and the constraint rows grow more slowly, so $\|R_{\text{std}}\|_2 = \Theta(N^{2k})$ and $\|H_{\text{std}}\| = \|R_{\text{std}}\|_2^2 = \Theta(N^{4k})$ (Corollary 1). The gap is bounded independently of $N$. On the coefficient vectors of the linear polynomials, $G$ and the lifts act exactly and every row evaluates the function, so by Parseval's identity (Equation (4.1)) the quotient $\|R_{\text{std}}b\|^2/\|b\|^2$ on them is a ratio of integrals and point values that does not depend on $N$; by Courant–Fischer, $\Delta \leq \lambda_1(H_{\text{std}}) \leq C$, with $C$ fixed by the equation and its data, for any polynomial coefficients (Proposition 14). That the gap does not fall either is measured. On R2a it is 7.04 at $n = 3$ and 7.04 at $n = 10$, and on $f'' - 2f' - 3f = 0$ (R2b, Wu et al.'s Fig. 3c) it is 0.415 at $n = 4$ and 0.415 at $n = 10$, unchanged to the figures shown at every $n$ between, while $\|H_{\text{std}}\|$ on R2a grows from $2.8 \times 10^5$ to $2.0 \times 10^{22}$. The same holds on the stiff $f'' + 5f' + 400f = 0$ (R2c), 27.4 at $n = 5$, its first trusted resolution, and 26.3 at $n = 10$, and on the decay R1, 4.95 and 4.95. Because no block encoding has $\alpha_R$ below $\|R_{\text{std}}\|_2$, the minimax edge filter's degree (Equation (7.6)) then satisfies

$$\|H_{\text{std}}\| = \Theta\big(N^{4k}\big), \quad \Delta \leq C \quad \Longrightarrow \quad d = \Omega\big(N^{2k}\log(1/\varepsilon)\big) \text{ for every block encoding of } R_{\text{std}}, \\ \text{(order } k\text{, constant coefficients, } a_k \neq 0\text{, constraints of order below } k\text{)}, \tag{4.8}$$

and $d = \Theta(N^{2k}\log(1/\varepsilon))$ for the structured encoding wherever the gap is also bounded below and $\alpha_R/\|R_{\text{std}}\|_2$ stays bounded, as both are on R2a. At second order, the order of Wu et al.'s examples, $\|H_{\text{std}}\| = \Theta(N^8)$ and no block encoding brings the degree below order $N^4\log(1/\varepsilon)$. This is the Hamiltonian's cost, and its growth can be measured without any encoding. Every block encoding of $R_{\text{std}}$ has $\alpha_R \geq \|R_{\text{std}}\|_2$, and the minimax filter's degree does not increase as $\alpha_R$ falls, so the degree at $\alpha_R = \|R_{\text{std}}\|_2$, which we call the *ideal-encoding bound*, is the least this filter needs on this residual through any encoding of it (Proposition 32). No polynomial filter of the walk reaches the suppression at a lower degree than the minimax edge filter (§ 2.4), so the floor holds for all of them; over all algorithms it holds only as far as Lin and Tong's optimality result supports [53].

The two costs predict different growth for the degree. With Wu et al.'s encoding the looseness, growing as $N^{3k}$, multiplies a degree that already grows as $N^{2k}$, so the degree grows as about $N^{5k}$; with the structured encoding, and at the ideal-encoding bound, it grows as $N^{2k}$. The difference between the first two is the encoder's cost and the common exponent the Hamiltonian's; § 8.2 measures all three. The conditioning of $R_{\text{std}}$ that the filter pays (§ 4.4) grows as $N^{2k}$ because each of the $k$ derivatives is eliminated through $G$.

The conclusion holds for constant coefficients only: there the structured encoding bounds $\alpha_H/\|H_{\text{std}}\|$, and for a lifted variable coefficient it is loose. On R3a the circuit pays for the product $\mathcal{M}_a G^2$ as $\|\mathcal{M}_a\|\,\|G^2\| \gg \|\mathcal{M}_a G^2\|$, because the coefficient $1 - x^2$ vanishes at the ends of the interval, where the derivatives of Chebyshev polynomials are largest. For the same reason the residual's norm is smaller than the constant-coefficient law gives: $\|H_{\text{std}}\|$ is $6.04 \times 10^8$ at $n = 7$ and $9.70 \times 10^9$ at $n = 8$, growing as $N^4$ where the constant-coefficient law gives $N^{4k} = N^8$. Any encoding assembled from one of $G^2$, as a product with the encoding of $\mathcal{M}_a$ or as a combination of terms each holding $G^2$, has a subnormalisation at least of order $\|G^2\|_2 = \Theta(N^4)$, against the $\|R_{\text{std}}\|_2 = \Theta(N^2)$ that the ideal-encoding bound would need, since the constant term $\mathcal{M}_1 G^2$ alone has norm $\sqrt{2}\,\|G^2\|_2$; reaching the ideal-encoding bound on R3a would therefore need $\mathcal{M}_a G^2$ encoded as one operator, which we have not built, and whether such an encoding exists at a cost polynomial in $n$ is open. On such a problem the encoder's cost reappears, and only the ideal-encoding bound separates it from the

Hamiltonian's; the convention of § 7.7 therefore quotes every cost both as built and at that bound, and the comparison with the descriptor on this problem is reported in § 8.2.

For constant coefficients the query count therefore has two parts, and only one belongs to the circuit. The encoder's cost, a factor growing as $N^{3k}$ in the residual's subnormalisation under Wu et al.'s encoding, is held to a constant by the structured encoding. The Hamiltonian's cost, $\Omega(N^{2k}\log(1/\varepsilon))$ queries, remains under every encoding, a factor $N^2$ for each derivative eliminated through $G$, and a gap bounded independently of $N$ cannot offset it. This is the cost of differentiation that Wu et al.'s Discussion raises, paid in the number of queries. Their polynomial gate count (§ 3.3), which the rebuilt circuits bear out (Appendix I.6.1), prices one query and so does not bound the method's cost, which grows exponentially in $n$. With the circuit already tight, the operator is what is left to change: § 5 keeps the derivatives as unknowns instead of eliminating them into the dense $G$.

# 5 The Descriptor Reformulation

In the standard form the state holds the field alone, and every derivative is represented through the dense differentiation matrix $G$, so that a term $f^{(j)}$ acts as $G^j$ on the field's coefficients. § 4.6 found the cost of that form in its operator: with the encoding already tight, $\|G^k\|_2 = \Theta(N^{2k})$ makes $\|H_{\text{std}}\| = \Theta(N^{4k})$ for constant coefficients, and the filter's degree grows as $N^{2k}$ against a gap that stays flat. A lower degree therefore needs a different residual, which leaves the question *can the residual be reformulated so that its Hamiltonian is banded and well conditioned without changing its solution, and what does the reformulation cost?*

## 5.1 The Banded Factorisation

Differentiation of a Chebyshev series is dense because the derivative is expanded in the basis of the function, which is also why the tau and collocation methods produce dense, ill-conditioned matrices [12, 34]; into the neighbouring basis, differentiation is banded. The coefficients of a function and of its derivative are tied by a three-term recurrence, each row of which ties one coefficient of the function to two of the derivative (Appendix A). Truncated to $N$ coefficients and written in the normalised basis of § 4.1, with $\hat{f}^{(0)}$ and $\hat{f}^{(1)}$ the coefficient vectors of $f$ and $f'$, it is the matrix identity

$$\tilde{B}\,\hat{f}^{(1)} = \tilde{D}\,\hat{f}^{(0)}. \tag{5.1}$$

$\tilde{D}$ has a single non-zero diagonal, the first superdiagonal, with entries $(2\sqrt{2}, 4, 6, \ldots, 2(N-1))$; $\tilde{B}$ has the diagonal $(2, 1, 1, \ldots)$ and the second superdiagonal $(-\sqrt{2}, -1, -1, \ldots)$, the irregular first entries coming from the normalisation of $T_0$. Both hold $\mathcal{O}(N)$ non-zero entries against the $N^2/4$ of $G$. Since $\tilde{B}$ is upper triangular with a non-zero diagonal, it is invertible, and

$$G = \tilde{B}^{-1}\tilde{D} \tag{5.2}$$

exactly at every $N$ (Appendix A). Away from its first row, $\tilde{B}^{-1}$ is a triangle of ones on each parity class: the density of $G$ is that inverse, and Equation (5.1) is the same map before the inverse is taken. The singular values of a single-diagonal matrix are its entries' magnitudes, so $\|\tilde{D}\|_2 = 2(N-1)$, while $\|\tilde{B}\|_2 \le 2 + \sqrt{2}$.

Up to the basis normalisation, $\tilde{B}$ and $\tilde{D}$ are twice the ultraspherical method's operators for conversion from first-kind to second-kind Chebyshev polynomials and for differentiation into the second kind [12], and the factorisation is that method's discrete shadow (§ 2.3). In the normalised basis the two factors return the coefficients of a function and of its derivative in the second-kind polynomials $U_m$ (Lemma 5), so Equation (5.1) equates two expansions of $f'$ in that basis. The ultraspherical method uses such operators to eliminate the derivatives and solves a banded system

in $f$ alone; that reduction is not built or compared here, because its conversions weight the rows, so that it poses a differently weighted least-squares problem, their entries would have to be encoded, and the gap mechanism of § 5.3 would not carry over to it without a measurement (Appendix A.5). What Olver and Townsend's conditioning bound leads one to expect of it is discussed in § 9.5.

The descriptor form keeps Equation (5.1) as a row and never forms the quotient Equation (5.2). It carries $f', \dots, f^{(k)}$ as unknowns beside $f$, ties consecutive derivatives by the factors, and writes the equation over all of them, so that one pair of factors serves every order, axis and coupled field. For order $k \geq 2$ these unknowns form the *chain*; a first-order equation or system needs none and takes the *mass form* (§ 5.2). Carrying derivatives as unknowns so as not to square the condition number of a least-squares functional is the idea of first-order-system least squares [13] (§ 2.3), and the physics-informed Hamiltonian is such a functional. Our increment is the transfer of both ideas to a block-encoded physics-informed Hamiltonian and the measurement of its consequences for ground-state preparation.

Figure 5.1 shows the effect on the running example $f'' + 4f' + 4f = 0$ (R2a). At $n = 7$ the standard form's stacked residual holds 8384 non-zero entries on 128 unknowns, and the descriptor's holds 1402 on 512: four times the unknowns, from the block register that § 5.2 counts, and six times fewer entries. With the kernel unchanged, the Hamiltonian's norm falls from $\Theta(N^{4k})$, which is $\Theta(N^8)$ at second order, to $\Theta(N^2)$ at fixed block weights; the price is a block register and a gap that the derivative blocks erode, which a weight per block, set from the equation and the resolution, restores on one axis without ancilla.

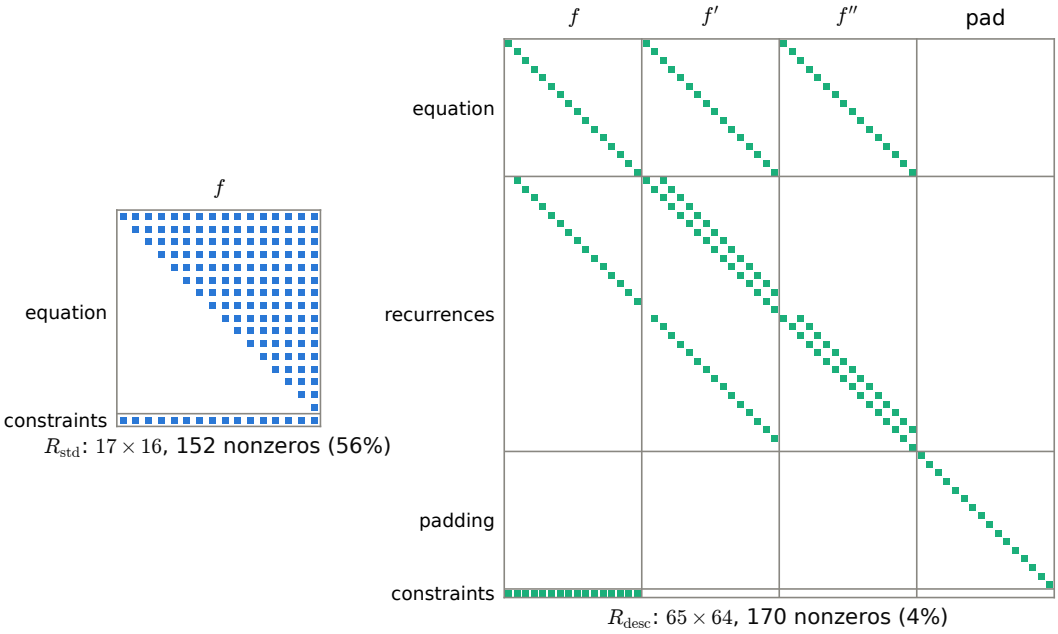



**Figure 5.1: Dense against banded: the two residuals of the same equation.** Sparsity of the stacked residuals of $f'' + 4f' + 4f = 0$ (R2a) at $n = 4$, constraint row included; the two panels share a scale, so each square is one non-zero entry. Left, the standard form's $R_{\text{std}}$, whose equation block $A_{\text{std}} = G^2 + 4G + 4I$ has a full upper triangle. Right, the descriptor's $R_{\text{desc}}$, whose recurrence blocks, equation rows and padding are banded; the only dense row is the constraint row $f(-1) = 0$, which both forms share.

## 5.2 The Descriptor Residual and Its Kernel

Consider the linear ODE of order $k$

$$\sum_{j=0}^{k} a_j(x)\, f^{(j)}(x) = r(x), \qquad x \in [-1, 1], \tag{5.3}$$

with the constraints of § 4.3. Every function here is truncated to $N$ normalised Chebyshev coefficients.

**The chain ($k \geq 2$).** Let $\hat{f}^{(j)} \in \mathbb{C}^N$ be the coefficient vector of $f^{(j)}$ for $j = 0, \ldots, k$, stacked into the state

$$w = (\hat{f}^{(0)};\ \hat{f}^{(1)};\ \ldots;\ \hat{f}^{(k)}). \tag{5.4}$$

Two kinds of row couple the blocks. The *recurrence rows*

$$-\tilde{D}\,\hat{f}^{(j)} + \tilde{B}\,\hat{f}^{(j+1)} = 0, \qquad j = 0, \ldots, k-1, \tag{5.5}$$

apply Equation (5.1) block by block. They contain no derivative operator, and on the kernel they are equivalent to $\hat{f}^{(j+1)} = G\hat{f}^{(j)}$. The *equation row* is the equation itself, $\sum_j \mathcal{M}_{a_j}\hat{f}^{(j)} - \mathcal{M}_r\mathbb{D}^{(0)}(x_s)\hat{f}^{(0)} = 0$, with $\mathcal{M}_{a_j}$ the multiplication operator of § 4.2, lifted exactly where the standard form lifts it, and the source entering through the collapse row as in Equation (4.3). The derivatives are unknowns of $w$ and are never replaced by $G^j\hat{f}^{(0)}$, so the dense quotient Equation (5.2) never appears. The name is borrowed from descriptor systems in control theory [67, 68], which likewise keep intermediate quantities as unknowns beside the state and tie them by algebraic rows.

**Definition 3 (Descriptor equation block):** Collecting the recurrence rows and the equation row, the descriptor's equation block $A_{\text{desc}}$ is the block matrix

$$A_{\text{desc}} = \begin{pmatrix} -\tilde{D} & \tilde{B} & & & \\ & -\tilde{D} & \tilde{B} & & \\ & & \ddots & \ddots & \\ & & & -\tilde{D} & \tilde{B} \\ \mathcal{M}_{a_0} - \mathcal{M}_r\mathbb{D}^{(0)} & \mathcal{M}_{a_1} & \cdots & \cdots & \mathcal{M}_{a_k} \end{pmatrix} \tag{5.6}$$

with $(k+1)N$ columns before padding, so that $A_{\text{desc}}\,w = 0$ is the equation.

**The padded block register.** The block index is held by $n_f = \lceil \log_2(k+1) \rceil$ qubits, so the state is padded to $2^{n_f}$ blocks, each block $b > k$ pinned to zero by its own row $\hat{f}^{(b)} = 0$. No other row reads them, so the padding decouples and leaves the physical spectrum unchanged (Proposition 15). It costs the state $2^{n_f}N$ amplitudes against the standard form's $N$, up to twice the unpadded $(k+1)N$, so a second-order equation carries $f$, $f'$ and $f''$ in four blocks. The row blocks, the several-axis case and the padded counts are in Appendix D.1.

**The mass form ($k = 1$).** A first-order system $z' = \Gamma z$ of $C$ components with a constant coupling matrix $\Gamma$ ($C = 1$ for a single equation) has in the standard form the equation rows $I_C \otimes G - \Gamma \otimes I_N$, one block per component. Its equation already gives each derivative, so the mass form carries no derivative block; it multiplies these rows on the left by $I_C \otimes \tilde{B}$ and uses $\tilde{B}G = \tilde{D}$,

$$(I_C \otimes \tilde{B})\big(I_C \otimes G - \Gamma \otimes I_N\big) = I_C \otimes \tilde{D} - \Gamma \otimes \tilde{B}. \tag{5.7}$$

which is banded and free of $G$; $\tilde{B}$ plays the mass matrix $M$ of $M\dot{y} = F(y)$ [69], hence the name. For a scalar equation with a constant coefficient it is, up to normalisation, the ultraspherical method's own first-order operator: R1, $f' + 2f = 0$, becomes $\tilde{D} + 2\tilde{B}$ where the standard form has the dense $G + 2I$ (both at $N = 4$, with the chain, in Appendix A). The mass form keeps the standard form's state, initial state and readout, so it adds no qubits and has no derivative block to rescale (§ 5.4). Coupled fields and Carleman lifts are such systems, and §§ 6.1 and 6.5 carry them in this form.

**Constraints.** A data constraint sits on the block that holds the quantity it constrains. A Dirichlet condition $f(x_z) = 0$ reads block 0 with the zero row $\mathbb{B}(x_z)$, as in the standard form. A Neumann condition $f'(x_m) = 0$ reads block 1 with the same row, $\mathbb{B}(x_m)\hat{f}^{(1)} = 0$, where the standard form must

write $\mathbb{B}(x_m)G$. The regular datum reads block 0, so the field is decoded and scaled exactly as in the standard form (§ 4.1), and the two forms differ only in how the derivatives are carried.

**Theorem 1** (Kernel equivalence)**.** *Let $R_{\text{std}}$ be the standard form's stacked residual (Definition 2) of a linear problem on one axis. Let $R_{\text{desc}}$ stack the descriptor's recurrence, equation and padding rows with the same constraint rows, each reading the block of the derivative it constrains, the equation row applying the standard form's multiplication operators. Let $R_{\text{mass}}$ be the mass form's stacked residual of a first-order system with a constant coupling matrix. Then*

$$\ker R_{\text{desc}} = \left\{ (p,\ Gp,\ \ldots,\ G^k p,\ 0)\ :\ p \in \ker R_{\text{std}} \right\}, \qquad \ker R_{\text{mass}} = \ker R_{\text{std}}. \tag{5.8}$$

*Proof.* Write $Ep = (p; Gp; \ldots; G^k p; 0)$. Chain: the padding rows force the padding blocks of a kernel vector $w$ to zero, and since $\tilde{B}$ is invertible each recurrence row gives $w_{j+1} = \tilde{B}^{-1}\tilde{D}\, w_j = G w_j$, so $w = Ew_0$. The recurrence rows vanish on $Ep$ for every $p$, because $(\tilde{B}G - \tilde{D})G^j p = 0$, and every other row of $R_{\text{desc}}$ acts on $Ep$ as the standard form's row of the same name acts on $p$: the equation row gives $\sum_j \mathcal{M}_{a_j} G^j p - \mathcal{M}_r \mathbb{D}^{(0)}(x_s)p$, the standard form's equation block, and a constraint row reads the block that holds $G^j p$. Hence $R_{\text{desc}}Ep$ vanishes exactly when $R_{\text{std}}p$ does, and $E$ is injective because its first block is the identity. Mass form: $R_{\text{mass}}$ is $R_{\text{std}}$ with its equation rows multiplied on the left by the invertible $I_C \otimes \tilde{B}$ (Equation (5.7)) and its constraint rows unchanged. □

**Several axes.** On several axes the field is block 0, and each axis $a$ owns a chain $\partial_a f, \ldots, \partial_a^{k_a} f$ up to the highest order that the equation or a constraint takes along it, tied by that axis's factors $\tilde{B}_a$ and $\tilde{D}_a$ tensored with the identity on the other axes. A mixed derivative would need a block of its own (which we do not build here). Initial and boundary functions enter both forms as the datum slices of Equation (4.4), read on the field block. Every axis adds its chain to the one block register, so the state grows with the total order: the register takes $n_f = \lceil \log_2(1 + \sum_a k_a) \rceil$ qubits, the one-axis count with the total order in place of $k$, and the state holds $2^{n_f} \prod_a N_a$ amplitudes against the standard form's $\prod_a N_a$ (Appendix D.2). The theorem holds on several axes, with $E$ carrying $G_a^j$ along each axis's chain, and for the block-rescaled residual of § 5.4, whose kernel is the image of this one under a diagonal map; both extensions are proved in Appendix D.3.

An exact kernel exists only where the solution lies in the span of the basis and meets every constraint exactly, as Legendre's $P_4$ does on R3a; otherwise both residuals have full rank, and their ground states are the smallest right singular vectors of different matrices. The descriptor measures a state's residual against a norm that carries its derivatives (§ 5.3), and the mass form weights the equation rows by $\tilde{B}$. We therefore claim equal kernels only. Both decoded fields converge to the solution, at rates measured in § 8.1 and not proved.

## 5.3 The Descriptor's Gap

While the kernel is the standard form's, the spectrum above it is not. With $E$ the map of Theorem 1, the descriptor's Rayleigh quotient on the states that satisfy the recurrences is

$$\frac{\|R_{\text{desc}} Ep\|^2}{\|Ep\|^2} = \frac{\|R_{\text{std}}\, p\|^2}{\sum_{m=0}^{k} \|G^m p\|^2}, \tag{5.9}$$

because the recurrence and padding rows vanish on $Ep$ and the remaining rows act as the standard form's. By the min–max principle the quotient bounds the descriptor's singular values from above (Proposition 17). Its numerator is the standard form's, but its denominator adds every carried derivative, so the gap stays flat in $N$ only where the equation controls each of them: where no state orthogonal to the kernel has a residual small against its largest derivative. On one axis the equation row recovers the top derivative from the lower ones by dividing by the leading coefficient, so the

control holds where that coefficient has no zero. Every later degree, gap and subnormalisation claim is scoped by the class this singles out.

**Definition 4 (Regular and singular problems):** A linear problem on one axis, an equation of order $k$ or a first-order system $z' = \Gamma z$ with a constant coupling matrix, is *regular* if its coefficients are polynomials and its leading coefficient ($a_k$, or the identity for a system) has no zero on $[-1, 1]$; its source and its solution $f$ are analytic on $[-1, 1]$; and its constraint rows read $f, \dots, f^{(k-1)}$ at fixed points, every $g \in C^k[-1, 1]$ on which its equation and constraint rows vanish being a multiple of $f$. An equation on one axis whose $a_k$ has a zero on $[-1, 1]$ is *singular*.

Of the benchmark problems, R1, R2, R4b–d, R7 and the truncated Carleman lifts (§ 6.5) are regular, and R3 and R4a singular.

**Theorem 2** (Single-axis gap)**.** *The gap of the descriptor residual of a regular problem, in the chain unscaled or block-rescaled at a fixed $\zeta$, or in the mass form for a first-order equation or system, is* $\Theta(1)$ *as* $N \to \infty$.

The proof (Appendix D.5) reads each recurrence row as the defect between block $j + 1$ and the derivative of block $j$, measured in the norm of the second-kind polynomials, integrates it, and closes by compactness, so it bounds the gap away from zero without giving the bound. For a homogeneous equation the condition on $g$ is the rank condition of Proposition 12. The size of the gap is left to the equation. The norm weighs block $m$ by the size of $f^{(m)}$, so where the solution's derivatives are large the derivative blocks dominate it, and the ground state's weight on the field block, $w_f = \|\hat{f}^{(0)}\|^2/\|w\|^2$, is small; for an exact kernel it is $\|b\|^2/\sum_m \|G^m b\|^2$, with $b$ the solution's coefficients. An initial state prepared on the field block then overlaps the ground state by $\gamma \le w_f^{1/2}$, and amplification costs $1/\gamma$. On R2a the unscaled gap is 0.458 at both $n = 3$ and $n = 10$. On the stiff equation $f'' + 5f' + 400f = 0$ (R2c, Wu et al.'s Fig. 3d), whose solution oscillates at an angular frequency near 20, the gap is $7.5 \times 10^{-6}$ at both $n = 5$ and $n = 10$, flat and small, and $w_f$ is $5.4 \times 10^{-6}$, against $4.3 \times 10^{-3}$ on R2a.

The gap falls with $N$ in two cases.

1. *A singular problem.* A polynomial of degree $N$ can concentrate within $\mathcal{O}(N^{-2})$ of an endpoint, where the Chebyshev points cluster, and each of its derivatives grows by a factor $\mathcal{O}(N^2)$ there, the extremal behaviour of Markov's inequality [54]. Where $a_k$ vanishes at that endpoint, the leading term $a_k p^{(k)}$ of such a state is small, the lower-order terms are small as well, and the quotient is small against the top block. The measured gap falls as about $N^{-4}$: on Legendre's equation $(1 - x^2)f'' - 2xf' + 20f = 0$ (R3a), whose leading coefficient vanishes at both ends, it is $7.6 \times 10^{-5}$ at $n = 5$ and $1.1 \times 10^{-10}$ at $n = 10$, and on $(x - 1)f'' - xf' + f = (x - 1)^2$ (R4a, Wu et al.'s Fig. 5a), whose leading coefficient vanishes at $x = 1$, $4.9 \times 10^{-5}$ and $4.3 \times 10^{-11}$. At an interior zero, where the resolution is $\mathcal{O}(N^{-1})$, Bernstein's inequality in place of Markov's gives $N^{-2}$ [54]; no benchmark problem has one, and Appendix D.4 measures the rate on a controlled equation.

2. *Two axes.* The equation constrains only a sum of top derivatives, $f_{xx} + f_{yy}$ or $f_t - \kappa f_{xx}$. A high-frequency state with $G_x^2 p \approx -G_y^2 p$, or with its weight in the block the equation weights least, has a small residual against large blocks, and only the datum rows penalise it. On the heat equation $\partial_t f - \frac{1}{25}\partial_x^2 f = 0$ (R5b, Wu et al.'s Fig. 6d) the gap is $3.2 \times 10^{-6}$ at $n = 4$ per axis, the first resolution at which both forms' references are trusted, and $2.5 \times 10^{-7}$ at $n = 5$, where the standard form's is 0.444 and 0.433.

The collapse is not an artefact of the lift, the padding, the arithmetic precision, the constraints, the basis or the classical reference: each was tested and ruled out, and a descriptor built in the ultraspherical basis collapses at the same rates (Appendix D.4).

## 5.4 Block Rescaling

The remedy is to weigh each carried derivative at the equation's own scale. Carrying block $m$ as $\zeta^{-m}\hat{f}^{(m)}$ replaces $R_{\text{desc}}$ by $R_{\text{desc}}S$, with $S$ diagonal, $\zeta^m$ on block $m$ and one on the padding. $S$ is invertible, so the kernel becomes $S^{-1}\ker R_{\text{desc}}$ and Theorem 1 survives; the field block keeps weight one, so the readout is unchanged, and the padding still decouples. Only the spectrum moves: the Hamiltonian becomes the congruence $SR_{\text{desc}}^{\dagger}R_{\text{desc}}S$, and the denominator of Equation (5.9) becomes $\sum_m \zeta^{-2m}\|G^m p\|^2$. From here on $R_{\text{desc}}$ and $H_{\text{desc}}$ denote the rescaled residual and its Hamiltonian, and the residual of § 5.3 is called *unscaled*.

In the circuit the weights multiply the bands before the block register's patterns are expanded in Pauli words, so they add no ancilla but lengthen the expansion wherever a pattern's entries were equal across its blocks: the terms, and with them the gates per query, grow by at most the $4^{n_f}$ words of the block register, a factor fixed by the equation and not by $n$ (Appendix D.6; § 8.2 counts the rescaled circuit). One weight per block is what keeps that factor bounded; a per-amplitude equilibration [70] would need a diagonal block encoding of its own, at $\Theta(\dim w)$ gates, and no circuit here uses one.

The weight $\zeta$ is computed from the equation's coefficients and the resolution, never from a solution or a spectrum. For a regular problem,

$$\zeta = 2\left(\bar{a}_j/\bar{a}_k\right)^{1/(k-j)}, \tag{5.10}$$

with $j < k$ the lowest order present and $\bar{a}$ the mean of $|a|$ over the interval. Half of $\zeta$ is the equation's own scale: for constant coefficients and $j = 0$ it is the geometric mean $\rho$ of the characteristic roots' magnitudes, a solution's derivatives grow as $\|f^{(m)}\| \approx \rho^m\|f\|$, and dividing block $m$ by $\zeta^m$ keeps every block at the field block's order. A singular problem needs a weight that grows with the resolution. The concentrated states of § 5.3 have a rescaled quotient of order $N^{-4}\zeta^{2k}$ against their top block (Appendix D.6), so bringing it to order one needs $\zeta \propto N^{2/k}$, which is $N$ at second order, the order of every singular benchmark problem; hence $\zeta = N/2$. A double zero shrinks those states' quotient by a further $N^{-4}$ and needs $N^{4/k}$, the case of item 3 below. On several axes $\zeta_a = N_a$, weighted on an axis of second order so that the top-order terms balance, as $(cN, N)$ for the wave equation $f_{tt} = c^2 f_{xx}$. The two constants, the 2 and the $\frac{1}{2}$, were selected by calibration on the benchmark problems, where the cost over a sweep of $\zeta$ is least, and then held fixed, never refitted; the check of the resulting weight against the best $\zeta$ on a grid is therefore in-sample and says nothing of a class of equation the set lacks. The cost stays within twice its least over a window of $\zeta$, so $\zeta$ need only be right to about a factor of two (Appendix D.6). The two classes are asymptotic, and the weight jumps from one to the other while the equation changes continuously: a regular leading coefficient that dips close to zero behaves as a singular one once the resolution sees the dip, and there the regular weight loses, its cost on $(\delta + x^2)f'' + f = 0$ at $\delta = 10^{-3}$ reaching 2.3 times the grid's best at $n = 5$ and 4.7 times at $n = 7$. Since the quotient already fixes how $\zeta$ must grow with $N$, deriving the weight from the equation, constants and crossover included, rather than calibrating it, should be possible; it is left to further work (§ 10.2).

On every one-axis problem the descriptor runs, the rescaled gap no longer falls with $N$, and the field block regains its weight. The gap is the physical subspace's, the one the filter meets, since the padding alone caps the whole spectrum's gap at one. At $n = 7$ on R2c the gap rises from $7.5 \times 10^{-6}$ to 0.715 and $w_f$ from $5.4 \times 10^{-6}$ to 0.774. On R3a the rescaled gap rises from 2.27 at $n = 5$ to 3.54 at $n = 7$ and levels off, 3.65 at $n = 10$, against $1.1 \times 10^{-10}$ unscaled.

**Remaining limits.**

1. *The norm grows with* $\zeta$. Block $m$ carries $\zeta^m$, so $\|H_{\text{desc}}\| \gtrsim \zeta^{2k}\bar{a}_k^2$. A fixed $\zeta$ changes only the constant; the $\zeta \propto N$ of a simple zero gives $\|H_{\text{desc}}\| = \Theta(N^{2k})$ and, at a flat gap, a degree of about $\tilde{\Theta}(N^2)$ at second order. A zero at both ends, where the Chebyshev polynomials' derivatives are

largest, lowers $\|H_{\text{std}}\|$ to the same $N^4$ (R3a, coefficient $1-x^2$; § 4.6), so there the descriptor gains no exponent at the ideal-encoding bound, only as built, where the structured encoding pays for the lifted multiplier in full; a zero at one end leaves $\|H_{\text{std}}\|$ at $N^8$ (R4a, coefficient $x-1$), and the lower exponent should survive the bound. § 8.2 measures both.

2. *Two axes.* The rescaled gap still falls with $N$, more slowly than unscaled: on R5b it is 0.139 at $n=4$ per axis and 0.0942 at $n=5$, against the unscaled values of § 5.3; § 8.3 measures the same on Laplace's equation (R5a).

3. *A double zero is beyond any block rescaling.* If $a_k$ vanishes to second order at an endpoint, balancing the concentrated states needs $\zeta \propto N^{4/k}$, which at $k=2$ makes $\|H_{\text{desc}}\| = \Theta(N^8)$, the constant-coefficient standard form's norm, and the measurement in Appendix D.6 finds no $\zeta$ that brings the descriptor's cost level with the standard form's. R3c and R3d have this double zero, so both forms run the equivalent problem for $v = f/\sqrt{1-x^2}$ (§ 4.3), whose zeros are simple; no claim is made for the double zero beyond that measurement.

Block rescaling is the descriptor's counterpart of the ultraspherical method's diagonal preconditioner [12], which bounds the condition number of its banded system; preconditioning a block-encoded system is established quantum practice [71, 72].

## 5.5 The Block Encoding

A norm of $\Theta(N^2)$ lowers the degree only as far as the subnormalisation follows it, so the encoding must keep $\alpha_R$ within a small factor of $\|R_{\text{desc}}\|_2$, or the encoder's cost of § 4 returns. Apart from a source's rank-one term, every block of $A_{\text{desc}}$ is a short sum of *bands*, each a diagonal followed by a truncated shift, the device of the explicit block encodings of banded circulant matrices [44]. Each diagonal is affine in the column index up to a few exceptions near the ends: $\tilde{D}$'s is $2\Lambda$, with $\Lambda = \operatorname{diag}(0, 1, \ldots, N-1)$ the index diagonal of Proposition 1, and the others are constant. The affine part is $n+1$ Pauli-$Z$ terms (Equation (C.1)) and each exception one reflection $Q_m = I - 2\,|m\rangle\langle m|$ (Lemma 6), so without a source $A_{\text{desc}}$ is one flat LCU of $\mathcal{O}(n)$ terms,

$$A_{\text{desc}} = \sum_{t=1}^{L} c_t\, U_t, \qquad U_t = P_t \otimes \text{SHIFT}^{\ell_t} D_t, \qquad \alpha_A = \sum_{t=1}^{L} |c_t|, \tag{5.11}$$

where the block word $P_t$, a Pauli word $X^x Z^z$ on the block register or $X^x \Pi_c$, with $\Pi_c$ realised by a flag qubit, for a term that reads block $c$ alone, places the term's row and column blocks, $\text{SHIFT}^{\ell_t}$ is the truncated shift by $\ell_t$, and $D_t$ is the identity, a $Z_j$ or a reflection. The truncated shift is not unitary on the axis register's $n$ qubits; it is applied as a cyclic addition on that register widened by one overflow qubit, where rows falling outside the interval land in the half that the final projection discards, so that qubit is the shift's only ancilla.

Two choices in the compilation set the cost of Figure 5.2. Bands whose axis factors agree are merged across the rows and columns they join, so the recurrences' common $\tilde{D}$ is paid for once, not $k$ times. The selectors are factored, one for the block word and, on each axis, one for the diagonal and one for the shift, each only as wide as the distinct values its terms use, so SELECT applies each distinct operation once rather than once per term and the ancilla stay $\mathcal{O}(\log n)$ (§ 5.6). Two pieces that R2 lacks enter apart: a source's rank-one term $|\hat{r}\rangle\langle\tau(x_s)|$, compiled through the feature map and a rotation tree that loads $\hat{r}$ and added by one further combination, and a system's coupling, a sum of matchings (§ 6.6). The algorithm, its compilation line by line and the check of the terms against the assembled rows are given in Appendix D.7.

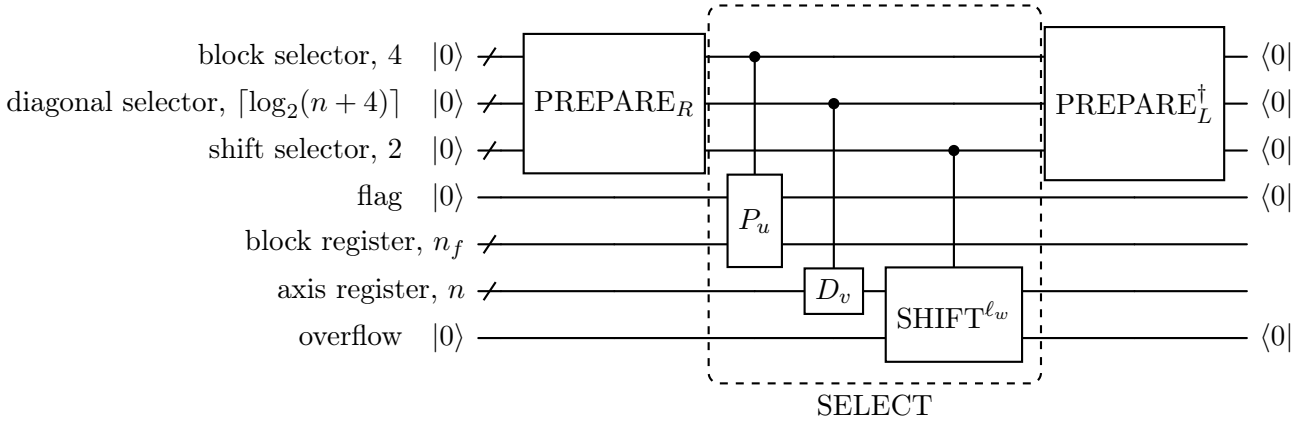



**Figure 5.2: R2's equation block is one preparation, one SELECT and one unpreparation.** The compiled circuit for R2 (second order, no source) on one axis of $n$ qubits, each register labelled with its width. $P_u$, $D_v$ and SHIFT$^{\ell_w}$ are a term's block word, diagonal and truncated shift (Equation (5.11)), each applied when its selector holds $u$, $v$ or $w$. PREPARE$_R$ loads $\mathrm{sgn}(c_t)\sqrt{|c_t|/\alpha_A}$ and PREPARE$_L$ loads $\sqrt{|c_t|/\alpha_A}$ at each term's joint selector index, so the signs survive. With every ancilla prepared in $|0\rangle$ and projected onto $\langle 0|$, the circuit applies $A_{\mathrm{desc}}/\alpha_A$ to the block and axis registers; the ancilla number $\lceil \log_2(n+4) \rceil + 8$ is Proposition 3's.

## 5.6 Cost of the Descriptor Encoding

**Proposition 3** (Cost of the descriptor encoding)**.** *For a descriptor residual of a fixed problem, the LCU of Equation* (5.11)*, with a source's rank-one term added by one further combination, is an exact block encoding of the system rows (the recurrence, equation and padding rows) at $\alpha_A = \sum_t |c_t|$, with $\mathcal{O}(\log n)$ ancilla. Without a source, or with a polynomial source of fixed degree, it has $\mathcal{O}(n^2)$ gates; a general source adds $\mathcal{O}(N)$. For the second-order family R2, which has no source,*

$$\lceil \log_2(n+4) \rceil + 8 \text{ ancilla}, \qquad \mathcal{O}(n^2) \text{ gates}. \tag{5.12}$$

*For the mass form of a scalar first-order equation, and for the chain of a regular second-order problem at a fixed $\zeta \geq 1$, $\alpha_A / \|A_{\mathrm{desc}}\|_2 = 1 + \mathcal{O}(N^{-1})$ without a source and $1 + \mathcal{O}(N^{-1/2})$ with one.*

The ancilla are the selector registers, the overflow qubit and a flag, and only the diagonal selector's width depends on $n$: for R2 it indexes the identity, the $n$ gates $Z_j$ and three reflections, $\lceil \log_2(n+4) \rceil$ qubits. The gate count follows because each distinct operation is applied once and the shift's multi-controlled ladder costs $\mathcal{O}(n^2)$; a source's rotation tree loads its $N$ coefficients with $\mathcal{O}(N)$ gates, as for any general state, and with poly$(n)$ when only a fixed number of them is non-zero. The subnormalisation tends to the norm because an affine diagonal whose end values share a sign is encoded at its own norm (Lemma 7), so the dominant terms, the recurrences' $\zeta^j \tilde{D}$, contribute $2\zeta^{k-1}(N-1)$ to $\alpha_A$, which is also a lower bound on $\|A_{\mathrm{desc}}\|_2$, while every other term is $\mathcal{O}(1)$ in $N$ at fixed $\zeta$ except a source's rank-one term, which grows as $N^{1/2}$. The proofs are given in Appendix D.8, and the ancilla count is checked on the compiled circuit of R2's system rows at $n = 2$, 4, 5 and 12.

The count is of the least order among exact LCU encodings that expand their diagonals in Pauli strings, since $\Lambda$ alone has $n+1$ non-zero Walsh coefficients and a combination of $L$ distinct operations needs $\lceil \log_2 L \rceil$ selector qubits [43]; it is not claimed against encodings built otherwise, by arithmetic on the index for instance. The structured encoding of the standard form needs $n-1$ selector qubits for $G$ alone, because it writes $U_{\mathrm{odd}}$ as one combination of its $N/2$ shifts (§ 4.5.3), so its ancilla grow as $n$ where the descriptor's grow as $\log n$, though an encoding of $U_{\mathrm{odd}}$ built otherwise is not excluded. At the sizes circuits reach the difference is not yet visible in the whole reflection's ancilla (§ 8.2).

The proposition concerns the system rows; for the whole stacked residual, constraints included, the ratio is computed exactly from the built circuit's $\alpha_R$ and the assembled residual's norm. It approaches one on regular problems, $\alpha_R / \|R_{\mathrm{desc}}\|_2 = 1.04$ on R2a at $n = 8$. It stays near two on singular ones, 2.12 on R3a at $n = 7$, because $\zeta\tilde{D}$, $\zeta^2\tilde{B}$ and $\zeta^2 a_2$ are then comparable and comparable

terms add in $\ell^1$ even when each is tight. On two axes it is 2.29 on R5b at $n = 4$ per axis. In the terms of § 4, the encoder's cost $\alpha_H/\|H\|$ on R2a at $n = 8$ is then 1.09 for the descriptor against 59.9 for the structured standard form, which is why an as-built comparison favours the descriptor (§ 7.7).

The gates per query need not follow the subnormalisation. A descriptor query applies more terms than a structured standard-form query, through a block register, so on constant-coefficient problems its advantage is expected in the filter's degree rather than in the gates per query; on a lifted variable coefficient, which the structured encoding pays for in full, its query can be the cheaper too. § 8.2 counts both on the rescaled circuit.

## 5.7 Consequences for the Spectrum and the Filter

Every block of the rescaled descriptor is $\mathcal{O}(N)$ at fixed $\zeta$, the recurrences' $\zeta^j \tilde{D}$ being the largest, and the block $\zeta^{k-1}\tilde{D}$ alone has norm $2\zeta^{k-1}(N-1)$, so $\|R_{\text{desc}}\|_2 = \Theta(N)$ and $\|H_{\text{desc}}\| = \Theta(N^2)$ whatever the order $k$: the derivative is banded, and its norm is that of $\tilde{D}$ rather than of $G^k$. Through Equation (2.2), with an encoding whose subnormalisation stays within a constant of that norm, the gap of Theorem 2 gives, for a regular problem,

$$\|H_{\text{desc}}\| = \Theta(N^2), \qquad \Delta = \Theta(1) \quad \Longrightarrow \quad d = \tilde{\Theta}(N), \tag{5.13}$$

against $\tilde{\Theta}(N^{2k})$ for the standard form with the best exact encoding (Equation (4.8)). On a singular problem and on two axes the rescaling's weights raise the norm, and the degree is about $\tilde{\Theta}(N^2)$ or grows more steeply; for a double zero no claim is made (Table 5.1). Neither form is polylogarithmic in $N$: $\tilde{\Theta}(N)$ is exponential in $n$, and the claim is a lower exponent only. The degrees are the minimax edge filter's, the least of any polynomial filter of the walk, with the hedge of § 4.6 for other algorithms; § 8.2 measures them, as built and at the ideal-encoding bound, and § 8.3 the gaps.

**Table 5.1: The descriptor lowers the degree's exponent where the equation controls every carried derivative.** Hamiltonian norm $\|H\|$, spectral gap and minimax filter degree of the standard form and the descriptor, by class of problem, the descriptor's block weight $\zeta$ given by Equation (5.10). Regular, singular: one axis, the leading coefficient without and with a zero on $[-1, 1]$ (Definition 4); $k$, equation order; $N_a$, resolution of axis $a$; $\tilde{\Theta}$, up to logarithmic factors; flat, falls: the gap's behaviour as $N$ grows; –, not applicable. Norms are proved; the regular gap is proved flat (Theorem 2), and the others are exact classical computations. A zero at both ends (R3) lowers the standard form's norm too, so there the simple-zero row is no advantage at the ideal-encoding bound $\alpha_R = \|R\|_2$.

| Case | $\zeta$ | $\|H\|$ | Gap | Degree |
|---|---|---|---|---|
| Standard, order $k$, constant coefficients | – | $\Theta(N^{4k})$ | flat | $\tilde{\Theta}(N^{2k})$ |
| Descriptor, regular | fixed | $\Theta(N^2)$ | flat | $\tilde{\Theta}(N)$ |
| Descriptor, singular, simple zero, $k = 2$ | $N/2$ | $\Theta(N^4)$ | flat | $\tilde{\Theta}(N^2)$ |
| Descriptor, two axes | $N_a$ per axis | $\Theta(N^4)$ | falls | steeper than $\tilde{\Theta}(N^2)$ |
| Descriptor, singular, double zero | $\propto N^2$ to balance | $\Theta(N^8)$ | – | not claimed |

On a regular problem the relative gap $\Delta/\|H\|$ is $\Theta(N^{-2})$ in the descriptor against $\Theta(N^{-4k})$ in the standard form: on R2a at $n = 8$ the descriptor's is $6.9 \times 10^{-7}$ and the standard form's $2.3 \times 10^{-17}$, below double precision. The classical reference forms neither Hamiltonian (§ 7.2), so a small relative gap costs the filter's degree, not the resolvability of the ground state. Per query the two encodings also differ in post-selected success, which compares the encodings, not the preparation (Appendix D.9).

## 5.8 Summary

Carrying the derivatives instead of eliminating them makes the residual banded with its solution unchanged, at a price.

1. The chain, for order two and above, and the mass form, for first-order equations and systems, keep the banded factors of $G$; the residual is banded apart from the rank-one constraint rows, and its kernel is the standard form's (Theorem 1).

2. Its block encoding is one flat LCU with $\lceil \log_2(n+4) \rceil + 8$ ancilla for R2's system rows, of the least order for an exact LCU that expands its diagonals in Pauli strings, with $\alpha_A / \|A_{\text{desc}}\|_2 \to 1$ on regular scalar problems of order one and two (Proposition 3). It pays with the chain's block register and, expected on constant-coefficient problems, more gates per query.

3. For a regular problem, $\|H_{\text{desc}}\| = \Theta(N^2)$ and $\Delta = \Theta(1)$ (Theorem 2), so the filter's degree is $\tilde{\Theta}(N)$ against the standard form's $\tilde{\Theta}(N^{2k})$ for constant coefficients. Neither is polylogarithmic.

4. *The limit.* The carried derivatives shrink the gap where the solution's derivatives are large, and make it fall with $N$ on singular problems and on two axes. Block rescaling restores it on one axis without ancilla, at the price of a larger norm on a singular problem; on two axes the gap still falls, and a double zero is beyond any block rescaling.

All of this concerns one real scalar field and a linear equation, the scope of Wu et al.'s construction. Coupled fields, complex-valued fields and nonlinear terms each change what the residual must carry, and the nonlinear route Wu et al. propose gives the filter no unique ground state to find. § 6 takes the reformulation there; §§ 8.2 and 8.3 test the cost and the gap on the benchmark problems.

# 6 Systems and Nonlinearity

Every residual so far has carried one real scalar field, the scope Wu et al. set themselves. They set complex-valued states aside, because with a Fourier basis, whose differentiation matrix is imaginary, the ground state of the resulting energy would be hard to find [1], and they carry a quadratic term in the doubled space of § 3.4. The doubled space's kernel is degenerate, so there is no gap above the ground value and Equation (2.2) gives no finite degree; as the degree grows, a filter returns the initial state's projection onto the kernel. A filter needs a unique ground state, which leaves the question *can the solution of a nonlinear equation be prepared as the unique ground state of a Hamiltonian, rather than selected by projection from a degenerate kernel, and does the reformulation's advantage survive there?*

Each extension that prepares is the same object, a coupled linear system on one state with one block per component: several fields directly, a complex field by realification as two real ones, and a polynomial nonlinearity by the Carleman lift, whose components are the field's monomials. The system is built first because the lift that answers the question is an instance of it; the doubled space, the one route that is not such a system, only projects. The constructions and proofs in full are in Appendix E.

## 6.1 Several Fields on One State

Consider a first-order system of $C$ fields with a constant coupling matrix $\Gamma$, $z_i' = \sum_j \Gamma_{ij} z_j$. A time interval $t \in [0, T]$ is carried on the Chebyshev axis $\xi = 2t/T - 1$, so $dz/d\xi = \frac{T}{2}\Gamma z$ and one ground state holds the whole trajectory, as in the Feynman–Kitaev clock of Barison et al. [29] and the space–time ground state that Pool et al. compute for Burgers' equation [22], both found variationally; here the time axis is a Chebyshev series and the ground state is prepared by a filter. The state holds one block of $N$ Chebyshev coefficients per field, the field index the most significant, on a register of $\lceil \log_2 C \rceil$ qubits; where $C$ is not a power of two, padding blocks decouple as the chain's do (Appendix E.1, with R7's rows). In the standard form (Definition 2) the state holds the fields alone and every derivative

is eliminated through the dense differentiation matrix $G$; the descriptor carries a system in the mass form of Equation (5.7). The equation rows are

$$\underbrace{I_C \otimes G_\xi - \tfrac{T}{2}\,\Gamma \otimes I_N}_{\text{standard form}}, \qquad \underbrace{I_C \otimes \tilde{D}_\xi - \tfrac{T}{2}\,\Gamma \otimes \tilde{B}_\xi}_{\text{descriptor (mass form)}}, \tag{6.1}$$

the second being the first multiplied on the left by $I_C \otimes \tilde{B}_\xi$, so by Theorem 1 their kernels coincide.

With the data rows of § 6.2, an initial-value problem (IVP) has a one-dimensional solution space (Proposition 25). Its solution is entire, so its residual on polynomials of degree below $N_t$ falls spectrally, and the rest of the discrete spectrum approximates the continuum operator's, which has no other kernel; the near-kernel is therefore expected to be one-dimensional in either form. No bound on the second singular value is proved, and the reference's trust test (§ 7.2) checks the near-kernel's dimension on every build.

The descriptor's chain could instead carry each field's derivative as a block, as it does for a field of higher order. Such a block doubles the state, needs the rescaling of § 5.4 and costs gap: on the harmonic oscillator $u' = v$, $v' = -\pi^2 u$ (R7) at $n = 8$, the mass form's gap is 0.132 and the block-rescaled chain's 0.0628, a ratio of 2.10, and between 2.10 and 2.13 from $n = 3$ upwards. The chain is kept only as a labelled ablation, and every system and lift of §§ 8.4 and 8.5 is compared in the mass form. A system of higher order enters by the usual reduction to first order, as R7 is $u'' = -\pi^2 u$ written as two fields; none of higher order than R7's was run.

## 6.2 Data for Several Fields: Ratio Rows

A single field's initial or boundary value is imposed by a homogeneous zero row, which leaves the field's scale free, and a regular datum, which fixes it (§ 4.3). Vector data $y = (y_1, \ldots, y_C)$ at a point $x^*$ need the same separation. With an anchor component $a$, $y_a \neq 0$, and $y_\perp = \sum_{c \neq a} y_c \,|c\rangle$ the rest of the data, the *ratio row* on the field register is

$$\mathbb{F} = y_a\big(I - |a\rangle\,\langle a|\big) - y_\perp\,\langle a|\,, \qquad (\mathbb{F}v)_c = y_a v_c - y_c v_a \;\; (c \neq a), \quad (\mathbb{F}v)_a = 0, \tag{6.2}$$

and the data rows are $(\mathbb{F}/y_a) \otimes \mathbb{B}(x^*)$. The kernel of $\mathbb{F}$ is exactly the line through $y$, so the rows impose $z_c(x^*) = (y_c/y_a)\, z_a(x^*)$: the direction of the data, with their scale free. The regular datum on the anchor fixes the scale, and a component whose datum is zero is held to zero exactly. Divided by $y_a$, the rows' non-zero singular values are $\|y\|/|y_a|$ and, for $C > 2$, one, each times the zero row's norm, so the anchor with the largest $|y_a|$ conditions them best (Proposition 20; their three-term circuit is in Appendix E.1). A system's initial state weights each field's block by the stated data $w_0/\|w_0\|$ (§ 7.4); equal weights would cancel exactly on the Taylor–Green vortex of § 6.7, whose two modes carry data of opposite sign.

## 6.3 Complex-Valued Equations

The Fourier bases that led Wu et al. to set complex-valued states aside are carried here on real bases. The fluids of § 6.7 use sine modes for Burgers' odd solution, and $\cos(\mathbf{k}\cdot\mathbf{x})$ and $\sin(\mathbf{k}\cdot\mathbf{x})$ for the vorticity, an isometric real embedding of the complex Fourier variables in which the truncated system is real (Appendix E.1.2). In these problems $k$ denotes a wavenumber and $\mathbf{k}$ a wavevector; elsewhere $k$ is an equation's order. A complex residual would not need a different filter either. With complex coefficients, data or sources the stacked residual $R$ is complex, yet $H = R^\dagger R$ is Hermitian and positive semi-definite with the smallest right singular vector of $R$ as its ground state, so the reflection and filter of § 7 act on it as before; the non-Hermitian operator is $R$, which the filter never sees. This is a statement about the formulation, since every residual prepared here is real.

A complex linear equation is carried instead by realification. Writing $z = u + \mathrm{i}v$, it becomes a real system of twice as many fields; for one mode $z' = cz$ with a complex coefficient $c$,

$$u' = \operatorname{Re} c\, u - \operatorname{Im} c\, v, \qquad v' = \operatorname{Im} c\, u + \operatorname{Re} c\, v, \tag{6.3}$$

a system of § 6.1 with one more field-register qubit. The map $z \mapsto (\operatorname{Re} z, \operatorname{Im} z)$ is a real-linear isomorphism that carries solutions to solutions, and real ratio rows on the realified data fix the phase with the direction, so an IVP keeps a one-dimensional solution space (Appendix E.1.2). R7 is the smallest instance: with $z = \pi u + \mathrm{i}v$, its equations are the real and imaginary parts of $z' = -\mathrm{i}\pi z$, a single-mode Schrödinger-type equation. The one natively complex encoding built is the tensor layout of the Carleman coupling (§ 6.6), whose LCU weights carry the phases $\mathrm{i}^{\sum_j a_j}$ of the Fourier symbols $\prod_j k_j^{a_j}$ it applies; it encodes the coupling alone and prepares no state. Native complex preparation would also need complex sources, data and coefficients in the circuits' loaders, which are not built, and is left to further work (§ 10.2).

## 6.4 The Doubled Space

Wu et al.'s doubled space is not a linear system on one state. We run it in the standard form only, the form in which they state it; the descriptor's route to a nonlinear equation is the Carleman lift, and Proposition 4 below applies to any Hamiltonian on the doubled space. Its encoding is exact, each product folded to one Chebyshev series by the nodal fold of § 6.6 and each linear term carried by a copy of the collapse row (Appendix E.2), and its kernel is large: the equation block has rank at most $2N$, the number of coefficients of the folded register, and each of the $m$ invariant constraints adds at most one, so

$$\dim \ker H_{\otimes} \;\geq\; N^2 - 2N - m, \tag{6.4}$$

above Wu et al.'s count of $2^{2n-1} = N^2/2$ from $n = 3$ for $m \leq 2$; the dimensions computed on R6 are in Table E.2.

**Proposition 4** (A filter projects onto the doubled kernel)**.** *Let $H_{\otimes} \succeq 0$ act on $\mathbb{R}^{N^2}$, $N \geq 2$, with $\Pi$ the projector onto its kernel.*

1. *The symmetric product states $\{u \otimes u\}$ do not form a linear subspace, and a kernel does. A kernel that contains two independent product states therefore contains states that are not products, and no Hamiltonian has the product states as its kernel.*

2. *Let $P_d$ be filters with $P_d(0) = 1$ whose largest magnitude on the rest of the spectrum tends to zero, as for the minimax filter as $d \to \infty$ or for $e^{-\beta H_{\otimes}}$ as $\beta \to \infty$. For any $\psi_{\mathrm{init}}$ with $\Pi\psi_{\mathrm{init}} \neq 0$,*

$$\lim_{d\to\infty} \frac{P_d(H_{\otimes})\,\psi_{\mathrm{init}}}{\left\|P_d(H_{\otimes})\,\psi_{\mathrm{init}}\right\|} = \frac{\Pi\,\psi_{\mathrm{init}}}{\left\|\Pi\,\psi_{\mathrm{init}}\right\|}. \tag{6.5}$$

3. *A penalty $P \succeq 0$ with $\ker(H_{\otimes} + P) = \operatorname{span}\{\psi_* \otimes \psi_*\}$ must vanish on $\psi_* \otimes \psi_*$ and on no other vector of $\ker H_{\otimes}$: a condition stated in terms of the solution $\psi_*$.*

The proof is given in Appendix E.3. A filter returns the initial state's projection onto a kernel that holds the solution's product state among exponentially many others, so $\psi_{\mathrm{init}}$ decides which state comes back, and a penalty that singles out the solution must be built from it, which makes it answer-dependent. Wu et al.'s selection of "a degenerate state that closely matches the analytical solution" is such a choice. We report the doubled space as a projection, never as a preparation.

## 6.5 The Carleman Lift in Space–Time

The Carleman lift of a polynomial system is a linear system on the system's monomials [4, 37, 38, 40] (§ 2.3), so everything in §§ 6.1 and 6.2 carries over: the components become monomials, the ratio rows carry the data's moments at $t = 0$, and $\Gamma$ becomes the lifted generator $\Gamma_K$ in Equation (6.1), in either form. For $u \in \mathbb{R}^M$,

$$\frac{du}{dt} = \sum_{p \geq 1} F_p \, u^{\otimes p} \quad \longrightarrow \quad \frac{dz}{dt} = \Gamma_K \, z, \qquad z = \big(u_\alpha\big)_{1 \leq |\alpha| \leq K}, \quad u_\alpha = \textstyle\prod_i u_i^{\alpha_i}, \tag{6.6}$$

where the state holds every monomial of degree 1 to $K$ once, level by level (the symmetric layout), and each row of $\Gamma_K$ differentiates a monomial and substitutes $u'$, dropping the terms beyond degree $K$ (Appendix E.4). The lift is carried in the scaled variable $u/s$, $s = \max(\|u_0\|, 1)$, a diagonal similarity of $\Gamma_K$ that keeps the data's moments on level $\ell$ from growing as $\|u_0\|^\ell$ (Proposition 22).

With $\lambda_1$ the eigenvalue of $F_1$ of largest real part and $\|F_2\|$ taken on symmetric tensors,

$$\rho = \frac{\|u_0\| \, \|F_2\|}{\big|\operatorname{Re} \lambda_1(F_1)\big|} \tag{6.7}$$

is the ratio of Liu et al., who prove for a dissipative quadratic system that $\rho < 1$ suffices for the truncation error at order $K$ to fall as $\mathcal{O}(\rho^K)$, uniformly in time [4]. The condition is sufficient and a worst case over all data of norm $\|u_0\|$; nothing in the bound says the lift diverges at $\rho \geq 1$, and other analyses give conditions of other forms: explicit time-dependent error bounds [73], a milder ratio in the $\ell^\infty$ norm for reaction–diffusion equations [74], and exponential convergence over all time for a class of stable systems [75]. For $y' = -y + y^2$ from $y(0) = \frac{1}{2}$ (R8), $\rho = \frac{1}{2}$; on Burgers it grows with the number of modes (§ 6.7). The convergence is therefore measured on every build, order by order against the exact solution of the truncated lift (§ 8.4).

The truncated lift is a linear IVP with its data fixed up to scale, so the argument of § 6.1 applies to it unchanged: its near-kernel is expected to be one-dimensional in both forms and for every $\Gamma_K$, including a lift whose truncation diverges (Appendix E.5). The uniqueness is the IVP's, which the lift carries; neither form contributes it. The decoded field's error splits accordingly,

$$\|\hat{u} - u\|_\infty \;\leq\; \underbrace{\big\|z_1^{(K)} - u\big\|_\infty}_{\text{the lift's}} + \underbrace{\big\|\hat{u} - z_1^{(K)}\big\|_\infty}_{\text{the time axis's}}, \tag{6.8}$$

with $z_1^{(K)}$ the first level of the truncated lift's exact solution. The lift's part does not depend on $n_t$ and the time axis's falls spectrally to rounding, so the field's error falls with $n_t$ until it meets the lift's floor and then holds; every build records both parts.

## 6.6 Couplings and Products on a Circuit

Everything in a system's or a lift's rows except its coupling is built from the bands and rank-one rows of § 5.5; the ratio rows are three LCU terms (Appendix E.1).

**The coupling as a sum of matchings.** A system's coupling $\Gamma$, or a lift's $\Gamma_K$ in the symmetric layout, is sparse and irregular. We split it into *matchings*, each with at most one entry in every row and column: a permutation of the field register's basis states followed by a diagonal of rotations on one flag qubit. Each is encoded at its largest entry, so the coupling's subnormalisation is the sum of those entries; the split is a greedy edge colouring (Appendix E.1.1). With $\delta$ the most entries in any row or column of the coupling and $\|\Gamma\|_{\max}$ its largest entry in magnitude, the colouring uses at most $2\delta - 1$ matchings, so $\alpha \leq (2\delta - 1)\|\Gamma\|_{\max}$, within a factor of two of the $\delta\|\Gamma\|_{\max}$ of a sparse-access encoding [41] and without that encoding's oracles, which would have to compute the position of each

row's entries from the symmetric layout's irregular monomial index; the alternative weighed was the expansion in Pauli words that encodes the other field-register matrices, which pays for every word with a non-zero coefficient. The factor is the same in both forms, so they pay the same for the coupling. On Burgers' lift on sixteen modes at $K = 3$ the matchings encode $\Gamma_K$ at $\alpha/\|\Gamma_K\| = 1.34$; on the Navier–Stokes lifts the ratio is larger, 7.34 on eighty modes at $K = 2$ (Table E.1).

**The tensor layout.** The number of matchings grows with $\Gamma_K$'s entries, and so with the number of modes. For a Fourier–Galerkin equation, whose linear operators are Fourier multipliers and whose products are convolutions, a second layout carries level $\ell$ as the whole tensor $u^{\otimes \ell}$ in $\ell$ slots of wavevectors, and the coupling is a sum of terms "$F_p$ on slots $j, \dots, j + p - 1$", $2K - 1$ of them for a quadratic field whatever the number of modes (Proposition 23). Its weights carry the complex phases of § 6.3, and its cost per query against the matchings' is reported in § 8.5.

**Products.** A fold transforms to node values, multiplies pointwise and transforms back. The doubled space's nodal fold does this with the Chebyshev–Gauss transform $U_{\mathrm{CG}}$, exactly and at $\alpha = 2$, within a factor 1.27 of its norm at $n = 2$ and 1.20 at $n = 12$: tight to a constant factor, and not optimal (Proposition 26). $U_{\mathrm{CG}}$ is the discrete cosine transform, for which quantum circuits with polylogarithmically many gates are known [76]; it is budgeted as a cited primitive and not built gate by gate, because the fold serves only the doubled space, which prepares no state. The tensor layout's Fourier fold uses the built quantum Fourier transform (QFT), at $\alpha = N^{1/2}$ per axis, the convolution's own norm.

## 6.7 Burgers and Navier–Stokes

The lift is tested on two periodic flows whose quadratic terms are convolutions. Truncated to finitely many Fourier modes on a real basis (§ 6.3) they are polynomial systems of the form of Equation (6.6). A build's error then has a third part beside the two of Equation (6.8), the truncation's, measured against the equation itself, while the lift's is measured against the Galerkin system's IVP. The systems, their couplings and the parameters of every run are given in Appendix E.7.

**Viscous Burgers (R9).** On $x \in [0, 2\pi)$,

$$u_t + u\, u_x = \nu\, u_{xx}, \qquad u(x, 0) = a \sin x, \tag{6.9}$$

whose solution stays odd and lives on the sine modes $\sin kx$, $k \le M$. The Galerkin system has $F_1 = \mathrm{diag}(-\nu k^2)$ and $F_2$ the projection of $-\frac{1}{2}(u^2)_x$. The least damped mode is $k = 1$, so $\rho = a\|F_2\|/\nu$, which grows with $M$ through $\|F_2\|$. The runs vary the strength $a/\nu$ and the number of modes $M$, and the Cole–Hopf solution [77, 78] measures the truncation's error.

**Two-dimensional Navier–Stokes (R10).** In vorticity form on the periodic square,

$$\omega_t + J(\psi, \omega) = \nu\, \nabla^2 \omega, \qquad \nabla^2 \psi = -\omega \ \text{ on } [0, 2\pi)^2, \qquad J(\psi, \omega) = \psi_x \omega_y - \psi_y \omega_x, \tag{6.10}$$

with $\nu = 1$, the vorticity on the real modes $\cos(\mathbf{k}\cdot\mathbf{x})$ and $\sin(\mathbf{k}\cdot\mathbf{x})$ of the half-plane with $0 < |\mathbf{k}|^2 \le k_c^2$, the stream function eliminated as $\hat{\psi}_{\mathbf{k}} = \hat{\omega}_{\mathbf{k}}/|\mathbf{k}|^2$, and every coupling rational and exact. Modes of equal $|\mathbf{k}|$ do not interact, and without viscosity the truncation conserves the energy and the enstrophy (Appendix E.7); a pseudo-spectral direct numerical simulation (DNS) measures the truncation's error. The Taylor–Green vortex $\omega_0 = 2 \sin x \sin y$ [79] occupies two modes of equal $|\mathbf{k}|$, so $\omega = \omega_0 e^{-2\nu t}$ exactly and its lift is exact at every order (Proposition 24): R10a tests the whole chain at a known answer and leaves the lift's convergence untested. A counter-rotating vortex pair of norm $\|w_0\| = a$ spans several $|\mathbf{k}|$, so its triads are active and its lift is tested.

## 6.8 Preparation Versus Projection

**Table 6.1: The Carleman lift prepares; the doubled space projects.** Each nonlinear route by the forms it is run in, its kernel, what a filter returns from an initial state $\psi_{\text{init}}$, and what a trusted kernel (one-dimensional, gap ratio above 20) certifies. $N = 2^n$; $m$, number of invariant constraints; IVP, initial-value problem. The lift's convergence is measured separately on every build.

| Route | Forms run | Kernel | A filter returns | A trusted kernel certifies |
|---|---|---|---|---|
| Doubled space (Wu et al.; R6) | standard only | degenerate, dimension $\geq N^2 - 2N - m$ (Equation (6.4)) | the projection of $\psi_{\text{init}}$ onto the kernel | nothing: the kernel is never one-dimensional, so never trusted |
| Carleman lift (R8–R10) | both | one-dimensional near-kernel, from the lifted IVP's uniqueness | the truncated lift's solution | the lifted *linear* problem, not the lift's convergence |

Table 6.1 sets the two routes side by side. Its last column carries the caveat: a trusted kernel certifies that the residual singles out the truncated lift's solution, whose distance from the nonlinear solution is the first term of Equation (6.8), measured separately. Past the convergence boundary of Burgers' lift the kernel stays trusted while the lift diverges (§ 8.4).

What the descriptor adds is the time axis. At one Carleman order the two forms differ only by the left factor $I \otimes \tilde{B}_\xi$ (Equation (6.1)), so the time axis contributes $\|\tilde{D}_\xi\| = 2(N_t - 1)$ to the mass form's norm against $\|G_\xi\| = \Theta(N_t^2)$ to the standard form's, while the coupling's term, of norm $\frac{T}{2}\|\Gamma_K\|$ in the standard form and $\|\tilde{B}_\xi\|_2 \in [2, 2+\sqrt{2}]$ times that in the mass form (§ 5.1), is of the same order in both and grows with the modes. The truncated lift is a regular first-order system (Definition 4), the class for which Equation (5.13) predicts $d = \tilde{\Theta}(N_t)$ at a fixed truncation, so the mass form's degree should grow more slowly with $N_t$; at fixed $n_t$ the advantage should narrow as the truncation widens and the coupling's term comes to dominate both norms.

For polynomial nonlinearities, then, a Hamiltonian with a unique ground state exists in either form, the Carleman lift's, which a filter prepares where the doubled space only projects; a preparation on a lift has two tests, the kernel's trust and the lift's convergence, and the reformulation's advantage survives on it through the time axis alone, expected to narrow as the truncation widens. The pipeline of § 7 turns the residuals of §§ 4 and 5 and this chapter into prepared states on equal terms, and §§ 8.4 and 8.5 test both halves of this answer, from the scalar R8 to two-dimensional Navier–Stokes.

# 7 Preparing and Verifying the Ground State

§§ 4 to 6 each end with a residual and a prediction: the filter's degree grows as $\sqrt{\alpha_H/\Delta}$ (Equation (2.2)), with a norm and a gap that each chapter derived for its own form. *Can both terms of that ratio be measured the same way for both forms?* Only if each form is judged against the same classical reference, filtered to the same suppression from the same answer-independent initial state, and costed with the encoder's share of $\alpha_H$ kept apart from the Hamiltonian's. One pipeline therefore serves every residual (Figure 7.1). The two forms enter only through the residual they hand to it, so a difference measured between them in § 8 is a difference between their Hamiltonians $H = R^\dagger R$ or between their encodings, which the design of the experiment separates (§ 7.7). The statements hold for any stacked residual, so $R$, $H$ and $\alpha_R$ carry no form subscript here.

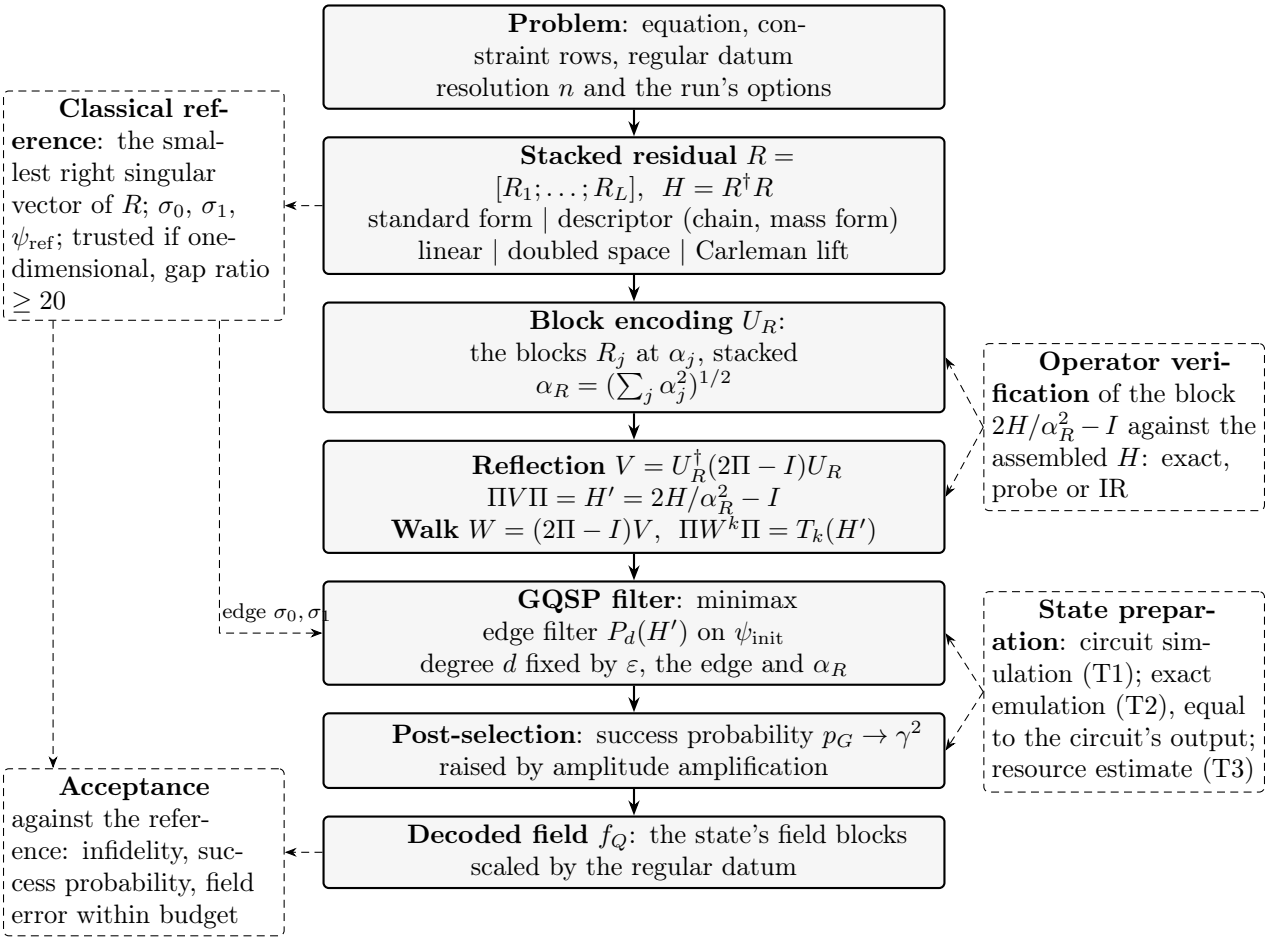



**Figure 7.1: One pipeline serves both forms, from a problem's data to a decoded field.** Centre (solid arrows): the forms differ only in the stacked residual $R$, block-encoded at subnormalisation $\alpha_R$; the reflection $V$ encodes $H' = 2H/\alpha_R^2 - I$ and the walk $W$ its Chebyshev polynomials; a GQSP circuit applies the minimax edge filter $P_d$, its degree fixed before the run, to an answer-independent $\psi_{\text{init}}$ of overlap $\gamma$ with the ground state; post-selection succeeds with probability $p_G \to \gamma^2$; the field $f_Q$ is decoded classically. Left (dashed): the classical reference $\psi_{\text{ref}}$, the smallest right singular vector of $R$, whose two least singular values $\sigma_0, \sigma_1$ fix the degree and against which every run is accepted. Right (dashed): operator verification and the three tiers of state preparation. $\Pi$ projects the ancilla onto zero; $\varepsilon$ is the target suppression; IR compares compiled terms with the assembled operator. Every result is a classical computation; nothing ran on hardware.

## 7.1 Problems and Model Inputs

A problem is data: its fields and axes; its equation, as a sum of terms, each a coefficient function, a derivative order on an axis, a field and possibly a source; its constraint rows; its regular datum; and its reference solution. Nothing downstream depends on a problem except through these. The benchmark problems R1–R10 of Table 4.1 are given in this way, each with its equation, reference, constraint rows, resolutions and parameters in Appendix H; the panels as printed are run only to test the published method (Appendix I). The data enter as rows of the four kinds of Definition 2, the model's boundary and initial conditions, each the same in both forms and listed with the problems that use them in Table H.3. Two kinds of datum are set up from the reference solution when the problem is built, for the reasons of § 4.3: the exact constraint points of R4 and R6b, and the heat equation's initial function at $t = -1$. Every other datum is the problem's statement.

A run is a problem at a resolution with a set of options: the form and, for the descriptor, its layout and block rescaling; the encoding of $G$; the treatment of a nonlinearity; the filter; the target $\varepsilon = 10^{-6}$; and the initial state. Their values and defaults are listed in Table H.4 (Appendix H.3). The defaults are the configuration every comparison of § 8 uses; the alternatives are controls, labelled ablations, or the printed choices kept to test the published method.

## 7.2 The Classical Reference

Every result is judged against a classical reference: the ground state of $H$, computed from $R$ and never from $H$, a departure from Wu et al.'s eigenvector of the formed Hamiltonian (Table 4.2). With the SVD $R = \sum_j \sigma_j \, u_j v_j^\dagger$, $\sigma_0 \le \sigma_1 \le \cdots \le \sigma_{\max}$, the reference is $|\psi_{\text{ref}}\rangle = v_0 = \arg\min_{\|\psi\|=1} \|R\psi\|_2$, the eigenvalues of $H$ are $\lambda_j = \sigma_j^2$ and its gap is $\Delta = \sigma_1^2 - \sigma_0^2$. Working from $R$ keeps its conditioning unsquared. A backward-stable SVD of $R$ resolves the ground state to about $\epsilon_{\text{mach}}\sigma_{\max}/(\sigma_1 - \sigma_0)$ by Wedin's theorem [80], an eigensolver on the formed $H$ only to about $\epsilon_{\text{mach}}\sigma_{\max}^2/(\sigma_1^2 - \sigma_0^2)$ by the Davis–Kahan theorem [81], larger by about the residual's effective condition number $\sigma_{\max}/\sigma_1$ and above one once the relative gap $\Delta/\|H\|$ falls below $\epsilon_{\text{mach}}$, as the standard form's does at moderate $n$ (§ 5.7, proposition 35, and appendix G.2). The dense SVD is used up to 8,192 physical columns, the descriptor's decoupled padding excluded (Proposition 15); beyond, shift-invert Lanczos iteration on an augmented system that never forms $R^\dagger R$ gives the smallest singular triplets, with an a posteriori bound on the ground state's error (Appendix G.3).

A computed spectrum always has a smallest value, and the reference must decide whether it is a kernel. The kernel is placed at the largest ratio $\sigma_k/\sigma_{k-1}$ among the lowest singular values, and the reference is trusted (§ 4.3) when that kernel is one-dimensional with gap ratio at least 20, an empirical threshold set above every ordinary spacing in the benchmark problems' spectra (Appendix G.4). Trust detects a kernel; the field's convergence is measured separately. A preparation is judged only where its reference is trusted, at three levels: the operator (each encoding against the assembled operator, § 7.5), the state (the infidelity $1 - |\langle\psi_{\text{ref}}|\psi\rangle|^2$ in the form's own space) and the field (the $L^\infty$ and relative $L^2$ errors of the decoded field against the reference solution on a uniform grid, § 7.6).

## 7.3 Block Encoding and the Quantum Walk

Each block $R_j$ of the residual, the equation's rows and each constraint's, is block-encoded by its form's circuits at a subnormalisation $\alpha_j$ (§ 4.5.3 and §§ 5.5 and 6.6). The blocks are stacked rather than summed: a row-select register of $\lceil \log_2 L \rceil$ qubits is prepared in $\sum_j (\alpha_j/\alpha_R)\,|j\rangle$ and $U_j$ is applied when it holds $j$, and unlike the selector of a LCU the register is never returned by an inverse preparation, so that

$$\big(\langle j| \otimes \langle 0_a|\big)\, U_R \,\big(|0\rangle \otimes |0_a\rangle\big) = \frac{R_j}{\alpha_R}, \qquad \alpha_R = \Big(\sum_{j=1}^{L} \alpha_j^2\Big)^{1/2}, \tag{7.1}$$

with $a$ the blocks' shared ancilla and the row select read as part of the output. The subnormalisations add in quadrature, $\alpha_R = \|(\alpha_j)\|_2$, the form of the bound $\|R\|_2 \le (\sum_j \|R_j\|_2^2)^{1/2}$ itself, so a stack of tight blocks is loose only by as much as $\|R\|$ falls below that bound. A reflection then turns the encoding of $R$ into one of $H$: the standard construction of $A^\dagger A$ from a block encoding of $A$, which underlies qubitisation and the QSVT [20, 41], applied here to a stacked residual and verified on the whole Hamiltonian it encodes (§ 7.5).

**Proposition 5** (Stacked-residual reflection)**.** *Let $U_R$ satisfy Equation* (7.1)*, let $\Pi_a$ project the shared ancilla onto $|0_a\rangle$, and let $\Pi_{\text{in}}$ project the shared ancilla and the row select onto zero. Then*

$$V = U_R^\dagger \big(2\Pi_a - I\big)\, U_R \quad \textit{satisfies} \quad V^\dagger = V, \quad V^2 = I, \quad \Pi_{\text{in}}\, V\, \Pi_{\text{in}} = \frac{2H}{\alpha_R^2} - I \equiv H', \tag{7.2}$$

*with $H = R^\dagger R$, and $V$ controlled on further qubits is $U_R^\dagger\, C(2\Pi_a - I)\, U_R$: only the reflection carries the control.*

The proof is given in Appendix F.1. The reflection block-encodes $H'$ at subnormalisation one and $H$ at $\alpha_H = \alpha_R^2$, at the cost of the stack, its inverse and one multi-controlled $Z$ gate. Its spectrum needs no rescaling: $H \succeq 0$ and $\|H\| = \|R\|^2 \le \alpha_R^2$, so $H'$ has its spectrum in $[-1, 1]$, its eigenvectors

are the right singular vectors of $R$, and its ground value sits at $x_0 = -1 + 2\sigma_0^2/\alpha_R^2$, at the edge of the interval when the kernel is exact. Its gap is $2\Delta/\alpha_R^2$: the encoding's subnormalisation divides the gap the filter must resolve, which is how the encoder's cost of § 4 enters the degree. Qubitisation [20] turns the reflection into a walk whose powers are Chebyshev polynomials of $H'$ (Appendix F.2),

$$W = \big(2\Pi_{\text{in}} - I\big)\, V, \qquad \Pi_{\text{in}}\, W^k\, \Pi_{\text{in}} = T_k(H'), \tag{7.3}$$

for one power of $W$ followed by one projection; controlling $W$ controls only its two reflections (Appendix F.3).

## 7.4 The Filter and Its Parameters

A filter is a polynomial $P$ of $H'$. GQSP [47] applies $d$ walks, each controlled on one signal qubit, between single-qubit rotations of that qubit, with no parity constraint (§ 2.4). On the signal's zero, restricted once by $\Pi_{\text{in}}$, the circuit block-encodes

$$\Pi_{\text{in}}\, P(W)\, \Pi_{\text{in}} = \sum_{k=0}^{d} q_k\, T_k(H'), \qquad P(z) = \sum_{k=0}^{d} q_k z^k, \qquad \sup_{|z|=1} |P(z)| \le 1, \tag{7.4}$$

by Equation (7.3), so a filter given by its Chebyshev coefficients $q_k$ is realised by the analytic polynomial with the same coefficients. The phases are computed from $P$ alone, and the polynomial they realise is checked against $P$ (Appendix F.7).

**The minimax edge filter.** The filter that prepares every comparison and extension is the polynomial of degree $d$ that is smallest on the excited spectrum relative to its value at the ground state: the Chebyshev polynomial of that spectrum, optimal by Chebyshev's inequality (§ 2.4). With $x_1$ the first excited value of $H'$ and $w$ the affine map of $[x_1, 1]$ onto $[-1, 1]$,

$$\begin{aligned} P_d(x) &= \frac{T_d\big(w(x)\big)}{T_d\big(w(-1)\big)}, \qquad w(x) = \frac{2x - (x_1 + 1)}{1 - x_1}, \\ \sup_{x \in [x_1, 1]} \frac{|P_d(x)|}{|P_d(x_0)|} &= \frac{1}{\cosh(d\theta_0)}, \qquad \theta_0 = \operatorname{arccosh}\big(-w(x_0)\big). \end{aligned} \tag{7.5}$$

Its Chebyshev coefficients alternate in sign and sum in magnitude to one, so Equation (7.4) realises it at full weight (Proposition 30); they are computed by exact interpolation, so nothing is truncated. We take the least degree that suppresses the excited spectrum to $\varepsilon$ relative to the ground state,

$$d = \left\lceil \frac{\operatorname{arccosh}(1/\varepsilon)}{\theta_0} \right\rceil \approx \frac{\alpha_R}{2\sqrt{\Delta}}\, \ln \frac{2}{\varepsilon}, \tag{7.6}$$

an instance of Equation (2.2). The approximation holds for $\sigma_1 \ll \alpha_R$, where $x_0$ and $x_1$ round to $-1$ in floating point, so $\theta_0$ is evaluated from the heights $1 + x_j = 2\sigma_j^2/\alpha_R^2$ above the edge (Appendix F.4). For an initial state of overlap $\gamma$ with a one-dimensional target, the prepared state's infidelity is at most $(1 - \gamma^2)\varepsilon^2/\gamma^2$ (Proposition 29).

**The parameters are a priori.** The degree needs $\varepsilon$, the circuit's own $\alpha_R$ and the spectrum's edge, $\sigma_0$ and $\sigma_1$ from the classical reference; it never reads the solution. The edge is assumed known, the standard assumption of filtering methods (§ 2.4), and is given exactly to both forms; finding it on the circuit, or replacing it by a lower bound on the gap at a degree proportional to the bound's inverse square root, is beyond our scope (§ 9.5). Wu et al.'s time rule, Equation (3.6), contains no $\varepsilon$ and falls short of it (Table 4.2); it is kept only as a tested variant (Appendix I.5).

**Imaginary time.** Wu et al.'s filter, imaginary time $e^{-\beta(1+x)}$ in the variable of $H'$, has modified Bessel functions for its Chebyshev coefficients, which also sum in magnitude to one, and a degree with the same factor $\alpha_R/\sqrt{\Delta}$, larger than the minimax filter's by a constant factor at equal suppression (Appendix F.5 and proposition 30). It runs only in the filter study, which measures that factor in both forms (§ 8.1), and in the reproduction of Wu et al. (Appendix I).

**The initial state.** The filter acts on the geometric product state,

$$|\psi_{\text{init}}\rangle \propto \bigotimes_{j=0}^{n-1}\Big(|0\rangle + r^{2^j}\,|1\rangle\Big) = \sum_{k=0}^{N-1} r^k\,|k\rangle\,, \qquad r = \pm\tfrac{1}{2}, \qquad \gamma = \big|\langle\psi_{\text{ref}}|\psi_{\text{init}}\rangle\big|, \tag{7.7}$$

on the field block and zero on every other block: $n$ single-qubit rotations and no entangling gate. Its rate and sign pattern guess the geometric decay of a smooth function's Chebyshev coefficients; neither is fitted. Both signs of $r$ are run and the one with the larger measured success probability is kept, a rule that needs only what a quantum computer observes. A system's fields each carry the state weighted by the problem's stated initial data, $w_0/\|w_0\|$ (§ 6.2). In a Carleman lift the fields are the level-one blocks, so the state sits there, and every higher level starts at zero, as the descriptor's derivative blocks and padding do. The uniform state and a seeded random state are controls, and $\gamma$ is reported for every run. Where the ground state has weight outside the field block, in the descriptor's derivative blocks or a lift's higher levels, $\gamma$ is at most the square root of the field block's weight (§ 5.3).

**Success and total cost.** The post-selection succeeds with probability $p_G = \|P_d(H')\psi_{\text{init}}\|^2 \geq \gamma^2 P_d(x_0)^2$, which tends to $\gamma^2$ as the ground value approaches the edge. Amplitude amplification [56] with $\lfloor \pi/(4\arcsin\gamma)\rfloor$ rounds raises it to at least one half, so with $g(W)$ the two-qubit gate count of one walk a preparation costs

$$\text{queries} = (2k_{\text{aa}}+1)\,d, \qquad \text{gates} = (2k_{\text{aa}}+1)\,d\,g(W), \qquad k_{\text{aa}} = \Big\lfloor \frac{\pi}{4\arcsin\gamma}\Big\rfloor, \tag{7.8}$$

linear in $1/\gamma$ for small $\gamma$. The count takes $\gamma$ from the classical reference, as the degree takes the edge, and leaves out the one-qubit rotations, the initial state, the amplification's reflections and the second sign of $r$ (Appendix F.8). The departures of this section from Wu et al.'s filter, phases, normalisation and initial state are collected with the others in Table 4.2.

## 7.5 Verification and the Error Budget

Two questions are kept apart: whether each circuit encodes the operator it claims to, and whether the filter prepares the ground state. The first, *operator verification*, targets the reflection's whole block $2H/\alpha_R^2 - I$, with $H$ formed from the residual assembled from the problem, never from the compiler that produced the circuit, and runs in three modes (Table G.1). The *exact* mode reconstructs the block, one statevector evolution per input column, and certifies the circuit completely at that size. The *probe* mode compares the circuit's action on twenty structured and Haar-random vectors with sparse products of the target; it estimates the root-mean-square error without bounding the worst entry (Proposition 34), and covers every circuit of at most 24 qubits that the exact mode does not. Where these two modes reach, the composed Hamiltonian is verified whole. The *intermediate representation (IR)* mode sums the compiled terms of a lift or coupling into a sparse matrix and compares it with the assembled operator at any size; the two sides are independent constructions from the problem, so the comparison is not a tautology, and the exact and probe modes close the step from terms to gates where a circuit can be simulated. Wider circuits are recorded for their subnormalisation and gate counts alone.

The second question is *state-preparation verification*, at three tiers. *Circuit simulation* (T1) runs the GQSP circuit on a statevector simulator, one controlled walk at a time. *Exact emulation* (T2) computes $\sum_k q_k T_k(H')\psi_{\text{init}}$ by the Chebyshev three-term recurrence with $H'v = 2R^\dagger(Rv)/\alpha_R^2 - v$, two sparse products a step and $H$ never formed, up to degree $2^{20}$. A *resource estimate* (T3) takes $\alpha_R$ and the walk's gate count from the circuit, the edge and $\gamma$ from the reference and the degree from Equation (7.6), and prepares nothing.

By Proposition 5, Equation (7.3) and Equation (7.4), the circuit's post-selected output is $\sum_k q_k T_k(H')\psi_{\text{init}}$, which depends on the encoding only through $\Pi_{\text{in}} V \Pi_{\text{in}} = H'$; two encodings with the same $H'$ give the same output for every filter. T2 therefore computes the circuit's output exactly, up to rounding, for the circuit's own $\alpha_R$, and T1 tests only that the circuit as built implements it, wherever both run. Each circuit simulation is preceded by an estimate of its memory and time, and a run beyond its budget is recorded with the estimate and not attempted (Appendix G.5). Circuit simulation therefore covers small widths only, a limit of the simulation budget and not of the method: a circuit that implements $V$ outputs T2's state at any size.

Each preparation is accepted against bars at every level (Table G.3): the state's infidelity at most $\varepsilon$ and within ten times its a priori bound, the circuit against its emulation within $10^{-10}$ before normalising, and the decoded field within the reference's own field error plus a budget for the prepared state's. The budget splits the state's distance from the reference into the filter's, the truncation's and the realised phases' terms, and carries their sum to the field by an exact readout bound (Equation (G.2)). Every encoding is exact and the field is decoded classically, so no block-encoding or sampling term enters.

## 7.6 Reading Out the Solution

The field is read out classically: it is decoded from the prepared state's field blocks and scaled by the regular datum. With $\ell_x(\psi) = \langle \tau(x) | D\psi \rangle$ the field's value functional, $D$ the map from the state to the field's coefficients (the field block; for a lift, its first level times the scale $s$),

$$f_Q(x) = f(x_s)\,\frac{\ell_x(\psi)}{\ell_{x_s}(\psi)}, \qquad \eta = \Big(\frac{f(x_s)}{\ell_{x_s}(\psi)}\Big)^2, \tag{7.9}$$

so the state's global phase and norm cancel and the decoded field carries the sign of the datum; $\eta$ is the linear model's scale of § 4.1. Wu et al. read the field out point by point instead, by an interferometric measurement (§ 3.3) whose identity assumes a deterministic preparation: after post-selection with probability $p_G < 1$ the field, a ratio, is unaffected and the scale is wrong by the factor $p_G$ (Equation (F.6)). The protocol is built and verified with this correction on R1, R2a and R5a at small $n$; its study at scale is further work (§ 10.2), not the method's readout.

## 7.7 Design of the Numerical Experiment

The study varies the resolution ($n$ per axis, $n_t$, the modes $M$ and the Carleman order $K$), the form, the encoding of $G$, the filter, the target $\varepsilon$ and the initial state, one at a time from the defaults of Table H.4. Within a comparison of the forms everything else is fixed: the problem and its constraint rows, $\varepsilon = 10^{-6}$, the minimax filter, the geometric initial state on the field block and the readout.

**Cost conventions.** A query is one application of the walk $W$; a preparation costs $d$ queries and an amplified one $(2k_{\text{aa}} + 1)d$ (Equation (7.8)). Gates per query are the two-qubit gates of one walk after transpiling to $\{\text{CX}, U\}$ [82], with one-qubit gates counted separately and dense primitives that are cited rather than built, the doubled space's discrete cosine transform, counted apart. Ancilla are counted on the composed reflection $V$: the row select, every block's ancilla and, in the descriptor's chain, nothing for the block register, which is part of the state.

**Fairness.** The standard form is costed with its structured encoding, the cheapest exact encoding of it built (§ 4.5.2), which is looser than the descriptor's (§§ 4.6 and 5.6). The degree is linear in $\alpha_R$ (Equation (2.2)), so an as-built comparison favours the descriptor by about the ratio of the two encodings' looseness, a factor of the encodings and not of the Hamiltonians. Every comparison is therefore quoted twice: as built, and at the ideal-encoding bound $\alpha_R = \|R\|_2$, the least degree the minimax filter needs through any encoding of the residual (Proposition 32). The difference between a form's two degrees is its encoder's cost; a difference between the forms at the ideal bound is a difference between their Hamiltonians, and only that difference says whether the cost is the Hamiltonian's and whether reformulating the residual lowers it. Where the order of the forms reverses at the ideal bound, the reversal is reported as a finding (§ 8.2).

**Tiers and fits.** Every number states its tier. Degrees beyond the emulation's reach are resource estimates: exact for this filter, given the spectrum and the circuit's $\alpha_R$, but with no state prepared, and compared with T2 where both exist. Every resource estimate is computed at its own resolution, from that resolution's circuit and reference, and none is extrapolated; the growth of a degree with $N$ is quoted as a measured exponent between stated resolutions.

## 7.8 Software and Computation

The pipeline is implemented in `pihm`, a Python package we wrote, on NumPy [83] and SciPy [84] for the classical linear algebra and Qiskit and Qiskit Aer for the circuits and their simulation [82]. Each concept has one implementation, shared by both forms and every problem, so a difference between the forms cannot come from different code for the same step. Each claim the thesis makes has a test that recomputes it. The production runs were made on the Setonix supercomputer of the Pawsey Supercomputing Research Centre [85] at one tagged version of the code, and every number and figure is regenerated from the archived records by one command. Its architecture, records and commands are given in Appendix J.

Within this pipeline the forms differ in their residual alone, and a comparison read at the ideal-encoding bound compares their Hamiltonians. Two things rest on assumption or argument: the spectral edge, supplied by the classical reference rather than found on the circuit, and the circuit's output beyond small widths, which the exact emulation supplies by the argument of § 7.5. § 8.1 tests the pipeline against its reference, its emulation and its error budget before § 8.2 measures the degree's two terms in both forms.

# 8 Numerical Study

The formulation chapters made their claims about operators. For constant coefficients the standard form's norm grows as $N^{4k}$ against a flat gap; the descriptor's falls to $\Theta(N^2)$ at fixed block weights, at the price of a gap its derivative blocks erode; and a Carleman lift prepares where the doubled space projects. Each claim predicts a filter degree through Equation (2.2), and the measurements below ask *whether the measured degrees follow those laws, and on which problems.* They use the benchmark problems of Table 4.1 with the pipeline and conventions of § 7, which is validated before any cost is compared. Every result is a classical computation and names its tier, T1, T2 or T3 (§ 7.5); a gap, a norm or a reference's error is an exact classical computation on the assembled residual. Every problem in both forms is tabulated in Appendix H.4, and the reproduction of Wu et al.'s panels, which tests the published method rather than our questions, in Appendix I.

## 8.1 Validation of the Pipeline

Every number in this chapter is read from one production of 3155 runs, 494 hours of computation in all, each of which left a record of its configuration, its outcome and the source code that produced it (Appendix J). Of these, 3144 completed; the other 11, all circuit simulations, were refused by their own estimates of time and recorded without being attempted. The checks of this section run over those records.

Solved classically through its reference (§ 7.2), the standard form converges spectrally on every linear benchmark problem: the smallest trusted error of the decoded field is $3.2 \times 10^{-13}$ on R2a at $n = 5$, $8.5 \times 10^{-12}$ on the inhomogeneous R4b at $n = 5$, and $4.6 \times 10^{-12}$ on the heat equation at $n = 5$ per axis. The descriptor, whose kernel is the standard form's (Theorem 1), reaches $3.3 \times 10^{-15}$ on R2a. Beyond its best resolution the standard form's error rises again, to $1.4 \times 10^{-6}$ on R2a at $n = 10$, where the descriptor's is $3.7 \times 10^{-13}$. The rise is consistent with rounding error amplified by the conditioning of a residual whose norm grows as $N^4$, an early sign of the Hamiltonian's cost of § 8.2.

**The circuits encode the operators.** All 252 verifications, 121 exact, 91 by probe and 40 by IR, meet their mode's tolerance (Table G.1) by at least four orders of magnitude, the least margin a factor of $1.1 \times 10^4$ (Figure G.1). The exact mode's worst entrywise error is $5.9 \times 10^{-15}$, and its widest verification, a whole reflection of the reproduction on 25 qubits, is accurate to $1.8 \times 10^{-15}$. The probe mode's worst is $9.1 \times 10^{-14}$, on a Navier–Stokes lift's reflection of 24 qubits, and the IR mode verifies the tensor layout's coupling on 316 modes to $3.6 \times 10^{-17}$. Every composed Hamiltonian of at most 24 qubits is therefore verified whole, and every lift and coupling by its terms at any size. The 57 wider reflections, those of the two-axis descriptor problems from 25 qubits and of the larger lifts, are built and counted but not independently verified (§ 7.5).

**The circuit computes the emulation.** On all 50 circuit simulations with the minimax filter the simulated output agrees with the exact emulation's before normalising, within the bar of $10^{-10}$. The worst difference is $1.9 \times 10^{-12}$ in the descriptor (R4a at $n = 3$, degree 1126), $1.4 \times 10^{-12}$ in the standard form (R4a at $n = 2$, degree 2093) and $5.8 \times 10^{-13}$ on a Carleman lift (Burgers on four modes, 19 qubits). Where both run, this confirms the argument of § 7.5 that the exact emulation is the circuit's output, which licenses the emulation beyond the widths simulated. The widest circuit simulated for a problem as run has 20 qubits. The two-axis descriptor circuits, of 26 to 28 qubits, are beyond the simulation budget, the narrowest estimated at $3.7 \times 10^5$ s against a budget of 86 387 s: a limit of the classical simulation, not of the circuit. Of the reproduction's 46 imaginary-time simulations, 3, at degrees of order $10^5$, miss the bar (Appendix I.9).

**Every prepared state meets its target.** Each of the 321 trusted preparations by exact emulation meets $\varepsilon$ and its a priori bound, in both forms, on every linear problem and every Carleman lift, and its decoded field meets the error budget of Equation (G.2) (Table H.5). The bound is tight: on R2a in the descriptor at $n = 8$ the infidelity is $3.8 \times 10^{-12}$ against the bound $(1 - \gamma^2)\varepsilon^2/\gamma^2 = 4.2 \times 10^{-12}$ of § 7.4. The field's error falls spectrally with $n$ until it meets the error the filter leaves, which is the same at every $n$, and then holds, as the plot of the error, $\varepsilon_{\text{disc}}$ and the budget's terms against $n$ in Figure G.2 shows: on R2a in the descriptor it is $1.8 \times 10^{-6}$ at $n = 4$ and $1.7 \times 10^{-6}$ at $n = 8$, while the discretisation's own error falls to $6.8 \times 10^{-14}$. From $n = 4$ the prepared field's accuracy is set by $\varepsilon$, which the degree pays for logarithmically (Equation (7.6)), and a larger $n$ gains nothing. The budget, $6.8 \times 10^{-5}$ there, is conservative by more than an order of magnitude, consistent with a readout bound that takes the state's worst direction rather than its actual one, and the margin widens with $n$: the bound's amplification grows from 3.61 at $n = 4$ to 14.7 at $n = 8$ while the prepared field's error does not.

**What answer independence costs.** The geometric initial state of Equation (7.7) assumes that the solution's coefficients decay geometrically, and overlaps well where they do: on R1 at $n = 3$, $\gamma^2 = 0.904$, against $6.4 \times 10^{-3}$ for the uniform state and 0.27 for a random one. On Legendre's equation R3a, whose solution is the polynomial $P_4$, the uniform state does better, 0.321 against the geometric state's 0.166 (exact emulation, T2). The degree does not depend on the initial state, so every control meets $\varepsilon$ at the same degree and a poor guess costs the amplification factor $1/\gamma$ of Equation (7.8). An underestimated spectral edge raises the degree as the inverse square root of the gap and keeps the target; the study on R2a, in which Wu et al.'s time rule falls short of $\varepsilon$, is given in Appendix H.5.

**The filter.** On synthetic spectra with the ground state at the edge, imaginary time needs 1.67–1.92 times the minimax filter's degree at equal suppression, whatever the gap, and on the benchmark problems compared the certified ratio lies in that range in both forms (exact emulation, T2). At equal measured infidelity the margin is smaller, and at $\varepsilon = 10^{-4}$ imaginary time needs fewer degrees on three of the eight cases (Appendix H.5). The choice of filter therefore sets the degree's constant and leaves its growth with $N$ unchanged.

The pipeline is therefore faithful wherever it can be checked, and the differences measured in the rest of this chapter are differences between residuals.

## 8.2 The Cost of the Two Forms

Table 8.1 separates the two costs of § 4 on R2a ($f'' + 4f' + 4f = 0$) at $n = 8$, the largest resolution estimated for every encoding. The first two columns encode one Hamiltonian, $H_{\mathrm{std}}$, so they share its norm, its gap and its ideal-bound degree, and differ only in how tightly they encode it: Wu et al.'s encoding exceeds the residual's norm by a factor of $6.1 \times 10^{13}$ and the structured encoding by 7.74, and the degree falls between them by the same factor. That factor is the encoder's cost. The descriptor's column is a different Hamiltonian with the same kernel, encoded within a few per cent of its norm. The distance between the standard form's ideal-bound degree and the descriptor's is the Hamiltonian's cost, which no encoding of $H_{\mathrm{std}}$ recovers (Proposition 32).

Figure 8.1 follows the three encodings in $N$. Between $n = 7$ and 8, $\|H_{\mathrm{std}}\|$ rises from $1.2 \times 10^{15}$ by a factor $2^8$ and $\|H_{\mathrm{desc}}\|$ from $1.0 \times 10^6$ by $2^2$, while both gaps hold (§§ 4.6 and 5.7); the degree's exponents in Table 8.1, two-point slopes of a degree with logarithmic factors, are consistent with $\tilde{\Theta}(N^{10})$, $\tilde{\Theta}(N^4)$ and $\tilde{\Theta}(N)$. The separation between Wu et al.'s curve and the structured one grows as $N^{3k} = N^6$, and an exact encoding of the same Hamiltonian removes it; that between the structured curve and the descriptor's grows as $N^3$, and only a different Hamiltonian removes it.

Table 8.2 extends the measurement to the classes of Table 5.1, with every linear problem and the coupled oscillator R7 in Table H.7. In the standard form on one axis, with constant coefficients, the degree grows as $N^{2k}$ as built and at the ideal bound alike: the structured encoding holds the encoder's cost to a constant, and what remains is the Hamiltonian's. In the descriptor, on every regular problem (Definition 4), it grows as about $N$, its exponents between 0.87 (R4d) and 0.98 (R1) as built and within a few per cent of one at the ideal bound. The stiff R2c is the exception as built, at $N^{0.66}$, because of its subnormalisation, not its gap. The gap is the same in both columns, and it is unchanged between $n = 7$ and $n = 8$, at 0.715 and 0.715, so the two exponents differ only through $\alpha_R$ against $\|R\|_2$. The subnormalisation is the rescaled recurrence's $2\zeta(N-1)$, with $\zeta = 40$ from the rule, added to a part independent of $N$ that is still the share 0.27 of $\alpha_R$ at $n = 8$, whereas the norm $\|R\|_2 = 20\,498$ there is already close to $\zeta\|\tilde{D}\|_2 = 2\zeta(N-1)$. The as-built exponent should therefore approach one only as $2\zeta(N-1)$ overtakes the constant, beyond the resolutions estimated, which is a prediction and not a measurement. The ideal-bound exponent, free of the constant, is already 0.97 (T3). On the singular problems (R3, R4a) the descriptor's degree grows as about $N^2$, the price of the rescaling's $\zeta \propto N$ (§ 5.4). On two axes it grows faster than $N^2$ per axis yet more slowly than the

**Table 8.1: The encoder's cost and the Hamiltonian's cost are separate, and only the second is the reformulation's to remove.** R2a ($f'' + 4f' + 4f = 0$) at $n = 8$, minimax filter at $\varepsilon = 10^{-6}$. The first two columns encode the same $H_{\text{std}}$, so their norm, gap and ideal-bound degree coincide; the third is the descriptor's $H_{\text{desc}}$, with the same kernel. $\|H\|$, spectral norm; $\Delta$, spectral gap; $\alpha_R/\|R\|_2$, looseness, the encoding's subnormalisation over the residual's norm; ancilla, those of the reflection $V$; query, one walk step; growth, the exponent of $N = 2^n$ between $n = 7$ and 8; $\alpha_R = \|R\|_2$, the ideal-encoding bound. Norms, gaps and looseness: exact classical computation. Degrees, qubits and gate counts: resource estimates (T3), the descriptor's degree confirmed by exact emulation (T2).

| | Standard form, $H_{\text{std}}$ | | Descriptor, $H_{\text{desc}}$ |
|---|---|---|---|
| | Wu et al.'s encoding | structured encoding | |
| $\|H\|$, its law | $\Theta(N^8)$ | $\Theta(N^8)$ | $\Theta(N^2)$ |
| $\|H\|$ | $3.0 \times 10^{17}$ | $3.0 \times 10^{17}$ | $4.2 \times 10^{6}$ |
| Gap $\Delta$ | 7.04 | 7.04 | 2.87 |
| Looseness $\alpha_R/\|R\|_2$ | $6.1 \times 10^{13}$ | 7.74 | 1.04 |
| Encoding | approximate, $\mathcal{O}(N^{-5})$ | exact | exact (no $G$) |
| Qubits | 26 | 25 | 24 |
| Ancilla | 17 | 16 | 13 |
| Two-qubit gates per query | 38 744 | 46 324 | 46 260 |
| Degree, as built | $9.1 \times 10^{22}$ | $1.2 \times 10^{10}$ | 9128 |
| Degree, $\alpha_R = \|R\|_2$ | $1.5 \times 10^{9}$ | $1.5 \times 10^{9}$ | 8736 |
| Degree's growth, as built | $N^{10}$ | $N^{4.0}$ | $N^{0.95}$ |
| Degree's growth, $\alpha_R = \|R\|_2$ | $N^{4.0}$ | $N^{4.0}$ | $N^{1.0}$ |
| Two-qubit gates, preparation | $1.1 \times 10^{28}$ | $1.6 \times 10^{15}$ | $1.3 \times 10^{9}$ |

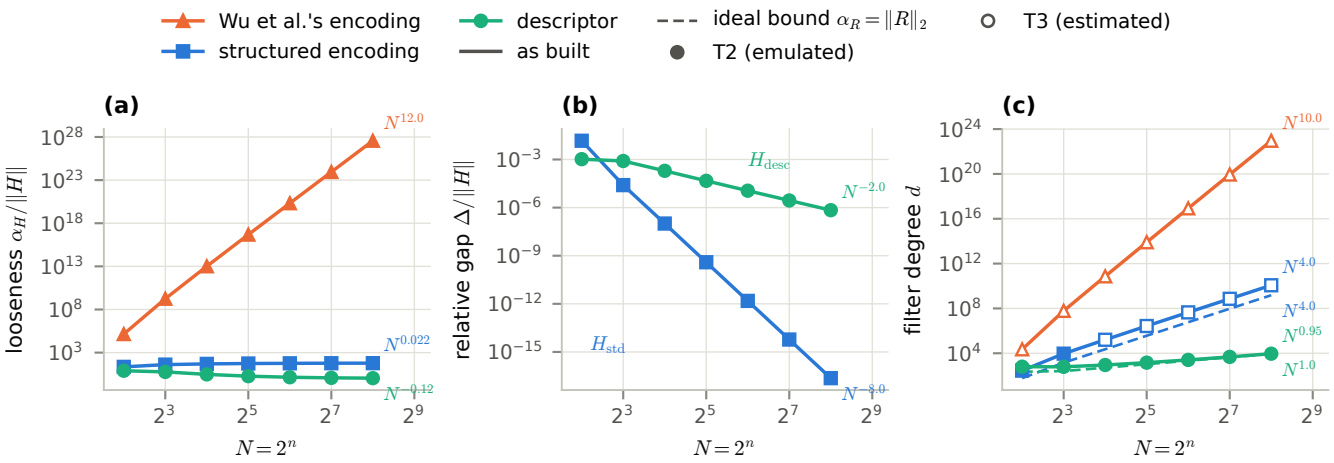



**Figure 8.1: The encoder's cost and the Hamiltonian's cost are two separate gaps on one plot.** R2a against $N = 2^n$, $n = 2$–8, log–log, for Wu et al.'s encoding, the structured encoding and the descriptor. (a) The encoding's looseness $\alpha_H/\|H\| = (\alpha_R/\|R\|_2)^2$: growing as $N^{12}$ for Wu et al.'s encoding, bounded for the structured, falling towards one for the descriptor. (b) The relative gap $\Delta/\|H\|$: $\Theta(N^{-8})$ for $H_{\text{std}}$, $\Theta(N^{-2})$ for $H_{\text{desc}}$. (c) The filter's degree as built (solid) and at the ideal-encoding bound $\alpha_R = \|R\|_2$ (dashed); it is about $\frac{1}{2}\sqrt{\alpha_H/\Delta}\,\ln(2/\varepsilon)$, so (c) is the square root of (a) over (b). The vertical distance between Wu et al.'s and the structured curves is the encoder's cost; that between the structured and the descriptor curves at the ideal bound is the Hamiltonian's cost. Exact emulation (T2, filled) on the resolutions emulated, the descriptor to $n = 8$ and the standard form to $n = 3$, the grid of its imaginary-time reproduction; resource estimates (T3, open) beyond. Norms and gaps are exact classical computations.

standard form's, between the least settled resolutions in the table, $n = 4$ and 5 per axis.

**Table 8.2: The descriptor's degree grows as about $N$ where the equation controls every carried derivative; the standard form's grows as $N^{2k}$ on one axis with constant coefficients.** Exponent of $N = 2^n$ in the minimax filter's degree between the two largest resolutions estimated ($n = 7 \to 8$ on one axis, $n = 4 \to 5$ per axis on two), as built and at the ideal-encoding bound $\alpha_R = \|R\|_2$, for representatives of each class of Table 5.1 (every problem: Table H.7). $k$, equation order; regular, singular: the leading coefficient without and with a zero on $[-1, 1]$ (Definition 4). The standard form's as-built exponent on R3a is its lifted multiplier's looseness; at the ideal bound both forms grow as $N^2$ there. Two-point slopes of a degree with logarithmic factors: consistent with the laws, not fits of them. Resource estimates (T3).

| | | Standard | | Descriptor | |
|---|---|---|---|---|---|
| Class | Problem | As built | Ideal | As built | Ideal |
| First order, $k = 1$ | R1 | 2.0 | 2.0 | 0.98 | 1.0 |
| Second order, regular | R2a | 4.0 | 4.0 | 0.95 | 1.0 |
| | R2c | 4.0 | 4.0 | 0.66 | 0.97 |
| Second order, singular | R3a | 4.0 | 2.0 | 2.0 | 2.0 |
| | R4a | 4.0 | 4.0 | 2.0 | 2.0 |
| Two axes, per axis | R5a | 4.1 | 4.0 | 3.3 | 3.3 |
| | R5b | 4.0 | 3.9 | 2.2 | 2.3 |

**The reversals.** Two comparisons reverse at the ideal bound, and both are findings (Figure 8.2). On the four Legendre problems R3 the standard form needs the smaller degree (Table H.6): on R3a at $n = 8$, $1.1 \times 10^5$ against the descriptor's $2.8 \times 10^5$. As built the order is the other way, $1.7 \times 10^{10}$ against $5.7 \times 10^5$, because the structured encoding pays for R3's lifted multiplier in full, so on R3 the descriptor's as-built advantage is the encoder's. Both forms prepare R3a's exact kernel $P_4$ from $n = 3$, and both ideal-bound degrees grow as $N^2$, while the standard form's as-built degree grows as $N^4$ (Table 8.2). The reversal is R3's and not that of every vanishing coefficient: R3a's $\|H_{\text{std}}\|$ grows only as $N^4$, from $6.0 \times 10^8$ at $n = 7$ to $9.7 \times 10^9$ at $n = 8$, while that of R4a, whose coefficient $x - 1$ vanishes at one end only, grows as $N^8$, from $9.2 \times 10^{15}$ to $2.4 \times 10^{18}$, and there the descriptor's ideal-bound degree is the lower, $4.0 \times 10^5$ against $6.7 \times 10^9$. The second reversal is the heat equation R5b, whose reference is trusted from $n = 4$ per axis. There the descriptor's degree is the higher, 29 044 against 27 692 as built and 12 706 against 4015 at the ideal bound; at $n = 5$ it is the lower as built, $1.4 \times 10^5$ against $4.5 \times 10^5$, and level at the ideal bound, $6.3 \times 10^4$ against $6.0 \times 10^4$. Its exponent per axis is the smaller (Table 8.2), so the order should settle in the descriptor's favour at larger $n$, an extrapolation. Two axes do not by themselves reverse the comparison: on the Laplace equation R5a the descriptor's ideal-bound degree is the lower at $n = 4$ and 5 per axis, $2.1 \times 10^5$ against $8.1 \times 10^5$ at $n = 5$.

**Queries and gates.** The degree counts queries, and a query's gates differ between the forms. On R2a one query of the descriptor costs more two-qubit gates than one of the structured standard form at every resolution built and verified, 25 914 against 8094 at $n = 2$ and 34 098 against 22 856 at $n = 5$, having more terms and a block register; its count grows more slowly, and the estimates at $n = 8$ put the two level (Table 8.1). On R3a, whose lifted multiplier the structured encoding pays for in full, the descriptor's query is the cheaper from $n = 4$: 38 864 against 43 108 at $n = 4$ and 48 542 against 52 038 at $n = 5$. The ancilla's growth, $\mathcal{O}(\log n)$ against $n$ (§ 5.6), first shows at $n = 8$ (Table 8.1); at $n = 2$ and 5 the descriptor's reflection on R2a needs 12 and 13 ancilla against the standard form's 10 and 13. On constant-coefficient problems the descriptor's advantage therefore lies in the number of queries, which dominates the preparation's gate count (Table 8.1). The standard form's degrees at $n = 8$ are resource estimates far beyond the $2^{20}$ that exact emulation reaches (§ 7.5), a limit of

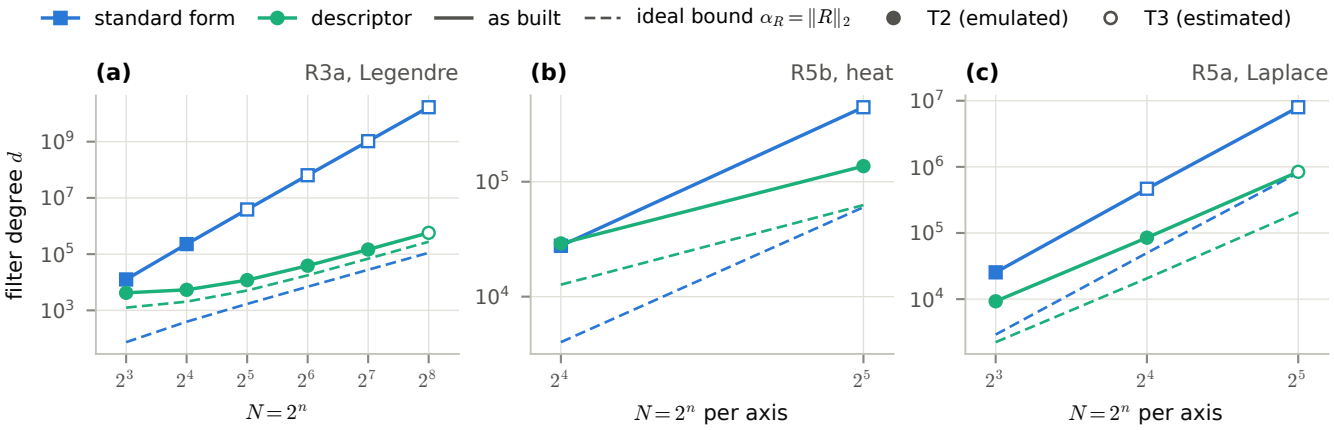



**Figure 8.2: At the ideal-encoding bound the standard form's degree is the lower on R3a at every $n$ and on the heat equation R5b at $n = 4$ per axis; on the Laplace equation R5a it is not.** The minimax filter's degree against $N$ (per axis on R5), standard (squares) and descriptor (circles), as built (solid) and at the ideal-encoding bound $\alpha_R = \|R\|_2$ (dashed), at the resolutions whose reference is trusted. (a) R3a, Legendre's equation: the descriptor lower as built, higher at the ideal bound. At the ideal bound both degrees grow as $N^2$; as built the standard form's grows as $N^4$, its encoding paying for R3a's lifted multiplier in full. (b) R5b, heat: higher at $n = 4$ per axis in both comparisons; at $n = 5$ lower as built and level at the ideal bound. (c) R5a, Laplace: lower in both comparisons. Exact emulation (T2, filled) where emulated, resource estimates (T3, open) beyond.

the classical emulation that does not bound the circuit; the descriptor's agree with exact emulation through $n = 8$.

The standard form's cost is therefore measured as well as argued: with Wu et al.'s encoding it is mostly the encoder's, an exact encoding removes that part, and what remains, $\tilde{\Theta}(N^{2k})$ in the degree for constant coefficients on one axis, is the Hamiltonian's. The descriptor removes the Hamiltonian's part on regular problems; on singular ones its degree still grows as $N^2$, and on R3 at the ideal bound the standard form's is the lower.

## 8.3 The Descriptor's Gap and Its Rescaling

§ 5.3 traced the descriptor's gap to the derivatives it carries: the gap shrinks where the solution's derivatives are large and closes on singular problems and on two axes, and block rescaling restores it on one axis only. Figure 8.3 measures this on every panel. Unscaled, the gap on the physical subspace at $n = 8$ is $0.458$ on R2a but $7.5 \times 10^{-6}$ on the stiff R2c and $2.7 \times 10^{-8}$ on the singular R3a, and the field block holds as little as $5.4 \times 10^{-6}$ of the ground state on R2c. Rescaled, the gaps are $2.87$, $0.715$ and $3.62$ and the field block's weights $0.384$, $0.774$ and $0.993$. Every reference here is trusted in both versions, so the rescaling changes the filter's work and not the prepared field. On two axes the Laplace equation's rescaled gap falls from $0.372$ at $n = 3$ per axis, where its reference is first trusted, to $1.1 \times 10^{-2}$ at $n = 5$, against $1.1 \times 10^{-4}$ and $1.1 \times 10^{-8}$ unscaled; at $n = 5$ per axis the heat and wave equations' rescaled gaps are $9.4 \times 10^{-2}$ and $2.0 \times 10^{-2}$. That is a negative finding: on two axes block rescaling slows the gap's fall and does not stop it. Rescaling also raises the ceiling $\gamma \leq w_f^{1/2}$ on the initial state's overlap (§ 7.4), though how near the geometric state comes to it depends on the guess: on R4b at $n = 8$, $w_f = 0.176$ and $\gamma^2 = 0.040$ (T2). The rule of Equation (5.10) fixes $\zeta$ before each run, but its two constants were set once on these problems, so its cost against the cheapest $\zeta$ on a grid, tabulated in Appendix D.6, is an in-sample check.

**The stiff problem.** R2c, $f'' + 5f' + 400f = 0$, has the smallest unscaled gap and field weight of the regular problems, and rescaling gains most there. Rescaled, the descriptor prepares it from $n = 5$, with infidelity $2.2 \times 10^{-12}$ at $n = 8$ by exact emulation. Its degree at $n = 8$ is $2.4 \times 10^5$ against the standard form's $6.0 \times 10^9$ as built, and $1.8 \times 10^5$ against $7.8 \times 10^8$ at the ideal bound (T3). Unscaled, the gap and the field block's weight are each about five orders of magnitude smaller;

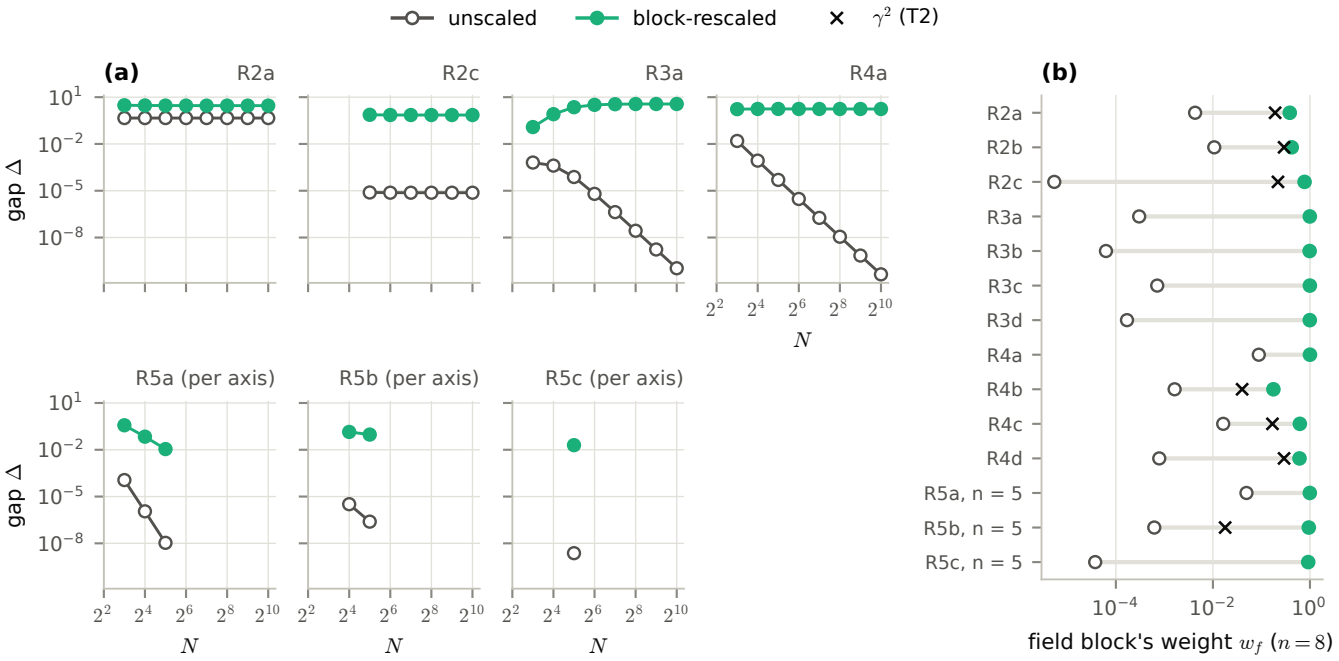



**Figure 8.3: Block rescaling restores the descriptor's gap and the field block's weight on one axis; on two axes the gap still falls.** (a) The gap $\Delta$ on the physical subspace against $N$, unscaled (open) and block-rescaled (filled), for R2a and R2c (regular, R2c stiff), R3a and R4a (singular: the leading coefficient vanishes on $[-1, 1]$) and R5a–R5c (two axes, per axis), at the resolutions whose reference is trusted. (b) For every panel at $n = 8$ ($n = 5$ per axis on R5), the field block's weight $w_f$ in the ground state, unscaled against rescaled, with the geometric initial state's overlap $\gamma^2 \leq w_f$ (crosses) where a preparation ran at that resolution: R2a–c, R4b–d and R5b; R3, R4a, R5a and R5c were prepared only at smaller $n$. Rescaling lifts the one-axis gaps to $\mathcal{O}(1)$ and $w_f$ towards one; on two axes it gains orders of magnitude but the gap still falls with $N$. Exact classical computations (the reference's spectrum); $\gamma^2$ from exact emulation (T2).

how much more degree and amplification an unscaled preparation would need was not estimated, since rescaling also changes $\|H_{\text{desc}}\|$ and $\alpha_R$. The other problems, in both forms and at every tier, are given in Appendix H.4.

Block rescaling therefore restores the gap the carried derivatives erode on every one-axis problem run, without ancilla; on two axes it only slows the gap's fall.

## 8.4 Systems and Nonlinear Problems

§ 6 claims that a Carleman lift is prepared where the doubled space only projects, and that the descriptor's lower degree survives on the lifted systems, where it enters through the time axis alone (§ 6.8). The coupled oscillator R7 comes first, because every lift is a coupled linear system of its kind.

**Two coupled fields.** The harmonic oscillator R7, $u' = v$, $v' = -\pi^2 u$, is the realification of $z' = -\mathrm{i}\pi z$ (§ 6.3), carried as two fields on one state with its data as ratio rows. Both forms converge to rounding, the descriptor's field error $2.0 \times 10^{-14}$ at $n = 5$, and both prepare it: from $n = 4$ the data-weighted geometric state has $\gamma^2 = 0.247$ in the descriptor and $0.247$ in the standard form, each at the largest $n$ it emulates, and every trusted exact emulation meets $\varepsilon$. The descriptor carries the system in its mass form, whose gap is about twice the block-rescaled chain's (§ 6.1) and which keeps the whole ground state on the field blocks, where the chain's derivative blocks take weight from them, $w_f = 0.852$. At $n = 8$ the mass form's degree is $1.1 \times 10^4$ against the standard form's $1.4 \times 10^6$ as built and $1.0 \times 10^4$ against $6.8 \times 10^5$ at the ideal bound (T3), growing as $N$ against $N^2$ (Table H.7). Simulated at $n = 2$ on 13 qubits, where the reference is not yet trusted, the circuit agrees with the emulation to $1.2 \times 10^{-15}$, a test of the circuit rather than of a preparation. What is prepared is the

realified real system, at the cost of one field-register qubit; no complex residual was prepared (§ 6.3).

**Projection against preparation.** The doubled space's kernel on R6a has dimension 10 at $n = 2$ and 962 at $n = 5$, in state spaces of 16 and 1024 (Table E.2), so a filter returns the initial state's projection onto it (Proposition 4), which is never a preparation. The Carleman lift of $y' = -y + y^2$ (R8) has a one-dimensional, trusted kernel at every order from $K = 2$ to 6 in both forms, so the filter prepares it.

**The lift on a time axis (R8).** The field's error splits as Equation (6.8) predicts. At each order it is flat in $n_t$ from $n_t = 3$, the time axis's part having fallen below the lift's, and it falls with the order, from $2.8 \times 10^{-2}$ at $K = 2$ to $7.8 \times 10^{-4}$ at $K = 6$, faster than the factor $\rho = \frac{1}{2}$ per order that the bound of § 6.5 allows. Both forms prepare every order to $\varepsilon$; the infidelity at $K = 4$ and $n_t = 8$ is $1.5 \times 10^{-13}$ in the descriptor and $1.7 \times 10^{-13}$ in the standard form. The descriptor's time axis lowers the degree, by a margin that widens with $n_t$: at $K = 4$ it is 233 against 1228 at $n_t = 4$ and 1901 against 319 510 at $n_t = 8$ by exact emulation, and 117 against 599 and 1772 against 151 492 at the ideal bound (T3).

**Burgers (R9): a trusted kernel can hide a diverging lift.** Figure 8.4 measures the lift's convergence on viscous Burgers from the data $a \sin x$, against the strength $a/\nu$ and the number of sine modes $M$, with $\nu = 1$. The lift's error, its exact solution against the Galerkin system's, falls by a factor of about $a/(5\nu)$ per order: at $a/\nu = 1$ from $8.7 \times 10^{-3}$ at $K = 2$ to $5.9 \times 10^{-5}$ at $K = 5$, and at $a/\nu = 3$ from 0.20 to 0.034. It does not depend on $M$: at $a/\nu = 1$ and $K = 2$ it is $8.7 \times 10^{-3}$ on sixteen modes, as on four, while the worst-case ratio of Equation (6.7) grows from $\rho = 1.86$ to 16.1. The lift therefore converges at both ratios, though both exceed the bound's condition $\rho < 1$, and its convergence follows $a/\nu$, not $\rho$. At $a/\nu = 10$ it diverges, its error growing from 4.56 at $K = 2$ to 24.4 at $K = 5$ against an amplitude of 10, while its kernel stays one-dimensional and trusted at every order, with gap ratio $5.3 \times 10^{12}$ at $K = 5$. Both forms prepare it, with infidelity $1.2 \times 10^{-12}$ in the descriptor and $2.2 \times 10^{-12}$ in the standard form by exact emulation, and the decoded field is wrong by 24.4. This negative finding is analysed in § 6.5: trust measures the kernel's isolation and says nothing of the lift's accuracy. The boundary lies between $a/\nu = 3$ and 10; that the factor $a/(5\nu)$ reaches one near $a/\nu = 5$ is an extrapolation from these strengths, not a measurement. As on R8 the descriptor's time axis lowers the degree once $n_t$ grows: at $a/\nu = 1$, $K = 2$ and $n_t = 5$ it is 747 against the standard form's 11 764 by exact emulation, and 286 against 3260 at the ideal bound (T3). From $n_t = 4$ it is the lower on every Burgers case estimated, 12 of 12 at $n_t = 4$ and 8 of 8 at $n_t = 5$; at $n_t = 3$ it is the lower on only 6 of 12 as built, the standard form winning at the stronger forcing and on more modes, as at $a/\nu = 10$, $K = 5$ on four modes, 46 832 against 71 365 (T3).

**The overlap.** The data-weighted initial state keeps the overlap of order one on every Carleman lift prepared, in both forms: $\gamma^2$ ranges from 0.038 to 0.84 across R8, R9 and R10, its minimum on R9e at $K = 5$, where the lift diverges (T2). The factor $1/\gamma$ of Equation (7.8) therefore does not grow with the lift at the sizes run.

A Carleman lift is therefore prepared in both forms on every problem run, from R8 to Burgers, where the doubled space only projects. A preparation certifies the lift; whether the lift is the nonlinear solution is measured order by order, and on Burgers it fails at $a/\nu = 10$ with no sign in the residual. From $n_t = 4$ the descriptor's degree is the lower on every system and lift estimated, as built and at the ideal bound; on the coarsest time axes the standard form's can be the lower. § 8.5 tests whether the advantage lasts as the coupling grows with the modes.

## 8.5 Two-Dimensional Navier–Stokes

Figure 8.5 carries the lift to two-dimensional incompressible flow, the vorticity equation of Equation (6.10) with $\nu = 1$, prepared on 12 to 48 Fourier modes and built classically to 80.

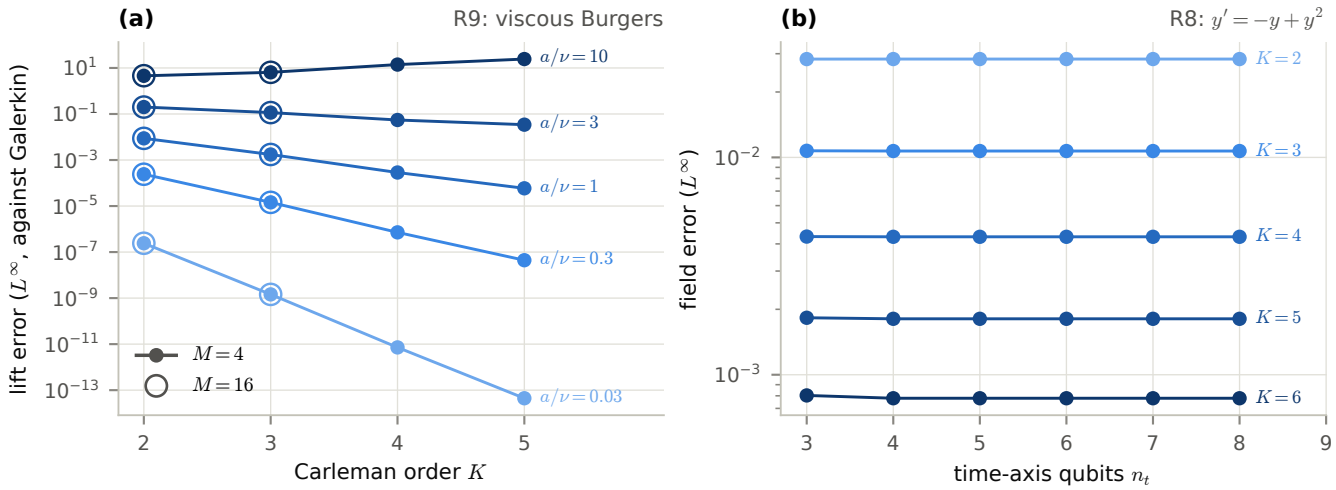



**Figure 8.4: The Carleman lift's convergence follows $a/\nu$, not $\rho$, and past it the kernel stays trusted while the lift diverges.** (a) Viscous Burgers (R9): the lift's error against the Galerkin system against the Carleman order $K$, at $a/\nu = 0.03$, 0.3, 1, 3 and 10, to $K = 5$ on $M = 4$ sine modes (filled) and to $K = 3$ on $M = 16$ (rings). Where both are built the two coincide at plot scale, although $\rho$ grows with $M$, from 1.86 to 16.1 at $a/\nu = 1$; at $a/\nu = 10$ the error grows with $K$. (b) R8 ($y' = -y + y^2$, $\rho = \frac{1}{2}$): the field's error against $n_t$ for $K = 2$–6, flat at the lift's floor from $n_t = 3$. Exact classical computations from the references; every trusted preparation of these lifts meets $\varepsilon$ by exact emulation (T2).

**Validation.** The Taylor–Green vortex (R10a), whose two modes do not interact, has an exact lift, and the pipeline returns it to rounding: the lift's error is $4.4 \times 10^{-16}$ on twelve modes and $1.1 \times 10^{-15}$ on eighty. This checks the Galerkin coefficients, the ratio rows on the stated data and the time axis against a known answer; it does not test the lift's convergence, which the vortex pair does.

**The vortex pair.** A counter-rotating pair of norm $\|w_0\| = a$ excites modes of several $|\mathbf{k}|$, so its triads are active. The truncation's own error against a direct simulation, about $4 \times 10^{-3}$, $4 \times 10^{-2}$ and 0.2 on twelve modes at $a/\nu = 3$, 10 and 30, is a property of the Galerkin system, not of the lift. The lift's error is set by the strength, as on Burgers (Figure 8.5); on twelve modes at $a/\nu = 10$ it falls from $7.1 \times 10^{-2}$ at $K = 2$ to $6.4 \times 10^{-3}$ at $K = 3$. Unlike Burgers', it also grows with the modes: at $a/\nu = 10$ and $K = 2$ it is 0.36 on 48 modes against 0.071 on twelve, though it still converges there, to 0.18 at $K = 3$, at a $\rho$ of 24. At $a/\nu = 30$ it converges on twelve modes, from 1.83 to 0.501, and diverges from 28, where the error grows from 5.28 to 6.07 against $\|w_0\| = 30$, as it does on 48 (Figure 8.5), with every kernel trusted. This is the Burgers finding again, on a flow, and trust does not detect it. A shorter horizon, $T = \frac{1}{8}$, restores convergence on 28 modes, from 3.65 to 2.36, as the lift's time-dependent radius of convergence leads one to expect [73]. Every trusted preparation by exact emulation meets $\varepsilon$, the worst at infidelity $6.6 \times 10^{-13}$.

**The time axis.** At one Carleman order the two forms differ only in the time axis (§ 6.8), and the descriptor's degree is the lower in all 20 pairs estimated at $n_t = 4$ and all 12 at $n_t = 5$, on 12 to 48 modes at $K = 2$ to 4 and $n_t = 4$ and 5, as built and at the ideal bound. A qubit of $n_t$ multiplies the standard form's degree by about four and the descriptor's by less than two: on the vortex pair at $a/\nu = 10$, twelve modes and $K = 2$, the degrees are 8604 and 1131 at $n_t = 4$, and 35 176 and 1970 at $n_t = 5$. At fixed $n_t$ the margin narrows as the modes grow, because the coupling's part of the norm is common to both forms (§ 6.8) and the descriptor's encoding of it loosens with its matchings: on 48 modes the degrees are 11 512 against 4647, and the descriptor's looseness $\alpha_R/\|R\|_2$ has grown from 3.83 to 9.92. The margin narrows at the ideal bound too, from 1577 against 296 to 1688 against 469, so part of the narrowing is the Hamiltonian's and part the encoder's.

**The overlap.** The estimated success probability of the preparation, $p_G$, stays between 0.46 and 0.66 across $K = 2$ to 4 and 12 to 48 modes, falling slowly with both, with at most 1 round of

amplification (T3). The overlap is not what limits the preparation of the vortex pair at these sizes; nothing is claimed beyond them.

**The tier, and the tensor layout.** No circuit was simulated on Navier–Stokes. The narrowest prepared circuit, the descriptor at $n_t = 4$ on twelve modes, has 27 qubits, wider than any circuit simulated here; the probe-verified reflections at $n_t = 2$ are not prepared. The fluid results are therefore exact emulations, which equal the circuit's output (§ 7.5), and resource estimates. The tensor layout's coupling, the one native complex encoding built, keeps $2K - 1$ terms whatever the number of modes and is verified by IR to 316 modes, but prepares no state (§ 6.3); its gates per query against the matchings' are given in Table E.1.

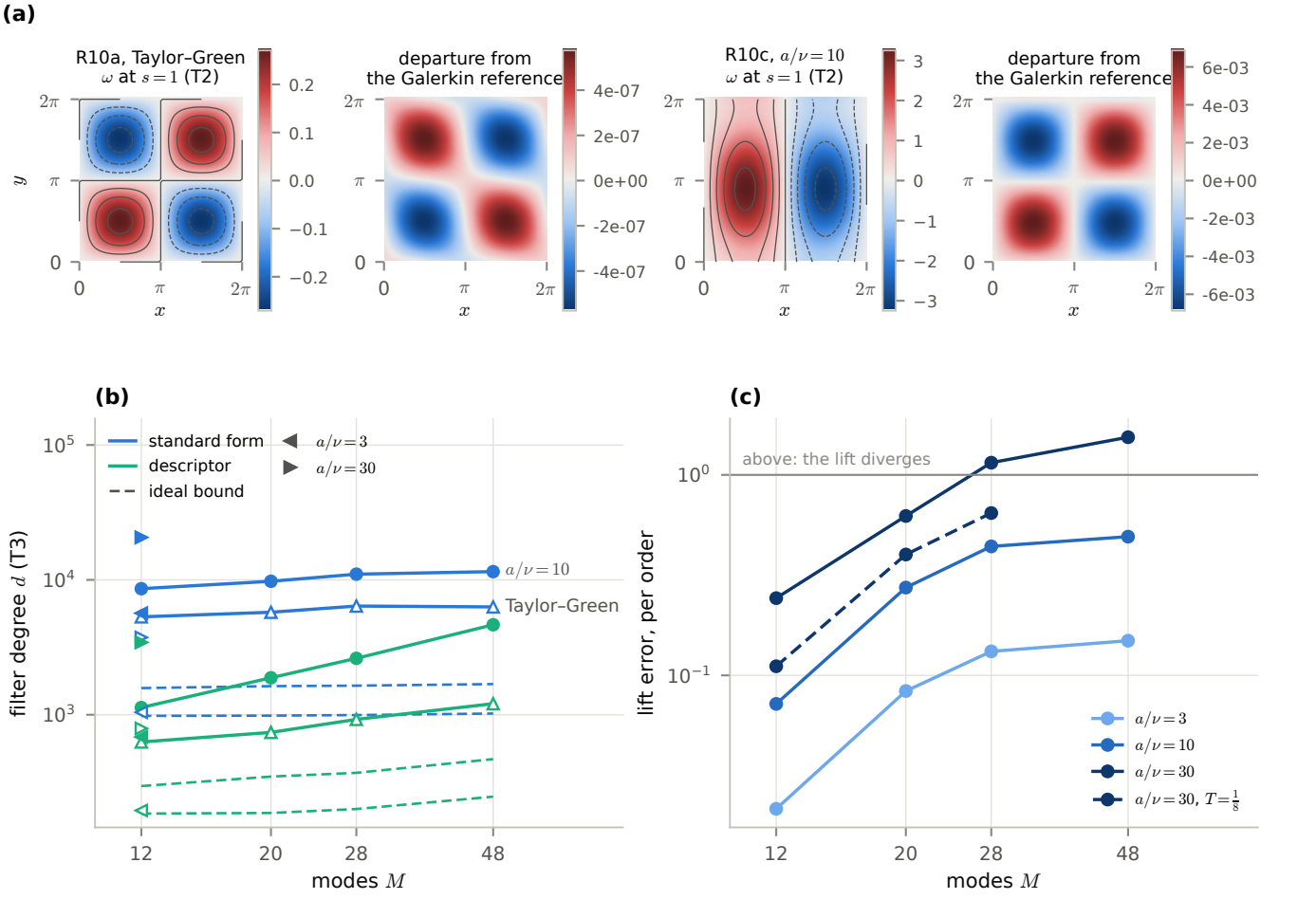



**Figure 8.5: Two-dimensional Navier–Stokes is prepared by exact emulation on up to 48 modes, and the descriptor's time axis lowers the degree in every pair.** (a) Vorticity at $t = T$: the field decoded from the prepared state against the Galerkin reference, for the Taylor–Green vortex (R10a, exact lift) and the vortex pair at $a/\nu = 10$ (R10c), twelve modes, $K = 3$, $n_t = 4$; exact emulation (T2). (b) The minimax filter's degree against the number of modes $M$ at $K = 2$, $n_t = 4$, the form by colour, as built (solid) and at the ideal-encoding bound (dashed), over $M$ for the vortex pair at $a/\nu = 10$ (circles) and the Taylor–Green vortex (open triangles), and on twelve modes, the only $M$ estimated at those strengths, for the pair at $a/\nu = 3$ and 30 (side-pointing triangles; filled as built, open at the ideal bound). The ratio of the two forms' degrees narrows as $M$ grows. (c) The lift's per-order error ratio against $M$ at $a/\nu = 3$, 10 and 30 over $T = 1$, and at $a/\nu = 30$ over the window $T = \frac{1}{8}$ on up to 28 modes, with the line of unit ratio marked: above it the lift diverges while its kernel stays trusted. Resource estimates (T3) in (b); exact classical computations in (c). No circuit simulation (T1) was run on this problem.

The lift therefore reaches two-dimensional incompressible flow by exact emulation on up to 48 modes, with the descriptor's degree the lower in every pair, and with Burgers' caveat: at $a/\nu = 30$ the lift diverges from 28 modes with its kernel trusted.

## 8.6 Summary of Findings

1. **The standard form's cost is its Hamiltonian's.** With Wu et al.'s encoding the degree on R2a grows as about $N^{10}$, mostly the encoder's cost; an exact encoding removes that part, and on one-axis problems with constant coefficients the degree still grows as $N^{2k}$, as built and at the ideal-encoding bound (Tables 8.1 and 8.2).

2. **The descriptor's degree advantage depends on the class of problem.** Its degree grows as about $N$ on regular problems, as about $N^2$ on singular ones, and faster than $N^2$ per axis on two axes; per query it costs more gates than the structured encoding at small $n$ and about as many at $n = 8$. *Negative:* at the ideal bound the standard form needs the lower degree on R3, and on the heat equation at $n = 4$ per axis, the two being level at $n = 5$ (§ 8.2).

3. **The descriptor's gap erodes where a carried derivative is uncontrolled, and block rescaling restores it on one axis only.** On every one-axis problem run the rescaled gap is $\mathcal{O}(1)$ and the field block regains its weight, without ancilla. *Negative:* on two axes the rescaled gap still falls with $N$ (§ 8.3).

4. **The pipeline computes what it claims.** Every composed Hamiltonian of at most 24 qubits is verified whole and every lift and coupling by its terms; every circuit simulation with the minimax filter equals the exact emulation within $10^{-10}$; every trusted preparation meets $\varepsilon$, its a priori bound and its field budget. Wider reflections are built and counted but not independently verified (§ 8.1).

5. **Coupled systems and Carleman lifts are prepared in both forms, but a trusted kernel can hide a diverging lift.** The mass form's gap is about twice the block-rescaled chain's on R7, and from $n_t = 4$ the descriptor's degree is the lower on every system and lift estimated. *Negative:* on Burgers at $a/\nu = 10$, and on Navier–Stokes at $a/\nu = 30$ from 28 modes, the lift diverges while its kernel stays one-dimensional and trusted (§§ 8.4 and 8.5).

6. **Navier–Stokes is reached by exact emulation and resource estimate only.** Its narrowest prepared circuit is wider than any simulated, and the tensor layout's coupling, verified to 316 modes, prepares nothing (§ 8.5).

Within its stated scope the central claim holds as measured: carrying the derivatives instead of eliminating them keeps the kernel and lowers the Hamiltonian's norm from $\Theta(N^8)$ to $\Theta(N^2)$ on a second-order equation with constant coefficients, and with it the degree from about $N^4$ to about $N$; the price is a block register and a gap the derivative blocks erode, which block rescaling restores on one axis without ancilla but not on two. § 9 gives the mechanism behind each finding and what the exponents imply beyond the benchmark problems.

# 9 Discussion

§ 8 found the measured degrees agree with the formulation chapters within their stated scope. They depart from them on singular problems (Definition 4), two axes and on a lift past its boundary. Exponents record how fast a degree grows, but they do not explain why carrying the descriptor form lowers the Hamiltonian's norm and endangers at the same time, nor what a lower degree is worth once gates, amplification, readout and the classical alternatives are counted, nor how far a study made entirely of classical computation can vouch for a quantum algorithm. This chapter asks *what mechanism produces the descriptor's advantage and its limits, and what the measurements imply beyond the benchmark problems.* The answer it develops is that ground-state preparation of a differential equation costs what the conditioning of its residual costs, and that the formulation, rather than the circuit, decides that conditioning.

## 9.1 Conditioning Moves from the Norm to the Gap

The structured encoding holds the standard form's subnormalisation to within a constant of its norm, and the degree still grows as $N^{2k}$ (Table 8.2), so the exponent must be from the residual itself. Consider factorisation $G = \tilde{B}^{-1}\tilde{D}$ of Equation (5.2). In $\|G\|_2$, there are two powers of $N$, one is $\|\tilde{D}\|_2 = 2(N-1)$ and the other comes from the inverse, $\tilde{B}^{-1}$ being a triangle of ones on each parity class. For an order $k$ equation, we compose $G$ $k$ times. So, $\|R_{\text{std}}\|_2 = \Theta(N^{2k})$ for constant coefficients, and the Gram product doubles the exponent again to $\|H_{\text{std}}\| = \Theta(N^{4k})$. The descriptor keeps the banded factors, which avoids the inverse, and carries each derivative as its own unknown, which avoids the composition. Every recurrence row applies $\tilde{D}$ and $\tilde{B}$ once, uninverted. For fixed $\zeta$ and any order equation, no block exceeds a $\mathcal{O}(N)$ norm, and $\|H_{\text{desc}}\| = \Theta(N^2)$ (§ 5.7).

The descriptor measures a state's residual against $\sum_m \|G^m p\|^2$ (Equation (5.9)), the same quantities the standard form's norm cost incorporates. A state with large carried derivatives compared to its residual has a small Rayleigh quotient. The reformulation therefore relocates the conditioning of differentiation rather than removing it: from a norm that grows with $N$ on every problem, to a gap that falls only where the equation fails to control one of the derivatives it carries. We prove that the control holds on a regular problem in Theorem 2, and both gap fall modes are failures of that control. Where the leading coefficient vanishes, a state concentrated near the zero has a top derivative that the equation does not penalise. On two axes, the equation constrains only a sum of top derivatives, so a state whose second derivatives along the two axes nearly cancel has a small residual (§ 5.3). The stiff case R2c shows how severe this can be even on a regular problem. Its unscaled gap at $n = 8$ being $7.5 \times 10^{-6}$ against 0.715 rescaled (§ 8.3). The mass form of a first-order equation removes the inverse without carrying any derivative, so its states are measured only by the field. This makes sense, since its gap exceeds a chain's on R7 (§ 6.1). At $k \geq 2$ no such option exists, we can only use the chain form. Lowering the norm but risking the gap.

Classical numerical analysis met the same trade from two directions. First-order-system least squares carries the derivatives to avoid squaring a high-order operator's condition number. Its theory rests on norm equivalence: the requirement that the least-squares functional bound every carried unknown in its own norm [13]. Theorem 2 is a discrete, one-axis instance coercivity, of the lower half of that equivalence. The collapse on two axes is what its failure would look like. On two axes a high-frequency near-solution is penalised only by the datum rows of § 4.3, which weigh it at the order of its field norm, while the carried derivatives that dominate the denominator of Equation (5.9) grow with $N$. That mismatch between the data on the boundary and the derivatives in the interior is the likely reason coercivity fails. We attempted weighting each datum row by a power of its tangential mode, so that the rows penalise high frequencies more. This raised the gap much more than the norm and lowered the cost by a modest factor at three and four qubits per axis, but the weighted gap still fell with every qubit, and a heavier weight performed worse (Appendix D.4). Thus, only reweighting the existing rows does not supply the missing coercivity. The ultraspherical method reaches the same banded factors and finds its systems well conditioned once diagonally preconditioned [12], which is the role block rescaling takes here. The increment of this thesis is therefore not the idea of carrying derivatives through banded factors but its transfer to a block-encoded Hamiltonian, and the measurement of what it costs and where it fails.

The block register multiplies the state by $2^{n_f}$. Four blocks at second order and eight for Laplace's equation and the wave equation (Appendix D.2). A query applies more terms through that register, so on constant-coefficient problems it costs more two-qubit gates than the structured standard form's at the sizes built and verified, 25 914 against 8094 on R2a at $n = 2$, the two converge to 46 260 against 46 324 at $n = 8$ (§ 8.2). Block rescaling adds no ancilla, but it lengthens the Pauli expansion of the register's patterns by a fixed factor depending on the equation. This cost for the reformulation is decided by the exponents, and they depend on the class of problem. On a singular problem the rule's $\zeta \propto N$ raises the descriptor's norm to $\Theta(N^4)$ at second order, and on two axes or at a double zero

no block weight restores the gap (§ 5.4).

## 9.2 End-to-End Complexity

Neither form is polylogarithmic in $N$. The degree of Equation (2.2), $d = \Theta\big(\sqrt{\alpha_H/\Delta}\,\log(1/\varepsilon)\big)$ with $\alpha_H \geq \|H\|$ for any block encoding, is the residual's effective condition number times $\log(1/\varepsilon)$. It has the form of the optimal linear-systems law, $\mathcal{O}(\kappa \log(1/\varepsilon))$ queries [18], which no algorithm improves to $\kappa^{1-\delta}$ (unless BQP = PSPACE [16]). So ultimately the ground-state route arrives at the same familiar conditioning problem from another path. Read this way, the standard form's $\tilde{\Theta}(N^{2k})$ for constant coefficients with the best exact encoding (Equation (4.8)) is the conditioning of an order-$k$ least-squares discretisation, and the descriptor's $\tilde{\Theta}(N)$ on a regular problem (Equation (5.13)) is that of a first-order one. This cost is an analogue of the order reduction preconditioning step that linear-systems solvers for ordinary equations typically take (§ 2.2). For singular problems and on two axes the descriptor's degree is about $\tilde{\Theta}(N^2)$ or steeper. Without a source, or with a polynomial source of fixed degree, a query costs $\mathcal{O}(n^2)$ two-qubit gates in both forms (Propositions 2 and 3), a general source adding $\mathcal{O}(N)$, and amplification multiplies the queries by $2k_{\mathrm{aa}}+1 = \mathcal{O}(1/\gamma)$ (Equation (7.8)). On a regular problem with constant coefficients a preparation therefore costs $\tilde{\Theta}(N\,\mathrm{poly}(n)/\gamma)$ gates in descriptor against $\tilde{\Theta}(N^{2k}\,\mathrm{poly}(n)/\gamma)$ in the standard form. The overlap $\gamma$ is measured rather than bounded, and it differs between the forms, the descriptor's capped by the field block's weight, which the rescaling restores (§ 8.3).

On R2a at $n = 8$ Wu et al.'s approximate encoding implies a degree of $9.1 \times 10^{22}$, and our exact encoding of the same residual, $1.2 \times 10^{10}$. The ratio is the encoder's looseness, and we remove most of it. The remaining $1.5 \times 10^{9}$ at the ideal bound is the Hamiltonian's. The descriptor's degrees are 9128 as built and 8736 at the ideal bound (Table 8.1). The descriptor's encoding is the tighter, within 1.04 of its residual's norm against 7.74 for the structured encoding. So the as-built comparison favours the descriptor and the ideal bound is a fairer comparison of the two Hamiltonians. Therefore the method's cost lives in the Hamiltonian it constructs and not in the circuits it reports. A gate count per query, which is how PIHM prices differentiation, is accurate but incomplete, because the number of queries isn't considered. The separation of the encoder's cost from the Hamiltonian's is an audit that any filter-based algorithm on a block-encoded operator must do.

Equation (4.8) holds for every block encoding of $R_{\mathrm{std}}$ under any polynomial filter of the walk, since by Chebyshev's extremal property none suppresses the excited spectrum at a lower degree than the minimax filter (§ 2.4). Over all algorithms the evidence is weaker. Lin and Tong prove by reduction from unstructured search, that their ground-state algorithm's dependence on the gap and the overlap is essentially optimal for an oracle that block-encodes $H$ [53]. The walk here is built from an encoding of $R$ which supplies the square root in Equation (2.2). So their bound does not apply as it stands, and whether their hard instances arise as stacked residuals is not shown. We therefore claim the floor for polynomial filters of the walk, and for second-order examples with constant coefficients it follows that $\|H_{\mathrm{std}}\| = \Theta(N^8)$ sets the number of queries under any block encoding, whatever circuit implements one query (§ 4.6). The descriptor lowers the floor by replacing $R$, which no statement about the encodings of a fixed $R$ excludes.

The GQSP circuit realises the minimax edge filter and the QITE filter from the same walk at the same normalisation (§ 2.4 and appendix F.7), so we can isolate the polynomial for comparison. At equal suppression QITE needs 1.67 to 1.92 times the minimax degree on synthetic spectra, regardless of gap, and the ratio does not drift with the resolution, 19 against 11 on R1 at $n = 2$ and 675 against 384 at $n = 8$ (T2, Appendix H.5). The reverse cases are instructive. At equal measured infidelity and a loose target, $\varepsilon = 10^{-4}$, QITE needs fewer degrees on three of the eight cases, 1784 against 2187 on R3a in the descriptor, because its exponential suppresses each excited state more the higher it lies, while the minimax filter holds all at the equiripple level. That dividend comes from the structure of the spectrum above the edge, which an a priori pipeline neither knows nor can certify. Therefore the

minimax filter is optimal for exactly the worst case that the edge alone determines (Proposition 28), and it is the filter that answer-independent parameters can guarantee. The lower degree also has a numerical benefit: three of the reproduction's QITE circuit simulations at degrees of order $10^5$ miss the agreement bar due to rounding accumulated over the walks, whereas every minimax record meets it (Appendix I.9). The filter is thus an engineering constant worth a factor below two, and Wu et al.'s open question about the gap is answered: the exponent lives in the residual.

Solved classically through its reference, the standard form converges spectrally. Its error falls to $3.2 \times 10^{-13}$ on R2a at $n = 5$ and to $4.6 \times 10^{-12}$ on the heat equation at $n = 5$ per axis, as the geometric decay of an analytic solution's coefficients predicts [32]. Beyond that resolution its error rises again, to $1.4 \times 10^{-6}$ at $n = 10$, where the descriptor's, with the same kernel, is $3.7 \times 10^{-13}$ (§ 8.1). The rise is consistent with rounding amplified by a residual whose norm grows as $N^4$, so the conditioning that sets the quantum degree is already visible in the classical solve. The prepared state shows the same saturation from the other side: on R2a in the descriptor the decoded field's error is $1.8 \times 10^{-6}$ at $n = 4$ and $1.7 \times 10^{-6}$ at $n = 8$, set by $\varepsilon$ while the discretisation's own error falls to $6.8 \times 10^{-14}$, so resolution beyond what the solution needs only improves the degree. For an analytic solution a field accurate to $\delta$ needs $N$ growing as $\log(1/\delta)$, both degrees are then polylogarithmic in the precision, and the comparison of exponents in $N$ implies no speed-up. On one axis a degree linear in $N$ is no asymptotic advantage in any case: the ultraspherical method solves such an equation classically in $\mathcal{O}(N)$ operations once its coefficients are resolved [12]. An advantage could appear only on many axes, where a classical solver's cost grows at least as the number of unknowns, $N^D$ on $D$ axes. However that is exactly where the descriptor's gap falls. Measured on R5b its estimated degree grows as $N^{2.2}$ per axis on the heat equation and $N^{3.3}$ on Laplace's (Table 8.2), faster than the $N^2$ unknowns. Whether a residual can keep its gap on several axes therefore decides whether this route can compete with classical spectral solvers at all. We leaves that question open.

## 9.3 Where the Construction Applies

Treatments for various problem classes are detailed in Table 9.1.

The split in the simple-zero row turns on where the coefficient vanishes. The standard form's norm is set by the top Chebyshev modes, whose derivatives are largest at the ends of the interval by Markov's inequality [54]. A coefficient that vanishes at both ends ($1 - x^2$ on R3) suppresses those modes where they peak. There $\|H_{\text{std}}\|$ grows only as $N^4$. On R4a, $x - 1$ vanishes at one end and the $N^8$ law holds (§ 8.2). The rescaled descriptor's norm grows as $N^4$ on both, so it gains on R4a but not on R3. This is an inference from two problems consistent with § 4.6 but not a theorem. It makes a testable prediction that no benchmark problem checks: an interior zero which suppresses neither end, should follow the standard form's $N^{4k}$ law, so the descriptor formulation is expected to scale better there.

On the nonlinear rows the guidance follows $a/\nu$ rather than $\rho$, and the reason is physical. The ratio $\rho = a\|F_2\|/\nu$ is a worst case over all data of norm $a$. $\|F_2\|$ grows with the modes because the advection term couples into a mode with a strength that rises with its wavenumber (§ 6.7). Burgers' data $a \sin x$ occupy one mode and reach the others only through the nonlinearity, so at moderate strength the high modes (where $F_2$ is largest) carry too little of the solution to move the lift's error. This would explain why that error is the same on four modes as on sixteen (§ 8.4). The vortex pair puts several wavenumbers into its data. Its triads are active from the start, and the error does grow with the modes (§ 8.5). Both readings are inferences from the measured errors. They agree with $\rho < 1$ being sufficient and far from necessary, and explain why a worst-case ratio is a poor guide to a particular flow.

A trusted kernel certifies that the lifted linear problem has an isolated ground state, which is what the filter needs. However it says nothing about whether that ground state is the nonlinear solution. On Burgers at $a/\nu = 10$ both forms prepare the lift to $\varepsilon$, and the decoded field is wrong

**Table 9.1: The descriptor is advantageous where the equation controls every derivative it carries and on nonlinear problems while the lift converges.** Shown is the descriptor's gain over the standard form for each class of problem, and the choice our evidence supports. $k$, order; regular, singular: one axis, the leading coefficient without and with a zero on $[-1, 1]$ (Definition 4); $n_t$, time-axis qubits; ideal bound, $\alpha_R = \|R\|_2$. Gains are proved (Theorem 2) or measured as resource estimates (Tables 8.2 and H.7) between stated resolutions; the complex-valued row is a statement about the formulation, with no complex residual prepared.

| Class (problems) | Descriptor advantage | What to do |
|---|---|---|
| Regular (R1, R2, R4b–d) | degree $\tilde{\Theta}(N)$ against $\tilde{\Theta}(N^{2k})$ for constant coefficients; gap proved $\Theta(1)$ | the chain with $\zeta$ from Equation (5.10) for $k \geq 2$, the mass form for $k = 1$; expect more gates per query at small $n$ |
| Singular, simple zero (R3, R4a) | degree about $\tilde{\Theta}(N^2)$; against $N^4$ where the zero is at one end (R4a), none at the ideal bound where it is at both (R3) | $\zeta = N/2$; judge at the ideal bound, since as built the gain on R3 is the encoder's |
| Singular, double zero | none claimed: balancing needs a norm above the standard form's own | substitute $f = \sqrt{1 - x^2}\, v$ to leave simple zeros, as for R3c and R3d |
| Two axes, no mixed derivative (R5) | degree's exponent per axis below the standard form's; reversed on the heat equation at $n = 4$ per axis; gap falls even when rescaled | $\zeta_a = N_a$; budget a state of four to eight blocks; compare at the ideal bound and expect no guaranteed gain |
| First-order systems (R7) | regular: about $N$ against $N^2$ | the mass form, with no block register and a larger gap than a chain's; data as ratio rows; the coupling as matchings |
| Complex-valued linear equations | that of the real system after realification (R7 is $z' = -\mathrm{i}\pi z$ realified) | realify, at one more field-register qubit; native complex preparation is not built |
| Polynomial nonlinearity, lift converging (R8; R9 to $a/\nu = 3$; R10 to $a/\nu = 10$, and at 30 on 12 modes) | through the time axis only, from $n_t = 4$, narrowing as the modes grow | the Carleman lift, mass form in time; raise $n_t$ with the truncation; measure the lift's error per order; never the doubled space, which projects |
| Past the lift's boundary (R9 at $a/\nu = 10$; R10 at $a/\nu = 30$ from 28 modes) | none: the kernel stays trusted while the lift diverges | shorten the horizon or lower the strength; read convergence from the lift's error, never from trust |

by 24.4 against an amplitude of 10. On Navier–Stokes at $a/\nu = 30$ from 28 modes the lift diverges with every kernel trusted. In neither case does anything in the residual's spectrum signal the failure (§§ 8.4 and 8.5). Thus every Carleman preparation needs the lift's measured error per order beside it. The shorter window that restores convergence on 28 modes is what the lift's time-dependent radius of convergence leads one to expect [73], and it is what Table 9.1 recommends.

At one Carleman order the two forms differ only in how the time axis and the coupling enter the norm. The time axis contributes $|\tilde{D}_\xi| = 2(N_t - 1)$ to the mass form, $|G_\xi| = \Theta(N_t^2)$ to the standard form. The coupling is of the same order in both but enters the mass form multiplied by $\tilde{B}_\xi$, with $|\tilde{B}_\xi|_2 \in [2, 2 + \sqrt{2}]$ (§ 6.8). On a coarse time axis, at strong forcing and on many modes, the coupling dominates and that factor makes the mass form's norm the larger one, which accounts for the standard form's wins at $n_t = 3$ on Burgers and for the descriptor's gain from $n_t = 4$ on. With the loosening of the descriptor's encoding of the coupling as its matchings multiply, the same balance narrows the Navier–Stokes margin as the modes grow at fixed $n_t$. It also predicts a reversal on enough modes, not reached on up to 48 (§ 8.5). This assumes comparable gaps in the two forms and was not checked term by term, but its recommendation to raise $n_t$ with the truncation does not depend on that assumption. The reformulation's advantage therefore survives nonlinearity, carried by the one axis on which the lifted system still differentiates.

## 9.4 Feasibility and the Reach of Classical Evidence

We assume a fault-tolerant machine for all cost analysis (§ 2.1). On R2a at $n = 8$ the descriptor's preparation is estimated at 24 logical qubits and $1.3 \times 10^9$ two-qubit gates, 3 amplified repetitions of a filter of degree 9128, against $1.6 \times 10^{15}$ for the structured standard form and $1.1 \times 10^{28}$ with Wu et al.'s encoding (Table 8.1). The qubit counts stay modest, with the widest circuits estimated, Navier–Stokes on twelve modes at $K = 4$ and $n_t = 5$, needing 38 qubits in the descriptor and 39 in the standard form. The obstacle is the gate count. For all the costs of R2a above, we solve one second-order equation on $N = 2^8$ coefficients, a problem of trivial classical difficulty. The reformulation divides the count by about the ratio of the two degrees, which grows as $N^{2k-1}$ on such a problem; that alone does not make the problem worth a quantum computer. The estimates count logical two-qubit gates after transpiling to $\{\mathrm{CX}, U\}$ (§ 7.7). They omit the non-Clifford gates from which an error-corrected machine synthesises the rotations $U$, whose count dominates its cost, along with the code's overhead and routing. They record no depth, which grows with the degree because the walks run in sequence. Jennings et al. cost a linear-systems route to linear ODEs in logical qubits and in queries to the block encoding of the system's matrix, with their constant factors [19]. Their counts and ours meet at the query, whose gates are supplied here for both forms, but their examples are time evolutions of stable systems and are not compared with ours.

Depth rather than width relegates the method to error-corrected hardware at every size run. A filter of 9128 sequential walks of 46 260 two-qubit gates each lies far beyond the depth that uncorrected processors execute reliably [15], and even the smallest preparation, the decay $f' + 2f = 0$ (R1) at $n = 2$ in the descriptor form, fits on 9 qubits but applies 24 sequential queries of 1898 two-qubit gates each (circuit simulation). No preparation here is a candidate for a noisy device, and the variational methods that target such devices answer a different question.

The production ran 3155 runs on the Setonix supercomputer [85]; 494 hours of computation in all. The ceiling for runs was set by circuit simulation, whose statevector memory doubles with every qubit and whose time grows with the degree (Appendix G.5). The widest circuit simulated had 20 qubits. The narrowest two-axis descriptor circuit, at 26 qubits, estimated its own run at $3.7 \times 10^5$ s against a budget of 86 387 s and was refused before it began, one of the 11 runs recorded that way (§ 8.1). The resolutions at which the method's open question lives, several axes and many modes, thus lie beyond direct simulation. This is a limit of the classical simulator rather than of the circuit. However, we show the exact emulation equals the circuit's post-selected output (§ 7.5), confirmed on all 50 minimax circuit simulations to within $10^{-10}$, which licenses the emulation to filter degrees of $2^{20}$ and we use a resource estimate beyond.

## 9.5 Limitations

Every result is a classical computation, and nothing ran on hardware. The claims are cost laws of operators and filters, which classical computation tests exactly. It cannot test noise, and the method's robustness to it is unknown. Exact emulation stands in for the circuit beyond 20 qubits by an argument that the agreement of the two confirms only where both run, the widest reflections, of the two-axis descriptor problems and the larger lifts, are built and counted but not verified whole, and Navier–Stokes is reached by emulation and estimate alone (§ 8.5). The growth exponents are two-point slopes between stated resolutions, consistent with the laws rather than fits of them, and the two-axis slopes, between $n = 4$ and 5 per axis, are the least settled. Where $N$ must grow with the problem, as on a stiff equation, the dependence of the gap's constant on the stiffness was measured on R2c alone. Beyond 8,192 physical columns the reference is iterative, and the error bound it reports is computed from the residual of the returned vector rather than proved, so it is reliable only if the iteration has converged and no singular value below $\tilde{\sigma}_1$ has been missed (Appendix G.3).

The descriptor's scope ends where its gap does. The gap is proved flat only on regular problems,

measured flat after rescaling on singular ones with a simple zero, and falls on two axes. A double zero needs the substitution of § 4.3, whose form presumes that the zero's order is known from the equation, and mixed derivatives have no block and are not supported (§ 5.2). The full ultraspherical reduction, a banded residual on $N$ unknowns with no block register, would remove the inverse without carrying any derivative, but it was neither built nor compared (§ 2.3). Its differentiation operator is a single diagonal whose norm grows linearly in $N$ at every order, and for a leading coefficient that does not vanish, as many boundary conditions as the order and a uniquely solvable problem, Olver and Townsend bound its condition number after a diagonal preconditioner, in the 2-norm for Dirichlet conditions, so that without one it grows linearly with the truncation [12]. On a regular problem this suggests that the reduction would reach the descriptor's $\tilde{\Theta}(N)$ degree on $N$ unknowns, without the block register, and the ultraspherical descriptor of Appendix D.4, which keeps a flat gap on regular problems, is consistent with that. The expectation is untested. The bound concerns a square system with conditions at the ends of the interval rather than a least-squares residual whose rows the conversions weight (Appendix A.5), and the reduction's encoding with variable coefficients and its gap on singular problems and on two axes were not measured.

Of the extensions, only first-order systems were run. Complex-valued fields are carried by realification alone; no complex residual was prepared. The tensor layout's native complex encoding is verified but prepares nothing (§ 6.3). The thesis presents no test of a lift's divergence that reads the residual alone (§ 8.4). Wu et al.'s partial differential and nonlinear examples, their Figs 6 and 7, were not reproduced under any hypothesis tested, and the authors were not contacted for clarification (Appendix I.4).

# 10 Conclusions and Future Work

## 10.1 Conclusions

Wu et al. priced their physics-informed effective Hamiltonian by the gates in one query and left open how its cost depends on the spectral gap and on the cost of differentiation [1]. Question (i) of § 3.5 asked how the number of queries, the filter's degree, grows with the resolution, and how much of that growth is the encoder's and how much the Hamiltonian's. Wu et al.'s circuits indicate the larger factor is the encoder's, but some unreducable amount is the Hamiltonian's; checked by an exact encoding as well as ideal bounds. In the standard form, whose state holds the field alone and represents every derivative through the dense Chebyshev differentiation matrix $G$, the degree grows as $\tilde{\Theta}(N^{2k})$ for an order-$k$ equation with constant coefficients under any block encoding and every polynomial filter of the walk.

Question (ii) asked whether the residual can be rewritten so that its Hamiltonian is banded and well conditioned without changing its solution, and at what cost. The descriptor form keeps apart the two banded factors whose product is $G$, carrying a higher-order equation's derivatives as unknowns and multiplying a first-order equation or system through by the conversion factor. Its kernel is equivalent to the standard form's. On a regular problem with one axis, polynomial coefficients, a leading coefficient without zeros on $[-1, 1]$ and analytic data, its degree grows as $\tilde{\Theta}(N)$ independent of order. The conditioning moves the norm's cost to the gap. On a singular problem whose leading coefficient has a simple zero, block rescaling restores a gap of order one on a single axis, without ancilla and at a cost in gates fixed by the equation, for a degree of about $\tilde{\Theta}(N^2)$. On two axes the gap falls with $N$ whatever the weights. The descriptor's other prices, a register of derivative blocks and, on constant-coefficient problems at the sizes built, more gates per query than the structured standard form, are factors independent of $N$, so its advantage lies wholly in the degree.

Nonlinear equations, the subject of question (iii), pose the same question at its limit. Wu et al.'s doubled space encodes a polynomial nonlinearity exactly, but its zero-energy eigenspace has dimen-

sion at least $2^{2n-1}$, so a filter can only project onto it. A truncated Carleman lift has an isolated ground state, which both forms prepare, from Burgers' equation to two-dimensional incompressible Navier–Stokes in exact emulation and resource estimate. The descriptor's advantage survives through the time axis — which the lift still differentiates — once that axis is resolved finely enough. The uniqueness that makes preparation possible certifies the truncated problem only; a lift past its boundary keeps an isolated ground state while its field departs from the nonlinear solution. The condition $\rho < 1$ under which Liu et al. prove that the truncation converges is a worst case over all data, which structured data need not meet [4].

The three answers describe one mechanism: ground-state preparation of a differential equation costs what the conditioning of its residual costs, and the formulation fixes that conditioning before circuit encodings are created. Neither form, on a linear equation or a lifted one, is polylogarithmic in $N$. Separating the encoder's cost from the Hamiltonian's is an audit that any filter-based algorithm on a block-encoded operator admits; because built circuits favour the form with the tighter encoding, a comparison speaks about the Hamiltonians only at the ideal-encoding bound $\alpha_R = \|R\|_2$. The degree has the form of the optimal law for linear systems, with the residual's effective condition number in place of the matrix's, so the ground-state route meets conditioning in another guise, visible already in the accuracy that the standard form's classical solve loses at high resolution. The tools for lowering it come from classical numerical analysis, two of which the descriptor transferred: the banded operators of the ultraspherical method [12] and the carried derivatives of first-order-system least squares [13].

No speed-up over classical solvers is claimed. An analytic solution needs a resolution growing only as $\log(1/\delta)$ for an accuracy $\delta$, and on one axis a degree linear in $N$ matches the ultraspherical method's classical operation count. Due to the reformulation, a preparation whose cost tracks the classical one in order on a regular problem, and nonlinear flows prepared as unique ground states are now possible. An advantage could appear only on several axes, where a classical solver's cost grows at least as the number of unknowns.

## 10.2 Future Work

Whether a residual can keep its gap there decides whether this route can compete with classical spectral solvers at all. The obvious repair, weighting the datum rows by a power of the tangential mode, has been measured: it raises the gap more than the norm and lowers the cost modestly, but the weighted gap still falls with every qubit. The question is now whether any reweighting, alone or with block rescaling, or rows of the kind first-order-system least squares adds to make its functional coercive, can make the residual bound every carried derivative on several axes. The most direct such rows impose the boundary data on the carried derivative blocks, through the data's tangential derivatives, as well as on the field. The solution satisfies them, so they would leave an exact kernel unchanged, and the gap and norm they give on Laplace's equation (R5a) at three to five qubits per axis would test the idea directly. The other candidate is the full ultraspherical reduction, a banded residual on the field alone that removes the inverse in $G$ without carrying any derivative. Olver and Townsend's bound on its condition number suggests that on a regular problem it would reach the descriptor's $\tilde{\Theta}(N)$ degree without a block register [12], but the bound concerns a square system rather than a least-squares residual, and the reduction's gap on singular problems and on two axes is unknown. Building it and measuring one problem of each class, R2a, Legendre's equation (R3) and the heat equation (R5b), would show which classes it serves.

Block rescaling sets its weight from the equation's coefficients and the resolution, but its two constants were selected by calibration on the benchmark problems and then held fixed, so its check is in-sample. The quotient of the states concentrated near a zero already fixes how $\zeta$ must grow with $N$, so deriving the weight from the equation, its constants and the crossover between the classes included, should be possible. A regular coefficient that dips towards zero, where the classes meet, is where a derived weight would first be tested against the best on a grid. Block rescaling weights the

kernel by one constant per block and so stays cheap, whereas an equilibration per amplitude would need a block encoding of its own at a cost linear in the dimension. Whether a diagonal structured in the Chebyshev index, cheap to encode where the general one is not, can do for the degree what diagonal preconditioning does for the ultraspherical method's banded systems [12], or whether a degree linear in $N$ is the least that a differentiating residual reaches on one axis, remains open.

Any condition that annihilates the solution enters the residual as rows, which is how coupled fields and realified complex-valued equations reached the same filter. A complex residual still gives a Hermitian Gram operator, so native complex preparation needs complex loaders and no new filter, and $z' = -\mathrm{i}\pi z$, which R7 carries realified, is the natural first comparison of the two encodings. An observation $f(x_i) = y_i$ would enter as the row $y_s f(x_i) - y_i f(x_s) = 0$. Adding observations of growing noise to a regular problem such as R2a, and following the ground value and the gap, would show how far a physics-informed Hamiltonian can become a data-informed one. Both directions change the rows and leave the filter and its verification as they are.

# 11 References


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# Appendices

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# A Chebyshev and Ultraspherical Identities

This appendix records the recurrence that defines the banded factors $\tilde{B}$ and $\tilde{D}$ of § 5.1, their entries, the exact factorisation $G = \tilde{B}^{-1}\tilde{D}$, its connection with the ultraspherical method, and the fully populated $N = 4$ instance of the first-order descriptor operator.

## A.1 The differentiation recurrence

For $f = \sum_{\ell\geq 0} a_\ell T_\ell$ with derivative $f' = \sum_{\ell\geq 0} p_\ell T_\ell$ on $[-1, 1]$, the Chebyshev coefficients satisfy

$$c_{\ell-1}\, p_{\ell-1} - p_{\ell+1} = 2\ell\, a_\ell, \qquad \ell \geq 1, \tag{A.1}$$

with $c_0 = 2$ and $c_\ell = 1$ for $\ell \geq 1$; the $\ell = 1$ row therefore carries the factor $c_0 = 2$ on $p_0$. Truncating to $N = 2^n$ coefficients and dropping the $p_N$ term, Equation (A.1) is the matrix identity $\tilde{B}_{\text{raw}}\, p = \tilde{D}_{\text{raw}}\, a$ with $\tilde{D}_{\text{raw}}$ a single superdiagonal, $(\tilde{D}_{\text{raw}})_{\ell,\ell+1} = 2(\ell + 1)$, and $\tilde{B}_{\text{raw}}$ two diagonals, $(\tilde{B}_{\text{raw}})_{\ell,\ell} = c_\ell$, $(\tilde{B}_{\text{raw}})_{\ell,\ell+2} = -1$.

The recurrence follows from two trigonometric identities. With $x = \cos\theta$, $T_k' = k \sin k\theta / \sin\theta$, and $\sin(k+1)\theta - \sin(k-1)\theta = 2\cos k\theta \sin\theta$ gives $T'_{k+1}/(k+1) - T'_{k-1}/(k-1) = 2T_k$ for $k \geq 2$, with $T_1 = T_2'/4$ and $T_0 = T_1'$. Integrating $f' = \sum_k p_k T_k$ term by term with these, the coefficient of $T_\ell$ in $f$ collects $c_{\ell-1}p_{\ell-1}/(2\ell)$ from $T_{\ell-1}$ and $-p_{\ell+1}/(2\ell)$ from $T_{\ell+1}$, which is Equation (A.1); the constant of integration affects $a_0$ alone, which no row of the recurrence reads. The truncation is exact: the derivative of a polynomial of degree below $N$ has degree at most $N - 2$, so the dropped coefficients $p_N$ and $p_{N+1}$ vanish, and the last row, $p_{N-1} = 0$, holds.

## A.2 The normalised basis

The construction of § 5.1 works in the normalised basis $\varphi_0 = T_0/2^{n/2}$, $\varphi_\ell = T_\ell/2^{(n-1)/2}$ $(\ell \geq 1)$, orthonormal under the discrete $N$-point Chebyshev inner product $\langle T_k, T_\ell\rangle_N = \frac{N}{2}\, c_\ell\, \delta_{k\ell}$ [86]: since $N = 2^n$, this gives $\|\varphi_0\|_N = \|\varphi_\ell\|_N = 1$ exactly, so the coefficient vector of a unit-norm function is a unit vector. Writing $S = \operatorname{diag}(\sqrt{c_\ell})$, the normalised factors are $\tilde{B} = S\,\tilde{B}_{\text{raw}}\,S^{-1}$ and $\tilde{D} = S\,\tilde{D}_{\text{raw}}\,S^{-1}$, that is

$$\tilde{D}_{\ell,\ell+1} = 2(\ell+1)\sqrt{c_\ell/c_{\ell+1}}, \qquad \tilde{B}_{\ell,\ell} = c_\ell, \qquad \tilde{B}_{\ell,\ell+2} = -\sqrt{c_\ell/c_{\ell+2}}, \tag{A.2}$$

so $\tilde{D}$ has the single non-zero diagonal $(2\sqrt{2},\, 4,\, 6,\, 8,\, \dots)$ and $\tilde{B}$ has main diagonal $(2,\, 1,\, 1,\, \dots)$ and second superdiagonal $(-\sqrt{2},\, -1,\, -1,\, \dots)$. Both hold $\mathcal{O}(N)$ non-zero entries at any $N$. The dense Chebyshev derivative matrix is their exact quotient,

$$G = \tilde{B}^{-1}\tilde{D}, \tag{A.3}$$

an algebraic identity rather than a numerical approximation. $\tilde{B}$ is upper triangular with the non-zero diagonal $(2, 1, \dots, 1)$, so it is invertible. For any coefficients $a$ the true derivative's coefficients satisfy the truncated rows of Equation (A.1), as shown above, and they are $Ga$; conjugating by $S$ gives $\tilde{B}G = \tilde{D}$, so the equality holds exactly at every $N$.

**Proposition 6** (The inverse of the mass factor)**.** *$\tilde{B}^{-1}$ is upper triangular and supported where the column index minus the row index is even. Its entries there are* 1 *in every row $k \geq 1$; in row* 0 *they are $\frac{1}{2}$ at column* 0 *and $1/\sqrt{2}$ at every even column $q \geq 2$.*

*Proof.* Away from row 0, $\tilde{B} = I - P$ with $P$ the shift by two, so on each parity class its inverse is $\sum_m P^m$, a triangle of ones. Row 0 of $\tilde{B}x = e_q$ reads $2x_0 - \sqrt{2}x_2 = \delta_{0q}$, which gives $x_0 = \frac{1}{2}$ for $q = 0$ and, with $x_2 = 1$, $x_0 = 1/\sqrt{2}$ for even $q \geq 2$. □

Multiplying out $\tilde{B}^{-1}\tilde{D}$ with this inverse returns the entries of Equation (4.2): $\tilde{D}$'s single diagonal forces the inner index to be $p-1$, the parity support of $\tilde{B}^{-1}$ makes $p-k$ odd, and the entry is $2p$ times one, or times $1/\sqrt{2}$ in row 0. The density of $G$ is the inverse of the banded $\tilde{B}$, which is what the descriptor never forms. The triangle of ones has norm $\Theta(N)$ and $\tilde{D}$ has norm $2(N-1)$; their product has the $\Theta(N^2)$ norm of $G$.

## A.3 The scale factor and the feature vector's norm

The basis functions $\varphi_0 = T_0/\sqrt{N}$ and $\varphi_k = T_k/\sqrt{N/2}$ $(1 \le k < N)$ are orthonormal under the discrete inner product on the $N$ Chebyshev nodes. Under the continuous Chebyshev weight $w(x) = (1-x^2)^{-1/2}$ they are orthogonal with a common norm: with $x = \cos\theta$, $\int_{-1}^{1} T_j T_k w\,dx = \int_0^{\pi} \cos j\theta \cos k\theta\,d\theta$, which is $\pi$ for $j = k = 0$, $\pi/2$ for $j = k \ge 1$ and zero otherwise, so $\int \varphi_j \varphi_k w\,dx = (\pi/N)\,\delta_{jk}$ for every $j, k < N$. For $f_N = \sum_k b_k \varphi_k$ Parseval's identity is therefore $\int_{-1}^{1} f_N^2 w\,dx = (\pi/N)\|b\|^2$, which is Equation (4.1).

**Proposition 7** (The feature vector's norm)**.** *With $x = \cos\theta$,*

$$\|\tau(x)\|^2 = 1 + \frac{1}{N}\Big(\frac{\sin(2N-1)\theta}{2\sin\theta} - \frac{1}{2}\Big), \qquad \|\tau(\pm 1)\|^2 = \frac{2N-1}{N}, \tag{A.4}$$

*and $\|\tau(x)\|^2 \le (2N-1)/N < 2$ on $[-1, 1]$.*

*Proof.* Since $2T_k^2 = 1 + \cos 2k\theta$,

$$\|\tau(x)\|^2 = \frac{1}{N}\Big(T_0^2 + 2\sum_{k=1}^{N-1} T_k^2\Big) = \frac{1}{N}\Big(N + \sum_{k=1}^{N-1} \cos 2k\theta\Big).$$

The sum is the Dirichlet kernel $\sin((2N-1)\theta)/(2\sin\theta) - \frac{1}{2}$, at most $N-1$, which it attains at $\theta = 0$ and $\pi$. □

The norm oscillates about one and is largest at the ends, so a constraint row at an end of the interval weighs most. That weight is part of Wu et al.'s construction and is kept (§ 4.1). Where the residual has no exact kernel its ground state is a least-squares compromise between the rows, so normalising $\tau(x)$ point by point, which gives Wu et al.'s $\tilde{\tau}(x)$, would change the rows' relative weights and with them the solution; the rows are therefore built, and block-encoded, from the unnormalised vector. Normalising $\tau(x)$ point by point would also break the consistency of the collapse rows: two collapses $\mathbb{B}(x_s)/f(x_s)$ and $\mathbb{B}(x'_s)/f(x'_s)$ map the solution to the same constant, because $\mathbb{B}(x)b = f_N(x)\sqrt{N}\,|0\rangle$, whereas with $\tau(x)/\|\tau(x)\|$ they would map it to constants differing by the factor $\|\tau(x'_s)\|/\|\tau(x_s)\|$.

## A.4 The structure and norm of the derivative matrix

**Proposition 8** (Structure and norm of $G$)**.** *For $n \ge 2$, $G$ of Equation (4.2) is upper triangular and nilpotent; $\ker G^j$ is the space of polynomials of degree below $j$; it has $N^2/4$ non-zero entries; it is orthogonally similar to the matrix that maps the values of a polynomial of degree below $N$ at the Chebyshev nodes to the values of its derivative there; and*

$$\frac{N^2}{4\sqrt{2}} \;\le\; \|G\|_2 \;\le\; \frac{N^2}{\sqrt{2}}. \tag{A.5}$$

*Proof.* The entries are non-zero only for $p > k$, so $G$ is strictly upper triangular and $G^N = 0$. $G$ represents differentiation exactly on polynomials of degree below $N$, so $G^j b = 0$ exactly when the

$j$-th derivative of the represented polynomial vanishes, that is when its degree is below $j$. Column $p$ has a non-zero entry in each row $k < p$ of the other parity, $\lceil p/2 \rceil$ of them, and $\sum_{p<N} \lceil p/2 \rceil = N^2/4$.

For the similarity, let $\Phi_{jk} = \varphi_k(\bar{x}_j)$ at the nodes $\bar{x}_j$. The discrete orthonormality of the $\varphi_k$ says that $\Phi$ is orthogonal, and $\Phi b$ is the vector of nodal values of the polynomial with coefficients $b$, so $\Phi G \Phi^\top$ maps nodal values to the nodal values of the derivative.

For the upper bound, $\|G\|_2 \le \|G\|_F$, and each of the $\lceil p/2 \rceil$ entries of column $p$ is at most $2p$ in magnitude, so

$$\|G\|_F^2 \le \sum_{p=0}^{N-1} 4p^2 \Big(\frac{p}{2} + \frac{1}{2}\Big) = \frac{(N-1)^2 N^2}{2} + \frac{(N-1)N(2N-1)}{3} = \frac{N^4}{2} - \frac{N^3}{3} - \frac{N^2}{2} + \frac{N}{3} \le \frac{N^4}{2}.$$

For the lower bound, let $u$ be the unit vector uniform over the $N/4$ even rows $k < N/2$ and $v$ the unit vector uniform over the $N/4$ odd columns $p > N/2$. Every pair $(k, p)$ has $p > k$ with $p - k$ odd, and its entry is $2p > N$ for $k \ge 2$ and $\sqrt{2}p > N/\sqrt{2}$ for $k = 0$, so

$$\|G\|_2 \ge u^\top G v \ge \frac{4}{N} \cdot \Big(\frac{N}{4}\Big)^2 \cdot \frac{N}{\sqrt{2}} = \frac{N^2}{4\sqrt{2}}. \qquad \square$$

The bounds give $\|G\|_2 = \Theta(N^2)$. The exact limit, $\|G\|_2/N^2 \to 1/(2j_{-3/4,1}) \approx 0.4724$, with $j_{-3/4,1}$ the first positive zero of the Bessel function $J_{-3/4}$, and the growth of every power, $\|G^k\|_2 = \Theta(N^{2k})$, are proved in Appendix C.4.

## A.5 The Ultraspherical Connection

The banded factorisation above is the first-derivative case of the ultraspherical spectral method of Olver and Townsend [12]. Stating the correspondence precisely is what fixes what we add to it.

Olver and Townsend represent a function's derivative not in the Chebyshev basis $T_k$ but in the ultraspherical (Gegenbauer) basis $C_k^{(\lambda)}$, the family of polynomials orthogonal with respect to the weight $(1 - x^2)^{\lambda - 1/2}$; at $\lambda = 1$ these are the Chebyshev polynomials of the second kind, $C_k^{(1)} = U_k$. Differentiation of a Chebyshev series obeys

$$\frac{\mathrm{d}T_k}{\mathrm{d}x} = k\, C_{k-1}^{(1)}, \qquad k \ge 1 \tag{A.6}$$

[12, Eq. 2.5], so for $u = \sum_k u_k T_k$ the coefficients of $u'$ in the $C^{(1)}$ basis are read off directly: $(\mathcal{D}_0 u)_k = (k+1)u_{k+1}$, a single superdiagonal, sparse at every truncation. Because $u'$'s coefficients now live in a different basis to $u$'s own, a second, purely algebraic operator is needed to convert a function's own Chebyshev coefficients into the $C^{(1)}$ basis without differentiating it: $\mathcal{S}_0$, with $(\mathcal{S}_0)_{k,k} = 1$ $(k = 0)$, $\frac{1}{2}$ $(k \ge 1)$, and $(\mathcal{S}_0)_{k,k+2} = -\frac{1}{2}$ [12, Eq. 2.8]. Comparing entries, $\tilde{D}_{\text{raw}} = 2\mathcal{D}_0$ and $\tilde{B}_{\text{raw}} = 2\mathcal{S}_0$ exactly: the recurrence $\tilde{B}_{\text{raw}}\, p = \tilde{D}_{\text{raw}}\, a$ of Equation (A.1) is literally $\mathcal{S}_0\, p = \mathcal{D}_0\, a$, doubled by the convention that folds the weight of $T_0$ into $c_0 = 2$ rather than into the polynomial itself. This is the discrete shadow of the framing of § 5.1 claims: the descriptor recurrence does not compute a new identity, it asserts an existing one – that $p$'s own Chebyshev coefficients, converted into the $C^{(1)}$ basis by $\mathcal{S}_0$, must equal the derivative computed directly by $\mathcal{D}_0$ – as one row of a linear system, rather than solving it for $p$ via $\mathcal{S}_0^{-1}$.

The generalisation to order $k \ge 1$ in [12] raises the basis one level per derivative: a $\lambda$-th derivative needs coefficients in $C^{(\lambda)}$, obtained from the analogous pair $\mathrm{d}C_k^{(\lambda)}/\mathrm{d}x = 2\lambda\, C_{k-1}^{(\lambda+1)}$ $(k \ge 1$, giving $\mathcal{D}_\lambda$, still a single superdiagonal) and a conversion operator $\mathcal{S}_\lambda : C^{(\lambda)} \to C^{(\lambda+1)}$, still two diagonals [12, Eqs. 3.2–3.3]. To assemble one equation of order $k$, every term is raised to the common top basis $C^{(k)}$ by a chain $\mathcal{S}_{k-1} \cdots \mathcal{S}_{\lambda+1}$ applied after each term's own $\mathcal{M}_\lambda[a^\lambda]\mathcal{D}_\lambda$, giving one almost-banded operator $\mathrm{L} = \mathcal{M}_k[a^k]\mathcal{D}_k + \sum_{\lambda=0}^{k-1} \mathcal{S}_{k-1} \cdots \mathcal{S}_\lambda \mathcal{M}_\lambda[a^\lambda]\mathcal{D}_\lambda$ that is solved once for the coefficients of $u^{(k)}$ [12, §3.2]

– so a $k$-th order equation needs $k$ distinct conversion operators $\mathcal{S}_0, \ldots, \mathcal{S}_{k-1}$ chained together and the lower-order unknowns eliminated.

The descriptor construction of § 5.2 takes a different route to the same sparsity. Rather than raising the basis and eliminating, it keeps every derivative's coefficients in the *same* normalised Chebyshev basis, as its own block $\hat{f}^{(j)}$ of $w$, and reuses the single $\lambda = 0$ pair $(\tilde{B}, \tilde{D})$ between every consecutive pair of blocks (Equation (5.5)). No elimination is performed, and no operator beyond $\tilde{B}$, $\tilde{D}$ and the multiplication operators of Appendix D.2 is required at any order $k$ – the price is the larger state $w$ that carries every intermediate derivative explicitly instead of eliminating it (Appendix D.2). What we add to [12] is not a new sparsification of differentiation – that is their result – but applying the *same* $\lambda = 0$ identity block-wise to build an unreduced, differential-algebraic residual for the physics-informed Hamiltonian of [1], whose block structure is exactly what § 5.5 block-encodes.

For a first-order equation the two coincide. The descriptor's mass form multiplies the standard rows by $\tilde{B}$ (Equation (5.7)), and for $f' = af$ with $a$ a number, $\tilde{B}(G - aI) = \tilde{D} - a\tilde{B}$, which in the raw basis is $2(\mathcal{D}_0 - a\mathcal{S}_0)$: the ultraspherical method's own operator in the basis $C^{(1)}$, on $N$ unknowns with no derivative block.

**A reduction the descriptor does not build.** For $k \geq 2$ the same left factor exists. In the raw basis $\mathcal{S}_1\mathcal{S}_0 G^2 = \mathcal{D}_1\mathcal{D}_0$ exactly, so $\mathcal{S}_{k-1} \cdots \mathcal{S}_0 \, A_{\text{std}}$ is the ultraspherical method's operator for a constant-coefficient equation: banded, with a bandwidth independent of $N$, on the $N$ coefficients of $f$, and, since the conversions are invertible, with the standard form's kernel, as in Theorem 1. It would remove the chain's block register and its recurrence and padding rows. It would also change three things we measure: the residual's rows are weighted by the conversions, so its least-squares ground state is that of a differently weighted problem; the conversions' entries would have to be encoded; and the gap mechanism of § 5.3 does not carry over without a measurement. This reduction was not built into the pipeline, and the thesis's descriptor is not compared against it, although Olver and Townsend's conditioning bound suggests that on a regular problem it would reach the descriptor's degree (§ 9.5).

## A.6 The $N = 4$ instance

At $n = 2$ ($N = 4$), Equation (A.2) gives

$$\tilde{B} = \begin{pmatrix} 2 & 0 & -\sqrt{2} & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \tilde{D} = \begin{pmatrix} 0 & 2\sqrt{2} & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad G = \begin{pmatrix} 0 & \sqrt{2} & 0 & 3\sqrt{2} \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{A.7}$$

The top-right entry $3\sqrt{2}$ of $G$ is the fill-in: coefficient 0 of the derivative depends on coefficient 3 of the input. Row $\ell$ of $G$ is non-zero at columns $\ell + 1, \ell + 3, \ldots$, so this is the extreme case – the lowest-order coefficient can depend on up to half the others, while the highest-order one (row 3 above) depends on none. The factorisation removes the fill-in entirely: no row of $\tilde{B}$ or $\tilde{D}$ touches more than two coefficients.

For the first-order equation $f' + 2f = 0$ (R1), the descriptor runs the mass form of Equation (5.7), on the field's own $N$ coefficients and with no derivative block:

$$\tilde{D} + 2\tilde{B} = \begin{pmatrix} 4 & 2\sqrt{2} & -2\sqrt{2} & 0 \\ 0 & 2 & 4 & -2 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 2 \end{pmatrix} = \tilde{B}\,(G + 2I), \qquad G + 2I = \begin{pmatrix} 2 & \sqrt{2} & 0 & 3\sqrt{2} \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 2 \end{pmatrix}. \tag{A.8}$$

The standard form's equation block $G + 2I$ carries the fill-in $3\sqrt{2}$; the mass form's has none, and

every row touches at most three coefficients. Its kernel is that of $G + 2I$, because $\tilde{B}$ is invertible.

The same equation can be carried as a chain, with its derivative as a second block, $w = (\hat{f}^{(0)};\ \hat{f}^{(1)})$, the recurrence $-\tilde{D}\,\hat{f}^{(0)} + \tilde{B}\,\hat{f}^{(1)} = 0$ and the equation row $2\hat{f}^{(0)} + \hat{f}^{(1)} = 0$:

$$A_{\text{desc}} = \left[\begin{array}{cccc|cccc} 0 & -2\sqrt{2} & 0 & 0 & 2 & 0 & -\sqrt{2} & 0 \\ 0 & 0 & -4 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & -6 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ \hline 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \end{array}\right]. \tag{A.9}$$

The top four rows are the recurrence and the bottom four the equation. This chain doubles the state for a block the equation already determines, which is why first-order equations take the mass form (§ 5.2); it is kept only as the mass form's ablation, whose kernel and decoded field agree with the mass form's.

# B The Standard Form in Full

This appendix gives the $N = 4$ instance of the standard-form residual of § 4.2 for a constant-coefficient equation, the entries of the lifts $\mathcal{M}_{x^p}$ that the same section uses for a variable coefficient, and the constructions and proofs that § 4.3 points to.

**Why a condition for all $x$ is a finite linear system.** A condition required at every $x \in [-1, 1]$, such as the differential equation, reaches the form $\langle\tau(x)|\,C\,|\psi\rangle = 0$ for all $x$. The components of $\langle\tau(x)|$ are linearly independent functions of $x$, so $\langle\tau(x)|\,\mathbf{v}$ vanishes for all $x$ only if $\mathbf{v} = 0$. Taking $\mathbf{v} = C\,|\psi\rangle$, the condition for all $x$ is therefore equivalent to the finite system $C\,|\psi\rangle = 0$, which no longer depends on $x$. This is the step that §§ 3.2 and 4.2 take without proof.

**The standard-form residual at $N = 4$.** For Wu et al.'s stiff pilot equation [1], $f''+5f'+400f = 0$, the equation block of Equation (4.3) is $A_{\text{std}} = G^2 + 5G + 400I$. At $N = 4$, with the entries of Equation (4.2),

$$A_{\text{std}} = \begin{pmatrix} 400 & 5\sqrt{2} & 4\sqrt{2} & 15\sqrt{2} \\ 0 & 400 & 20 & 24 \\ 0 & 0 & 400 & 30 \\ 0 & 0 & 0 & 400 \end{pmatrix}. \tag{B.1}$$

It is upper triangular, as every polynomial in the upper-triangular $G$ is. At this size the largest off-diagonal entry, 30, is about one order of magnitude below the diagonal; the entries of $G^2$ grow as $N^4$, so at larger $N$ the derivative terms dominate the constant one.

**The lifting operator $\mathcal{M}_{x^p}$.** $\mathcal{M}_{x^p}$ is exact and *rectangular*: multiplying a degree-$(N-1)$ model by $x^p$ can raise its degree by as much as $p$, so the result is represented one register wider, $2N \times N$, with no truncation. At $n = 1$ ($N = 2$, so $p \in \{1, 2\}$),

$$\mathcal{M}_{x^1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 0 & \frac{1}{\sqrt{2}} \\ 0 & 0 \end{pmatrix}, \qquad \mathcal{M}_{x^2} = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{3}{2\sqrt{2}} \\ \frac{1}{2} & 0 \\ 0 & \frac{1}{2\sqrt{2}} \end{pmatrix}, \tag{B.2}$$

which reproduce Wu et al.'s printed matrices. The general entry follows from the product-to-sum identity conjugated into the normalised bases of $|\tau(x)\rangle_n$ and $|\tau(x)\rangle_{n+1}$ (Appendix B.5).

## B.1 The standard form on several axes

§ 3.2 writes the two-variable case. For $\omega$ variables $\mathbf{x} = (x_1, \ldots, x_\omega)$, axis $u$ carries $n_u$ qubits and the first $N_u = 2^{n_u}$ Chebyshev polynomials of $x_u$. The product functions $\varphi_{k_1}(x_1) \cdots \varphi_{k_\omega}(x_\omega)$ span the polynomials of degree below $N_u$ in each variable, and their coefficients fill an array of $\prod_u N_u = 2^{\sum_u n_u}$ entries, so the state lives on $\sum_u n_u$ qubits, $\omega n$ when every axis has $n$. The feature vector is the product $|\tau(\mathbf{x})\rangle = |\tau(x_1)\rangle_{n_1} \otimes \cdots \otimes |\tau(x_\omega)\rangle_{n_\omega}$, and $f_q(\mathbf{x}) = \langle \tau(\mathbf{x}) | \psi \rangle$. The derivative along axis $u$ is $G_u = I \otimes \cdots \otimes G \otimes \cdots \otimes I$, with $G$ on factor $u$. Operators on different factors commute, so a derivative of multi-index $\boldsymbol{\chi} = (\chi_1, \ldots, \chi_\omega)$ is $\partial^{\boldsymbol{\chi}} \mapsto G^{\boldsymbol{\chi}} = \bigotimes_u G^{\chi_u}$ whatever the order of differentiation, and each $G_u$ has the norm $\Theta(N_u^2)$ of $G$.

**Terms and lifts.** Each term $t$ of the equation has a constant $c_t$, a multi-index $\boldsymbol{\chi}_t$ and a coefficient $m_t$ that is a polynomial in a single variable $x_{u_t}$; a coefficient that depends on several variables is not built. On axis $u_t$ the product $m_t f$ raises the degree, so it is represented exactly only one qubit wider, by the lift $\mathcal{M}_{m_t}$ of Proposition 13. The terms of $A_{\text{std}}$ are summed, so every term must reach the same output register, and an axis on which any term needs a lift is therefore *lifted* for all of them: a term whose coefficient on that axis is constant passes through $\mathcal{M}_1$, which places the same function on the wider register. A source is entered, as Wu et al. enter it, as the rank-one operator $\mathcal{M}_r \mathbb{D}^{(0)}(\mathbf{x}_s) = |\hat{r}\rangle \langle \tau(\mathbf{x}_s)| / y_s$ with $|\hat{r}\rangle$ the coefficients of $r$ on the lifted register, so an equation with a source lifts every axis. Writing $\mathcal{M}_t^{(u)}$ for $\mathcal{M}_{m_t}$ if $u = u_t$, $\mathcal{M}_1$ on any other lifted axis and the identity on an unlifted one,

$$A_{\text{std}} = \sum_t c_t \bigotimes_{u=1}^{\omega} \mathcal{M}_t^{(u)} \, G^{\chi_{t,u}} \; - \; \mathcal{M}_r \, \mathbb{D}^{(0)}(\mathbf{x}_s), \tag{B.3}$$

which for $\omega = 1$ is Equation (4.3). The input register has $\sum_u n_u$ qubits and the output register $\sum_u n_u + \omega_{\text{lift}}$, with $\omega_{\text{lift}}$ the number of lifted axes, so $A_{\text{std}}$ has $2^{\omega_{\text{lift}}} \prod_u N_u$ rows.

**Proposition 9** (The residual on several axes)**.** *Let every coefficient $m_t$ have degree at most $N_{u_t}$ in its variable, and let $r$ stand for the polynomial whose coefficients $\hat{r}$ holds on the lifted register. For every coefficient vector $b$ of a polynomial $p$ of degree below $N_u$ in each $x_u$, $A_{\text{std}} b$ holds the Chebyshev coefficients of $\sum_t c_t m_t \partial^{\boldsymbol{\chi}_t} p - r\, p(\mathbf{x}_s)/y_s$ on the output register, without truncation. In particular $A_{\text{std}} b = 0$ exactly when that function vanishes on $[-1, 1]^\omega$.*

*Proof.* On a product function $\varphi_{k_1}(x_1) \cdots \varphi_{k_\omega}(x_\omega)$, $m_t \partial^{\boldsymbol{\chi}_t}$ acts on each factor separately: $\partial_u^{\chi_{t,u}}$ on every factor, which $G^{\chi_{t,u}}$ represents exactly, and the multiplication by $m_t$ on factor $u_t$ alone, which $\mathcal{M}_{m_t}$ represents exactly when $\deg m_t \le N_{u_t}$ (Proposition 13). On the other lifted axes $\mathcal{M}_1$ re-expresses the same factor on the wider register, and on an unlifted axis the factor is unchanged. The Kronecker product of these factor maps is therefore the term's action on the product function, and by linearity on every $p$. The source term maps $b$ to $\hat{r}\, \langle \tau(\mathbf{x}_s) | b \rangle / y_s = \hat{r}\, p(\mathbf{x}_s)/y_s$. The sum is the coefficient vector of the stated function, and since the product functions on the output register are linearly independent it vanishes exactly when the function does, the argument of Appendix B. □

Applied to $cf_N$, with $f_N$ a solution represented on the input register and $c$ any scalar, the source term gives $c\,\hat{r}$ and the equation terms $c$ times the coefficients of $\sum_t c_t m_t \partial^{\boldsymbol{\chi}_t} f_N$, so the residual is homogeneous in the scale, as on one axis. Whether the truncated solution is represented exactly, and whether the kernel is one-dimensional, are computed (§ 4.3).

**Constraints on several axes.** An invariant constraint $\partial_u^j f = 0$ on the face $x_u = x_z$ is the row block $I \otimes \cdots \otimes \mathbb{B}_{n_u}(x_z) G^j \otimes \cdots \otimes I$, or $\mathbb{B}_{n_u+1}(x_z) \mathcal{M}_1 G^j$ on its own axis if that axis is lifted. It has one row for each basis state of the other axes, $\prod_{v \neq u} N_v$ in all, and acts on the input register only:

its identity factors are on $n_v$ qubits whether or not axis $v$ is lifted, since only the equation block's output needs the wider register. A function imposed on a face is the datum slice of Appendix B.2, and the regular datum at $\mathbf{x}_s$ is the rank-one row $\mathbb{D}^{(0)}(\mathbf{x}_s) = \bigotimes_u \mathbb{B}_{n_u}(x_{s,u})/y_s$.

**What was run.** The benchmark problems use at most two axes, the heat, wave and Laplace equations (R5a–R5c), each with constant coefficients and no source, so no two-axis problem here is lifted. Equation (B.3) and the constraint rows are written for any $\omega$ and any $n_u$, but every problem run has the same $n$ on each axis, and no problem on three or more axes was built or verified, and none is claimed.

## B.2 Datum-slice rows

An inhomogeneous initial or boundary function $g$ on the slice $x_a = x^*$ of a problem on several Chebyshev axes is imposed by the rows of Equation (4.4),

$$\mathbb{S}_a(x^*) - \frac{\hat{g}\ \langle\tau(x_s)|}{f(x_s)}, \qquad \mathbb{S}_a(x^*) = I \otimes \cdots \otimes \mathbb{B}(x^*) \otimes \cdots \otimes I \quad (\mathbb{B} \text{ on axis } a),$$

with $\hat{g}$ the coefficients of $g$ on the other axes broadcast as a constant along axis $a$: its mode 0 on that axis, $\sqrt{N_a}$ times the coefficients of $g$, as $\mathbb{B}$ emits. $\langle\tau(x_s)|$ is the feature row at a datum point $x_s$ of the full domain, where the value $f(x_s)$ is known.

**Proposition 10** (Datum slices)**.** *On $cf$, with $f$ the solution and $c$ any scalar, the rows return $c\,(\hat{f}|_{x_a=x^*} - \hat{g})$, with $\hat{f}|_{x_a=x^*}$ the broadcast slice of $f$. They are therefore homogeneous in the scale and vanish on the solution and its multiples.*

*Proof.* $\mathbb{S}_a(x^*)$ contracts axis $a$ with $\tau(x^*)$, which gives the slice function's coefficients on the other axes times $\sqrt{N_a}\,|0\rangle$ on axis $a$: applied to $cf$, the broadcast slice $c\hat{f}|_{x_a=x^*}$. The second term applied to $cf$ is $\hat{g}\,c\,f(x_s)/f(x_s) = c\,\hat{g}$. If $f|_{x_a=x^*} = g$ the difference is zero. □

The block is square on $\prod_a N_a$ coefficients but non-zero only on the $\prod_{b\neq a} N_b$ rows with mode 0 on axis $a$, which carry the slice; the rest are dropped from $R$ without changing $R^\dagger R$. On one axis the slice is a point, and the rows reduce to the single row $\mathbb{B}(x^*) - y^*\mathbb{D}^{(0)}(x_s)$ of a point datum $f(x^*) = y^*$, or to $\mathbb{B}(x^*)G^j - y^*\mathbb{D}^{(0)}(x_s)$ for a datum on a derivative (Definition 2). The rows are built from the problem's statement, the function $g$, the slice and one regular value, so they serve both forms and every resolution; whether the kernel is then one-dimensional is computed (§ 4.3).

**Why the heat data sit at $t = -1$.** Wu et al.'s heat problem, $\partial_t f = \frac{1}{25}\partial_x^2 f$ with $f(t, \pm 1) = 0$, states its initial function $\sin(2\pi x)$ at $t = 0$, inside the Chebyshev time domain $[-1, 1]$. Imposed there, it fixes the solution on $[-1, 0]$ by a backward heat problem: going back a time $s$, a mode $\sin\frac{\ell\pi(x+1)}{2}$ of the data is multiplied by $\exp\big(\frac{1}{25}(\ell\pi/2)^2 s\big)$, a factor unbounded in $\ell$, so the problem is ill posed in Hadamard's sense. The standard form states the problem forward from $t = -1$ instead, with the initial function $e^{4\pi^2/25}\sin(2\pi x)$, Wu et al.'s solution $e^{-4\pi^2 t/25}\sin(2\pi x)$ evaluated there. The solution is the same; the data are set up from it, as R4's constraint points are.

## B.3 The substitution for R3c and R3d

Wu et al.'s R3c and R3d are the associated Legendre equation with $m = 1$, multiplied through by $1 - x^2$,

$$(1 - x^2)^2 f'' - 2x(1 - x^2)f' + \big(l(l + 1)(1 - x^2) - 1\big)f = 0,$$

with $l = 5$ and $l = 6$. Its leading coefficient has a double zero at each end, and its solution $P_l^1 = -\sqrt{1 - x^2}\,P_l'$ has a branch point there, so its Chebyshev coefficients decay only algebraically:

$\sqrt{1-x^2} = |\sin\theta|$ has cosine coefficients of order $k^{-2}$. At the printed $n = 5$ neither problem has a trusted ground state, and the error on R3c is $8.7 \times 10^{-1}$.

**Proposition 11** (The substitution). *With $f = \sqrt{1-x^2}\,v$ the equation becomes*

$$(1-x^2)\,v'' - 4x\,v' + \big(l(l+1) - 2\big)v = 0, \tag{B.4}$$

*whose leading coefficient has simple zeros at $\pm 1$ and whose solution $v = -P_l'$ is a polynomial of degree $l-1$.*

*Proof.* Write $s = \sqrt{1-x^2}$, so that $s' = -x/s$ and $s'' = -1/s^3$. Then $f' = sv' - xv/s$ and $f'' = sv'' - 2xv'/s - v/s^3$. Substituting, $(1-x^2)^2 f'' = s^5 v'' - 2xs^3 v' - sv$, $-2x(1-x^2)f' = -2xs^3 v' + 2x^2 sv$ and $\big(l(l+1)s^2 - 1\big)sv = l(l+1)s^3 v - sv$. The sum is $s^5 v'' - 4xs^3 v' + \big(l(l+1)s^3 - 2s(1-x^2)\big)v = s^3\big[(1-x^2)v'' - 4xv' + (l(l+1)-2)v\big]$, and $s^3 > 0$ on $(-1,1)$. The solution is $v = f/s = -P_l'$. □

The data carry through: $f(\pm 1) = 0$ holds for every $v$ and is dropped; R3c's $f'(0) = 0$ is $v'(0) = 0$, since $s(0) = 1$ and $s'(0) = 0$; R3d's $f(0) = 0$ is $v(0) = 0$; and the regular datum becomes $v(x_s) = f(x_s)/\sqrt{1-x_s^2}$, at $x_s = \frac{1}{2}$. The coefficients of Equation (B.4) are polynomials of degree at most two, inside the lift's range, and $v$ is represented exactly once $N > l - 1$, so the field error falls to rounding at the first such $n$ (§ 4.3). The substitution removes this double zero because the branch point is a known factor of the solution; it is not a remedy for a double zero in general. The simple zeros it leaves are what the descriptor's block rescaling can treat (§ 5.4).

## B.4 The uniqueness count for an ordinary equation

**Proposition 12** (Uniqueness count). *Let $\sum_{j=0}^{k} a_j f^{(j)} = 0$ on $[-1,1]$ with smooth coefficients and $a_k$ nowhere zero, and let $c_1, \ldots, c_m$ be invariant constraints (each sets a derivative of $f$ at a point to zero). The solutions that satisfy all of them form a space of dimension $k - r$, where $r$ is the rank of the $c_i$ restricted to the $k$-dimensional solution space $S$. The ground state of the constraint energy is unique up to scale exactly when $r = k - 1$.*

*Proof.* Since $a_k$ has no zero, the equation is a regular linear system of order $k$ and $S$ has dimension $k$. The map $f \mapsto (c_i(f))_{i \le m}$ is linear on $S$ with rank $r$, so its kernel, the solutions that satisfy every constraint, has dimension $k - r$. A ground state of zero energy is a nonzero element of that kernel, unique up to scale exactly when the kernel is one-dimensional. For $r = k$ the kernel is $\{0\}$ and no nonzero state has zero energy. For $r < k - 1$ it has dimension at least two. □

The rank is that of the constraints restricted to $S$, not of the functionals, so $k - 1$ constraints chosen without regard to the equation can fail. The regular datum fixes the scale of the selected solution and requires that it not vanish at the datum point. The proposition concerns the continuous problem: the assembled residual follows the count only as far as the truncation resolves $S$, and the dimension and gap of its kernel are computed, not inferred (§ 4.3). For an inhomogeneous equation, or one whose leading coefficient vanishes (R3a), no count is claimed here and uniqueness is judged by the computed kernel.

**The exact constraint points.** The standard form keeps Wu et al.'s invariant row and moves it to the exact root of the solution's derivative, rather than imposing the problem's stated data, so that the residual keeps Wu et al.'s rows and differs from the printed one only in the defect. The root is computed from the reference solution when the problem is built, which makes the set-up answer-dependent. The stated data could enter without the answer, as the point-datum rows $\mathbb{B}(x^*)G^j - y^*\mathbb{D}^{(0)}(x_s)$ of Definition 2; none of our runs uses them in place of an invariant row.

## B.5 Lifts and folds

For $a(x) = \sum_m a_m T_m(x)$ the identity $T_m T_k = \frac{1}{2}\big(T_{m+k} + T_{|m-k|}\big)$ gives the Chebyshev coefficients of $a\,T_k$, $X_{jk} = \frac{1}{2}\sum_m a_m\big([\,j = m+k\,] + [\,j = |m-k|\,]\big)$. In the normalised bases of $n$ input and $n+1$ output qubits, with $\sigma_k$ the normalisation of $T_k$ on each register ($\varphi_k = T_k/\sigma_k$), the lift is $(\mathcal{M}_a)_{jk} = (\sigma_j^{\text{out}}/\sigma_k^{\text{in}})\,X_{jk}$; a monomial coefficient is converted to its Chebyshev series first.

**Proposition 13** (The lift)**.** *$\mathcal{M}_a \in \mathbb{R}^{2N\times N}$ represents the product $a\,f_N$ without truncation for every $f_N$ of degree below $N$ if and only if $\deg a \le N$. A constant enters as $a_0\mathcal{M}_1 = a_0\sqrt{2}\,[\,I;\,0\,]$, and $\|\mathcal{M}_a\|_2 \le \sqrt{2}\,\sup_{[-1,1]}|a|$.*

*Proof.* The product has degree at most $N-1+\deg a$, and $2N$ coefficients hold degrees up to $2N-1$, so $\deg a \le N$ suffices; if $\deg a > N$, a series of degree $N-1$ gives a product of degree at least $2N$, which does not fit. Each normalisation $\sigma_k$ on $n+1$ qubits is $\sqrt{2}$ times that on $n$, since $\varphi_k = T_k/\sqrt{N/2}$ on $n$ qubits and $T_k/\sqrt{N}$ on $n+1$ for $k \ge 1$, and likewise for $T_0$, so the coefficients of a function are multiplied by $\sqrt{2}$ on the wider register, which gives $\mathcal{M}_1$. For the norm, Parseval's identity of Appendix A.3 on $n+1$ qubits, where the product is exact, gives $\|\mathcal{M}_a b\|^2 = (2N/\pi)\int a^2 f_N^2 w\,dx \le 2\sup|a|^2\,(N/\pi)\int f_N^2 w\,dx = 2\sup|a|^2\,\|b\|^2$. □

The lift of $x$ has the non-zero entries $(\mathcal{M}_x)_{10} = (\mathcal{M}_x)_{01} = 1$ and $(\mathcal{M}_x)_{k\pm1,k} = 1/\sqrt{2}$ otherwise, from $xT_0 = T_1$ and $xT_k = \frac{1}{2}(T_{k+1} + T_{k-1})$, which is Equation (B.2) at $n = 1$. On a circuit the lift of $x$ is a combination of truncated shifts, built by the post-selected subtraction of Appendix C.2, and a polynomial coefficient is the sum of the powers of that block weighted by its monomial coefficients, so its subnormalisation is $\sum_p |a_p|\,\alpha_{\mathcal{M}_x}^p$. For a coefficient that vanishes where the derivatives are largest, such as the $(1-x^2)$ of R3a, this sum far exceeds what the product $\mathcal{M}_a G^2$ needs, which is the looseness of the standard form's encoding on R3 (§ 4.6). The folds $\mathcal{N}_1$ and $\mathcal{N}_x$ of the doubled space follow from the same product-to-sum identity applied to a product of two series (Equation (3.7)); their circuits and norms are given in Appendix E.2.

# C The Structured Exact Encoding of the Differentiation Matrix

## C.1 Proof of Proposition 1, and the two diagonal factors

*Proof of Proposition 1.* $R_0 = \operatorname{diag}(1/\sqrt{2}, 1, \ldots, 1)$ and $\Lambda = \operatorname{diag}(0, 1, \ldots, N-1)$ are diagonal, so $\big(R_0(2U_{\text{odd}})\Lambda\big)_{kp} = (R_0)_{kk}\cdot 2[\,p > k,\ p-k \text{ odd}\,]\cdot p$. This is $2p$ for $k \ge 1$ and $\sqrt{2}\,p$ for $k = 0$ when $p > k$ and $p-k$ is odd, and zero otherwise, which is Equation (4.2). $U_{\text{odd}} = \sum_{d\ \text{odd}} S_d$ with $(S_d)_{kp} = [\,p-k = d\,]$, the sum running over the $N/2$ odd $d \in [1, N-1]$. □

**Lemma 1** ($\Lambda$ as a sum of Pauli strings)**.**

$$\Lambda = \frac{N-1}{2}\,I - \sum_{j=0}^{n-1} 2^{j-1} Z_j, \tag{C.1}$$

*so its LCU has $n+1$ unitary terms, $\lceil \log_2(n+1)\rceil$ selector qubits and $\alpha_\Lambda = N-1 = \|\Lambda\|_2$.*

*Proof.* On $|p\rangle$ with bits $p_j$, $Z_j$ has the eigenvalue $z_j = 1-2p_j$, so $p = \sum_j 2^j p_j = \sum_j 2^j(1-z_j)/2 = \frac{N-1}{2} - \sum_j 2^{j-1} z_j$. The weights' magnitudes sum to $\frac{N-1}{2} + \sum_j 2^{j-1} = N-1$. The negative weights are phases on SELECT's branches, so no value is loaded by rotation. □

**Lemma 2** ($R_0$ as two terms)**.** *$R_0 = a\,I + b\,(I - 2\,|0\rangle\langle 0|)$ with $a = \frac{1}{2}(1+1/\sqrt{2})$ and $b = \frac{1}{2}(1-1/\sqrt{2})$, both positive, so $\alpha_{R_0} = a+b = 1 = \|R_0\|_2$, with one selector qubit.*

*Proof.* On $|0\rangle$ the right side is $a-b = 1/\sqrt{2}$; on every other basis state it is $a+b = 1$. □

## C.2 The arithmetic SELECT

The ripple-carry adder of Cuccaro et al. [66], on registers $a$ and $b$ of $n$ qubits with a carry-in $c$ and an overflow $z$, acts as $(c, a, b, z) \mapsto \big(c, a, (a+b+c) \bmod 2^n, z \oplus [a+b+c \geq 2^n]\big)$ on every input, since its MAJ and UMA identities do not use $c = 0$. Its inverse therefore maps $b$ to $(b-a-c) \bmod 2^n$ and flips $z$ exactly when $b < a + c$: with $c = z = 0$ it is the in-place subtraction $b \leftarrow (b-a) \bmod 2^n$ with the borrow $z = [\, b < a \,]$. The circuit of Figure 4.1 uses an offset register $d$ of $n$ qubits, the carry and the borrow, $n+2$ ancilla in all: PREPARE applies $X$ to the lowest qubit of $d$ and Hadamards to the others, SELECT is the subtraction with $a = d$ and $b$ the system register, and PREPARE$^\dagger$ follows.

**Lemma 3** (The odd shifts by subtraction)**.** *The circuit block-encodes $U_{\text{odd}}$ exactly at $\alpha = N/2$.*

*Proof.* PREPARE maps $|0\rangle$ to $\sqrt{2/N} \sum_{\ell \text{ odd}} |\ell\rangle$. From $|p\rangle\,|0\rangle$, PREPARE and SELECT give $\sqrt{2/N} \sum_{\ell \text{ odd}} |(p$ Projecting the borrow on zero keeps the terms with $\ell \leq p$, for which $(p-\ell) \bmod N = p-\ell$; projecting the offset register on zero after PREPARE$^\dagger$ weights each by $\langle 0|\,\text{PREPARE}^\dagger\,|\ell\rangle = \sqrt{2/N}$. The block's entry $(k, p)$ is therefore $\frac{2}{N}\,\#\{\ell \text{ odd} : \ell = p - k \geq 1\} = \frac{2}{N}[\, p > k,\ p - k \text{ odd} \,]$. □

The terms that wrap around carry a borrow of one and vanish on post-selection, and SELECT is a permutation, so one PREPARE serves all $N/2$ terms. The subtractor costs $n$ MAJ and $n$ UMA gates, each two CNOTs and a Toffoli, and one CNOT: $16n + 1$ CNOTs with a six-CNOT Toffoli, which is the transpiled count at every $n$ computed.

## C.3 Proof of Proposition 2

*Proof of Proposition 2. Subnormalisation.* A product of block encodings is a block encoding of the product at the product of the subnormalisations, so $\alpha_G = \alpha_{R_0}\,\alpha_{2U_{\text{odd}}}\,\alpha_\Lambda = 1 \cdot N \cdot (N-1)$ by the two lemmas above and Lemma 3 (a block encoding of $U_{\text{odd}}$ at $N/2$ is one of $2U_{\text{odd}}$ at $N$). Its ratio to $\|G\|_2$ tends to $2j_{-3/4,1}$ by Theorem 3, since $N(N-1)/N^2 \to 1$.

*Ancilla.* The encodings are composed as $R_0 \cdot \big((2U_{\text{odd}}) \cdot \Lambda\big)$, $\Lambda$ applied first, each product sharing one ancilla register between its factors and adding one flag that marks the right factor's failure branch. On the right factor's success branch its ancilla are back at zero, the state the left factor needs, and on its failure branch the flag is raised and the branch never reaches the post-selected output, so the product stays exact. The inner product shares $\max\big(n+2, \lceil \log_2(n+1) \rceil\big) = n+2$ qubits and adds a flag; the outer shares $\max(1, n+3)$ with $R_0$'s selector and adds a flag: $n+4$ ancilla in all, the count of the built circuit for every $n$ from 2 to 8. Of these, the $n-1$ qubits that carry the uniform superposition are forced by the $N/2$ terms of $U_{\text{odd}}$'s combination, since an LCU of $L$ distinct terms needs $\lceil \log_2 L \rceil$ selector qubits.

*Gates.* $U_{\text{odd}}$'s PREPARE and PREPARE$^\dagger$ are $2n$ one-qubit gates and its subtractor $O(n)$ gates. $\Lambda$'s PREPARE is a rotation tree on $\lceil \log_2(n+1) \rceil$ qubits, and its SELECT is $n$ gates $Z_j$ and the negative weights' phases, each controlled on the selector. $R_0$ is $2n$ CNOTs and one $n$-controlled $Z$, and the two flags are an $X$ controlled on $\Lambda$'s selector and one controlled on the inner product's $n+3$ ancilla. An $m$-controlled gate costs $O(m^2)$ CNOTs in an ancilla-free synthesis, so the $n$-controlled gates of $R_0$ and the flags give $O(n^2)$; with a linear-cost synthesis that borrows an idle qubit [87] they cost $O(n)$, and $\Lambda$'s SELECT, $O(n \log n)$, dominates. □

The transpiled two-qubit gate counts of the encoding of $G$ alone are 109 at $n = 2$, 471 at $n = 5$ and 1365 at $n = 8$: consistent with the bound, though the transpiler's synthesis of the multi-controlled gates makes them too irregular to fit an exponent.

**Lemma 4** (The norm of the odd shifts)**.** $\|U_{\text{odd}}\|_2 = 1/\big(2 \sin \frac{\pi}{2N+2}\big)$*, so* $\alpha_{U_{\text{odd}}}/\|U_{\text{odd}}\|_2 = N \sin \frac{\pi}{2N+2}$*, which rises to* $\pi/2$*.*

*Proof.* Ordering rows and columns by parity, $U_{\text{odd}}$ maps odd columns to even rows and even columns to odd rows, and its two blocks are the $M\times M$ upper triangle of ones with its diagonal, $M = N/2$, and the $(M-1)\times(M-1)$ one, padded with zeros. For the $m\times m$ triangle of ones $L$, $L^{-1} = I-S$ with $S$ the shift above the diagonal, and $(LL^\top)^{-1} = (I-S^\top)(I-S)$ is tridiagonal with $1, 2, \dots, 2$ on the diagonal and $-1$ beside it, whose eigenvalues are $4\sin^2\frac{(2i-1)\pi}{2(2m+1)}$, $i = 1, \dots, m$. So $\|L\|_2 = 1/\big(2\sin\frac{\pi}{4m+2}\big)$, increasing in $m$, and $m = M$ gives the lemma. □

$R_0$ and $\Lambda$ are encoded at their norms, so of the limiting ratio $2j_{-3/4,1}$ the factor $\pi/2$ is $U_{\text{odd}}$'s and the remaining $4j_{-3/4,1}/\pi \approx 1.35$ is the product's: $\|R_0(2U_{\text{odd}})\Lambda\|_2 < \|R_0\|_2\,\|2U_{\text{odd}}\|_2\,\|\Lambda\|_2$, because $\Lambda$ is largest on the last columns while the leading singular vectors of $U_{\text{odd}}$ are spread over all of them.

## C.4 The norms of the derivative and its powers

**Theorem 3** (The norm of $G$ in the limit)**.** $\|G\|_2/N^2 \to \|K\| = 1/(2j_{-3/4,1}) \approx 0.4724$ *as* $N \to \infty$, *where* $(Kf)(s) = \int_s^1 t\,f(t)\,dt$ *on* $L^2(0,1)$ *and* $j_{-3/4,1}$ *is the first positive zero of the Bessel function* $J_{-3/4}$.

*Proof.* Order the rows and columns by parity. $G$ maps odd columns to even rows and even columns to odd rows, so in that order it is $\left(\begin{smallmatrix} 0 & E \\ O & 0 \end{smallmatrix}\right)$ and $\|G\|_2 = \max(\|E\|_2, \|O\|_2)$, with, for $M = N/2$ and $0 \le i, j < M$,

$$E_{ij} = \frac{2(2j+1)}{\sqrt{c_{2i}}}\,[\,j \ge i\,], \qquad O_{ij} = 4j\,[\,j > i\,]$$

(row $2i$, column $2j+1$ for $E$; row $2i+1$, column $2j$ for $O$; $c_0 = 2$, $c_{k\ge1} = 1$). An $M \times M$ matrix with entries $\frac{1}{M}\kappa(i,j)$ has the operator norm of the integral operator on $L^2(0,1)$ with the step kernel $\kappa(\lfloor Ms\rfloor, \lfloor Mt\rfloor)$, since $J : v \mapsto \sqrt{M}\sum_i v_i \mathbf{1}_{[i/M,(i+1)/M)}$ is an isometry onto the step functions. Here $E/N^2 = \frac{1}{M}\kappa_E$ with $\kappa_E(i,j) = \frac{2j+1}{2M}[\,j \ge i\,]/\sqrt{c_{2i}}$, and $O/N^2 = \frac{1}{M}\kappa_O$ with $\kappa_O(i,j) = \frac{j}{M}[\,j > i\,]$. Both step kernels are bounded and converge to $k(s,t) = t\,[\,t \ge s\,]$ except on a set of measure $O(1/M)$ (the diagonal cells and the first row), and to within $O(1/M)$ elsewhere, so they converge in $L^2([0,1]^2)$. The Hilbert–Schmidt norm bounds the operator norm, so $\|E\|_2/N^2$ and $\|O\|_2/N^2$ both converge to $\|K\|$.

$K$ is Hilbert–Schmidt, so $\|K\|^2$ is the largest eigenvalue $\lambda$ of $K^*K$, with $(K^*g)(t) = t\int_0^t g(s)\,ds$. Let $\lambda f = K^*Kf$ and $h(t) = \int_0^t (Kf)(s)\,ds$. Then $\lambda f = t\,h$, $h(0) = 0$, $h'(1) = (Kf)(1) = 0$ and $h'' = (Kf)' = -tf = -t^2h/\lambda$. With $a = \lambda^{-1/2}$, $h'' + a^2t^2h = 0$, whose solutions are $\sqrt{t}\,J_{\pm1/4}(at^2/2)$, and $h(0) = 0$ selects $J_{1/4}$. With $z = at^2/2$, the identity $zJ_\nu'(z) = zJ_{\nu-1}(z) - \nu J_\nu(z)$ at $\nu = \frac14$ gives

$$h'(t) = \frac{1}{2\sqrt{t}}\big(J_{1/4}(z) + 4zJ_{1/4}'(z)\big) = \frac{2z}{\sqrt{t}}\,J_{-3/4}(z),$$

so $h'(1) = 0$ exactly when $J_{-3/4}(a/2) = 0$. The eigenvalues are $\lambda = 1/a^2$ with $a/2$ the positive zeros of $J_{-3/4}$, the largest at the first, and $\|K\| = 1/(2j_{-3/4,1})$. □

**Theorem 4** (The powers of $G$)**.** *For every* $k \ge 1$, $\|G^k\|_2/N^{2k} \to \|K^k\| > 0$.

*Proof.* In the parity ordering $G^k$ is block diagonal or block anti-diagonal, with blocks that are alternating products of $k$ factors $E$ and $O$. Under the isometry $J$ of the previous proof, with $J^*J = I$, each block of $G^k/N^{2k}$ is unitarily equivalent to a product of $k$ operators that converge in norm to $K$, and so converges to $K^k$. $K$ is injective, since $\int_s^1 tf(t)\,dt = 0$ for all $s$ forces $f = 0$, so $K^k \ne 0$. □

**Corollary 1** (The standard form's norm)**.** *For an equation* $\sum_{j\le k} a_j f^{(j)} = r$ *of order* $k$ *with constant coefficients and* $a_k \ne 0$, $\|R_{\text{std}}\|_2 = |a_k|\,\|K^k\|\,N^{2k}(1+o(1)) = \Theta(N^{2k})$ *and* $\|H_{\text{std}}\| = \Theta(N^{4k})$.

*Proof.* The equation block $\sum_j a_jG^j$ has norm $|a_k|\,\|K^k\|\,N^{2k}(1+o(1))$ by Theorem 4, its lower orders being $O(N^{2j})$ with $j < k$. A source block $\mathcal{M}_r\mathbb{D}^{(0)}(x_s)$ is $O(\sqrt{N})$, and a constraint row $\mathbb{B}(x)G^j$ is

rank one with norm $\sqrt{N}\|\tau(x)^\top G^j\| \le \sqrt{2N}\,\|G\|_2^j = O(N^{2j+1/2})$ by Proposition 7, with $j \le k-1$, so every constraint row is $O(N^{2k-3/2})$. Since $\|R\|_2^2 = \|A^\dagger A + \sum_i C_i^\dagger C_i\|$ lies between $\|A\|^2$ and $\|A\|^2 + \sum_i \|C_i\|^2$, the rows do not move the leading term. □

The same limits fix the looseness of a power encoded as a product: $\alpha_{G^k} = \alpha_G^k = (N(N-1))^k$, so $\alpha_{G^k}/\|G^k\|_2 \to 1/\|K^k\|$, a constant for each $k$, equal to $2j_{-3/4,1}$ at $k=1$. For R2a, whose equation block is $G^2$ to leading order, this is the constant towards which the measured $\alpha_R/\|R_{\mathrm{std}}\|_2$ of § 4.6 levels off, from 4.72 at $n=2$ to 7.57 at $n=6$ and 7.74 at $n=8$; the theorem does not by itself bound the whole residual's ratio.

## C.5 The standard form's gap and the degree it implies

Write $\|g\|_w^2 = \int_{-1}^{1} g(x)^2(1-x^2)^{-1/2}\,dx$. On a register of $N'$ terms, a polynomial $g$ of degree below $N'$ has a coefficient vector $\hat{g}$ in the normalised basis with $\langle\tau(x)|\hat{g}\rangle = g(x)$ and, by the Parseval identity of Equation (4.1), $\|\hat{g}\|^2 = (N'/\pi)\|g\|_w^2$.

**Proposition 14** (The standard form's gap is bounded)**.** *Let $\sum_{j\le k} a_j f^{(j)} = r$ be an ODE on one axis with polynomial coefficients $a_j$ and a source $r$ continuous on $[-1,1]$, with regular datum $f(x_s) = y_s \neq 0$, and let its standard form (Definition 2) carry zero rows $\mathbb{B}(x^*)G^j$ and point-datum rows $\mathbb{B}(x^*)G^j - y^*\mathbb{D}^{(0)}(x_s)$, each at weight one. There is a constant $C$, fixed by the equation, its source and its rows and independent of $N$, such that for every $N \ge \max(2, \max_j \deg a_j)$*

$$\Delta = \lambda_1(H_{\mathrm{std}}) - \lambda_0(H_{\mathrm{std}}) \le \lambda_1(H_{\mathrm{std}}) \le C.$$

*If moreover the coefficients are constant, $a_k \neq 0$ and every row has order below $k$, then for every block encoding of $R_{\mathrm{std}}$ the degree of the minimax edge filter (Equation (7.6)) satisfies, once $\|R_{\mathrm{std}}\|_2^2 \ge 2C$,*

$$d \;\ge\; \frac{\|R_{\mathrm{std}}\|_2}{2\sqrt{2C}}\,\operatorname{arccosh}\frac{1}{\varepsilon} \;=\; \Omega\big(N^{2k}\log(1/\varepsilon)\big).$$

*Proof.* Let $P_1$ be the polynomials of degree at most one and $b(p)$ the coefficient vector of $p \in P_1$ on the field register. The map $p \mapsto b(p)$ is linear and injective, so $S = b(P_1)$ is two-dimensional, and $\|b(p)\|^2 = (N/\pi)\|p\|_w^2$.

*The equation block.* $G$ differentiates exactly on the represented space, and each lift multiplies exactly since $\deg a_j \le N$ (Proposition 13), so $\sum_j \mathcal{M}_{a_j} G^j b(p)$ is the coefficient vector of $\sum_j a_j p^{(j)}$ on the equation block's register of $N_A \in \{N, 2N\}$ terms. The source block gives $\mathcal{M}_r \mathbb{D}^{(0)}(x_s) b(p) = \hat{r}\,p(x_s)/y_s$, with $\hat{r}$ the first $N_A$ Chebyshev coefficients of the interpolant $I_m r$ of $r$ at $m \ge N_A$ Chebyshev–Gauss nodes, as the source is built. So $A_{\mathrm{std}} b(p)$ is the coefficient vector of $g_p = \sum_j a_j p^{(j)} - r_{N_A}\,p(x_s)/y_s$, with $r_{N_A}$ the truncation of $I_m r$. Truncation is an orthogonal projection in $\|\cdot\|_w$, and the $m$-point Gauss–Chebyshev rule integrates $(I_m r)^2$, of degree $2m-2$, exactly, so $\|r_{N_A}\|_w^2 \le \|I_m r\|_w^2 = (\pi/m)\sum_i r(x_i)^2 \le \pi \sup|r|^2$. Hence

$$\frac{\|A_{\mathrm{std}} b(p)\|^2}{\|b(p)\|^2} = \frac{N_A}{N}\,\frac{\|g_p\|_w^2}{\|p\|_w^2} \le \frac{2}{\|p\|_w^2}\Big(\sum_j \|a_j p^{(j)}\|_w + \sqrt{\pi}\,\sup|r|\,\frac{|p(x_s)|}{|y_s|}\Big)^2.$$

*The rows.* $\mathbb{B}(x^*) = \sqrt{N}\,|0\rangle\langle\tau(x^*)|$, so $\mathbb{B}(x^*)G^j b(p) = \sqrt{N}\,p^{(j)}(x^*)\,|0\rangle$ and a point-datum row gives $\sqrt{N}\big(p^{(j)}(x^*) - y^* p(x_s)/y_s\big)\,|0\rangle$. Each row $C_i$ is therefore $\sqrt{N}\,\ell_i(p)\,|0\rangle$ for a linear functional $\ell_i$ on $P_1$ that does not depend on $N$, and $\|C_i b(p)\|^2/\|b(p)\|^2 = \pi\,\ell_i(p)^2/\|p\|_w^2$.

*The bound.* Summing, $\langle b|H_{\mathrm{std}}|b\rangle \,/\, \langle b|b\rangle = \big(\|A_{\mathrm{std}} b\|^2 + \sum_i \|C_i b\|^2\big)/\|b\|^2 \le Q(p)$ for $b = b(p)$, where $Q$ is the sum of the two right-hand sides above. $Q$ does not depend on $N$, is continuous on $P_1\setminus\{0\}$ and is unchanged by scaling $p$, so it attains a finite maximum $C$ on the unit sphere of the two-dimensional

space $P_1$. By the Courant–Fischer theorem $\lambda_1(H_{\text{std}}) = \min_{\dim V=2} \max_{v\in V} \langle v|H_{\text{std}}|v\rangle \,/\, \langle v|v\rangle \le \max_{v\in S} \langle v|H_{\text{std}}|v$ $C$, and $\lambda_0 \ge 0$ because $H_{\text{std}} = R_{\text{std}}^\dagger R_{\text{std}}$.

*The degree.* In Equation (7.6), written in the heights $h_j = 2\lambda_j/\alpha_R^2$ of Appendix F.4, $\theta_0 = \operatorname{arccosh}(1+e)$ with $e = 2(h_1 - h_0)/(2 - h_1)$. Since $\cosh\sqrt{2e} \ge 1 + e$, $\theta_0 \le \sqrt{2e}$. Every block encoding has $\alpha_R \ge \|R_{\text{std}}\|_2$, so $h_1 \le 2C/\|R_{\text{std}}\|_2^2 \le 1$, whence $e \le 2(h_1 - h_0) = 4\Delta/\alpha_R^2$ and $\theta_0 \le \sqrt{8\Delta}/\alpha_R \le \sqrt{8C}/\|R_{\text{std}}\|_2$. Then $d \ge \operatorname{arccosh}(1/\varepsilon)/\theta_0$ gives the bound, and Corollary 1 gives $\|R_{\text{std}}\|_2 = \Theta(N^{2k})$, with $\operatorname{arccosh}(1/\varepsilon) \ge \ln(1/\varepsilon)$. □

The bound uses nothing about the solution and holds for variable coefficients, R3's included: two smooth trial functions, on which the residual is $O(1)$ at every resolution, already bound the first excited value. It is an upper bound only. The matching lower bound on $\Delta$, which the degree's upper bound $O(N^{2k}\log(1/\varepsilon))$ for the structured encoding needs, is not proved; it is measured on R2a and R2b (§ 4.6). The degree bound is the minimax edge filter's; for polynomial filters in general it holds as far as Lin and Tong's optimality supports [53].

## C.6 Subnormalisation across $n$

**Table C.1: The structured encoding rises to $2j_{-3/4,1} \approx 2.117$ times $\|G\|_2$; Wu et al.'s prefactor grows as $j_{-3/4,1}N^3$.** Subnormalisation over $\|G\|_2$ of the structured encoding (Proposition 2) and of Wu et al.'s prefactor (Equation (3.5)) for the differentiation matrix $G$, $n = 2$–$10$, $N = 2^n$. $j_{-3/4,1}$, first positive zero of the Bessel function $J_{-3/4}$. Exact classical computation.

| $n$ | $\|G\|_2$ | Structured | Wu et al.'s |
|---|---|---|---|
| 2 | 7.39 | 1.623 | 49.9 |
| 3 | 30.1 | 1.858 | 468 |
| 4 | 121 | 1.986 | $4.05 \times 10^3$ |
| 5 | 484 | 2.051 | $3.36 \times 10^4$ |
| 6 | 1935 | 2.084 | $2.73 \times 10^5$ |
| 7 | 7739 | 2.100 | $2.20 \times 10^6$ |
| 8 | 30 957 | 2.109 | $1.77 \times 10^7$ |
| 9 | 123 827 | 2.113 | $1.42 \times 10^8$ |
| 10 | 495 308 | 2.115 | $1.14 \times 10^9$ |

# D The Descriptor Construction in Full

## D.1 Block Rows in Full

Table D.1 lists every row block of the chain. On one axis the state is $w = (\hat f^{(0)}; \dots; \hat f^{(k)}; 0; \dots; 0)$, one block of $N$ amplitudes per derivative, padded with zero blocks to a power of two so that the block index is a register of $n_f$ qubits, the most significant index of the state. On several axes the field is block 0 and each axis $a$ owns the chain $\partial_a f, \dots, \partial_a^{k_a} f$ up to the highest order that the equation or a constraint takes along it; a mixed derivative would need a block of its own and is not built. $\tilde B_a$ and $\tilde D_a$ are the banded factors on axis $a$, tensored with the identity on the other axes.

A Neumann condition reads the derivative block with the zero row $\mathbb{B}(x)$, the same row that a Dirichlet condition applies to the field block, and no $G$. A source enters the equation row as in the standard form, through the collapse row on the field block. A first-order equation or system takes the mass form of Equation (5.7) instead of a chain, with the standard form's state and constraint rows. The system rows of several fields and of a Carleman lift, and their ratio rows, are given in Appendix E.1.

**Table D.1: The chain's row blocks: one equation row, banded recurrences, padding and constraints.** Each row block of the descriptor's stacked residual, the state blocks it reads and its operator. $\hat{f}^{(m)}$, coefficients of the $m$-th derivative; $\mathcal{M}_{a_j}$, multiplication by the coefficient $a_j$ (the standard form's, lifted where it lifts); $\mathcal{M}_r$, multiplication by the source $r$; $\mathbb{D}^{(0)}(x_s)$, collapse row at the datum point $x_s$; $\tilde{B}_a$, $\tilde{D}_a$, the banded factors of $G = \tilde{B}^{-1}\tilde{D}$ along axis $a$; $c_{\text{pad}}$, padding weight; $\mathbb{B}(x^*)$, zero row at $x^*$. Constraint rows are the standard form's (Definition 2) with each power of $G$ replaced by the block that carries the derivative.

| Row block | Reads | Content |
|---|---|---|
| The equation | the field and every carried derivative | $\sum_j \mathcal{M}_{a_j} \hat{f}^{(j)} - \mathcal{M}_r \mathbb{D}^{(0)}(x_s) \hat{f}^{(0)}$ |
| Recurrence $m \to m+1$ along axis $a$ | blocks $m$ and $m+1$ of $a$'s chain | $-\tilde{D}_a \hat{f}^{(m)} + \tilde{B}_a \hat{f}^{(m+1)}$ |
| Padding block $b$ | block $b$ | $c_{\text{pad}} I$, $c_{\text{pad}} = 1$ |
| A constraint on $\partial_a^m f$ | block $m$ of $a$'s chain | a zero row $\mathbb{B}(x^*)$, a collapse row, a datum slice (Equation (4.4)) or a ratio row (§ 6.2) |

**Proposition 15** (The padding decouples)**.** *After ordering the columns,*

$$R_{\text{desc}} = \begin{pmatrix} R_{\text{phys}} & 0 \\ 0 & c_{\text{pad}} I \end{pmatrix},$$

*so $H_{\text{desc}}$ is block diagonal, every polynomial in $H_{\text{desc}}$ maps the physical blocks to themselves, and the padding's singular values all equal $c_{\text{pad}}$.*

*Proof.* No equation, recurrence or constraint row reads a padding block, and each padding row block is $c_{\text{pad}} I$ on its own block. □

The reference, the gap, the filter and the initial state are therefore the physical subspace's, and $c_{\text{pad}} = 1$ adds one to $\alpha_R^2$ and nothing to the gap the preparation meets. Over the whole spectrum the padding's singular value can lie below the physical subspace's first excited one, but an initial state on the field block never meets it.

## D.2 Variable Coefficients and the Padded Block Count

A variable coefficient $a_j(x)$ enters the equation row through the standard form's multiplication operator $\mathcal{M}_{a_j}$, the lift of Paine et al. (Proposition 13), on exactly the axes where the standard form lifts it: an axis with a non-constant coefficient or a source. The descriptor's equation row therefore applies the same operators as the standard form's equation block, which Theorem 1 needs. A multiplication by a polynomial is banded, with bandwidth the polynomial's degree, so it compiles to the same bands as $\tilde{B}$ and $\tilde{D}$ (Appendix D.7). The square Galerkin form, which truncates the exact product to the first $N$ modes and never lifts, is used only as a control in the gap study of Appendix D.4.

With $K = 1 + \sum_a k_a$ physical blocks, the padded state has

$$\dim w = 2^{\lceil \log_2 K \rceil} \prod_a N_a, \qquad n_f = \lceil \log_2 K \rceil, \tag{D.1}$$

amplitudes, against the standard form's $\prod_a N_a$. On one axis $K = k + 1$:

| Order $k$ | Blocks $k+1$ | Field qubits $n_f$ | Padded blocks $2^{n_f}$ | Padding blocks |
|---|---|---|---|---|
| 2 | 3 | 2 | 4 | 1 |
| 3 | 4 | 2 | 4 | 0 |
| 4 | 5 | 3 | 8 | 3 |
| 7 | 8 | 3 | 8 | 0 |

A first-order equation takes the mass form and carries no derivative block. The heat equation has $K = 4$, and Laplace's and the wave equation $K = 5$, padded to eight. The padded count equals the unpadded one exactly when $K$ is a power of two and is less than twice it otherwise; the chain's state is at most $2K$ times the standard form's. This is the qubit cost that accompanies the gate counts of § 5.6.

## D.3 Kernel Equivalence on Several Axes and Under Rescaling

The proof of Theorem 1 in the body treats one axis without rescaling. Both extensions follow from one identity. Let $E$ map a field's coefficients $p$ to the state whose block of order $m$ along axis $a$ is $G_a^m p$, with $G_a$ the differentiation matrix on axis $a$ tensored with the identity on the others, and whose padding blocks are zero. Let $S$ be the diagonal of block weights of § 5.4, equal to $\prod_a \zeta_a^{m_a}$ on a block of order $m_a$ along each axis and to one on the field and padding blocks.

**Proposition 16** (Substitution identity)**.** $R_{\mathrm{desc}} S\,(S^{-1}Ep) = \big(R_{\mathrm{std}}\,p;\,0\big)$*, after ordering the rows: the recurrence and padding rows vanish on* $Ep$*, and every other row block acts on* $Ep$ *as the standard form's row block of the same name acts on* $p$*.*

*Proof.* The recurrence of order $m$ along axis $a$ gives $-\tilde{D}_a G_a^m p + \tilde{B}_a G_a^{m+1} p = (\tilde{B}_a G_a - \tilde{D}_a) G_a^m p = 0$, and the padding blocks of $Ep$ are zero. The equation row gives $\sum_j \mathcal{M}_{a_j} G^j p - \mathcal{M}_r \mathbb{D}^{(0)}(x_s) p$ term by term, each term's derivative being read from the block that holds it, which is the standard form's equation block. A constraint on $\partial_a^m f$ applies its row to $G_a^m p$, which is the standard form's row. The factor $S$ multiplies the columns of each block by its weight and $S^{-1}$ divides the block of $Ep$ by the same weight, so they cancel. □

**Theorem 5** (Kernel equivalence, general form)**.** $\ker(R_{\mathrm{desc}} S) = S^{-1} E \ker R_{\mathrm{std}}$*, and* $E$ *is injective, so the two kernels have the same dimension.*

*Proof.* If $p \in \ker R_{\mathrm{std}}$, Proposition 16 gives $(R_{\mathrm{desc}} S)(S^{-1}Ep) = 0$. Conversely, let $(R_{\mathrm{desc}} S)v = 0$ and $w = Sv$. The padding rows force the padding blocks of $w$ to zero. Along each axis's chain each recurrence gives $w_{(a,m+1)} = \tilde{B}_a^{-1} \tilde{D}_a w_{(a,m)} = G_a w_{(a,m)}$, since $\tilde{B}_a$ is invertible, so $w = E w_0$, and Proposition 16 gives $R_{\mathrm{std}} w_0 = 0$. $E$ is injective because its first block is the identity. □

**What it does not say.** The theorem says that an exact kernel is present in both forms or in neither, with the same dimension; a one-dimensional kernel is one-dimensional in both. It does not say that near-kernels are equally sharp. The ratio of the first excited to the lowest singular value, which the trust threshold of § 7.2 reads, is each form's own and differs between them. For a Carleman lift the kernel is the truncated lift's in both forms, and its uniqueness is the initial-value problem's, not a property of either form (§ 6.5).

## D.4 The Gap Mechanism and the Artefacts Ruled Out

**Proposition 17** (The quotient on represented states)**.** *On the represented states the Rayleigh quotient of the rescaled descriptor is*

$$\frac{\|R_{\mathrm{desc}} S\,(S^{-1}Ep)\|^2}{\|S^{-1}Ep\|^2} = \frac{\|R_{\mathrm{std}}\,p\|^2}{\sum_{m=0}^{k} \zeta^{-2m} \|G^m p\|^2}, \tag{D.2}$$

*and the* $j$*-th smallest singular value of* $R_{\mathrm{desc}} S$ *is at most the least, over* $(j+1)$*-dimensional subspaces* $V$ *of coefficient vectors, of the largest square root of this quotient on* $V$*.*

*Proof.* Proposition 16 gives the numerator, and the denominator is the norm of the represented state, whose block $m$ is $\zeta^{-m} G^m p$. By the Courant–Fischer theorem the $j$-th smallest singular value is the

least, over $(j+1)$-dimensional subspaces of the state space, of the largest quotient on the subspace; restricting the subspaces to the range of $S^{-1}E$ can only raise it. □

The stationary values of the quotient are the generalised singular values of the pair $(R_{\text{std}}, S^{-1}E)$ [88]. The bound is not tight. States that break the recurrences cost little where $\tilde{B}$'s least singular value is small, and it falls with $N$, as about $\pi/N$, with its singular vector at the ends of the interval. Measured against the bound at $n = 8$, the descriptor's lowest non-zero singular value is 0.99 to 1.0 times it on the controlled equations of Table D.2 with a single zero and on the first-order one without, 0.74 and 0.91 times it on the second-order ones without a zero, 0.04 times it on the equation with zeros at both ends, and 0.35 to 0.72 times it on the two-axis problems at $n = 4$ per axis.

**The concentrated states.** Let $a_k$ vanish simply at an endpoint. A polynomial of degree $N$ can concentrate within $\mathcal{O}(N^{-2})$ of the endpoint, where the Chebyshev points cluster, and each of its derivatives then grows by a factor $\mathcal{O}(N^2)$. On such a state $\|a_k p^{(k)}\| = \mathcal{O}(N^{-2})\|p^{(k)}\|$ and the lower orders are of the same size, so the square root of the quotient is $\mathcal{O}(N^{-2}\zeta^k)$ against the top block. Unscaled, the lowest non-zero singular value is $\mathcal{O}(N^{-2})$ and the gap $\mathcal{O}(N^{-4})$. At an interior zero, where the resolution is $\mathcal{O}(N^{-1})$, the gap is $\mathcal{O}(N^{-2})$. At a double zero at an endpoint, $\|a_k p^{(k)}\| = \mathcal{O}(N^{-4})\|p^{(k)}\|$ and the gap is $\mathcal{O}(N^{-8})$. These are heuristics; the rates are measured. Table D.2 gives the unscaled gap on controlled equations whose solutions are polynomials where the coefficient vanishes, so that their kernels are exact, built with the square multiplication.

**Table D.2: A zero of the leading coefficient closes the unscaled descriptor's gap at the rate the Chebyshev resolution at the zero predicts.** Unscaled descriptor gap at $n = 5$ and $n = 9$, its factor of decrease per added qubit, and the standard form's gap, on seven equations whose leading coefficient is bounded away from zero, has a simple zero, or vanishes at both ends, each solved on the square multiplication with no lift. Exact classical computation.

| Equation | Leading coefficient | Descriptor, $n = 5 \to 9$ | Per qubit | Standard |
|---|---|---|---|---|
| $f'' + 4f' + 4f$ | 1 | $0.458 \to 0.458$ | flat | 7.04 |
| $(2+x)f'' + f' - f$ | $\geq 1$ | $0.434 \to 0.432$ | flat | 0.777 |
| $(2+x)f' + f$ | $\geq 1$ | $1.01 \to 1.00$ | flat | 4.90 |
| $(1-x)f'' - xf' + 3f$ | zero at $x = 1$ | $1.5 \times 10^{-5} \to 2.2 \times 10^{-10}$ | $\div 16$ | 2.02 |
| $(1-x)f' + f$ | zero at $x = 1$ | $2.4 \times 10^{-5} \to 3.5 \times 10^{-10}$ | $\div 16$ | 1.59 |
| $(1-x^2)f'' - 2xf' + 20f$ | zeros at $x = \pm 1$ | $9.1 \times 10^{-7} \to 8.0 \times 10^{-13}$ | $\div 33$ | 20.5 |
| $xf'' + (1-x)f' + 3f$ | zero at $x = 0$ | $9.3 \times 10^{-3} \to 3.8 \times 10^{-5}$ | $\div 4.0$ | 1.85–2.02 |

On the equations whose leading coefficient vanishes, the first excited state holds 0.98 to 1.0 of its weight in the top-derivative block from $n = 5$ to 9, and at $n = 9$ a share of 0.86 to 0.99 of that block's function lies within 0.02 of the coefficient's zero. Variable coefficients alone do not close the gap; a zero does.

**Two axes.** The equation constrains a sum of top derivatives. Its lowest modes are equal and opposite in two top blocks (transport, Laplace's equation) or sit in the block the equation weights least (the heat equation's $\partial_x^2 f$ block, weighted 1/25), and only the slice rows penalise them: by $\mathcal{O}(1)$ against $\|f\|$, which is small against the derivative blocks. With R5b's constraints and resolution, the lowest singular value that is not a near-solution falls from 0.29 to 0.033 for transport, from 0.081 to 0.0018 for the heat equation and from 0.10 to 0.0011 for Laplace's equation as $n$ goes from 2 to 4 per axis, while for $f_t + f$, which has one top block, it stays at 0.87–0.86.

**Weights on the slice rows.** If the slice rows are too weak against the derivative blocks, rows that penalise a high-frequency datum more are the obvious repair. Each slice row was weighted by $(m+1)^s$, with $m$ its tangential mode, a discrete trace norm of order $s$ that leaves an exact kernel unchanged, for $s$ from 0 to 2 in steps of $\frac{1}{2}$, on the unscaled descriptor of Laplace's equation (R5a) and

of the heat equation (R5b) at three and four qubits per axis (exact classical computation). The cost compared is $\sqrt{\|H\|/\Delta}/\gamma$, as in Appendix D.6, with $\gamma$ the geometric product state's overlap with the whole state. Table D.3 gives the result. The least cost lies at the same weight on every panel, and there the gap rises by a larger factor than the norm, so the cost falls. A heavier weight does worse: at $s = 1$ the cost is no better than unweighted on R5a and a smaller gain than the best weight on R5b. The gain does not restore the gap. From three to four qubits per axis the unweighted gap falls by a factor of order a hundred on R5a and of order ten on R5b, and the weight's factor on the gap grows far less over the same step, so the weighted gap still falls several-fold or more per qubit. Two sizes per problem fix no rate, and the weights were not combined with block rescaling.

**Table D.3: Weighting the slice rows raises the gap more than the norm, but the cost falls only modestly and the gap still falls with $n$.** Unscaled descriptor of Laplace's equation (R5a) and the heat equation (R5b) at $n$ qubits per axis, with each slice row weighted by $(m+1)^s$, $m$ its tangential mode. $\Delta$ is the unweighted gap ($s = 0$); $s^*$ is the weight of least cost $\sqrt{\|H\|/\Delta}/\gamma$ among $s \in \{0, \frac{1}{2}, \ldots, 2\}$; the next three columns are the gap, the norm $\|H\|$ and the cost at $s^*$ over their values at $s = 0$; the last is the cost at $s = 1$ over that at $s = 0$. A cost ratio below one is a gain. Exact classical computation.

| Problem | $n$ | $\Delta$ | $s^*$ | Gap | Norm | Cost | Cost at $s = 1$ |
|---|---|---|---|---|---|---|---|
| R5a | 3 | $1.1 \times 10^{-4}$ | 0.5 | 2.4 | 1.3 | 0.73 | 1.2 |
| R5a | 4 | $1.1 \times 10^{-6}$ | 0.5 | 3.4 | 1.2 | 0.58 | 0.99 |
| R5b | 3 | $3.2 \times 10^{-5}$ | 0.5 | 3.2 | 1.2 | 0.63 | 0.78 |
| R5b | 4 | $3.2 \times 10^{-6}$ | 0.5 | 7.0 | 1.2 | 0.41 | 0.64 |

**The field block's weight.** Let $P_0$ project onto the field block and let the initial state be supported there. Then $\gamma = |\langle\psi_g|\psi_{\text{init}}\rangle| = |\langle P_0\psi_g|\psi_{\text{init}}\rangle| \le \|P_0\psi_g\| = w_f^{1/2}$ by the Cauchy–Schwarz inequality, with $w_f = \|P_0\psi_g\|^2$ the field block's weight that the build records hold. For an exact kernel, $\psi_g \propto S^{-1}Eb$ with $b$ the solution's coefficients, so $w_f = \|b\|^2 / \sum_m \zeta^{-2m}\|G^m b\|^2$. Amplitude amplification then costs a factor $1/\gamma \ge w_f^{-1/2}$ (§ 7.4).

**What it is not.** Each candidate artefact was tested:

| Candidate | Test | Outcome |
|---|---|---|
| The lift | R3a with the square multiplication | collapses faster: $2.6 \times 10^{-11}$ at $n = 8$, against $2.7 \times 10^{-8}$ lifted |
| The padding | lowest singular value against $c_{\text{pad}}$; the excited state's padding weight | at most 0.72, below $c_{\text{pad}} = 1$, on every unscaled padded record; no padding weight |
| The constraints | controlled equations with none | they collapse too |
| Precision | lowest singular value against the decomposition's floor $\varepsilon\sigma_{\max}$ | six orders of magnitude above it at the smallest |
| The basis | a descriptor carrying $f^{(j)}$ in the ultraspherical basis $C^{(j)}$ | the same rates where $a_k$ vanishes |
| The reference | dense decomposition against the iterative reference, on R2a at $n = 8$ to 10 | agree to $0.68\,\varepsilon\sigma_{\max}$ |

The ultraspherical descriptor keeps a flat gap on regular problems with a norm 4.0 times smaller at $n = 9$, and on Legendre's equation its gap is 169 times the Chebyshev descriptor's there, the share of the recurrences, but it still collapses.

## D.5 The Gap on One Axis

This section proves Theorem 2. Write $\|g\|_T^2 = \int_{-1}^{1} g^2(1-x^2)^{-1/2}\,dx$ and $\|g\|_U^2 = \int_{-1}^{1} g^2(1-x^2)^{1/2}\,dx$, the norms under which the Chebyshev polynomials of the first and the second kind are orthogonal.

By Appendix A.3, a polynomial $g$ of degree below $N$ with coefficients $\hat{g}$ in the normalised basis has $\|\hat{g}\| = \sqrt{N/\pi}\,\|g\|_T$.

**Lemma 5** (The factors in the second kind)**.** *Let $\psi_0 = U_0/(2\sqrt{N})$ and $\psi_m = U_m/\sqrt{2N}$ for $1 \le m < N$. For a polynomial $g$ of degree below $N$, $\tilde{B}\hat{g}$ and $\tilde{D}\hat{g}$ are the coefficients of $g$ and of $g'$ in the basis $(\psi_m)$, and a polynomial $h = \sum_m c_m\psi_m$ has $\|c\| \ge 2\sqrt{N/\pi}\,\|h\|_U$.*

*Proof.* With $\varphi_0 = T_0/\sqrt{N}$ and $\varphi_\ell = T_\ell\sqrt{2/N}$, the identities $T_0 = U_0$, $T_1 = U_1/2$ and $T_\ell = (U_\ell - U_{\ell-2})/2$ for $\ell \ge 2$ give $\varphi_0 = 2\psi_0$, $\varphi_1 = \psi_1$, $\varphi_2 = \psi_2 - \sqrt{2}\,\psi_0$ and $\varphi_\ell = \psi_\ell - \psi_{\ell-2}$ for $\ell \ge 3$, the columns of $\tilde{B}$ (Equation (A.2)). The identity $T'_\ell = \ell U_{\ell-1}$ (Equation (A.6)) gives $\varphi'_1 = 2\sqrt{2}\,\psi_0$ and $\varphi'_\ell = 2\ell\,\psi_{\ell-1}$ for $\ell \ge 2$, the columns of $\tilde{D}$. With $x = \cos\theta$, $U_m = \sin(m+1)\theta/\sin\theta$, so $\int_{-1}^{1} U_m U_{m'}(1-x^2)^{1/2}dx = \int_0^\pi \sin(m+1)\theta\,\sin(m'+1)\theta\,d\theta = \frac{\pi}{2}\delta_{mm'}$: the $\psi_m$ are orthogonal under $\|\cdot\|_U$ with $\|\psi_m\|_U^2 \le \pi/(4N)$, which gives the bound. □

A recurrence row therefore measures the defect $F_{j+1} - F'_j$ between the functions of consecutive blocks in $\|\cdot\|_U$, a norm weaker than the state's $\|\cdot\|_T$ near the ends of the interval; this is why $\tilde{B}$'s least singular value falls as $N$ grows. Integration recovers what the weaker norm loses. For integrable $g$ and $-1 \le y \le x \le 1$ the Cauchy–Schwarz inequality gives

$$\Big|\int_y^x g\Big| \le \|g\|_T\,|x-y|^{1/2}, \qquad \Big|\int_y^x g\Big| \le \sqrt{\pi}\,\|g\|_U, \tag{D.3}$$

since $\int_y^x (1-t^2)^{1/2}dt \le x - y$ and $\int_{-1}^{1}(1-t^2)^{-1/2}dt = \pi$.

*Proof of Theorem 2. Reductions.* The padding decouples (Proposition 15), so only the physical columns count. The rescaled residual is $R_{\text{desc}}S$ with $S$ diagonal, invertible and independent of $N$ at a fixed $\zeta$; writing $\|R_{\text{desc}}Sv\|/\|v\| = (\|R_{\text{desc}}u\|/\|u\|)(\|u\|/\|v\|)$ with $u = Sv$, the Courant–Fischer theorem places each singular value of $R_{\text{desc}}S$ between $\min_m \zeta^m$ and $\max_m \zeta^m$ times the unscaled one, so it suffices to treat $R = R_{\text{desc}}$ unscaled. Divide every row by $\sqrt{N/\pi}$. A state $w$ whose blocks are the coefficients of polynomials $F_0, \dots, F_k$ of degree below $N$ then has $\|w\|^2 = \sum_j \|F_j\|_T^2$. By Lemma 5, by Proposition 13, which makes the lifted equation row exact, and by $\mathbb{B}(x)b = \sqrt{N}f_N(x)\,|0\rangle$ (Appendix A.3),

$$\|Rw\|^2 \ge 4\sum_{j<k}\|F_{j+1} - F'_j\|_U^2 + \|E\|_T^2 + \pi\sum_i |\ell_i(w)|^2, \tag{D.4}$$

with $E = \sum_j a_j F_j - r_N F_0(x_s)/f(x_s)$ the equation row's function, $r_N$ the source as represented, and $\ell_i(w)$ the $i$-th constraint row's combination of the values $F_j(x^*)$, $j < k$, and $F_0(x_s)$.

*The lower bound.* Let $V_N$ be the states with $\langle F_0, f\rangle_T = 0$, a subspace of codimension one, so that $\sigma_2 \ge \min_{w\in V_N}\|Rw\|/\|w\|$ by the Courant–Fischer theorem. Suppose no $c > 0$ bounds this minimum below for every large $N$. Then there are $N \to \infty$ and $w^N \in V_N$ with $\|w^N\| = 1$ and $\|Rw^N\| \to 0$; write $F_j^N$ for their blocks. For $j < k$,

$$F_j^N(x) = F_j^N(0) + \int_0^x F_{j+1}^N - \int_0^x \big(F_{j+1}^N - (F_j^N)'\big).$$

By Equation (D.3) the middle term is bounded by 1 and equicontinuous, since $\|F_{j+1}^N\|_T \le 1$, and the last tends to zero uniformly, since its integrand's $\|\cdot\|_U$ does by Equation (D.4). The constants are bounded as well, $\sqrt{\pi}\,|F_j^N(0)| \le \|F_j^N\|_T + \sqrt{\pi}\sup|F_j^N - F_j^N(0)|$. By the Arzelà–Ascoli theorem a subsequence has $F_j^N \to F_j$ uniformly for every $j < k$. On it the equation row gives

$$a_k F_k^N = E^N - \sum_{j<k} a_j F_j^N + r_N\frac{F_0^N(x_s)}{f(x_s)} \;\longrightarrow\; -\sum_{j<k} a_j F_j + r\,\frac{F_0(x_s)}{f(x_s)}$$

in $\|\cdot\|_T$, because $\|E^N\|_T \to 0$ and $r_N \to r$ uniformly for an analytic source [32, Ch. 8]. Since $|a_k|$ is bounded below, $F_k^N$ converges in $\|\cdot\|_T$ to the continuous function $F_k$ that the limit defines. Letting $N \to \infty$ in the integral identity, with the first bound of Equation (D.3) applied to $F_{j+1}^N - F_{j+1}$, gives $F_j(x) = F_j(0) + \int_0^x F_{j+1}$, so $F_0 \in C^k[-1,1]$ with $F_0^{(j)} = F_j$. The limit of the display says that the equation row vanishes on $F_0$, and $\ell_i(w^N) \to 0$ with uniform convergence says that every constraint row does. By hypothesis $F_0 = tf$, and $\langle F_0, f\rangle_T = \lim\langle F_0^N, f\rangle_T = 0$ forces $t = 0$, so every $F_j$ vanishes. The convergence is in norm, however, so $\sum_j \|F_j\|_T^2 = \lim \|w^N\|^2 = 1$, a contradiction.

*The upper bound and the ground value.* Let $b$ be the coefficients of the Chebyshev truncation $f_N$ and $\hat{q}$ those of a fixed polynomial $q$ that is not a multiple of $f$. For $p = \alpha b + \beta\hat{q}$, Proposition 16 gives $\|REp\| = \|R_{\text{std}}p\|$, and $\|Ep\| \geq \|p\|$. For $N > \deg q$ the rows of $R_{\text{std}}$ act on $q$ through its derivatives and values, which do not depend on $N$, and through $r_N$; they act on $f_N$ through $f_N^{(j)}$, which converge to $f^{(j)}$ uniformly for $j \leq k$, since the coefficients of an analytic function decay geometrically [32, Ch. 8] and $\|T_n^{(j)}\|_\infty \leq n^{2j}$ by Markov's inequality [54]. So $\|R_{\text{std}}p\|$ is bounded for $|\alpha|^2 + |\beta|^2 = 1$, while $\|p\|$, which tends to $\|\alpha f + \beta q\|_T$, stays away from zero, and the Courant–Fischer theorem gives $\sigma_2 \leq C$. Because $f$ satisfies its equation and constraints exactly, the same convergence gives $\sigma_1 \leq \|R_{\text{std}}b\|/\|b\| \to 0$. Hence $\Delta = \sigma_2^2 - \sigma_1^2$ lies between $c^2/2$ and $C^2$ for large $N$.

*The mass form.* Its rows are the standard form's equation rows multiplied by $I_C \otimes \tilde{B}$, so by Lemma 5 they are the coefficients of $z' - \Gamma z$ in the basis $(\psi_m)$, and Equation (D.4) holds with $4\|z' - \Gamma z\|_U^2$ in place of the recurrence and equation terms. With $\delta^N = (z^N)' - \Gamma z^N$, the identity $z^N(x) = z^N(0) + \int_0^x \Gamma z^N + \int_0^x \delta^N$ and the same argument give the lower bound. Its state is the standard form's, and $\|R_{\text{mass}}p\| \leq (2+\sqrt{2})\|R_{\text{std}}p\|$ since $\|\tilde{B}\|_2 \leq 2+\sqrt{2}$, so the upper bound and the ground value follow as above. □

The argument gives no value for the bound, because it closes by compactness. It needs the leading coefficient's lower bound at one step only, to recover the top block from the equation row; where $a_k$ vanishes that step fails, which is the mechanism of Appendix D.4. On two axes the equation row controls only a sum of top derivatives, and no single block can be recovered from it.

## D.6 Block Rescaling

The rule of § 5.4 returns each axis's weight from the equation's coefficients and the resolution alone:

| Problem | $\zeta$ |
|---|---|
| One axis, regular ($a_k$ with no zero on $[-1,1]$) | $2(\bar{a}_j/\bar{a}_k)^{1/(k-j)}$, $j$ the lowest order present, $\bar{a}$ the mean of $\lvert a\rvert$ |
| One axis, singular ($a_k$ with a zero on $[-1,1]$) | $N/2$ |
| Several axes | $N_a$, times $\sqrt{\max_b \bar{c}_b/\bar{c}_a}$ on an axis of top order two, $\bar{c}_a$ the mean magnitude of its top coefficient |

Whether $a_k$ vanishes is decided from its roots, to a tolerance loose enough to find a double root. On R2a the rule gives $\zeta = 2\sqrt{4/1} = 4$, on R2c $\zeta = 2\sqrt{400} = 40$, and on R3a, whose leading coefficient vanishes at both ends, $\zeta = N/2$.

**Why each.** For a regular problem (Definition 4) with constant coefficients, $(\bar{a}_0/\bar{a}_k)^{1/k}$ is the geometric mean of the characteristic roots' magnitudes, the product of the roots being $\pm a_0/a_k$; a solution $e^{\lambda x}$ has $\|f^{(m)}\| \approx |\lambda|^m\|f\|$, so dividing block $m$ by $\zeta^m$ balances the blocks at the equation's own scale. The factor two is where the cost's minimum over $\zeta$ sits on the regular benchmark problems. On a singular problem the concentrated states of Appendix D.4 have quotient $\mathcal{O}(N^{-4}\zeta^{2k})$, so for second-order problems $\zeta \propto N$ makes it of order one; half the resolution is the constant the sweep found. At an interior simple zero the same argument with Bernstein's inequality in place of Markov's, states

concentrated within $\mathcal{O}(N^{-1})$ and derivatives growing by $\mathcal{O}(N)$ each, gives a quotient of order $N^{-2}\zeta^{2k}$ and so $\zeta \propto N^{1/k}$, which is $N^{1/2}$ at second order; the rule, which tests only whether $a_k$ vanishes on $[-1, 1]$, would give such a problem $N/2$, a weight larger than this balance needs and so a larger norm than necessary. No benchmark problem has an interior zero and this case was not run. On several axes the resolution is used per axis, as for a singular problem, and an axis of top order two is weighted so that its top-order term $\bar{c}_a\zeta_a^2$ balances the others: the wave equation $f_{tt} = c^2 f_{xx}$, with wave speed $c$, takes $(cN, N)$, which costs at most 1.06 times the best pair on a grid at three and four qubits per axis, where $(N/2, N/2)$ costs $1.2 \times 10^4$ and $5.8 \times 10^6$ times it.

**The calibration and its check.** The cost measured is $\sqrt{\|H\|/\Delta}/\gamma$, the walk's queries at an ideal encoding times amplitude amplification, up to the logarithms both forms share. The two constants were selected by calibration: $\zeta$ was swept over powers of two on the benchmark problems, each constant set where the cost is least, and both were then held fixed and never refitted. The same problems then checked the resulting weights, so the check is in-sample. Table D.4 gives the window of $\zeta$ over which the cost stays within twice its least, and Table D.5 the rule's cost against the grid's best.

**Table D.4: The block weight need only be right to about a factor of two.** Range of the block weight $\zeta$ over which the cost $\sqrt{\|H\|/\Delta}/\gamma$ stays within twice its least, at $n = 7$ on one axis and $n = 4$ per axis on two (one weight shown per two-axis problem). $\|H\|$, Hamiltonian norm; $\Delta$, spectral gap; $\gamma$, the initial state's overlap; the cost counts the walk's queries at an ideal encoding times amplitude amplification. Exact classical computation.

| | R2a | R2c | R3a | R4a | R4d | R5a | R5b |
|---|---|---|---|---|---|---|---|
| Window of $\zeta$ | 2.0–16 | 16–64 | 32–91 | 23–91 | 4.0–16 | 8.0–32 | 8.0–16 |

**Table D.5: The block weight of Equation (5.10) lands near the best on a grid.** Cost $\sqrt{\|H\|/\Delta}/\gamma$ at the rule's $\zeta$, over the least cost on a grid of powers of two; $\Delta$, gap on the physical subspace; $\gamma$, overlap of the geometric initial state on the field block. A ratio below one is the grid's spacing. Exact classical computation, in-sample: the rule's two constants were calibrated on these problems.

| Problems | $n$ | Rule's cost / grid's best |
|---|---|---|
| R1, R2a–c, R4b–d (regular) | 5, 7 | 0.99–1.44 at $n = 5$, the highest R2c's; 0.99–1.13 at $n = 7$ |
| R3a, R3b, R4a (a simple zero) | 5, 7 | 1.00–1.04 |
| R3c, R3d (solved for $v = f/\sqrt{1-x^2}$) | 5, 7 | 1.00–1.18 at $n = 5$; 2.07 and 2.22 at $n = 7$ |
| R5a, R5b, R5c | 3, 4 per axis | 1.00–1.17 |

**A coefficient that dips.** $(\delta + x^2)f'' + f = 0$ has no zero but dips to $\delta$. The rule is within 1.3 times the grid's best for $\delta \geq 0.01$ at $n = 5$ and 7, and 2.3 and 4.7 times at $\delta = 10^{-3}$, where the resolution sees the dip and it behaves like a zero, so that the regular weight is the wrong one. The two classes are asymptotic, and the weight jumps between them while the coefficient changes continuously. None of the benchmark problems dips so.

**A double zero.** Wu et al.'s R3c and R3d multiply the associated Legendre operator by $1 - x^2$, so their leading coefficient $(1-x^2)^2$ vanishes twice at each end. Unscaled, the descriptor's gap there falls as $N^{-8}$. Balancing the concentrated states needs $\zeta \propto N^2$, by which $\|H_{\text{desc}}\|$ has grown as $\zeta^4 = N^8$. On a grid of $\zeta$ the best cost of the descriptor grows as about $N^{3.7}$ on R3d and $N^{3.8}$ on R3c over $n = 5$ to 8, against the standard form's $N^{2.2}$ and $N^{2.3}$, and at $n = 8$ it is 758 and 856 times the standard form's; no block weight brings the two level. Solved for $v = f/\sqrt{1-x^2}$, whose leading coefficient has simple zeros, the two forms' costs grow alike, as $N^{1.8}$ and $N^{1.9}$ at the rule's weight against the standard form's $N^{2.0}$. The ground state stays accurate in both forms at the double zero, its error set by the solution's branch point; it is the cost that the double zero raises.

**What it changes in the circuit.** The weights multiply the bands before the block register's patterns are expanded in Pauli words (Algorithm 1, line 3), so they change the coefficients $c_t$ and $\alpha_A = \sum_t |c_t|$ and, where a merged pattern's entries were equal across its blocks, the length of its expansion: the recurrences' $\mathrm{diag}(1,1,0,0)$, which is $\frac{1}{2}(I+Z)$ on the block register's leading qubit, becomes $\mathrm{diag}(1,\zeta,0,0)$, which needs all four words of the two qubits. No register changes width, since a pattern on $n_f$ qubits has at most $4^{n_f}$ words whatever its entries, so the rescaling adds no ancilla; the terms, and the preparations and block-register operations that apply them, grow by at most that factor, which the equation fixes and $n$ does not. A general diagonal equilibration, a different weight for each amplitude, would need its own diagonal block encoding, with $\Theta(\dim w)$ gates, and is excluded from every circuit.

## D.7 The Term IR and Its Compilation

The system rows of the descriptor (the equation, the recurrences and the padding; the rank-one constraint rows are encoded apart) form one square operator on the block register and the axes' registers. Each row block sits at a block index of its own: a recurrence at the index of its lower block, the equation at the lowest index left, and each padding row at its own. The term IR writes this operator as

$$A_{\mathrm{sys}} = \sum_t c_t \, P_t \otimes \bigotimes_a F_{t,a}, \tag{D.5}$$

with the block word $P_t$ a Pauli word $X^x Z^z$ on the block register (or, for a system's coupling, a matching, § 6.6), and each $F_{t,a}$ a *band*: a diagonal $D$ followed by the truncated shift that takes column $q$ to row $q-\ell$ and discards it where that row does not exist, with $D$ the identity, one $Z_j$ or a reflection $Q_m$.

**Lemma 6** (An affine band with exceptions)**.** *A diagonal whose entries are $v_q = a + bq$ except at a few columns $m$, where $v_m = a + bm + d_m$, is*

$$\mathrm{diag}(v) = \Big(a + b\,\frac{N-1}{2} + \sum_m \frac{d_m}{2}\Big) I - b \sum_{j=0}^{n-1} 2^{j-1} Z_j - \sum_m \frac{d_m}{2}\, Q_m :$$

*$n+1$ terms and one reflection for each exception.*

*Proof.* $\Lambda = \frac{N-1}{2} I - \sum_j 2^{j-1} Z_j$ (Equation (C.1)), and $|m\rangle\langle m| = (I - Q_m)/2$. □

**Lemma 7** (The affine part is tight)**.** *If $a$ and $a+b(N-1)$ have one sign, the combination of Lemma 6 without exceptions has $\alpha = \max(|a|, |a+b(N-1)|) = \|aI + b\Lambda\|_2$.*

*Proof.* $\alpha = |a + b\frac{N-1}{2}| + |b|\frac{N-1}{2}$, the mean of the end values plus half their range, which is the larger end value in magnitude when the two share a sign. □

Every band of the descriptor has this form. $\tilde{D}$'s diagonal is $2\Lambda$ except at column 1, where the normalisation of $T_0$ makes it $2\sqrt{2}$; $\tilde{B}$'s and a constant multiplication's are constant except at their ends; a variable coefficient's multiplication bands and their truncation at the register's edge give the other exceptions. The columns a shift discards are free, and are fitted to keep the combination shortest.

**The block register.** Terms whose axis factors agree are merged across the row and column blocks they join, so that the block register carries their pattern as one small matrix $F$, paid for once; the recurrences share $\tilde{D}$ in this way. $F$ is expanded in Pauli words, $F = \sum_{x,z} f_{xz} X^x Z^z$ with $f_{xz} = 2^{-n_f} \sum_b (-1)^{z\cdot b} F_{b\oplus x,\, b}$. Where the merged pattern is a single matrix unit on one block, its $2^{n_f}$ Pauli words are replaced by one term with a flag qubit post-selected to keep that block, at the same $\alpha$. A large irregular coupling, a Carleman generator above all, is split into matchings instead (§ 6.6).

**Proposition 18** (Compilation). *Let PREPARE$_R$ load* $\operatorname{sgn}(c_t)\sqrt{|c_t|/\alpha_A}$ *and PREPARE$_L$ load* $\sqrt{|c_t|/\alpha_A}$ *at each term's joint index of the factored selectors, with* $\alpha_A = \sum_t |c_t|$. *Then the circuit that applies PREPARE$_R$, then SELECT, then PREPARE$_L^\dagger$, post-selected on the selectors, the overflow qubits and the flag, block-encodes* $A_{\text{sys}}$ *at* $\alpha_A$.

*Proof.* On the joint index of term $t$, the factored SELECT applies the tensor product of the operations its registers hold: the term's block-register word and, on each axis, its diagonal and then its shift, which is the band. A joint index that no term uses has zero amplitude in both preparations. The post-selected block is $\sum_t \sqrt{|c_t|/\alpha_A}\cdot\operatorname{sgn}(c_t)\sqrt{|c_t|/\alpha_A}\,U_t = \sum_t c_t U_t/\alpha_A$. With one preparation and its own inverse the signs would square away, and the block would be $\sum_t |c_t|U_t/\alpha_A$. □

**The truncated shift.** On the output register widened by one overflow qubit, initially zero, add $-\ell$ modulo twice the register's size. A row $q-\ell$ inside $[0, N)$ lands in the lower half, and one outside it lands in the upper half, which the projection of the overflow qubit on $|0\rangle$ discards. The addition is an increment of the qubits from $j$ upwards for each set bit $j$ of $|\ell|$, each a ladder of multi-controlled $X$ (MCX) gates, and the overflow qubit is the shift's only ancilla.

**Input:** the blocks of $A_{\text{desc}}$ as bands, the block weights $\zeta$
**Output:** PREPARE$_R$, SELECT and PREPARE$_L^\dagger$; the subnormalisation $\alpha_A$
Write every block as a sum of bands: a diagonal and a truncated shift;
Fit each diagonal as affine in the column index plus a few exceptions (Lemma 6);
Weight each band by $\zeta^m$ for its column block $m$;
Merge bands with equal axis factors across blocks; expand each resulting block-register pattern in Pauli words, giving the terms $c_t U_t$;
$\alpha_A \leftarrow \sum_t |c_t|$; on the joint selector index of term $t$, PREPARE$_R$ loads $\operatorname{sgn}(c_t)\sqrt{|c_t|/\alpha_A}$ and PREPARE$_L$ loads $\sqrt{|c_t|/\alpha_A}$;
SELECT: controlled on each selector register, apply the block word, then on each axis the diagonal and the shift;
**return** *PREPARE$_R$, SELECT, PREPARE$_L^\dagger$,* $\alpha_A$;
**Algorithm 1:** Compiling the rows of $A_{\text{desc}}$ into two preparations and one SELECT.

**Rank-one terms.** A source $|\hat{r}\rangle\langle\tau(x_s)|/f(x_s)$ is compiled apart and summed with the bands by a further combination: a matrix unit on the block register, the feature map reading $\tau(x_s)$, and a rotation tree loading $\hat{r}$, with two ancilla and no arithmetic. The constraint rows are encoded the same way, each as its own rows, and stacked into the reflection of § 7.3.

**The IR check.** The terms are built from the problem, the operator library, the residual's layout, its lifts and its block weights, never from the assembled matrices. Summing them back and comparing the sum with the independently assembled system rows is the IR mode of operator verification, possible at any $n$ (Table G.1); the compiled circuit is then compared with the terms exactly, or by probe states where it is too large.

## D.8 Proof of the Cost Proposition, and the Norms

**Ancilla.** The registers are those of Appendix D.7: one overflow qubit for each axis that shifts, at most one flag, and the factored selectors. On axis $a$ the diagonal selector indexes the identity, the $n_a$ gates $Z_j$ and the distinct reflections $r_a$, so it has $\lceil\log_2(1+n_a+r_a)\rceil$ qubits. The reflections are bounded in number by the problem's bands, and the block-register words and the shifts depend on the equation's structure, not on $n$, so every other selector has a width fixed by the problem, and the total is $\mathcal{O}(\log n)$. For R2 the reflections are three ($\tilde{B}$'s diagonal and second superdiagonal at their first entries and $\tilde{D}$'s first entry), so the diagonal selector has $\lceil\log_2(n+4)\rceil$ qubits; the shifts are 0, 1

and 2, two qubits; the block-register selector has four; with the overflow qubit and the flag the count is $\lceil \log_2(n+4) \rceil + 8$. It is verified on the compiled circuit of R2's system rows at $n = 2$, 4, 5 and 12.

**Gates.** SELECT applies each distinct operation once. A $Z_j$ is one gate controlled on $\mathcal{O}(\log n)$ selector qubits; a reflection is one phase controlled on $n_a$ qubits; a shift by a fixed $\ell$ is $\mathcal{O}(\log |\ell|)$ increments of at most $n_a + 1$ qubits, each $n_a + 1$ MCX gates of up to $n_a$ controls, $\mathcal{O}(n^2)$ gates with a linear-cost decomposition of each. The block-register operations act on $n_f$ qubits, fixed by the problem, and the preparations are rotation trees over $\mathcal{O}(n)$ joint indices. There are $\mathcal{O}(n)$ distinct operations, the $Z_j$ costing $\mathcal{O}(\log n)$ each and the rest $\mathcal{O}(n^2)$ in all, so the total is $\mathcal{O}(n^2)$. A source's rank-one term adds the feature map, of poly$(n)$ gates, and a rotation tree over its $N$ coefficients, which has $\mathcal{O}(N)$ gates for a general source, as for any general state, and poly$(n)$ when only a fixed number of the coefficients is non-zero, as for a polynomial source of fixed degree; the further combination that adds it costs $\mathcal{O}(1)$ selector qubits.

**Exactness.** Proposition 18; every term is a unitary, so no part of the encoding is approximate.

**The subnormalisation.** Take first the chain of a regular second-order problem at a fixed $\zeta \geq 1$. The recurrences' $\tilde{D}$ terms share the block-register pattern $\mathrm{diag}(1, \zeta, 0, 0)$, whose Pauli expansion has $\alpha = \frac{1}{2}(|1+\zeta| + |1-\zeta|) = \max(1, \zeta) = \zeta$, its largest entry. With Lemma 7, $\tilde{D}$'s affine part is encoded at $2(N-1)$, so these terms contribute $2\zeta(N-1)$ to $\alpha_A$. Every other term is independent of $N$: $\tilde{D}$'s exception at column 1, the $\tilde{B}$ bands weighted by $\zeta$ and $\zeta^2$, the coefficients' multiplication bands weighted by at most $\zeta^2$, and the padding. A source's rank-one term has coefficient $\|\hat{r}\| \, \|\tau(x_s)\| / |f(x_s)|$, and $\|\hat{r}\|^2$ grows as $N$ for a fixed smooth source (Equation (4.1)), while $\|\tau(x_s)\|^2 < 2$ (Proposition 7). On the other side, $\zeta\tilde{D}$ is a block of $A_{\mathrm{desc}}$, so $\|A_{\mathrm{desc}}\|_2 \geq 2\zeta(N-1)$. Hence $\alpha_A / \|A_{\mathrm{desc}}\|_2 \leq 1 + C/N$ without a source and $1 + C'/\sqrt{N}$ with one, with constants that grow with $\zeta$ and the coefficients. For the mass form of a scalar equation $f' = af$, $\alpha_A \leq 2(N-1) + \mathcal{O}(1)$ and $\|\tilde{D} - a\tilde{B}\|_2 \geq 2(N-1) - |a|(2+\sqrt{2})$, with the same conclusion. For a chain of order three or more the recurrences' pattern need not be encoded at its largest entry, and no limit is claimed; no benchmark problem has one. On a singular problem $\zeta = N/2$ makes $\zeta\tilde{D}$, $\zeta^2\tilde{B}$ and $\zeta^2 a_2$ all of order $N^2$ in one row block, and terms of comparable size add in $\ell^1$ in $\alpha_A$, so the ratio tends to a constant, measured near two (§ 5.6).

**Optimality.** An exact combination whose SELECT indexes $L$ distinct operations needs $\lceil \log_2 L \rceil$ selector qubits [43]. $\Lambda$ has exactly $n+1$ non-zero Walsh coefficients, so a combination of Pauli-$Z$ strings needs $n+1$ terms for it and $\Omega(\log n)$ selector qubits. The descriptor's count is therefore of the least order among exact combinations that expand their diagonals in Pauli strings, and is not claimed against encodings built otherwise.

**The norms of Table 5.1.** Each row block of $R_{\mathrm{desc}}$ is a finite sum of blocks weighted by powers of $\zeta$: $\zeta^m\tilde{D}$, of norm $2\zeta^m(N-1)$; $\zeta^{m+1}\tilde{B}$, of norm at most $(2+\sqrt{2})\zeta^{m+1}$; $\zeta^j\mathcal{M}_{a_j}$, of norm at most $\sqrt{2}\,\zeta^j \sup|a_j|$ (Proposition 13); the padding, of norm one; and the rank-one constraint and source rows, of norm $\mathcal{O}(\sqrt{N})$, since $\mathbb{B}(x) = \sqrt{N}\,|0\rangle\langle\tau(x)|$ with $\|\tau(x)\|^2 < 2$. The norm of $R_{\mathrm{desc}}$ is at most the sum of these and at least the largest of them. At fixed $\zeta$ the largest is $2\zeta^{k-1}(N-1)$, so $\|R_{\mathrm{desc}}\|_2 = \Theta(N)$ and $\|H_{\mathrm{desc}}\| = \Theta(N^2)$. With $\zeta = N/2$ and $k = 2$, $\zeta\tilde{D}$, $\zeta^2\tilde{B}$ and $\zeta^2\mathcal{M}_{a_2}$ are each $\Theta(N^2)$, so $\|H_{\mathrm{desc}}\| = \Theta(N^4)$; on two axes with $\zeta_a = N_a$ and $k_a = 2$ the same count gives $\Theta(N^4)$; and with $\zeta \propto N^2$ the block $\zeta^2\tilde{B}$, of norm at least $2\zeta^2$, gives $\Theta(N^8)$.

## D.9 One Query's Success

**Proposition 19** (One query on a Haar-random state)**.** *One application of a block encoding of $R$ at $\alpha_R$, post-selected, succeeds on a Haar-random state of dimension $D$ with mean probability $\|R\|_F^2/(D\alpha_R^2)$.*

*Proof.* The success probability on $\psi$ is $\|R\psi\|^2/\alpha_R^2$, and $\mathbb{E}\,\psi\psi^\dagger = I/D$ for Haar-random $\psi$, so its mean is $\mathrm{tr}(R^\dagger R)/(D\alpha_R^2)$. □

In the structured standard form the dense $G^k$ dominates: for $k=1$, $\|G\|_F^2 \sim N^4/2$ (Proposition 8) against $\alpha_G^2 \sim N^4$ on $D=N$ amplitudes, so the mean falls as $1/(2N)$. In the descriptor the dominant $\tilde{D}$ has $\|\tilde{D}\|_F^2 = \sum_{q<N}(2q)^2 \sim 4N^3/3$, up to its first entry, against $\alpha^2 \sim 4N^2$, so the mean stays $\Theta(1)$. The pipeline post-selects no single query, its walk being unitary, so this compares the encodings and is not a cost of the preparation.

# E Systems and the Nonlinear Route in Full

## E.1 Systems in Full

**The rows of R7.** R7 is $u' = v$, $v' = -\pi^2 u$ on $x \in [-1,1]$ with $(u,v)(-1) = (0,\pi)$, so $C=2$, $\Gamma = \left(\begin{smallmatrix} 0 & 1 \\ -\pi^2 & 0 \end{smallmatrix}\right)$ and the interval needs no rescaling ($T/2 = 1$). With the field index the most significant, the equation rows of Equation (6.1) are

$$\begin{pmatrix} G & -I \\ \pi^2 I & G \end{pmatrix} \text{ (standard form)}, \qquad \begin{pmatrix} \tilde{D} & -\tilde{B} \\ \pi^2 \tilde{B} & \tilde{D} \end{pmatrix} \text{ (mass form)}, \tag{E.1}$$

the second being the first multiplied on the left by $I_2 \otimes \tilde{B}$. Two fields fill the one-qubit field register, so there is no padding. In general a padding block $b \geq C$ carries the row $\hat{f}^{(b)} = 0$ of weight one and no other row reads it, so the Hamiltonian is block diagonal between the physical and the padding blocks; the padding's eigenvalue is one, and the gap, the reference and the initial state are those of the physical blocks.

**R7's data.** The anchor is $v$, with $y_v = \pi$, and $y_\perp = 0$ because $u(-1) = 0$, so the ratio row divided by $y_a$ is $|u\rangle\langle u| \otimes \mathbb{B}(-1)$: the zero datum holds $u(-1)$ to zero exactly. The regular datum is $v(-1) = \pi$, a collapse row on the $v$ block at $x_s = -1$, which fixes the scale and the readout.

**The ratio rows' spectrum.** For $C$ components and $\mathbb{F}$ of Equation (6.2):

**Proposition 20** (Ratio rows)**.** $\ker \mathbb{F} = \mathrm{span}(y)$*, and the singular values of* $\mathbb{F}$ *are* $\|y\|$ *once,* $|y_a|$ *with multiplicity* $C-2$*, and* $0$ *once.*

*Proof.* If $\mathbb{F}v = 0$ then $v_c = (v_a/y_a)y_c$ for every $c \neq a$, and trivially for $c = a$, so $v \parallel y$; and $\mathbb{F}y = 0$. The $C-1$ non-zero rows, indexed by $c \neq a$, are $[\, y_a I \mid -u \,]$ with $u = (y_c)_{c\neq a}$ (columns $c \neq a$, then $a$), so $\mathbb{F}\mathbb{F}^\dagger$ restricted to them is $|y_a|^2 I + uu^\dagger$, with eigenvalues $|y_a|^2$ on $u^\perp$ (multiplicity $C-2$) and $|y_a|^2 + \|u\|^2 = \|y\|^2$. □

The rows therefore fix the ratios of the components and leave the scale free, as a zero row does, and their weight against the equation is set by the data: the largest singular value is $\|y\|$ and the smallest non-zero one $|y_a|$, so the anchor with the largest $|y_a|$ conditions them best. For a Carleman lift $y$ is the vector of the data's scaled moments (Appendix E.5).

**The ratio rows' circuit.** Divided by $y_a$, the rows are

$$\frac{\mathbb{F}}{y_a} \otimes \mathbb{B}(x^*) = \left( I - |a\rangle\langle a| - \frac{\|y_\perp\|}{y_a} |\hat{y}_\perp\rangle\langle a| \right) \otimes \mathbb{B}(x^*), \qquad \hat{y}_\perp = y_\perp/\|y_\perp\|, \tag{E.2}$$

three LCU terms, each a field-register operation times the zero row's encoding: the identity; the projector $|a\rangle\langle a|$, a matrix unit placed by $X$ gates around the zero projector; and the rank-one term, the matrix unit $|0\rangle\langle a|$ followed by a rotation tree that prepares $\hat{y}_\perp$ from $|0\rangle$. The subnormalisation is $(2 + \|y_\perp\|/|y_a|)$ times the zero row's, and a datum with $y_\perp = 0$ drops the third term.

### E.1.1 The coupling as a sum of matchings

**Construction.** A matching $\mathcal{P}$ is a set of entries of $\Gamma$ with at most one in each row and column. It acts as a partial permutation of the field register's basis states followed by a diagonal: column $c$ goes to row $\pi(c)$ with weight $v_{\pi(c)}$. The permutation is completed to a permutation of all basis states and built from transpositions, each a cascade of CNOTs around one MCX, the only gate that takes the selector's control. The weights, divided by the matching's largest magnitude $\alpha_{\mathcal{P}}$, lie in $[-1, 1]$ and are applied as $R_Y(2 \arccos v_r)$ on a flag qubit, whose zero then carries $v_r$, with weight zero on a row the matching does not reach. The rotations of every matching form one uniformly controlled $R_Y$ on the field register and the matching selector, $2^{n_f+w}$ CNOTs for all of them, $w$ the selector's width. A matching whose weights share one value is a bare permutation, its sign moved into the coefficient. The coupling is $\sum_{\mathcal{P}} \alpha_{\mathcal{P}} U_{\mathcal{P}}$, block-encoded at $\alpha = \sum_{\mathcal{P}} \alpha_{\mathcal{P}}$.

**How many.** The matchings are the colour classes of an edge colouring of the bipartite graph whose two sides are $\Gamma$'s rows and columns and whose edges are its non-zero entries. The greedy colouring takes the entries in row order and gives each the least colour that its row and its column do not yet hold. Let $\delta$ be the largest number of entries in a row or column. An entry meets at most $\delta - 1$ others in its row and $\delta - 1$ in its column, so at most $2\delta - 1$ colours are used. $\delta$ is the least possible, since the entries of one row need distinct colours, and it suffices for a bipartite graph by König's theorem [89]; on every Carleman lift built, the greedy colouring reached $\delta$. A Pauli-word expansion of the same coupling, $\Gamma = \sum_{x,z} f_{xz} X^x Z^z$, pays for every word with a non-zero coefficient, and on a Carleman lift these are a sizeable fraction of the $4^{n_f}$ words, against $\delta$ matchings.

Table E.1 sets the matchings against the tensor layout of Appendix E.4 on the lifts of Burgers' and Navier–Stokes' couplings, each encoded alone.

**Table E.1: The matchings are cheaper only on small lifts; the tensor layout's term count does not grow.** The Carleman lift's coupling $\Gamma_K$, encoded alone, for Burgers (R9c, $a/\nu = 1$) and the Navier–Stokes vortex pair (R10c, $a/\nu = 10$): in the symmetric layout as a sum of matchings, in the tensor layout as $2K - 1$ terms on its physical states. $M$, Galerkin modes; $K$, Carleman order; $\alpha/\|\Gamma_K\|$, subnormalisation over the coupling's norm; CNOTs, per query; –, symmetric layout not built. Every entry verified by IR to $10^{-10}$. Classical construction and verification.

| | | | Symmetric layout (matchings) | | | Tensor layout | | |
|---|---|---|---|---|---|---|---|---|
| Problem | $M$ | $K$ | matchings | $\alpha/\|\Gamma_K\|$ | CNOTs | terms | $\alpha/\|\Gamma_K\|$ | CNOTs |
| R9c | 4 | 3 | 7 | 1.29 | 4656 | 5 | 1.24 | 7998 |
| R9c | 16 | 3 | 31 | 1.34 | 525 436 | 5 | 1.12 | 17 218 |
| R9c | 16 | 4 | 45 | 1.50 | 6 370 468 | 7 | 1.13 | 26 596 |
| R10c | 12 | 2 | 5 | 2.25 | 4022 | 3 | 14.8 | 23 208 |
| R10c | 20 | 3 | 25 | 3.94 | 636 214 | 5 | 11.4 | 89 408 |
| R10c | 48 | 2 | 33 | 5.89 | 326 444 | 3 | 19.9 | 79 044 |
| R10c | 80 | 2 | 61 | 7.34 | 883 336 | 3 | 17.1 | 78 724 |
| R10c | 316 | 2 | – | – | – | 3 | 23.5 | 288 120 |

The matchings' count and cost grow with the entries of $\Gamma_K$, the tensor layout's term count stays at $2K - 1$ and its CNOTs grow slowly with $M$, so the matchings are cheaper per query only on the smallest lifts. The tensor layout's subnormalisation is close to the norm on Burgers and loose in two dimensions, where the Fourier fold's $\alpha$ is $N$.

### E.1.2 Realification and the real embedding

**Realification.** Let $\iota : \mathbb{C}^m \to \mathbb{R}^{2m}$, $z \mapsto (\operatorname{Re} z, \operatorname{Im} z)$, a real-linear isomorphism, and let a complex residual be $R = R_r + \mathrm{i}R_i$ with $R_r$, $R_i$ real. Then $\iota(Rz) = R_\iota\, \iota(z)$ with

$$R_\iota = \begin{pmatrix} R_r & -R_i \\ R_i & R_r \end{pmatrix}, \tag{E.3}$$

so $Rz = 0$ if and only if $R_\iota \iota(z) = 0$, and $\ker R_\iota = \iota(\ker R)$, of real dimension twice the complex one: the solution's real scale and its phase. The data close both. Imposed as real ratio rows on the realified components (Equation (6.2)), they fix the direction of $\iota(z(x^*))$ in $\mathbb{R}^{2m}$, which fixes the phase as well as the ratios, so an IVP has a one-dimensional real near-kernel, as a real system does. For one mode, $z' = cz$ realifies to Equation (6.3). R7 is $z' = -\mathrm{i}\pi z$ with $z = \pi u + \mathrm{i}v$: the realified state is $(\pi u, v)$, and R7 carries $(u, v) = D^{-1}(\pi u, v)$ with $D = \operatorname{diag}(\pi, 1)$, so its coupling $\left(\begin{smallmatrix} 0 & 1 \\ -\pi^2 & 0 \end{smallmatrix}\right) = D^{-1}\left(\begin{smallmatrix} 0 & \pi \\ -\pi & 0 \end{smallmatrix}\right) D$ is the realification's under a diagonal change of variables, and its kernel the image of the realification's under $D^{-1} \otimes I$.

**The real embedding of the Fourier variables.** For the fluids, with $\hat{u}_{\pm\mathbf{k}} = (a \mp \mathrm{i}b)/2$ the complex coefficients of $a\cos(\mathbf{k}\cdot\mathbf{x}) + b\sin(\mathbf{k}\cdot\mathbf{x})$, the variables $v = \sqrt{2}\,\hat{u}$ on the wavevectors of both signs are $v = Ew$, with $w$ the real coefficients and $E$ putting $1/\sqrt{2}$ on $\pm\mathbf{k}$ for a cosine and $\mp\mathrm{i}/\sqrt{2}$ for a sine. $E$ is an isometry, $E^\dagger E = I$, and unitary when the real basis holds every kept wavevector, so the complex Fourier–Galerkin operators act on real data through $E$, and the real system's coupling is $E^\dagger(\cdot)E$. Burgers' sine modes are the case of one axis and sines only.

## E.2 The Doubled Space

**The doubled residual.** On $|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle$, let a term have coefficient $c$ and $k \le 2$ field factors, the $i$-th carrying the derivative $G^{j_i}$, and a multiplier of degree at most one; it becomes $c\mathcal{N}_1\big(\mathbb{D}^{(0)}(x_s)^{\otimes(2-k)} \otimes G^{j_1} \otimes \cdots\big)$ for a constant multiplier and the same with $\mathcal{N}_x$ for the multiplier $x$. A linear term is thereby carried by one copy of the collapse row, which on $\psi$ is the constant $f_q(x_s)/f(x_s)$, and a source by two. An invariant constraint becomes $\mathcal{N}_1\big(\mathbb{D}^{(0)}(x_s) \otimes \mathbb{B}(x^*)G^j\big)$, and the readout is $\mathcal{N}_1(\mathbb{D}^{(0)} \otimes I)$, with $f_Q = \eta f_q$ ($p = 1$ in $f_Q = \eta^{1/p} f_q$). The equation block maps the $N^2$ amplitudes of $\Psi$ to the $2N$ coefficients of the folded register on $n+1$ qubits.

**The kernel bound.** The equation block has $2N$ rows, so rank at most $2N$, and each constraint block is a fold of a Kronecker product of two rank-one rows, so rank at most one. With $m$ invariant constraints, $\operatorname{rank} R \le 2N + m$ and $\dim\ker H_\otimes = \dim\ker R \ge N^2 - 2N - m$, which is at least $N^2/2 = 2^{2n-1}$ once $N \ge 8$ for $m \le 2$. At $n = 2$ the bound is weaker than Wu et al.'s count, which the records meet there by measurement. Table E.2 gives the measured dimensions; at $n = 2$ to 4 the equation blocks have rank $2N - 3$ (R6a) and $2N - 1$ (R6b), each problem with one constraint.

**Table E.2: The doubled space's kernel grows as $N^2$.** Kernel dimension of the doubled-space Hamiltonian $H_\otimes$ for R6a ($4f'' + 2(f')^2 + f = 0$) and R6b ($f'' - 2f^2 + x = 0$), against the doubled state's dimension $N^2$ and Wu et al.'s lower bound $2^{2n-1}$; $N = 2^n$. Classical computation (SVD of the stacked residual).

| $n$ | $N^2$ | $2^{2n-1}$ | R6a | R6b |
|---|---|---|---|---|
| 2 | 16 | 8 | 10 | 8 |
| 3 | 64 | 32 | 50 | 48 |
| 4 | 256 | 128 | 226 | 224 |
| 5 | 1024 | 512 | 962 | 960 |

**Higher degree.** § 3.4 states the published construction as quadratic only. A term of degree $p$ in the unknown needs $p$ copies, $|\Psi\rangle = |\psi\rangle^{\otimes p}$, of dimension $N^p$, its factors folded back into one Chebyshev series by the $p$-fold product of Proposition 26 (a cubic term by a tripled space, a quartic by a quadrupled one). The register therefore grows with the degree, a limitation also noted in the concurrent work of Paine [63]. The argument of Proposition 4 should carry over, since the corresponding kernel is again a subspace and the $p$-fold product states are not; we do not use the doubled space beyond $p = 2$. Our alternative for every degree is the Carleman lift of § 6.5.

## E.3 Proof of Proposition 4

*Proof.* (1) As a matrix, $u \otimes u$ is $uu^\top$, of rank one. For independent $u$ and $v$, $uu^\top + vv^\top$ has rank two, so the sum of two product states is not of the form $w \otimes w$, and the product states are not closed under addition. A kernel is a linear subspace, so a kernel containing $u \otimes u$ and $v \otimes v$ contains their sum, and no kernel is exactly the set of product states.

(2) Write $H_\otimes = \sum_\lambda \lambda \Pi_\lambda$ with $\Pi_0 = \Pi$. Then $P_d(H_\otimes)\psi_{\text{init}} = \Pi\psi_{\text{init}} + \sum_{\lambda>0} P_d(\lambda)\Pi_\lambda\psi_{\text{init}}$, and the second term has norm at most $\max_{\lambda>0} |P_d(\lambda)|\, \|\psi_{\text{init}}\| \to 0$. Since $\Pi\psi_{\text{init}} \neq 0$, normalising gives Equation (6.5). The imaginary-time filter has $P(0) = 1$ and $\max_{\lambda>0} e^{-\beta\lambda} = e^{-\beta\lambda_1} \to 0$; the minimax filter takes the value one at the ground state and at most $\varepsilon$ on the excited spectrum, and $\varepsilon \to 0$ as its degree grows at a fixed edge.

(3) For $H_\otimes, P \succeq 0$, $\langle\phi, (H_\otimes + P)\phi\rangle = 0$ if and only if $\langle\phi, H_\otimes\phi\rangle = \langle\phi, P\phi\rangle = 0$, so $\ker(H_\otimes + P) = \ker H_\otimes \cap \ker P$. For this to be $\operatorname{span}\{\psi_* \otimes \psi_*\}$, $P$ must vanish on $\psi_* \otimes \psi_*$ and on no other direction of $\ker H_\otimes$, a condition that refers to $\psi_*$. □

The scope is the one stated: a penalty built without knowledge of the solution. A penalty that knows the solution isolates it, and is answer-dependent.

## E.4 The Carleman Lift

**The symmetric layout.** A polynomial system $u' = \sum_{p\geq 1} F_p u^{\otimes p}$ on $M$ components, $F_p$ of shape $M \times M^p$, is carried on the monomials $z_\alpha = \prod_i u_i^{\alpha_i}$ with $1 \leq |\alpha| \leq K$, once each: $\sum_{k=1}^{K} \binom{M+k-1}{k}$ components, level by level, the fields first. Differentiating a monomial,

$$z'_\alpha = \sum_i \alpha_i u^{\alpha - e_i} u'_i = \sum_i \alpha_i \sum_p \sum_{j_1,\ldots,j_p} (F_p)_{i,j_1\ldots j_p}\, z_{\alpha - e_i + e_{j_1} + \cdots + e_{j_p}}, \tag{E.4}$$

and $\Gamma_K$ keeps every term whose monomial has degree at most $K$. A linear system is its own lift, $\Gamma_1 = F_1$. For R8, $y' = -y + y^2$, before the time axis's factor $T/2$,

$$\Gamma_3 = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -2 & 2 \\ 0 & 0 & -3 \end{pmatrix}. \tag{E.5}$$

**Proposition 21** (The lift's spectrum)**.** *For a quadratic system with $F_1$ diagonalisable, $\Gamma_K$ is block upper triangular in its levels, and its eigenvalues are the sums $\lambda_{i_1} + \cdots + \lambda_{i_k}$, $1 \leq i_1 \leq \cdots \leq i_k \leq M$, $1 \leq k \leq K$, of the eigenvalues of $F_1$.*

*Proof.* $F_1$ maps a monomial to monomials of its own degree and $F_2$ to degree one higher, so level $k$ reads levels $k$ and $k+1$ only. In $F_1$'s eigencoordinates the diagonal block of level $k$ acts on a monomial $v^\alpha$ as $\sum_i \alpha_i \lambda_i v^\alpha$. The change of coordinates is linear and maps the monomials of degree $k$ to polynomials of degree $k$, so it is a similarity on each level. □

**Proposition 22** (The level scale)**.** *Carrying $w = u/s$ multiplies $F_p$ by $s^{p-1}$, and the lifted generator becomes $D^{-1}\Gamma_K D$, $D$ the diagonal equal to $s^k$ on level $k$. The truncated solution, the truncation error and $\rho$ are unchanged.*

*Proof.* $z(u) = Dz(w)$, so $z(w)' = D^{-1}\Gamma_K Dz(w)$, whose entry $(\alpha, \beta)$ is multiplied by $s^{|\beta|-|\alpha|} = s^{p-1}$ for the terms of $F_p$. A similarity maps solutions to solutions, and $\rho$ gains $1/s$ from the data and $s$ from $F_2$. □

With $s = \max(\|u_0\|, 1)$, level $k$ of the data's moments has norm $(\|u_0\|/s)^k \leq 1$. Unscaled, data of norm above one grow as $\|u_0\|^k$ with the level, the ground state's weight moves from the fields to the top level, where no initial state independent of the answer reaches it, and the moments inflate the ratio rows' subnormalisation. The readout is scaled back to the physical datum.

**The tensor layout.** Level $k$ is carried as the whole tensor $u^{\otimes k}$, $M^k$ components in $k$ slots, slot 0 the most significant.

**Proposition 23** (The tensor layout's terms)**.** *In the tensor layout the block of $\Gamma_K$ from level $k+p-1$ to level $k$ is*

$$\left(\Gamma_K^{\otimes}\right)_{k,\,k+p-1} = \sum_{j=0}^{k-1} I^{\otimes j} \otimes F_p \otimes I^{\otimes(k-1-j)}, \tag{E.6}$$

*a sum of terms "$F_p$ on slots $j, \ldots, j+p-1$", one per degree $p$ and first slot $j$ with $j+p \leq K$: $\sum_p (K-p+1)$ terms, $2K-1$ for a quadratic field, whatever $M$. With $V$ the embedding of the symmetric layout in the tensor layout, $\Gamma_K^{\otimes} V = V\Gamma_K$.*

*Proof.* $\frac{d}{dt}u^{\otimes k} = \sum_j u^{\otimes j} \otimes u' \otimes u^{\otimes(k-1-j)}$ and $u' = \sum_p F_p u^{\otimes p}$, so slot $j$'s factor $F_p$ contracts slots $j$ to $j+p-1$ of level $k+p-1$ into slot $j$, and one term $(p, j)$ serves every level that occupies its slots. For the embedding, both sides map the moments $z(u)$ to the truncated derivative of $(u, u^{\otimes 2}, \ldots, u^{\otimes K})$; the moments over all $u$ span the monomial space, distinct monomials being linearly independent functions, so the identity holds as matrices. □

The individual terms are not equal on symmetric tensors; only their sum is the symmetric layout's. A native complex preparation in this layout would also need a phase on the collapse row and a complex state loader for the data's moments, neither of which is built.

**The tensor layout's circuit.** It is built for a Fourier–Galerkin equation in structured form, $u' = L_1 u + \sum \mathrm{out}\big((\mathrm{left}\, u) \cdot (\mathrm{right}\, u)\big)$, every operator a Fourier multiplier, in the variables $v = \sqrt{2}\,\hat{u}$ of Appendix E.1.2. The register is $K$ slots of $d\,m$ qubits, each holding a wavevector's components in two's complement with $2^m \geq 3 \max|k_d| + 1$, so that the product of two kept modes never aliases onto a kept mode, and a level register of $\lceil \log_2(K+1) \rceil$ qubits. A level-$k$ state occupies slots 0 to $k-1$ with kept wavevectors and leaves the others at the zero wavevector, which no truncation keeps: these are the physical states. $F_1$ on slot $j$ is its multiplier's magnitudes on the kept wavevectors, as one uniformly controlled $R_Y$ on a flag ($\alpha$ the largest magnitude), times the projector onto the levels above $j$; a symbol $c \prod_d k_d^{a_d} |\mathbf{k}|^{2e}$ has the phase $\mathrm{i}^{\sum_d a_d}$, which joins the term's weight. $F_2$ on slots $j$ and $j+1$ is the projector onto the levels from $j+2$, the Fourier fold of its input multipliers (Appendix E.6), the output multiplier, which is the Galerkin truncation, and a cyclic shift of the slots above $j$ with the level lowered by one. The terms are combined with weights $T/2$ and, for $F_2$, $s/\sqrt{2}$. The multipliers vanish off the truncation, so the encoding maps physical states to physical states only, with no leakage. It is verified exactly on Burgers' lift on four modes at $K = 2$, to $1.7 \times 10^{-15}$ with leakage $1.1 \times 10^{-14}$, and by IR on every truncation where IR was run, to 316 modes at $K = 2$ on 100 172 physical states; at $K = 3$ on 148 and 316 modes it is a construction record only.

**Where the lift is exact.**

**Proposition 24** (An exact lift)**.** *If the data lie in a subspace $S$ that $F_1$ maps into itself and on which $F_2(u \otimes u) = 0$, the truncated lift's first level is the solution exactly, at every $K$.*

*Proof.* The solution is $u(t) = e^{F_1 t} u_0 \in S$, with $F_2(u \otimes u) = 0$ throughout. In the tensor layout, which embeds the symmetric one (Proposition 23), the top level evolves by the Kronecker sum of $F_1$ alone, so from $u_0^{\otimes K}$ it is $u(t)^{\otimes K}$. If level $k+1$ is $u(t)^{\otimes(k+1)}$, the $F_2$ terms feeding level $k$ apply $F_2$ to $u \otimes u$ in two adjacent slots, which is zero, so level $k$ evolves by $F_1$'s Kronecker sum and is $u(t)^{\otimes k}$. Down to the first level, $z_1 = u(t)$. □

The Taylor–Green vortex is this case (Appendix H.2).

## E.5 The Time Axis, Uniqueness and the Error

**Rows and data.** A system or a lift on $t \in [0, T]$ is carried on $\xi = 2t/T - 1$, and its equation rows are Equation (6.1) with $\Gamma$, or $\Gamma_K$, holding the factor $T/2$. Its data are the ratio rows $(\mathbb{F}/y_a) \otimes \mathbb{B}(-1)$ at $\xi = -1$ with $y$ the scaled moments $z(u_0/s)$, imposing $z_c(-1) = (y_c/y_a)\, z_a(-1)$. On a padding block $\mathbb{F}/y_a$ is the identity, whose rows meet only that block, so the padding still decouples. The two forms share these rows and differ by the invertible $I \otimes \tilde{B}_\xi$ on the equation rows, so they share their kernel (Theorem 1).

**Proposition 25** (A one-dimensional near-kernel)**.** *On $C^1([-1,1])^C$ the solutions of $z' = \Gamma z$ with $z(-1) \in \operatorname{span}(y)$ are $\{c\, e^{\Gamma(\xi+1)} y : c \in \mathbb{R}\}$, a one-dimensional space.*

*Proof.* A linear IVP has exactly one solution for each initial value, and the ratio rows leave the initial value one degree of freedom. □

The residual is this operator restricted to polynomials of degree below $N_t$. Its smallest singular value is the polynomial residual of the true solution, which falls spectrally in $N_t$ for an analytic solution, and the rest of its spectrum approximates the continuum operator's, which has no other kernel. The near-kernel is therefore one-dimensional in both forms and for every $\Gamma$, including a lift whose truncation does not converge: the uniqueness is the IVP's, which the lift carries. Every Carleman build in the records has a one-dimensional kernel; the few not trusted lie at $n_t = 3$, where the time axis does not yet resolve the solution.

**The error in two parts.** With $u$ the reference, $z^{(K)}$ the exact solution of the truncated lift from the data's moments, and $\hat{u}$ the field decoded from the prepared state, the triangle inequality gives Equation (6.8), both terms on the comparison grid. The lift's term is computed by exponentiating $\Gamma_K$ and does not depend on $n_t$; the time axis's falls spectrally with $n_t$. The field's error therefore falls with $n_t$ until the lift's floor and then holds.

## E.6 Product Folds

A product of $p$ fields on $n$ qubits has degree at most $p(N-1) < 2^\ell N$, $\ell = \lceil \log_2 p \rceil$, so it is represented exactly on $n + \ell$ qubits, where it is the pointwise product of the fields' values at the nodes. Let $U_{\mathrm{CG}}$ be the Chebyshev–Gauss transform on $n + \ell$ qubits (coefficients to node values, orthogonal), $E$ the zero-padding from $n$ to $n + \ell$ qubits, and $\mathcal{D}_p = \sum_i |i\rangle\langle i|^{\otimes p}$ the map that keeps the diagonal.

**Proposition 26** (The nodal fold is exact and tight)**.** *The fold*

$$\mathcal{N}_1^{(p)} = 2^{p\ell/2}\, U_{\mathrm{CG}}^\top \mathcal{D}_p \left(U_{\mathrm{CG}} E\right)^{\otimes p} \tag{E.7}$$

*satisfies* $\langle\tau(x)|_n^{\otimes p} = \langle\tau(x)|_{n+\ell}\,\mathcal{N}_1^{(p)}$ *exactly. For* $p = 2$ *it is block-encoded at* $\alpha = 2$*, and* $\|\mathcal{N}_1^{(2)}\|_2 = 2\sqrt{\lambda_{\max}(Q \circ Q)} \le 2$*, with* $Q = U_{\mathrm{CG}}E(U_{\mathrm{CG}}E)^\top$ *the projector onto the padded fields' node values and* $\circ$ *the entrywise product.*

*Proof.* The normalised bases on $n$ and $n+\ell$ qubits differ by $\varphi_k^{(n)} = 2^{\ell/2}\varphi_k^{(n+\ell)}$, so a field with coefficients $b$ on $n$ qubits has the values $2^{\ell/2}\,U_{\mathrm{CG}}Eb$ at the $2^\ell N$ nodes. A product of $p$ such fields is represented exactly on $n+\ell$ qubits and its node values are the pointwise product $2^{p\ell/2}\mathcal{D}_p \bigotimes_i (U_{\mathrm{CG}}Eb_i)$; $U_{\mathrm{CG}}$ is orthogonal, so $U_{\mathrm{CG}}^\top$ returns the coefficients, which is the first claim. (The same map follows from repeated product-to-sum, $T_jT_k = \frac{1}{2}(T_{j+k} + T_{|j-k|})$.) For $p = 2$, $\ell = 1$, the circuit applies $U_{\mathrm{CG}}$ to both copies, each padded by one qubit, copies the first into the second with $n+1$ CNOTs, so that the second is zero exactly on the diagonal, post-selects it on zero, and applies $U_{\mathrm{CG}}^\dagger$ to the first: a block encoding of $\mathcal{N}_1/2$, so $\alpha = 2$. For the norm, $U_{\mathrm{CG}}^\top$ is orthogonal, so $\|\mathcal{N}_1\| = 2\|Y\|$ with $Y = \mathcal{D}_2(U_{\mathrm{CG}}E \otimes U_{\mathrm{CG}}E)$, and $YY^\top = \mathcal{D}_2(Q \otimes Q)\mathcal{D}_2^\top$ has entry $(i,j)$ equal to $Q_{ij}^2$. $Q \circ Q$ has non-negative entries and row sums $\sum_j Q_{ij}^2 = Q_{ii} \le 1$, so its largest eigenvalue is at most one. □

The norm rises with $n$, from 1.58 at $n = 2$ to 1.665 at $n = 12$, so $\alpha = 2$ exceeds it by a factor that falls from 1.27 to 1.20: the encoding is tight to a constant and not optimal. $\mathcal{N}_x$, the fold with the multiplier $x$, multiplies by the node values $\bar{x}_m = \cos\theta_m$ before $U_{\mathrm{CG}}^\top$; since $\theta_m$ is linear in $m$, $\mathrm{diag}(\bar{x}) = (D + D^\dagger)/2$ for the diagonal phase $D = \mathrm{diag}(e^{\mathrm{i}\theta_m})$, one phase gate per qubit combined on one ancilla, and $\alpha$ stays 2. The circuit's ancillas are the two padding qubits and $n-1$ spare qubits that return the register to zero for stacking. $U_{\mathrm{CG}}$ is the discrete cosine transform (DCT), a dense gate counted apart and budgeted as a cited primitive [76]; it is not built gate by gate.

**The Fourier fold.** The tensor layout's fold is the circular convolution $(u * v)_{\mathbf{k}} = \sum_{\mathbf{p}+\mathbf{q}\equiv\mathbf{k}} u_{\mathbf{p}}v_{\mathbf{q}}$, which is $\mathrm{QFT}^\dagger\mathcal{D}_2(\mathrm{QFT} \otimes \mathrm{QFT})$ times $N^{1/2}$ for each axis: the QFT takes both copies to grid values, a CNOT from each qubit of the first copy to the second leaves the second at zero exactly where the grid points agree, the second copy is post-selected, and the inverse QFT returns the first to coefficients. Its $\alpha$, $N^{1/2}$ per axis, is the convolution's own norm on the register, since nothing restricts its input. The QFT is built.

## E.7 Fourier–Galerkin Operators

**Burgers.** On $x \in [0, 2\pi)$, Equation (6.9) from $u(x,0) = a\sin x$ keeps an odd solution odd ($u_x$ is even, $uu_x$ and $u_{xx}$ odd), so $u = \sum_k b_k(t)\sin kx$.

**Proposition 27** (Burgers' Galerkin system)**.** *Truncated to* $k \le M$*,*

$$b_k' = -\nu k^2 b_k - \frac{k}{4}\Big[\sum_{j+l=k} b_jb_l - \sum_{|j-l|=k} b_jb_l\Big], \qquad j,l \le M, \tag{E.8}$$

*the sums over ordered pairs.*

*Proof.* $u^2 = \frac{1}{2}\sum_{j,l} b_jb_l\big(\cos(j-l)x - \cos(j+l)x\big)$, so $-(u^2/2)_x = \frac{1}{4}\sum_{j,l} b_jb_l\big((j-l)\sin(j-l)x - (j+l)\sin(j+l)x\big)$. Its coefficient of $\sin kx$ takes $-k/4$ from each pair with $j+l=k$ and $+k/4$ from each pair with $j-l = \pm k$, since $(-k)\sin(-kx) = k\sin kx$. With $\nu u_{xx} \mapsto -\nu k^2 b_k$ and the projection onto $k \le M$, this is the system. □

So $F_1 = \mathrm{diag}(-\nu k^2)$ and $F_2$ is the bracket, each times $T/2$ on the time axis. The data $b(0) = (a, 0, \ldots, 0)$ enter as ratio rows anchored on $b_1$, which also carries the regular datum. The least damped mode is $k = 1$, so $\rho = a\|F_2\|/\nu$, the factors $T/2$ cancelling, and $\|F_2\|$ grows with $M$. The reference for the lift and the time axis is the Galerkin system's IVP, integrated with its exact sparse

Jacobian, which is stiff at sixteen modes ($\nu k^2$ up to $256\nu$). The truncation's own error is measured against the Cole–Hopf solution [77, 78]: with $\beta_B = a/(2\nu)$,

$$u(x,t) = \frac{4\nu \sum_{m\geq 1} m\, I_m(\beta_B)\, e^{-\nu m^2 t} \sin mx}{I_0(\beta_B) + 2\sum_{m\geq 1} I_m(\beta_B)\, e^{-\nu m^2 t} \cos mx}, \tag{E.9}$$

from $u = -2\nu\,\partial_x \ln\varphi$, which turns Burgers' equation into the heat equation with $\varphi(x,0) = e^{-\beta_B} e^{\beta_B \cos x}$.

**Navier–Stokes.** Equation (6.10) is carried on $\cos(\mathbf{k}\cdot\mathbf{x})$ and $\sin(\mathbf{k}\cdot\mathbf{x})$ over the half-plane $k_x > 0$, or $k_x = 0 < k_y$, with $0 < |\mathbf{k}|^2 \leq k_c^2$: $k_c^2 = 4$, 5, 9, 16 and 25 keep $M = 12$, 20, 28, 48 and 80 modes, and 49 and 100, used for the tensor layout's coupling alone, keep 148 and 316. The mean vorticity is invariant and zero, and the stream function is eliminated, $\hat{\psi}_{\mathbf{k}} = \hat{\omega}_{\mathbf{k}}/|\mathbf{k}|^2$. With $\varphi_a$ a basis function of wavevector $\mathbf{p}$, $\varphi_b$ one of $\mathbf{q}$, and $\mathbf{p}\times\mathbf{q} = p_x q_y - p_y q_x$,

$$J\big(\varphi_a/|\mathbf{p}|^2, \varphi_b\big) = \sigma_a\sigma_b\, \frac{\mathbf{p}\times\mathbf{q}}{|\mathbf{p}|^2}\, h_a(\mathbf{p}\cdot\mathbf{x})\, h_b(\mathbf{q}\cdot\mathbf{x}), \tag{E.10}$$

with $\sigma = -1$, $h = \sin$ for a cosine and $\sigma = 1$, $h = \cos$ for a sine, since $\nabla\varphi_a = \sigma_a \mathbf{p}\, h_a$ and $J(f,g) = \nabla f \times \nabla g$. The product $h_a h_b$ is a sum of the modes $\mathbf{p}\pm\mathbf{q}$, each folded into the half-plane, and the truncation keeps those in the set; every coefficient is rational and computed in exact arithmetic. Mode $c$'s equation is $w_c' = -\nu|\mathbf{k}_c|^2 w_c + \sum_{a\leq b} c_{cab} w_a w_b$, each coefficient times $T/2$, with $\nu = 1$; the data enter as ratio rows anchored on the mode with the largest datum, which carries the regular datum.

One property follows. *Modes of equal $|\mathbf{k}|$ do not interact.* In the complex basis the Jacobian's coefficients are $\widehat{J}_{\mathbf{k}} = -\sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} (\mathbf{p}\times\mathbf{q})\,\hat{\psi}_{\mathbf{p}}\hat{\omega}_{\mathbf{q}}$, and with $\hat{\psi}_{\mathbf{p}} = \hat{\omega}_{\mathbf{p}}/|\mathbf{p}|^2$, symmetrised over $\mathbf{p}\leftrightarrow\mathbf{q}$,

$$\widehat{J}_{\mathbf{k}} = -\frac{1}{2}\sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} (\mathbf{p}\times\mathbf{q})\big(|\mathbf{p}|^{-2} - |\mathbf{q}|^{-2}\big)\hat{\omega}_{\mathbf{p}}\hat{\omega}_{\mathbf{q}}, \tag{E.11}$$

which vanishes for $|\mathbf{p}| = |\mathbf{q}|$; the real basis is a unitary change of the complex one.

# F The Shared Pipeline: Identities and Formulas

Throughout, $R = [R_1; \ldots; R_L]$ is a stacked residual whose block $R_j$ has a block encoding $U_j$ at subnormalisation $\alpha_j$ on one system register with a shared ancilla register $a$, $H = R^\dagger R$, and $\sigma_0 \leq \sigma_1 \leq \cdots \leq \sigma_{\max}$ are the singular values of $R$.

## F.1 The Stacked Residual and Its Reflection

Let a row-select register of $\lceil \log_2 L \rceil$ qubits sit beside $a$. PREPARE takes it from $|0\rangle$ to $\sum_j (\alpha_j/\alpha_R)\,|j\rangle$ with $\alpha_R = (\sum_j \alpha_j^2)^{1/2}$, and SELECT applies $U_j$ when it holds $j$. Write $\Pi_a$ for the projector of $a$ onto $|0_a\rangle$ and $\Pi_{\text{in}}$ for that of $a$ and the row select together onto zero, and $U_R = \text{SELECT}\cdot\text{PREPARE}$.

**Lemma 8** (The stack)**.** $\big(\langle j|\otimes\langle 0_a|\big)U_R\big(|0\rangle\otimes|0_a\rangle\big) = R_j/\alpha_R$*; that is,* $\Pi_a U_R \Pi_{\text{in}} = R/\alpha_R$ *with the row select read as output.*

*Proof.* $\langle j|\,\text{PREPARE}\,|0\rangle = \alpha_j/\alpha_R$, and SELECT on $|j\rangle$ is $U_j$, whose block is $R_j/\alpha_j$. □

Unlike a linear combination's selector, the row select is never returned by an inverse preparation: the blocks are stacked, not summed, and their weights $\alpha_j^2$ add in quadrature.

*Proof of Proposition 5.* $2\Pi_a - I$ is a Hermitian unitary, so its conjugate $V$ by the unitary $U_R$ is Hermitian and unitary, and $V^2 = I$. Using $\Pi_a^2 = \Pi_a$ and $\Pi_{\text{in}} U_R^\dagger U_R \Pi_{\text{in}} = \Pi_{\text{in}}$,

$$\Pi_{\text{in}} V \Pi_{\text{in}} = 2\big(\Pi_a U_R \Pi_{\text{in}}\big)^\dagger \big(\Pi_a U_R \Pi_{\text{in}}\big) - \Pi_{\text{in}} = \frac{2R^\dagger R}{\alpha_R^2} - I$$

by Lemma 8. For the controlled version: where the controls hold, $U_R^\dagger C(2\Pi_a - I)U_R$ is $V$; where they fail it is $U_R^\dagger U_R = I$. □

$U_R$'s own gates are never controlled, so controlling $V$ adds a control to one multi-controlled $Z$ and nothing wider than $U_R$'s widest gate. Since $H \succeq 0$ and $\|H\| = \|R\|^2 \le \alpha_R^2$, the spectrum of $H'$ lies in $[-1, 1]$; a singular value $s$ of $R$ sits at $x = -1 + 2s^2/\alpha_R^2$, at the *height* $h = 1 + x = 2s^2/\alpha_R^2$ above the edge. The filters below work with heights, which keep their relative precision where $x$ itself would round to $-1$. Numerically, $V$ is verified exactly against $2H/\alpha_R^2 - I$ on every exact-mode record, the worst entrywise error $5.9 \times 10^{-15}$.

## F.2 Qubitisation

Let $\Pi = \Pi_{\text{in}}$ and $W = (2\Pi - I)V$.

**Theorem 6** (Qubitisation [20])**.** *For every $k \ge 0$, $\Pi W^k \Pi = T_k(H')$ with $H' = \Pi V \Pi$.*

*Proof.* Let $|\lambda\rangle$ be an eigenvector of $H'$ in the image of $\Pi$ with eigenvalue $\lambda$. Then $\Pi V |\lambda\rangle = \lambda |\lambda\rangle$ and $\|V |\lambda\rangle\| = 1$, so for $|\lambda| < 1$, $V |\lambda\rangle = \lambda |\lambda\rangle + s |\lambda^\perp\rangle$ with $s = \sqrt{1 - \lambda^2}$ and $|\lambda^\perp\rangle$ a unit vector orthogonal to the image of $\Pi$. From $V^2 = I$, $V |\lambda^\perp\rangle = s |\lambda\rangle - \lambda |\lambda^\perp\rangle$. On the plane of $|\lambda\rangle$ and $|\lambda^\perp\rangle$, $2\Pi - I = \text{diag}(1, -1)$, so $W$ is the rotation

$$W = \begin{pmatrix} \cos\vartheta & \sin\vartheta \\ -\sin\vartheta & \cos\vartheta \end{pmatrix}, \qquad \vartheta = \arccos\lambda,$$

and $\langle\lambda| W^k |\lambda\rangle = \cos k\vartheta = T_k(\lambda)$. $W^k |\lambda\rangle$ stays in the plane, which meets the image of $\Pi$ in $|\lambda\rangle$'s line, so $\Pi W^k |\lambda\rangle = T_k(\lambda) |\lambda\rangle$. For $\lambda = \pm 1$, $V |\lambda\rangle = \lambda |\lambda\rangle$ and $W^k |\lambda\rangle = \lambda^k |\lambda\rangle$. The eigenvectors span the image of $\Pi$. □

The theorem projects once, after $k$ steps. Applying $W$ and projecting after every step gives other polynomials: the branches with a non-zero ancilla carry $|\lambda^\perp\rangle$'s component, which later steps rotate back into the image of $\Pi$. Every verification of a walk or filter circuit therefore restricts once, at the end.

## F.3 The Controlled Walk

With $Z_0 = I - 2 |0\rangle \langle 0|$ on a register, $2\Pi - I = -Z_0$, so $W = Z_0^{(\text{in})} U_R^\dagger Z_0^{(a)} U_R$, the two signs cancelling. Controlled on further qubits, $W$ is the same circuit with the two reflections controlled, since where the controls fail both reflections are the identity and $U_R^\dagger U_R = I$. A controlled walk thus holds no gate wider than $U_R$'s widest, and simulating it is a statevector problem of $2^q$ amplitudes, never an operator problem of $4^q$. A circuit's global phase becomes a relative phase once it is controlled: the phases of the linear combination's terms, and GQSP's global phase when a whole preparation is controlled (Appendix F.9), are therefore carried as phase gates on the controls. A dropped phase $\lambda$ would scale an overlap by $\cos\lambda$.

## F.4 The Minimax Edge Filter

Let $x_0 < x_1$ be the ground and first excited values of $H'$, $w(x) = (2x - (x_1 + 1))/(1 - x_1)$ the affine map of $[x_1, 1]$ onto $[-1, 1]$, and $P_d = T_d(w(x))/T_d(w(-1))$ as in Equation (7.5).

**Proposition 28** (Suppression and optimality)**.** *$|P_d| \le 1$ on $[-1, 1]$ with equality at $-1$, and*

$$\sup_{x \in [x_1, 1]} \frac{|P_d(x)|}{|P_d(x_0)|} = \frac{1}{\cosh(d\theta_0)}, \qquad \theta_0 = \operatorname{arccosh}\big(-w(x_0)\big).$$

*No polynomial of degree $d$ achieves a smaller ratio.*

*Proof.* On $[x_1, 1]$, $|T_d(w)| \le 1$. Below $x_1$, $w < -1$ and $|T_d(w)| = \cosh\big(d \operatorname{arccosh}(-w)\big)$, which increases as $x$ falls to $-1$; so $|P_d| \le 1$ with equality at $-1$, and the ratio is $1/|T_d(w(x_0))| = 1/\cosh(d\theta_0)$. Optimality is Chebyshev's extremal property: among polynomials of degree $d$ that take the value 1 at a point $z$ outside $[-1, 1]$, $T_d/T_d(z)$ has the least maximum on $[-1, 1]$, namely $1/|T_d(z)|$, which is Chebyshev's inequality [54]; Lin and Tong use the same property for eigenstate filtering [11]. Apply it in the variable $w$ at $z = w(x_0)$. □

**The degree from heights.** The degree of Equation (7.6) is the least $d$ with $1/\cosh(d\theta_0) \le \varepsilon$. With the heights $h_j = 1 + x_j = 2\sigma_j^2/\alpha_R^2$,

$$-w(x_0) - 1 = \frac{2(h_1 - h_0)}{2 - h_1},$$

so $\theta_0 = \operatorname{arccosh}(1 + e)$ with $e = 2(h_1 - h_0)/(2 - h_1)$, evaluated as $\log\big(1 + e + \sqrt{e(2 + e)}\big)$ without forming $1 + e$. For small $e$, $\operatorname{arccosh}(1 + e) \approx \sqrt{2e}$ and $h_1 - h_0 = 2\Delta/\alpha_R^2$, so $\theta_0 \approx 2\sqrt{\Delta}/\alpha_R$ and

$$d \approx \frac{\alpha_R}{2\sqrt{\Delta}} \ln \frac{2}{\varepsilon}. \tag{F.1}$$

**The coefficients.** $P_d$ is sampled at $2^{\lceil \log_2 2(d+1) \rceil} > d$ Chebyshev–Gauss nodes, and its Chebyshev coefficients are read off by a type-II discrete cosine transform, exact for a polynomial of degree below the number of nodes: interpolation, not a fit. Nothing is truncated.

**Proposition 29** (Guarantee)**.** *Let $\psi_{\text{init}}$ have overlap $\gamma > 0$ with a one-dimensional target, the ground state. At the degree of Equation (7.6) the normalised output $P_d(H')\psi_{\text{init}}/\|P_d(H')\psi_{\text{init}}\|$ has infidelity at most*

$$\Big(\frac{\sqrt{1 - \gamma^2}}{\gamma \cosh(d\theta_0)}\Big)^2 \le \frac{1 - \gamma^2}{\gamma^2}\, \varepsilon^2.$$

*Proof.* Write $\psi_{\text{init}} = \gamma u + \sqrt{1 - \gamma^2}\, u_\perp$ with $u$ the ground state and $u_\perp$ in the excited span. The filter maps $\gamma u$ to $\gamma P_d(x_0)u$ and $\sqrt{1 - \gamma^2}u_\perp$ to a vector in the excited span of norm at most $\sqrt{1 - \gamma^2}\,|P_d(x_0)|/\cosh(d\theta$ by Proposition 28. The infidelity is the squared sine of the angle to $u$, at most its squared tangent, the off-target norm over the on-target one. □

For a target of $t > 1$ singular vectors the same argument holds with the ratios of the angles of the $t$-th and lower heights. The suppression is measured relative to the ground state's own filtered value, so there is no absolute tail for that value to fall beneath, and the bound is finite whenever $\gamma > 0$; imaginary time's can fail (Appendix F.5). The success probability is $p_G = \|P_d(H')\psi_{\text{init}}\|^2 \ge \gamma^2 P_d(x_0)^2$, with $P_d(x_0) = \cosh(d\theta_0)/\cosh(d\theta_{-1})$, $\theta_{-1} = \operatorname{arccosh}(-w(-1))$: one when the kernel is exact, $x_0 = -1$, and close to one wherever $h_0 \ll h_1$, as for a trusted kernel.

## F.5 The Imaginary-Time Filter

On a singular value $s$ of $R$, $f_\beta(x) = e^{-\beta(1+x)} = e^{-2\beta s^2/\alpha_R^2}$, so $f_\beta(H') = e^{-tH}$ with $t = 2\beta/\alpha_R^2$: imaginary time $\beta$ in the units of $H'$. From the generating function of the modified Bessel functions [90], $e^{z\cos\vartheta} = I_0(z) + 2\sum_{k\geq 1} I_k(z)\cos k\vartheta$, at $z = -\beta$ and $x = \cos\vartheta$, with $I_k(-\beta) = (-1)^k I_k(\beta)$,

$$e^{-\beta x} = I_0(\beta) + 2\sum_{k=1}^{\infty}(-1)^k I_k(\beta)\, T_k(x), \tag{F.2}$$

which, multiplied by $e^{-\beta}$, gives the filter and its coefficients,

$$f_\beta(x) = e^{-\beta(1+x)}, \qquad f_\beta(H') = e^{-2\beta H/\alpha_R^2}, \qquad q_k = \big(2 - \delta_{k0}\big)(-1)^k e^{-\beta} I_k(\beta). \tag{F.3}$$

**Certified truncation.** Truncated at degree $d$, the series errs on $[-1, 1]$ by at most its tail $\sum_{k>d}|q_k|$, since $|T_k| \leq 1$, and by Proposition 30 the tail is $1 - \sum_{k\leq d}|q_k|$, computed without summing the infinite series. The degree used is the least whose tail is below the tolerance $\varepsilon_t = \varepsilon/10$: a certified degree. For large $\beta$ and $k = \mathcal{O}(\sqrt{\beta})$, $e^{-\beta} I_k(\beta) \approx e^{-k^2/(2\beta)}/\sqrt{2\pi\beta}$, so the tail is about $\operatorname{erfc}(d/\sqrt{2\beta})$ and $d \approx \sqrt{2\beta \ln(2/\varepsilon_t)}$.

**The time.** With $\Delta = \sigma_1^2 - \sigma_0^2$ and

$$\beta = \frac{\alpha_R^2}{2\Delta}\ln\frac{1}{\varepsilon}, \qquad d \approx \frac{\alpha_R}{\sqrt{\Delta}}\Big(\ln\frac{1}{\varepsilon}\,\ln\frac{2}{\varepsilon_t}\Big)^{1/2}, \tag{F.4}$$

the untruncated filter holds every excited state's amplitude to at most $\varepsilon$ times the ground state's, since $f_\beta(x_1)/f_\beta(x_0) = e^{-2\beta\Delta/\alpha_R^2} = \varepsilon$ and $f_\beta$ falls with $s$. Against Equation (F.1) the factor $\alpha_R/\sqrt{\Delta}$ is the same, and the logarithms make imaginary time's degree approach twice the minimax filter's as $\varepsilon \to 0$ with $\varepsilon_t = \varepsilon/10$.

**The a priori bound.** For a one-dimensional target and tail $T$, the argument of Proposition 29, with the truncation an operator of norm at most $T$, gives the infidelity bound

$$1 - F \leq \Big(\frac{\sqrt{1-\gamma^2}\, f(\sigma_1) + T}{\gamma f(\sigma_0) - T}\Big)^2, \qquad f(s) = e^{-2\beta s^2/\alpha_R^2},$$

wherever the denominator is positive. The tail is absolute while the ground state's filtered value is $f(\sigma_0) = \varepsilon^{\sigma_0^2/\Delta}$ under the time rule, so where the residual has no near-kernel the tail can reach it and the bound certifies nothing. With a trusted kernel, $\sigma_1/\sigma_0 \geq 20$, $\sigma_0^2/\Delta < 1/399$ and $f(\sigma_0) > \varepsilon^{1/399}$.

## F.6 Unit Weight of Both Filters

**Proposition 30** (Unit weight of both filters)**.** *For every $\beta > 0$ the Chebyshev coefficients $q_k$ of $e^{-\beta(1+x)}$ (Equation* (F.3)*), and for every $d$ and $x_1 \in (-1, 1)$ those of $P_d$ in Equation* (7.5)*, alternate in sign and satisfy*

$$\sum_k |q_k| = 1, \tag{F.5}$$

*so $|P(z)| \leq \sum_k |q_k| = 1$ on the unit circle, with equality at $z = -1$.*

Each filter is the reflection, about the spectrum's edge, of a function whose Chebyshev coefficients are non-negative and sum to its value of one there.

*Proof of Proposition 30. Imaginary time.* $I_k(\beta) > 0$ for $\beta > 0$, so the $q_k$ alternate in sign and $\sum_k |q_k| = e^{-\beta}\big(I_0(\beta) + 2\sum_{k\geq 1} I_k(\beta)\big)$, the generating function at $\vartheta = 0$ times $e^{-\beta}$, which is $e^{-\beta}e^{\beta} = 1$.

*The minimax filter.* Let $\tilde{P}(y) = P_d(-y)$. With $a = 2/(1-x_1)$ and $b = (1+x_1)/(1-x_1) \geq 0$, $w(-y) = -(ay+b)$ and $a = 1+b$, so $ay+b = y+b(1+y)$ and $\tilde{P}(y) = T_d\big(y+b(1+y)\big)/T_d(1+2b)$, the factors $(-1)^d$ cancelling. Taylor's theorem, exact for a polynomial, gives

$$T_d\big(y+b(1+y)\big) = \sum_{m=0}^{d} \frac{b^m}{m!}\,(1+y)^m\,T_d^{(m)}(y).$$

Every derivative of $T_d$ has non-negative Chebyshev coefficients, by induction from $T_j' = jU_{j-1}$ and $U_{j-1} = 2\sum_i T_i$ over $0 < i < j$ with $i \equiv j-1 \pmod 2$, plus $T_0$ when $j$ is odd; $(1+y)^m = (T_0+T_1)^m$ has non-negative coefficients; and so does a product of two such series, since $T_iT_j = \frac{1}{2}(T_{i+j}+T_{|i-j|})$. With $b^m/m! \geq 0$ and $T_d(1+2b) > 0$, $\tilde{P}$ has non-negative coefficients $\tilde{q}_k$, and $\sum_k \tilde{q}_k = \tilde{P}(1) = P_d(-1) = 1$. Since $T_k(-x) = (-1)^k T_k(x)$, the coefficients of $P_d$ are $(-1)^k\tilde{q}_k$.

In both cases $|\sum_k q_k z^k| \leq \sum_k |q_k| = 1$ on the unit circle, with equality at $z = -1$, where the alternating signs align. □

## F.7 GQSP and Its Phases

A signal qubit is added to the walk's register. With

$$R(\theta,\phi) = \begin{pmatrix} e^{\mathrm{i}\phi}\cos\theta & e^{\mathrm{i}\phi}\sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$$

on the signal and $C_0W = |0\rangle\langle 0| \otimes W + |1\rangle\langle 1| \otimes I$, the circuit is

$$U_\Phi = e^{\mathrm{i}\lambda} R(\theta_d,\phi_d)\, C_0W\, R(\theta_{d-1},\phi_{d-1}) \cdots C_0W\, R(\theta_0,\phi_0).$$

On an eigenvector of $W$ with eigenvalue $z$, $C_0W$ acts on the signal as $\mathrm{diag}(z,1)$. Motlagh and Wiebe [47] show that for every $P(z) = \sum_{k\leq d} q_k z^k$ with $|P| \leq 1$ on the unit circle there are phases with $e^{\mathrm{i}\lambda}\langle 0|\,R(\theta_d,\phi_d)\,\mathrm{diag}(z,1)\cdots R(\theta_0,\phi_0)\,|0\rangle = P(z)$. $W$ is unitary, hence normal, so the signal's zero block of $U_\Phi$ is $P(W)$ on the walk's register, and restricting once by $\Pi_{\mathrm{in}}$ and applying Theorem 6 gives Equation (7.4). The circuit is $d$ controlled walks and $d+1$ one-qubit rotations on one more qubit than the walk, uses $W$ but never $W^\dagger$, and needs no parity.

Before the phases are found, $P$ is divided by $\sup|P|/(1-\mu)$, $\mu = 10^{-2}$, so that $1-|P|^2 > 0$ on the circle and its logarithm is finite. Both filters have $\sup|P| = 1$, so the circuit realises $(1-\mu)P$ and succeeds with probability $(1-\mu)^2 p_G = 0.9801\,p_G$, a constant that the cost estimates leave out. The phases then follow without root-finding, which is unstable well below the degrees needed here: spectral factorisation of $\log(1-|P|^2)$ gives the outer polynomial $Q$ of degree $d$ with $|P|^2+|Q|^2 = 1$ on the circle [48], the pair $(Q,P)$ is the SU(2) NLFT of a sequence $(F_k)$ that a fast inverse transform recovers stably because $Q$ is outer [49, 50], and the moduli and arguments of the $F_k$ give the angles $\theta_k$ and $\phi_k$. The polynomial the angles realise is rebuilt from them alone, in coefficient space, and compared with the target on $4(d+1)$ points of the circle. Acceptance is $10^{-12}$, in double precision throughout. The realised polynomials meet it to degrees of order $10^6$ in the standard form; in the descriptor's filter study the two filters of R3c at $n = 7$, at degrees of order $2\times10^5$, reach $1.1\times10^{-12}$ and $1.3\times10^{-12}$.

## F.8 Amplification, the Total Count and the Ideal-Encoding Degree

**Proposition 31** (Amplification). *With $\vartheta = \arcsin\gamma$ and $k_{\mathrm{aa}} = \lfloor\pi/(4\vartheta)\rfloor$ rounds of amplitude amplification [56], a preparation that succeeds with probability $\gamma^2$ succeeds with probability at least $\max(\gamma^2, 1-\gamma^2) \geq \frac{1}{2}$, at $2k_{\mathrm{aa}}+1$ applications of the preparation or its inverse.*

*Proof.* $k$ rounds rotate the success amplitude from $\sin\vartheta$ to $\sin(2k+1)\vartheta$. If $\gamma \geq \sin(\pi/4)$, then $k_{\text{aa}} = 0$ and the probability is $\gamma^2 \geq \frac{1}{2}$. Otherwise $k_{\text{aa}} \leq \pi/(4\vartheta) < k_{\text{aa}}+1$ gives $\pi/2-\vartheta < (2k_{\text{aa}}+1)\vartheta \leq \pi/2+\vartheta$, so $\sin^2(2k_{\text{aa}}+1)\vartheta \geq \cos^2\vartheta = 1-\gamma^2 > \frac{1}{2}$. □

For small $\gamma$, $k_{\text{aa}} \approx \pi/(4\gamma)$: a poor initial state costs linearly in $1/\gamma$. Exact amplification needs $\gamma$, which the resource estimate takes from the reference, as the degree takes the spectral edge; without it, fixed-point amplification costs a constant factor more. The geometric initial state is run for both signs of $r$ and the one with the larger success probability kept, which doubles the preparations; the count leaves that factor out, as it leaves out the margin below. The total count of Equation (7.8) takes $p_G = \gamma^2$ and the transpiled two-qubit count $g(W)$ of one walk, and leaves out what grows more slowly: the $d+1$ rotations of GQSP, the initial state's $n$ rotations, the amplification's reflections and the factor $(1-\mu)^{-2}$. A resource estimate computes every quantity at the target resolution itself: $\alpha_R$ and $g(W)$ from the circuit, $\sigma_0$, $\sigma_1$ and $\gamma$ from the classical reference, the minimax degree exactly from Equation (7.6).

**Proposition 32** (The degree at the ideal-encoding bound)**.** *Every block encoding of $R$ has $\alpha \geq \|R\|_2$, and the degree of Equation* (7.6) *does not increase as $\alpha$ falls. The degree at $\alpha = \|R\|_2$ is therefore the least that the minimax edge filter needs on $R$ through any block encoding of it. The same holds for a block encoding of $H$ itself, at $\alpha_H \geq \|H\| = \|R\|_2^2$.*

*Proof.* $\alpha \geq \|R\|_2$ is a block encoding's defining condition. The heights are $h_j = 2\sigma_j^2/\alpha^2$; as $\alpha$ falls, $h_1$ and $h_1 - h_0 = 2\Delta/\alpha^2$ both grow, so $\theta_0 = \operatorname{arccosh}\big(1 + 2(h_1-h_0)/(2-h_1)\big)$ grows and $d = \lceil \operatorname{arccosh}(1/\varepsilon)/\theta_0 \rceil$ does not. For an encoding of $H$ the heights are $2\lambda_j/\alpha_H$, the same function of $\alpha_H = \alpha^2$. □

Any difference between a form's degree as built and its ideal degree is its encoder's cost; any difference between two forms at the ideal degree is their Hamiltonians'. With Equation (F.1) and $\alpha_R \geq \|R\|$, $d \gtrsim \frac{1}{2}\sqrt{\|H\|/\Delta}\,\ln(2/\varepsilon)$, a property of $H$ that no encoding removes.

## F.9 The Readout Identity for Probabilistic Preparation

Let $\hat{U}_{\text{prep}}$ be the preparation, the initial state's rotations followed by the GQSP circuit, whose output is $|0_a\rangle\, g + |\text{junk}\rangle$, with $g = \sqrt{p_G}\,\psi$ the system part on the all-zero ancilla and $|\text{junk}\rangle$ supported on non-zero ancilla. Let $\hat{U}_\tau(x)$ be Wu et al.'s feature map, one per axis, and $u(x) = \tau(x)/2^{m/2}$ the first column of their product on $m$ axes, $\tau(x)$ the Kronecker product of the axes' feature vectors.

- $G$: $\hat{U}_{\text{prep}}$, then $\hat{U}_\tau^\dagger(x)$; $P_G(x)$ is the probability that every qubit reads zero.
- $C$: a selector qubit in $|+\rangle$, $\hat{U}_{\text{prep}}$ controlled on it, a Hadamard on it, then $\hat{U}_\tau^\dagger(x)$; $P_C(x)$ is the all-zero probability, the selector included.

**Lemma 9.** *The all-zero amplitude of $\hat{U}_\tau^\dagger(x)$ applied to $|v\rangle\,|0\rangle$ is $u(x)\cdot v$.*

*Proof.* It is $\langle 0|\,\hat{U}_\tau^\dagger\,|v,0\rangle = \overline{\langle v,0|\,\hat{U}_\tau\,|0\rangle}$, and the feature maps' all-zero column restricted to their ancilla's zero is the real vector $u(x)$. □

**Proposition 33** (The readout identity)**.** *With $N$ the field's dimension on $m$ axes,*

$$4P_C(x) - P_G(x) - \frac{1}{2^m N} = \frac{2}{2^m}\sqrt{\frac{p_G}{N}}\;\tau(x)\cdot\psi,$$

*On one axis, $m = 1$,*

$$4P_C(x) - P_G(x) - \frac{1}{2N} = \sqrt{\frac{p_G}{N}}\;\langle\tau(x)|\psi\rangle\,. \tag{F.6}$$

*Proof.* The junk has a non-zero ancilla and contributes nothing to an all-zero outcome, so by Lemma 9 $P_G = (u \cdot g)^2$. In C, before the closing Hadamard the state is $(|0\rangle\,|0\rangle + |1\rangle\,\hat{U}_{\text{prep}}\,|0\rangle)/\sqrt{2}$; after it the selector's zero carries $(|0\rangle + \hat{U}_{\text{prep}}\,|0\rangle)/2$, whose all-zero-ancilla system part is $(e_0 + g)/2$. So $P_C = (u_0 + u \cdot g)^2/4$ and $4P_C - P_G = u_0^2 + 2u_0\,u \cdot g$. Every feature vector has $\tau_0 = 1/\sqrt{N}$, so $u_0 = 1/\sqrt{2^m N}$ and

$$2u_0\,u \cdot g = \frac{2}{2^m}\sqrt{\frac{p_G}{N}}\;\tau \cdot \psi. \qquad \square$$

The interference with $e_0$ recovers the overlap's sign, which one probability alone cannot. For one axis and a deterministic preparation, $p_G = 1$, this is Wu et al.'s Eq. 36 [1]. The model is $f_Q(x) = \sqrt{\eta}\,\tau(x) \cdot \psi$, so with $f^*$ the left-hand side the field follows from a ratio against the regular datum and needs no $p_G$, while the scale does:

$$f_Q(x) = f(x_s)\,\frac{f^*(x)}{f^*(x_s)}, \qquad \eta_e = \Big(\frac{2}{2^m}\Big)^2\,\frac{f(x_s)^2\,p_G}{N\,f^*(x_s)^2}.$$

Taking $p_G = 1$ misstates $\eta_e$ by the factor $p_G$. C needs $\hat{U}_{\text{prep}}$ controlled on the selector: the rotations take the control, each controlled walk needs only its two reflections controlled on the selector and the signal together (Appendix F.3), and GQSP's global phase becomes a phase gate on the selector, whose loss would scale $f^*$ by $\cos\lambda$ and $\eta_e$ by $1/\cos^2\lambda$. Simulated, the two circuits reproduce the exact overlap, the field and $\eta_e$ on R1, R2a and R5a, the first two on one axis and the third on two.

# G Verification and the Classical Reference

## G.1 Modes, Sizes and Tolerances

**Table G.1: Every circuit is compared with an independently assembled operator.** Each verification mode by what it compares, where it is used and its tolerance. $X$, the target: the reflection's block $2H/\alpha_R^2 - I$ or an atom's operator; $\alpha$, the circuit's subnormalisation; $\psi$, an input state; IR, the compiled term list. The approximation mode applies only to Wu et al.'s approximate encoding of $G$, which enters resource estimates alone.

| Mode | Comparison | Used on | Tolerance |
|---|---|---|---|
| Exact | the whole block, one statevector evolution per input column, entry by entry | circuits whose input columns it can afford, to 25 qubits | max-abs $\le$ $10^{-10}\|X\|_2$ |
| Probe | 4 structured and 16 Haar-random inputs against sparse products $X\psi/\alpha$ | every other circuit of at most 24 qubits | relative $\le 10^{-9}$ per input |
| IR | the compiled terms, summed to a sparse matrix, against the assembled operator | lifts and couplings at any size | relative max-abs $\le 10^{-10}$ |
| Approx. | Wu et al.'s $G$: entry error against its bound | the sizes the exact mode reaches | reported, not gated |

A block encoding $(U, \alpha)$ of an operator $X$ is exact when $\alpha\,\Pi_{\text{out}} U \Pi_{\text{in}} = X$, the projectors fixing the ancilla at zero. Every mode of Table G.1 compares that block, or its action on vectors, with a target built from the problem by the operator library and the residual's assembly, never from the compiler that produced the circuit; a target rebuilt from the circuit's own parts would check the compiler against itself. The pipeline's usual target is the stacked reflection's block $2H/\alpha_R^2 - I$ with $H$ formed from the assembled residual; the atoms (the derivative, a lift, a feature-map row) are compared with their operators in the same way.

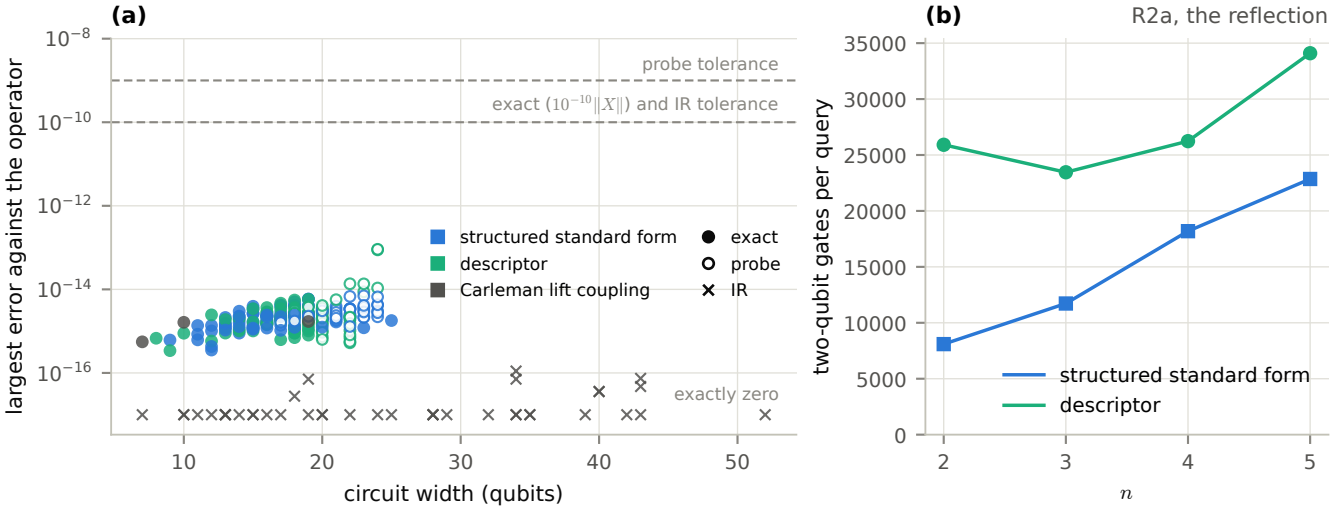



**Figure G.1: Every verified circuit encodes its operator to rounding, in every verification mode.** (a) Largest error of each verified circuit against its independently assembled operator, against the circuit's width in qubits: exact mode (filled markers; the reflection's whole block $2H/\alpha_R^2 - I$, entrywise), probe mode (open markers; twenty structured and Haar-random inputs) and IR mode (crosses; a lift's or coupling's compiled terms), for the structured standard-form and descriptor encodings and the Carleman lifts' couplings, with each mode's tolerance dashed (Table G.1). Wu et al.'s encoding, approximate by construction, is not shown: only its encoding of $G$ is compared, against its error bound, and it enters resource estimates alone. (b) Transpiled two-qubit gates per query of the reflection against $n$ on R2a, structured standard and descriptor forms. Every error lies at least four orders of magnitude below its tolerance; the widest exact verification, 25 qubits, is from the reproduction. Exact classical computation from the built circuits.

**Exact.** The whole block is reconstructed, one statevector evolution per input basis vector, and its largest entrywise error $\|E\|_{\max}$ is compared with $10^{-10}\|X\|_2$. An $r \times c$ block within that tolerance entry by entry has $\|E\|_2 \le \sqrt{rc}\,\|E\|_{\max}$, so the mode certifies the circuit completely at that size. Its cost is $c$ evolutions of $2^q$ amplitudes.

**Probe.** The circuit is applied to 4 structured vectors and 16 Haar-random ones, each one evolution at any size, and compared with a sparse product of the target, which is never formed densely; a probe $v$ passes when $\|Ev\| \le 10^{-9}\max(1, \|Xv\|)$, the random probes seeded by the run.

**Proposition 34** (What random probes see)**.** *For $\psi$ Haar-distributed on dimension $D$,*

$$\mathbb{E}\,\|E\psi\|^2 = \frac{\|E\|_F^2}{D}.$$

*If the error lies in one column, $E = \delta\, w\, e_j^\dagger$ with $\|w\| = 1$, then $\|E\psi\| = \delta|\psi_j|$ and $D|\psi_j|^2$ has mean one.*

*Proof.* $\mathbb{E}\,\psi\psi^\dagger = I/D$, so $\mathbb{E}\|E\psi\|^2 = \mathrm{tr}(E^\dagger E)/D$; for the column error $\|E\psi\|^2 = \delta^2|\psi_j|^2$. □

Random probes therefore estimate the root-mean-square error $\|E\|_F/\sqrt{D}$, and an error of size $\delta$ confined to one column is seen at about $\delta/\sqrt{D}$: caught with certainty only when $\delta \gtrsim 10^{-9}\sqrt{D}$. The probe mode is evidence of correctness, not a proof of it. Unlike the exact mode, whose cost grows with the number of input columns, it costs twenty evolutions at any size, so it runs on every circuit of at most 24 qubits that the exact mode does not cover.

**IR.** The compiled terms of a descriptor residual, a coupling or a lift are summed back into a sparse matrix and compared with the operator assembled from the problem, entry by entry relative to its norm, at tolerance $10^{-10}$, at any size and with no simulation (Appendix D.7). The two sides are independent constructions, so the comparison is not a tautology. What it leaves open is the step from

terms to gates, which the exact and probe modes check where a circuit can be simulated. Together they chain: the assembled operator equals the terms at every $n$, and the terms equal the circuit at every simulable size. The compiler is structural, one atom per term, so the circuit at a larger size is the same construction, but it is not independently verified there.

**Approximation.** Wu et al.'s approximate encoding of $G$ (Appendix I.6.1) is compared with $G$ and its entrywise error reported beside the bound of Proposition 38; the mode never fails, since the encoding is approximate by design and enters resource estimates only.

Circuits wider than the probe ceiling are construction-only records: their subnormalisation, register split and gate counts are recorded, with no verdict.

The four structured inputs of the probe mode are placed where a defect confined to one column arises, because a random input sees such a defect only at about $\delta/\sqrt{D}$ (Proposition 34), while the basis vector that selects its column sees it at full strength. The first basis vector is the $T_0$ mode and the start of every chain of shifts, and the last is where a truncated shift must not wrap round to the start of the register. The middle one, index $D/2$, sets only the most significant qubit, so it sits on the edge between the two halves of a register and between the blocks of a padded state, where block offsets and axis orders go wrong; the uniform vector tests the sum of all columns at once. A defect in a column that none of them selects is left to the random probes. Every atom is also verified alone before it is composed: the structured factorisation of $G$ (Appendix C), the lifts and folds, the feature-map rows, each band of the descriptor (Appendix D.7) and the matchings of a coupling, each exactly at small $n$ and by probe or IR beyond; the banded factorisation $\tilde{B}G = \tilde{D}$ holds to rounding for $n = 1$ to 10. The composed reflection is then verified whole, so that correct atoms cannot hide an error made in composing them, such as a wrong block offset or a mis-ordered register.

## G.2 The Classical Reference: Eigensolver Against SVD

The reference is the smallest right singular vector of $R$, computed by a backward-stable SVD (LAPACK's divide-and-conquer driver, with the QR-based driver where it fails to converge), in double precision only; $H$ is never formed.

**Proposition 35** (The two bounds)**.** *A backward-stable SVD returns the exact singular vectors of $R + \delta R$ with $\|\delta R\|_2 \le c\,\epsilon_{\mathrm{mach}}\sigma_{\max}$, and the computed ground state makes with the exact one an angle with*

$$\sin\vartheta_{\mathrm{SVD}} \le \frac{\|\delta R\|_2}{\sigma_1 - \hat{\sigma}_0} \approx \frac{c\,\epsilon_{\mathrm{mach}}\sigma_{\max}}{\sigma_1 - \sigma_0}, \tag{G.1}$$

*$\hat{\sigma}_0$ the perturbed value. A backward-stable symmetric eigensolver applied to the formed $H$ returns the eigenvectors of $H + \delta H$ with $\|\delta H\|_2 \approx c\,\epsilon_{\mathrm{mach}}\sigma_{\max}^2$, and*

$$\sin\vartheta_{\mathrm{eig}} \lesssim \frac{c\,\epsilon_{\mathrm{mach}}\sigma_{\max}^2}{\sigma_1^2 - \sigma_0^2}.$$

*Proof.* The first is Wedin's $\sin\Theta$ theorem for singular subspaces [80] applied to the perturbation $\delta R$; the second is the Davis–Kahan theorem [81] applied to $\delta H$, with the gap $\lambda_1 - \lambda_0 = \sigma_1^2 - \sigma_0^2$. □

The ratio of the two is $\sigma_{\max}/(\sigma_1 + \sigma_0)$, the residual's condition number on the complement of its near-kernel: the eigensolver loses that factor. When $\sigma_0 \ll \sigma_1$ the SVD's bound is about $\epsilon_{\mathrm{mach}}\sqrt{\|H\|/\Delta}$, the square root of the eigensolver's. Both require $\sigma_0$ to be isolated, the trusted-kernel condition of Appendix G.4. On the standard form the eigensolver's bound grows with $n$ by the factor that $\sigma_{\max}/\sigma_1$ grows, $2^{2k}$ per qubit at order $k$, until it saturates, and Wedin's, smaller by that factor, stays below one; on R2a the eigensolver's angle reaches order one by $n = 8$, where the SVD's is still small (Table G.2). The bounds are what a backward-stable algorithm can promise: on the standard form the smallest singular value is returned at the floor $\epsilon_{\mathrm{mach}}\sigma_{\max}$ rather than at its true,

smaller value, which is why the kernel is detected by a ratio against that floor. A single-precision residual has $\epsilon_{32}\sigma_{\max}/(\sigma_1 - \sigma_0) > 10^{-3}$ on R2a from $n = 5$, and is refused.

**Table G.2: The eigensolver loses the residual's condition number; the SVD does not.** Sine of the angle between the classical reference and the state of the exact coefficients on R2a ($f'' + 4f' + 4f = 0$): the standard form by a symmetric eigensolver on $R_{\text{std}}^{\top} R_{\text{std}}$ and by SVD of $R_{\text{std}}$, and the descriptor by SVD, on its physical columns. At $n = 2$ and 3 the truncation dominates both routes; from $n = 4$ the angle is the solver's. Classical computation.

| $n$ | Standard, eigensolver | Standard, SVD | Descriptor |
|---|---|---|---|
| 2 | $7.0 \times 10^{-1}$ | $7.0 \times 10^{-1}$ | $3.0 \times 10^{-1}$ |
| 3 | $6.4 \times 10^{-4}$ | $6.4 \times 10^{-4}$ | $4.2 \times 10^{-3}$ |
| 4 | $2.6 \times 10^{-11}$ | $1.8 \times 10^{-12}$ | $2.0 \times 10^{-10}$ |
| 5 | $3.7 \times 10^{-8}$ | $4.3 \times 10^{-13}$ | $2.6 \times 10^{-14}$ |
| 6 | $2.2 \times 10^{-6}$ | $4.7 \times 10^{-12}$ | $2.7 \times 10^{-13}$ |
| 7 | $9.2 \times 10^{-2}$ | $3.7 \times 10^{-11}$ | $3.1 \times 10^{-12}$ |
| 8 | $9.7 \times 10^{-1}$ | $6.3 \times 10^{-9}$ | $4.1 \times 10^{-11}$ |

## G.3 The Iterative Reference

Beyond 8,192 physical columns the dense SVD takes too long and too much memory, and the smallest singular triplets are found iteratively. Lanczos iteration finds the largest eigenvalues of a symmetric operator, and the smallest singular values of $R$ are the largest eigenvalues of $(R^{\dagger}R + \mu I)^{-1}$, which is applied without forming $R^{\dagger}R$.

**Proposition 36** (The augmented system)**.** *The solution of* $\left(\begin{smallmatrix} I & R \\ R^{\dagger} & -\mu I \end{smallmatrix}\right)\left(\begin{smallmatrix} r \\ x \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 \\ c \end{smallmatrix}\right)$ *has* $x = -(R^{\dagger}R + \mu I)^{-1}c$.

*Proof.* The first row gives $r = -Rx$; substituting, $-R^{\dagger}Rx - \mu x = c$. □

One sparse LU factorisation of the augmented matrix serves every Lanczos step. It does not fill as the normal matrix does: a dense constraint row of $R$, with $N$ entries, appears once in $R$ and once in $R^{\dagger}$, where it would put a dense $N \times N$ block into $R^{\dagger}R$.

**Upper bounds and the a posteriori estimate.** For $V$ with orthonormal columns, the $j$-th smallest singular value of $RV$ is at least the $j$-th smallest of $R$ (Cauchy interlacing), so the computed $\tilde{\sigma}_0$ and $\tilde{\sigma}_1$ never fall below the true ones. For the returned $v$, with $r = \|Rv\| \geq \sigma_0$, the infidelity against the true ground state $v_\star$ satisfies $1 - |\langle v|v_\star\rangle|^2 \leq (r^2 - \sigma_0^2)/(\sigma_1^2 - \sigma_0^2) \leq r^2/\sigma_1^2$, and the reported bound is $r^2/(\tilde{\sigma}_1^2 - r^2)$, which is at least $r^2/\sigma_1^2$ when no singular value below $\tilde{\sigma}_1$ is missed. It is therefore an estimate, exact to rounding once the iteration has converged and nothing below $\tilde{\sigma}_1$ is missing; it floors at about $(\sigma_0/\sigma_1)^2$ where $R$ has no kernel, and is sharp where it has one.

**The start vector.** Both Lanczos runs start from a Gaussian vector of fixed seed: deterministic, answer-independent and with a component on every singular vector. A uniform start would not do: if a permutation $P$ of the columns commutes with $R^{\dagger}R$ and $Pu = u$, every Krylov vector is $P$-fixed, so the iteration never meets a singular vector of another symmetry sector. On the Taylor–Green vortex, whose residual has such a symmetry, a uniform start misses most of the lowest singular values.

**Checks.** The iterative reference reaches residuals of 524 288 columns, on the Navier–Stokes lifts. Where both run, it reproduces the dense one's ground state to the dense one's own error. Past the standard form's moderate $n$, where the dense SVD's backward error $\epsilon_{\text{mach}}\sigma_{\max}$ is no longer small against $\sigma_1$, the iterative reference is the more accurate: on R2a from $n = 8$ to 10 its $\sigma_1$ varies by $4 \times 10^{-15}$ relative, where the dense value varies by $4 \times 10^{-7}$, and on the descriptor there the two agree to $0.68\,\epsilon_{\text{mach}}\sigma_{\max}$.

## G.4 The Trust Threshold

The ascending singular values are floored at $10^{-300}$, so that an exact zero gives a finite ratio, and the ratios $\sigma_k/\sigma_{k-1}$ are formed for $k = 1, \ldots, 16$. The kernel's dimension is the $k$ of the largest ratio, the *gap ratio* is that ratio, and the reference is trusted when the dimension is one and the gap ratio at least 20. An exact kernel of dimension $k$ is found because its computed values are at most $c\,\epsilon_{\text{mach}}\sigma_{\max}$ by Weyl's inequality, so its ratio exceeds 20 whenever $\sigma_k/\sigma_{\max}$ exceeds about $4 \times 10^{-15}$; beyond that the floor is the gap's own size and nothing in the computed spectrum separates them. For a near-kernel, if the truncated solution $\hat{b}$ nearly minimises the residual, $\|R\hat{b}\| \approx \sigma_0$, its angle to the ground state is at most about $\sigma_0/\sigma_1$: a gap ratio of 20 puts the ground state within 0.05 of it in sine.

**Why** 20**.** A spectrum with no kernel still has ratios, and the detector reports the largest as a kernel's. The threshold must exceed the largest ordinary spacing in the benchmark problems' singular spectra, so that a ratio above it cannot come from spacing alone. Over the classical builds of R1 to R8 the largest such spacing is the stiff R2c's structural $\sigma_2/\sigma_1$, just under 20, with the median far below; the only larger ratios are those of three of the reproduction's panels (Appendix I), whose second near-kernel is not a spacing. The value is empirical, a property of these spectra and not a theorem.

**What it does not decide.** A kernel of dimension above one is never trusted; the doubled space's is the example (§ 6.4). A second near-kernel higher up, with a larger ratio, moves the reported dimension above one and is untrusted, which is conservative. A one-dimensional kernel with a modest gap ratio, an under-resolved problem found by luck, is not trusted. An iterative reference counts among its smallest seventeen values with the Ritz upper bounds above, and can report a ratio too small by their overestimate. Trust means that a kernel is detected, not that the field has converged; the convergence in $n$ is measured separately.

## G.5 The Circuit-Simulation Memory and Time Model

A circuit simulation applies the GQSP circuit one controlled walk at a time on a statevector simulator [82], with gate fusion off, on the walk's qubits and the signal, $q = q_W + 1$; it holds a statevector, never an operator, so its memory grows as $2^q$ and not $4^q$ (Appendix F.3).

**Memory.** A statevector of $q$ qubits is $S = 16 \cdot 2^q$ bytes of complex double-precision amplitudes, and a dense $k$-qubit gate matrix $16 \cdot 4^k$ bytes, totalling $G$ over a step's dense gates. The peak resident memory is estimated as

$$M_{\text{peak}} = M_0 + 8S + 4G,$$

with $M_0$ the interpreter and libraries, eight statevector-sized buffers at the simulation's peak and four copies of each gate matrix. The multipliers are measured, not derived, from 20 to 24 qubits, and the model's peak is within 0.85 to 1.05 of the memory resident in a fresh process at 22 qubits. The statevector term alone is $2^{q-23}$ GiB: 8 GiB at 26 qubits and 128 GiB at 30. Memory is the hard limit of circuit simulation; time is the soft one.

**Time.** Once the memory fits, one step is simulated untimed to warm the caches, then five times, and the median is $t_{\text{step}}$; a single cold step on a shared node can read many times its steady cost, and the median of warm steps does not move for one slow outlier. The run's estimate is (candidates) $\cdot$ $(d+1) \cdot t_{\text{step}}$, $d+1$ steps for the degree and the opening rotation and one simulation per initial-state candidate. Which resolutions circuit simulation reaches is therefore a joint statement of $q$ and $d$: the degree is set by the spectrum, and one more qubit roughly doubles both the memory and the step's time.

**Refusal.** The budget is the job's allocation of memory and time. The estimate is logged against it whatever the verdict, and a run that exceeds either is refused and recorded as beyond the simulation budget, with the estimate and the resource estimate's numbers beside it; if the memory does not fit, the step is never timed.

## G.6 Acceptance and the Error Budget

**Table G.3: Every preparation is judged at five levels, each against a bar it can be measured to.** Applied to every preparation whose reference is trusted; a run that fails a bar is reported as a finding, never re-judged. $\psi$, $\psi_{\mathrm{ref}}$, prepared and reference states; $\varepsilon$, target suppression; $p_G$, post-selection success probability; $d$, filter degree; $\varepsilon_{\mathrm{disc}}$, discretisation error; $B(d_{\mathrm{total}})$, the error budget. The state bar's a priori bound follows from the spectral edge, $\gamma$ and the filter (Proposition 29); the circuit bar is judged before normalising, which divides rounding by $\sqrt{p_G}$, and on normalised states only where $p_G \geq 10^{-12}$ (Lemma 10).

| Level | Measured | Bar |
|---|---|---|
| State | infidelity $1 - \lvert\langle\psi_{\mathrm{ref}}\rvert\psi\rangle\rvert^2$ in the form's own space | $\leq \varepsilon = 10^{-6}$, and $\leq 10\times$ its a priori bound |
| Circuit | circuit simulation against exact emulation from a unit input, before normalising | $\leq 10^{-10}$ |
| Probability | the two tiers' $p_G$ | relative $\leq 10^{-8}$ |
| Phases | realised GQSP polynomial against its target, on $4(d+1)$ points of the unit circle | $\leq 10^{-12}$ |
| Field | decoded field against the reference solution, $L^\infty$ | $\leq \varepsilon_{\mathrm{disc}} + B(d_{\mathrm{total}})$, Equation (G.2) |

**Lemma 10** (Normalising amplifies differences)**.** *For non-zero vectors $x$ and $y$, $\left\lVert x/\lVert x\rVert - y/\lVert y\rVert\right\rVert \leq 2\lVert x-y\rVert/\lVert x\rVert$, and $\left\lvert\lVert x\rVert^2 - \lVert y\rVert^2\right\rvert \leq \lVert x-y\rVert(\lVert x\rVert + \lVert y\rVert)$.*

*Proof.* $\left\lVert x/\lVert x\rVert - y/\lVert y\rVert\right\rVert \leq \lVert x-y\rVert/\lVert x\rVert + \lVert y\rVert\left\lvert 1/\lVert x\rVert - 1/\lVert y\rVert\right\rvert = (\lVert x-y\rVert + \lvert\lVert y\rVert - \lVert x\rVert\rvert)/\lVert x\rVert$, and the reverse triangle inequality bounds the second term by $\lVert x-y\rVert$. The second inequality factorises a difference of squares. □

The circuit's output has norm $\sqrt{p_G}$, so a rounding error of about $10^{-16}$ in the unnormalised output becomes about $10^{-16}/\sqrt{p_G}$ in the normalised state; at $p_G = 10^{-12}$ that is the bar of $10^{-10}$. The circuit and its emulation are therefore compared before normalising, from a unit input, and the normalised states only where $p_G \geq 10^{-12}$; below it the comparison would measure rounding, not the circuit. The probability bar follows from the circuit bar by the lemma's second part only for $p_G \gtrsim 4\times10^{-4}$, and is a separate measurement below. A run that fails a bar is reported as a finding, with its degree and the growth of the difference, and never re-judged against an adjusted bar.

**The state metric.** For unit vectors the distance between states, $d(a,b) = \min_\phi \lVert a - e^{\mathrm{i}\phi}b\rVert = \sqrt{2 - 2\lvert\langle a\rvert b\rangle\rvert}$, is a metric, and it is related to the infidelity $\epsilon = 1 - \lvert\langle a\rvert b\rangle\rvert^2$ exactly by $\epsilon = d^2(1 - d^2/4)$; so $d \approx \sqrt{\epsilon}$, and distances add by the triangle inequality where infidelities do not. Each distance is computed as the norm of a difference of phase-aligned vectors, never by the overlap formula, which cancels and cannot resolve distances below about $10^{-8}$, the size of the terms themselves.

**The chain.** A prepared state passes through three states on its way from the target, the ground state: the untruncated filter's output $\psi_0$, T2's state $\psi_2$, and the circuit's state $\psi_c$, which the realised phases prepare. The filter term $d(\mathrm{target}, \psi_0)$ is computed exactly from the physical subspace's SVD: with $c_i = \langle v_i\rvert\psi_{\mathrm{init}}\rangle$ and $w_i$ the filter's value on the $i$-th singular value,

$$\epsilon_{\mathrm{filter}} = \frac{\sum_{i\geq 1} w_i^2 |c_i|^2}{\sum_{i\geq 0} w_i^2 |c_i|^2}, \qquad d_{\mathrm{filter}} = \Big(\frac{2\epsilon_{\mathrm{filter}}}{1 + \sqrt{1 - \epsilon_{\mathrm{filter}}}}\Big)^{1/2}.$$

It depends on the edge, the overlaps and the filter, never on the solution. The truncation and phase terms are bounded a priori: if two polynomials of $H'$ differ by at most $\eta$ in the sum of the absolute differences of their Chebyshev coefficients, their outputs differ by at most $\eta$ since $\|T_k(H')\| \le 1$, and by Lemma 10 the normalised states by at most $2\eta/\sqrt{p_G}$. For truncation $\eta$ is imaginary time's certified tail, and for the minimax filter, which truncates nothing, the rounding of the Chebyshev recurrence, up to about $5 \times 10^{-8}$ at degrees beyond $10^4$; for the phases it is the difference between the realised and target coefficients. Every encoding is exact and the field is decoded classically, so no further term enters.

**Proposition 37** (The readout bound)**.** *Write the decoded field as $f = w\,u$ with weight $w$ and $u(x) = u_s\,\ell_x(\psi)/\ell_s(\psi)$, where $\ell_x(\psi) = \langle\tau(x)|D\psi\rangle$ is the field functional at $x$, $\ell_s = \ell_{x_s}$ and $u_s$ the regular datum. Let $\psi_{\mathrm{ref}}$ and $\psi$ be unit vectors, $\delta = \psi - \psi_{\mathrm{ref}}$ with $\psi$'s phase chosen so that $d = \|\delta\|$ is their distance, $s = u_s/\ell_s(\psi_{\mathrm{ref}})$, $u_{\mathrm{ref}} = s\,\ell_x(\psi_{\mathrm{ref}})$, $K = |s|\max_x \|D^\top\tau(x)\|$, $U = \max|u_{\mathrm{ref}}|$ and $W = \max|w|$, the maxima over the comparison grid. Then, wherever $Kd < |u_s|$,*

$$\max_x |f - f_{\mathrm{ref}}| \le B(d) = \frac{WK\,(1 + U/|u_s|)\,d}{1 - Kd/|u_s|}.$$

*Proof.* With $a = \ell_x(\psi_{\mathrm{ref}})$ and $a_s = \ell_s(\psi_{\mathrm{ref}})$, $u_{\mathrm{ref}} = sa$, $u_s = sa_s$ and $u = s(a + \ell_x\delta)/(1 + \ell_s\delta/a_s)$, so

$$u - u_{\mathrm{ref}} = s\,\frac{\ell_x\delta - (u_{\mathrm{ref}}/u_s)\,\ell_s\delta}{1 + \ell_s\delta/a_s}.$$

$|s\,\ell_x\delta| \le Kd$, and $|s\,\ell_s\delta| \le Kd$ because the datum point's functional is no larger than the grid's largest (on one axis it is a grid point, and on two axes the functional's norm is largest at the corners, which are). The numerator is therefore at most $Kd(1 + U/|u_s|)$ and the denominator at least $1 - Kd/|u_s|$; the weight multiplies by at most $W$. □

The bound is an inequality, not a first-order estimate, and it holds on states from $10^{-8}$ to $10^{-1}$ from the reference with random phases. The field's error is then bounded by the error budget

$$\|f_Q - f\|_\infty \;\le\; \underbrace{\|f_{\mathrm{ref}} - f\|_\infty}_{\varepsilon_{\mathrm{disc}}} + B\big(d_{\mathrm{total}}\big), \qquad d_{\mathrm{total}} \le d_{\mathrm{filter}} + d_{\mathrm{trunc}} + d_{\mathrm{phase}}, \tag{G.2}$$

where $f_{\mathrm{ref}}$ is the classical reference's own field, so that $\varepsilon_{\mathrm{disc}}$ is the discretisation's error, and $B(d)$ is linear in $d$ for small $d$. The budget is taken on the physical columns, since the descriptor's padding decouples and every state is zero on it.

# H Benchmark Problems and Results by Problem

## H.1 Linear problems

## H.2 Nonlinear and coupled problems

**The fluid runs.** Burgers' equation (Equation (6.9)) is run from $u_0 = a\sin x$ over $T = 1$ at five ratios $a/\nu = 0.03$, 0.3, 1, 3 and 10 (R9a to R9e), each truncated to $M = 4$, 8 and 16 sine modes; its Galerkin system, and the Cole–Hopf solution against which the truncation's own error is measured, are derived with the other Fourier–Galerkin operators in Appendix E.7.

**NS Initial conditions.** The Taylor–Green vortex $\omega_0 = 2\sin x\sin y = \cos(x - y) - \cos(x + y)$ occupies two modes with $|\mathbf{k}|^2 = 2$, whose coupling vanishes, so $\omega = \omega_0 e^{-2\nu t}$ exactly in any truncation that holds them, and in the equation itself. Its data lie in a subspace that $F_1$ preserves and on which $F_2$ vanishes, so its lift is exact at every order (Proposition 24): R10a validates the lift, the time axis

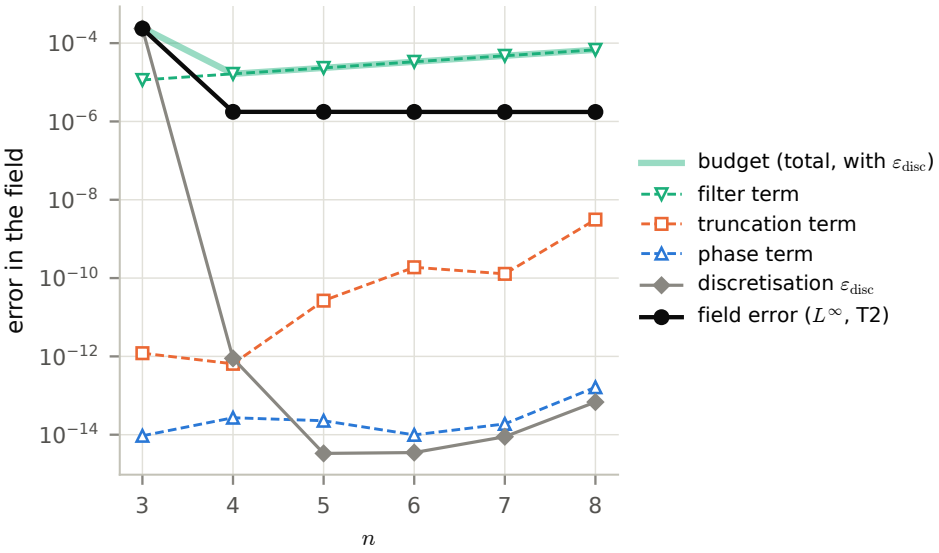



**Figure G.2: From $n = 4$ the filter, not the resolution, sets the prepared field's error, and the error budget bounds it with a margin that widens with $n$.** Against $n$ for R2a in the descriptor: the $L^\infty$ error of the field decoded classically from the prepared state, the discretisation's error $\varepsilon_{\text{disc}}$, and the budget of Equation (G.2), whole (with $\varepsilon_{\text{disc}}$) and as its filter, truncation and phase terms, each carried to the field by the readout bound (Proposition 37); minimax filter, $\varepsilon = 10^{-6}$. The error follows $\varepsilon_{\text{disc}}$ to $n = 3$; from $n = 4$ it holds at $1.7 \times 10^{-6}$, because the filter leaves the same infidelity at every $n$, while $\varepsilon_{\text{disc}}$ falls to rounding. The filter term, which carries that infidelity through the bound's worst direction, lies above the error and grows with $n$ as the bound's amplification $K$ does, from 3.61 at $n = 4$ to 14.7 at $n = 8$; the truncation and phase terms are negligible. Exact emulation (T2).

**Table H.1: The linear benchmark problems R1–R5, as run.** Equation, constraint rows, datum point $x_s$ and reference solution. $x_z$, the root of the reference (or of its derivative) nearest the value Wu et al. print, set up from the reference; $P_l$, $P_l^m$, Legendre and associated Legendre functions; $v = f/\sqrt{1-x^2}$, the variable R3c and R3d are solved for. Every source enters by its Chebyshev coefficients; the heat equation's initial function is imposed at $t = -1$. Wu et al.'s printed rows, run only in the reproduction: Appendix I.

| | Equation | Constraint rows | $x_s$ | Reference |
|---|---|---|---|---|
| R1 | $f' + 2f = 0$ | – (regular datum $f(-1) = 1$) | $-1$ | $e^{-2(x+1)}$ |
| R2a | $f'' + 4f' + 4f = 0$ | $f(-1) = 0$ | 0 | $\frac{1}{2}(1+x)e^{-2x}$ |
| R2b | $f'' - 2f' - 3f = 0$ | $f(0) = 0$ | $\frac{1}{2}$ | $\frac{1}{4}\left(e^{3x} - e^{-x}\right)$ |
| R2c | $f'' + 5f' + 400f = 0$ | $f'(0) = 0$ | 0 | $e^{-5x/2}\left(\cos\omega x + \frac{\sqrt{7}}{21}\sin\omega x\right)$, $\omega = \frac{15\sqrt{7}}{2}$ |
| R3a | $(1-x^2)f'' - 2xf' + 20f = 0$ | $f'(0) = 0$ | 1 | $P_4$ |
| R3b | $(1-x^2)f'' - 2xf' + 30f = 0$ | $f(0) = 0$ | 1 | $P_5$ |
| R3c | $(1-x^2)v'' - 4xv' + 28v = 0$ | $v'(0) = 0$ | $\frac{1}{2}$ | $f = P_5^1 = -\sqrt{1-x^2}\,P_5'$ |
| R3d | $(1-x^2)v'' - 4xv' + 40v = 0$ | $v(0) = 0$ | $\frac{1}{2}$ | $f = P_6^1 = -\sqrt{1-x^2}\,P_6'$ |
| R4a | $(x-1)f'' - xf' + f = (x-1)^2$ | $f'(x_z) = 0$, $x_z = -0.19598$ | 0 | $\frac{3}{2}e^x - \frac{1}{8}x(8x+13) - 1$ |
| R4b | $f'' + 4f' + 4f = e^{-2x}$ | $f(x_z) = 0$, $x_z = 1 - \sqrt{3}$ | 0 | $\frac{1}{2}e^{-2x}(x^2 - 2x - 2)$ |
| R4c | $f'' - 5f' + 6f = xe^{2x}$ | $f(x_z) = 0$, $x_z = 0.90542$ | 0 | $\frac{1}{6}\left(4e^{3x} - e^{2x}(3x^2 + 6x + 2)\right)$ |
| R4d | $f'' + 4f' + 20f = xe^{-2x}\sin 4x$ | $f'(x_z) = 0$, $x_z = -0.74681$ | 0 | $\frac{1}{64}e^{-2x}\left((x-48)\sin 4x - 4(x^2+16)\cos 4x\right)$ |
| R5a | Laplace, $f_{xx} + f_{yy} = 0$ | $f(\pm1, y) = f(x, -1) = 0$; datum slice $f(x, 1) = \sin\frac{\pi(x+1)}{2}$ | $(0, 1)$ | $\cos\frac{\pi x}{2}\,\sinh\frac{\pi(y+1)}{2}/\sinh\pi$ |
| R5b | heat, $f_t = \frac{1}{25}f_{xx}$ | $f(t, \pm1) = 0$; datum slice $f(-1, x) = e^{4\pi^2/25}\sin 2\pi x$ | $(-1, \frac{1}{4})$ | $e^{-4\pi^2 t/25}\sin 2\pi x$ |
| R5c | wave, $f_{tt} = 4f_{xx}$ | $f(t, \pm1) = 0$; datum slice $f(0, x) = \sin 2\pi x$; $\partial_t f(0, x) = 0$ | $(0, \frac{1}{4})$ | $\cos 4\pi t\,\sin 2\pi x$ |

**Table H.2: The nonlinear and coupled benchmark problems R6–R10.** Equation, data and reference. $x_s$, datum point; $x_z$, exact root nearest the printed value; $\rho$, Carleman ratio; $a$, initial amplitude; $\nu$, viscosity; $k \le M$, wavenumbers of the $M$ sine modes; $\omega_0$, initial vorticity; $J$, the advective nonlinearity; $w_0$, initial mode vector; IVP, initial-value problem; DNS, direct numerical simulation; (truncation), the reference for the Galerkin truncation's error.

| | Equation | Data | Reference |
|---|---|---|---|
| R6a | $4f''+2f'^2+f=0$ (Wu et al. Eq. 29) | $f'(0)=0$, $x_s=0$ | $1-x^2/8$ |
| R6b | $f''-2f^2+x=0$ (Wu et al. Eq. 34) | $f(x_z)=0$, $x_z=0.026147$ (printed 0.02615) | boundary-value solve to $10^{-12}$ |
| R7 | $u'=v$, $v'=-\omega^2 u$, $\omega=\pi$ | $u(-1)=0$, $v(-1)=\omega$ | $u=\sin\omega(x+1)$ |
| R8 | $y'=-y+y^2$, $t\in[0,3]$ | $y(0)=\frac{1}{2}$ ($\rho=0.5$) | $1/(1+e^t)$ |
| R9 | viscous Burgers, periodic, $u_0=a\sin x$ | sine modes $k\le M$, $M=4,8,16$; $a/\nu=0.03$–$10$ | Galerkin IVP; Cole–Hopf (truncation); exact lift |
| R10a | 2-D vorticity Navier–Stokes, $\omega_0=2\sin x\sin y$ | $J\equiv 0$ (Taylor–Green) | exact viscous decay |
| R10b–d | 2-D vorticity Navier–Stokes, counter-rotating vortex pair | $\|w_0\|=a$, $a/\nu=3,10,30$; active triads | Galerkin IVP; DNS (truncation); exact lift |

and the preparation at a known answer. The counter-rotating pair is $\pm e^{-|\mathbf{x}-\mathbf{c}|^2/(2\sigma^2)}$ with $\sigma^2=\frac{1}{4}$ at $\mathbf{c}=(3\pi/4,\pi)$ and $(5\pi/4,\pi)$, periodised and projected exactly onto the modes: it has no cosine part, and its sine coefficients are proportional to $(-1)^{k_x+k_y+1}\sin(\pi k_x/4)\,e^{-|\mathbf{k}|^2/8}$, so modes with $k_x\equiv 0 \pmod 4$ have zero datum. It spans several $|\mathbf{k}|$, so its triads are active. It is scaled to $\|w_0\|=a$ and run at $a/\nu=3$, 10 and 30 (R10b to R10d) over $T=1$, and at $a/\nu=30$ also over $T=1/8$. The reference for the lift and the time axis is the Galerkin system's IVP, with its exact sparse Jacobian; the truncation's own error is measured against a pseudo-spectral DNS of Equation (6.10) from the same data, whose slope inside the truncation is the Galerkin system's.

## H.3 Run Options

A problem's data enter as the four kinds of row of Definition 2, listed with the problems that use them in Table H.3.

Every run is a problem of Table 4.1 at a resolution, with the options of Table H.4. The defaults are the configuration every comparison of § 8 uses; the alternatives are controls, labelled ablations, or the printed choices kept to test the published method.

## H.4 Results by problem

The linear benchmark problems and the coupled oscillator R7 in both forms, with the minimax edge filter at $\varepsilon=10^{-6}$ and the geometric initial state, under the conventions of § 7.7. Table H.5 gives the largest resolution at which each form's reference is trusted and its preparation was emulated, with the prepared state's infidelity against the reference and the filter's degree. Every one of these preparations meets $\varepsilon$, its a priori bound and its field budget (§ 8.1). The standard form's emulations stop at smaller $n$ than the descriptor's because its degree grows as $N^{2k}$; a dash marks a problem with no trusted reference at the resolutions emulated, R2b and R2c in the standard form and the wave equation R5c in both, whose references are trusted only at larger $n$. Below $n=5$ per axis the wave equation's near-kernel is several-dimensional, 5 at $n=4$, consistent with a solution, $\cos 4\pi t\,\sin 2\pi x$, not yet resolved in $t$, and no preparation is claimed there. Table H.6 gives the resource estimates at the largest resolution estimated for every problem, as built and at the ideal-encoding bound $\alpha_R=\|R\|_2$, and Table H.7 the growth of these degrees with $N$, of which Table 8.2 carries a representative of each class.

**Table H.3: Every constraint row either leaves the solution's scale free or ties data to it.** The four rows that carry a problem's data, grouped by that role. Each enters the stacked residual at weight one, is the same in both forms, and acts on the block holding the quantity it constrains: a derivative's zero row is $\mathbb{B}(x^*)G^j$ in the standard form and $\mathbb{B}(x^*)$ on the $j$-th derivative block of the descriptor's chain. $x^*$, the point or slice of the data; $x_s$, the datum point, where $f(x_s) \neq 0$; $\tau(x)$, the Chebyshev polynomials evaluated at $x$; $\mathbb{S}_a$, slice along axis $a$; $\hat{g}$, coefficients of $g$; $a$, the anchor component of a vector datum $y$ and $y_\perp$ the rest; $\mathcal{M}_r$, multiplication by the source $r$; $G$, differentiation matrix. Definitions: §§ 4.3 and 6.2.

| Row | Data it carries | Operator | Problems |
|---|---|---|---|
| *Vanishes on the solution and its multiples: fixes the shape, leaves the scale free* | | | |
| Zero | $f^{(j)}(x^*) = 0$ at a point: Dirichlet ($j = 0$), Neumann ($j = 1$), or the same on a boundary of a PDE | $\mathbb{B}(x^*)G^j$, with $\mathbb{B}(x) = \sqrt{N}\,\lvert 0\rangle\langle\tau(x)\rvert$ | R2–R6 |
| Datum slice | an initial or boundary *function* $g$ on the slice $x_a = x^*$ of a PDE | $\mathbb{S}_a(x^*) - \hat{g}\,\langle\tau(x_s)\rvert\,/f(x_s)$ | R5 |
| Ratio | a vector datum $y$ at $x^*$: the initial values of a coupled system or of a Carleman lift | $\big(\mathbb{F}/y_a\big) \otimes \mathbb{B}(x^*)$, with $\mathbb{F} = y_a(I - \lvert a\rangle\langle a\rvert) - y_\perp\langle a\rvert$ fixing the direction of $y$ | R7–R10 |
| *Maps the solution to a constant, whatever its scale: ties data to the scale* | | | |
| Collapse | the regular datum $f(x_s)$, which fixes the scale after preparation; through it a source $r$, a non-zero point datum $y^*$, and the linear terms of the doubled space | $\mathbb{D}^{(0)}(x_s) = \mathbb{B}(x_s)/f(x_s)$, entering as $-\mathcal{M}_r\mathbb{D}^{(0)}(x_s)$ for a source and as $\mathbb{B}(x^*)G^j - y^*\mathbb{D}^{(0)}(x_s)$ for a datum | every problem (the scale); R4 (sources); R6 |

**Table H.4: The model's input options and the values used.** Every run is a problem of Table 4.1 at a resolution with these options, at the default unless a result states otherwise. $n$, qubits per Chebyshev axis ($N = 2^n$); $n_t$, qubits on a time axis; $M$, Galerkin modes; $\zeta$, block weight; $K$, Carleman order; $\varepsilon$, target suppression; $r$, ratio of the geometric product state.

| Option | Values | Default and use |
|---|---|---|
| Resolution | $n$ qubits per Chebyshev axis, $N = 2^n$; $n_t$ on a time axis; modes $M$ (R9, R10) | swept per problem (Tables H.1 and H.2) |
| Form | standard; descriptor | both, compared |
| Descriptor layout | chain (order $\geq 2$); mass form (first order) | by the equation's order; the chain on a first-order system as an ablation |
| Block rescaling | $\zeta$ from Equation (5.10); none | $\zeta$; unscaled as an ablation |
| Encoding of $G$ | structured, exact; Wu et al.'s | structured; Wu et al.'s in resource estimates only |
| Nonlinearity | doubled space; Carleman order $K$, symmetric or tensor layout | as the problem requires; the doubled space in the standard form only |
| Filter | minimax edge filter; imaginary time | minimax; imaginary time in the filter study and the reproduction |
| Target $\varepsilon$ | suppression of the excited spectrum relative to the ground state | $10^{-6}$; also $10^{-4}$ and $10^{-8}$ in the filter study |
| Initial state | geometric product state, $r = \pm\frac{1}{2}$; uniform; random | geometric; uniform and random as controls |
| Tier | circuit simulation (T1); exact emulation (T2); resource estimate (T3) | every tier the run's size allows (§ 7.5) |

**Table H.5: Every trusted preparation meets its target, in both forms.** Per problem and form: the largest resolution $n$ (per axis on R5) with a trusted reference and an exact emulation, the prepared state's infidelity against the reference, and the minimax filter's degree. –, no trusted reference at the resolutions emulated. Exact emulation (T2).

| | Standard | | | Descriptor | | |
|---|---|---|---|---|---|---|
| Problem | $n$ | Infidelity | Degree | $n$ | Infidelity | Degree |
| R1 | 5 | $1.0 \times 10^{-13}$ | 3240 | 8 | $8.0 \times 10^{-14}$ | 860 |
| R2a | 3 | $9.9 \times 10^{-13}$ | 9198 | 8 | $3.8 \times 10^{-12}$ | 9128 |
| R2b | | – | | 8 | $2.2 \times 10^{-12}$ | 23 279 |
| R2c | | – | | 8 | $2.2 \times 10^{-12}$ | 240 658 |
| R3a | 4 | $1.2 \times 10^{-12}$ | 227 610 | 7 | $3.4 \times 10^{-12}$ | 146 002 |
| R3b | 4 | $5.2 \times 10^{-12}$ | 190 516 | 7 | $1.5 \times 10^{-11}$ | 149 490 |
| R3c | 4 | $1.2 \times 10^{-12}$ | 493 783 | 7 | $8.2 \times 10^{-13}$ | 377 948 |
| R3d | 4 | $5.8 \times 10^{-12}$ | 493 555 | 6 | $5.8 \times 10^{-12}$ | 119 478 |
| R4a | 3 | $2.1 \times 10^{-13}$ | 41 951 | 7 | $2.6 \times 10^{-13}$ | 226 204 |
| R4b | 4 | $3.0 \times 10^{-12}$ | 247 819 | 8 | $2.3 \times 10^{-11}$ | 8892 |
| R4c | 4 | $2.6 \times 10^{-12}$ | 232 606 | 8 | $2.8 \times 10^{-12}$ | 18 331 |
| R4d | 4 | $1.0 \times 10^{-12}$ | 54 837 | 8 | $1.7 \times 10^{-12}$ | 9412 |
| R5a | 3 | $3.5 \times 10^{-13}$ | 25 421 | 4 | $3.5 \times 10^{-13}$ | 84 913 |
| R5b | 4 | $3.2 \times 10^{-11}$ | 27 692 | 5 | $2.8 \times 10^{-11}$ | 136 658 |
| R5c | | – | | | – | |
| R7 | 7 | $1.4 \times 10^{-12}$ | 356 379 | 9 | $2.5 \times 10^{-12}$ | 21 143 |

**Table H.6: As built the descriptor's degree is the lower on every problem; at the ideal-encoding bound the standard form's is the lower on R3.** Minimax edge filter degree at $\varepsilon = 10^{-6}$, $n = 8$ on one axis and $n = 5$ per axis on R5, as built and at the ideal-encoding bound $\alpha_R = \|R\|_2$ ($\alpha_R$, the residual's subnormalisation; $\|R\|_2$, its norm). At the ideal bound the standard form's degree is the lower on R3a–R3d and the two are close on R5b. Resource estimates (T3).

| | | As built | | $\alpha_R = \|R\|_2$ | |
|---|---|---|---|---|---|
| Problem | $n$ | Standard | Descriptor | Standard | Descriptor |
| R1 | 8 | $2.1 \times 10^{5}$ | $8.6 \times 10^{2}$ | $1.0 \times 10^{5}$ | $8.5 \times 10^{2}$ |
| R2a | 8 | $1.2 \times 10^{10}$ | $9.1 \times 10^{3}$ | $1.5 \times 10^{9}$ | $8.7 \times 10^{3}$ |
| R2b | 8 | $4.8 \times 10^{10}$ | $2.3 \times 10^{4}$ | $6.2 \times 10^{9}$ | $2.2 \times 10^{4}$ |
| R2c | 8 | $6.0 \times 10^{9}$ | $2.4 \times 10^{5}$ | $7.8 \times 10^{8}$ | $1.8 \times 10^{5}$ |
| R3a | 8 | $1.7 \times 10^{10}$ | $5.7 \times 10^{5}$ | $1.1 \times 10^{5}$ | $2.8 \times 10^{5}$ |
| R3b | 8 | $1.4 \times 10^{10}$ | $5.8 \times 10^{5}$ | $9.3 \times 10^{4}$ | $2.8 \times 10^{5}$ |
| R3c | 8 | $3.6 \times 10^{10}$ | $1.5 \times 10^{6}$ | $4.1 \times 10^{5}$ | $7.1 \times 10^{5}$ |
| R3d | 8 | $3.6 \times 10^{10}$ | $1.7 \times 10^{6}$ | $4.1 \times 10^{5}$ | $8.4 \times 10^{5}$ |
| R4a | 8 | $5.8 \times 10^{10}$ | $9.0 \times 10^{5}$ | $6.7 \times 10^{9}$ | $4.0 \times 10^{5}$ |
| R4b | 8 | $1.8 \times 10^{10}$ | $8.9 \times 10^{3}$ | $2.3 \times 10^{9}$ | $8.1 \times 10^{3}$ |
| R4c | 8 | $1.7 \times 10^{10}$ | $1.8 \times 10^{4}$ | $2.2 \times 10^{9}$ | $1.6 \times 10^{4}$ |
| R4d | 8 | $4.0 \times 10^{9}$ | $9.4 \times 10^{3}$ | $5.2 \times 10^{8}$ | $8.4 \times 10^{3}$ |
| R5a | 5 | $8.0 \times 10^{6}$ | $8.4 \times 10^{5}$ | $8.1 \times 10^{5}$ | $2.1 \times 10^{5}$ |
| R5b | 5 | $4.5 \times 10^{5}$ | $1.4 \times 10^{5}$ | $6.0 \times 10^{4}$ | $6.3 \times 10^{4}$ |
| R5c | 5 | $4.7 \times 10^{7}$ | $1.6 \times 10^{6}$ | $5.4 \times 10^{6}$ | $5.8 \times 10^{5}$ |
| R7 | 8 | $1.4 \times 10^{6}$ | $1.1 \times 10^{4}$ | $6.8 \times 10^{5}$ | $1.0 \times 10^{4}$ |

**Table H.7: The degree's growth with $N$ on every linear problem and on R7.** Exponent of $N = 2^n$ in the minimax filter's degree between the two largest resolutions estimated ($n = 7 \to 8$ on one axis, $n = 4 \to 5$ per axis on two), as built and at the ideal-encoding bound $\alpha_R = \|R\|_2$, grouped by class: $k$, equation order; regular, singular: the leading coefficient without and with a zero on $[-1, 1]$ (Definition 4). †Reference not trusted at $n = 4$ per axis; slope indicative only. Two-point slopes of a degree with logarithmic factors (§ 8.2). Resource estimates (T3).

| | | Standard | | Descriptor | |
|---|---|---|---|---|---|
| Class | Problem | As built | Ideal | As built | Ideal |
| First order, $k = 1$ | R1 | 2.0 | 2.0 | 0.98 | 1.0 |
| | R7 | 2.0 | 2.0 | 0.95 | 0.99 |
| Second order, regular | R2a | 4.0 | 4.0 | 0.95 | 1.0 |
| | R2b | 4.0 | 4.0 | 0.95 | 1.0 |
| | R2c | 4.0 | 4.0 | 0.66 | 0.97 |
| | R4b | 4.0 | 4.0 | 0.92 | 1.0 |
| | R4c | 4.0 | 4.0 | 0.88 | 1.0 |
| | R4d | 4.0 | 4.0 | 0.87 | 1.0 |
| Second order, singular | R3a | 4.0 | 2.0 | 2.0 | 2.0 |
| | R3b | 4.0 | 2.0 | 2.0 | 2.0 |
| | R3c | 4.0 | 2.0 | 2.0 | 2.0 |
| | R3d | 4.0 | 2.0 | 2.0 | 2.0 |
| | R4a | 4.0 | 4.0 | 2.0 | 2.0 |
| Two axes, per axis | R5a | 4.1 | 4.0 | 3.3 | 3.3 |
| | R5b | 4.0 | 3.9 | 2.2 | 2.3 |
| | R5c† | 2.9 | 2.9 | 2.8 | 2.8 |

## H.5 Filter studies

**Comparing the two filters at like error.** A degree means something only at a stated error, so the filter study of § 8.1 holds both filters to the same one, at three strengths. *Equal suppression* $\eta$ bounds the excited interval relative to the ground state, $\sup_{x\in[x_1,1]} |P(x)|/|P(x_0)| \le \eta$; it bounds every initial state's infidelity by $\eta^2(1-\gamma^2)/\gamma^2$, depends only on $x_0$ and $x_1$, and is the strength at which the minimax filter is optimal (Proposition 28). *Equal certified infidelity* $\varepsilon$ is the same bound at the problem's own overlap, $\eta = \gamma\sqrt{\varepsilon/(1-\gamma^2)}$. *Equal measured infidelity* $\varepsilon$ is reached on the problem's own spectrum and initial state, and a degree counts only if every larger degree up to the certified one also reaches $\varepsilon$, so that neither filter's oscillation lands on $\varepsilon$ by coincidence. Imaginary time's $\beta$ is free: each of its degrees is the least over $\beta$, found by a grid search refined about its best point, and so an upper bound on the true least to the refinement's spacing. Both filters are anchored at the spectrum's edge with unit weight (Proposition 30), so GQSP realises either at the same rescaling, and a filter whose success probability falls below $10^{-12}$ is not counted.

On synthetic spectra with the ground state at the edge, gaps from $10^{-8}$ to $10^{-1}$ on the scale of $H'$ and suppressions from $10^{-2}$ to $10^{-10}$, imaginary time needs 1.67–1.92 times the minimax filter's degree at equal suppression, varying with the suppression and hardly with the gap; above the edge the ratio grows, since imaginary time's value on the ground state then falls with its time. Table H.8 gives the study on the problems whose references both forms trust. At equal certified infidelity both filters' degrees come from the same worst-case bound, so the ratio is the equal-suppression ratio at the problem's own $\gamma$; every trusted reference has its ground state at the edge, and the ratio lies within the synthetic range in both forms. It does not change with the resolution: on R1 in the descriptor it is 19 / 11 at $n = 2$ and 675 / 384 at $n = 8$, so the filter changes the degree's constant and not its growth with $N$. At equal measured infidelity the ratio is smaller, because imaginary time suppresses each excited state more the higher it lies, while the bound holds every one at the equiripple level. At $\varepsilon = 10^{-4}$ imaginary time needs fewer degrees on R1 in the standard form and on R3a and R5a

in the descriptor, and the same on R1 in the descriptor; at $10^{-6}$ the minimax filter needs fewer on every case.

**Table H.8: The minimax edge filter needs a lower degree than imaginary time at equal error, except at loose $\varepsilon$ on some problems.** Each cell: imaginary-time (QITE) degree / minimax degree, both forms, at one resolution per problem ($n$ per axis on R5a) whose reference both forms trust. Three equal-error criteria: certified infidelity at $\varepsilon = 10^{-6}$ (both degrees from the worst-case bound at the problem's own $\gamma$), and measured infidelity at $\varepsilon = 10^{-6}$ and $10^{-4}$ on the problem's own spectrum from the geometric initial state. $\gamma^2$, that state's overlap with the ground state. Exact emulation (T2).

| Problem | Form | $\gamma^2$ | Certified, $10^{-6}$ | Measured, $10^{-6}$ | Measured, $10^{-4}$ |
|---|---|---|---|---|---|
| R1, $n = 4$ | standard | 0.904 | 620 / 353 | 528 / 352 | 198 / 227 |
| | descriptor | 0.904 | 49 / 28 | 41 / 27 | 18 / 18 |
| R2a, $n = 3$ | standard | 0.499 | 8603 / 4820 | 6422 / 4796 | 5459 / 3317 |
| | descriptor | 0.191 | 636 / 354 | 533 / 353 | 400 / 255 |
| R3a, $n = 3$ | standard | 0.166 | 13 127 / 7295 | 9247 / 7135 | 7236 / 5189 |
| | descriptor | $8.0 \times 10^{-3}$ | 5294 / 2904 | 3072 / 2862 | 1784 / 2187 |
| R5a, $n = 3$ | standard | 0.587 | 23 175 / 13 011 | 18 055 / 12 611 | 10 248 / 8650 |
| | descriptor | 0.526 | 8612 / 4829 | 4776 / 4617 | 2285 / 3253 |

**An underestimated edge.** The degree is set a priori from the spectral edge (§ 7.4), and a study on R2a at $n = 3$ in the standard form measures what an underestimate of it costs, with the imaginary-time filter, whose time the gap sets directly (Equation (F.4)). At the true gap the degree is 17 750 and the infidelity $9.7 \times 10^{-13}$. With the gap halved the degree is 25 103, and with it divided by ten 56 131: the factors $\sqrt{2}$ and $\sqrt{10}$ of an inverse square root, with every preparation still within $\varepsilon$. Wu et al.'s own time rule, run in the same study, gives degree 3336 and infidelity 0.24: its time is shorter than the gap rule's and its state does not reach $\varepsilon = 10^{-6}$. The time rule is examined with the rest of the reproduction in Appendix I. Exact emulation (T2).

# I Reproduction and Audit of Wu et al.

Wu et al.'s method [1] is tested here against its own record. Each panel of their Figs 3–7 is rebuilt with their choices as printed and with the standard form's, and each claim we tested, their circuits and filter included, is given a verdict with the computation or derivation behind it. Unless a tier is named, every number is an exact classical computation on the assembled residual: the ground state is the smallest right singular vector of the stacked residual (§ 7.2), errors are $L^\infty$ on 1001 points ($101 \times 101$ on two axes), and a dagger marks a ground state that is not trusted.

## I.1 The paper-faithful variant

The paper-faithful variant is a panel with every choice Wu et al. print, and it runs only in this appendix. It shares with the standard form the conventions that fix Wu et al.'s matrices: the state holds the solution's coefficients in the normalised Chebyshev basis; every rank-one row uses the feature vector $\tau(x)$ unnormalised; $G$ is the upper-triangular matrix acting on coefficients, the transpose of the $\mathbb{G}_n$ Wu et al. print; a polynomial coefficient or source enters through the lifts $\mathcal{M}_{x^p}$ from $n$ to $n + 1$ qubits; an invariant constraint on a derivative is the row $\mathbb{B}(x_m)G^j$; and $H$ is the Gram sum of Equation (3.4), which equals $R^\dagger R$ for the residual that stacks its operators. With these conventions the construction reproduces entry by entry the matrices $G$, $\mathcal{M}_{x^p}$, $\mathcal{N}_1$ and $\mathbb{B}$ that Wu et al.'s supplementary material prints at $n = 2$ (claim A-01), and the exact solution's $\eta_e = \|b\|^2$ (Equation (4.1)) gives the printed scale factor of every ordinary panel (Table I.1).

What the variant alone keeps is the data as printed. Its constraint points are rounded to two decimal places, $x_m = -0.2$, $x_z = -0.73$, 0.91 and $x_m = -0.75$ on Figs 5a–5d and $x_z = 0.02615$

on Fig. 7c, against the exact roots of Tables H.1 and H.2. Its sources on Figs 5b–5d are Maclaurin polynomials of orders $\bar{p} = 3$, 5 and 7. R3c and R3d keep the associated Legendre equation multiplied through by $1 - x^2$, without the substitution of Appendix B.3. The partial differential panels keep the invariant sets of Table I.3, and the heat equation keeps its initial function at $t = 0$. Where the printed choice is already exact (Fig. 3, Figs 4a and 4b, Fig. 7b) the variant is the problem as run, and Fig. 7's quadratic panels are carried by the doubled space in both (§ 6.4). Two choices that Wu et al. leave open are made alike in both variants. The datum point, not printed for Figs 3–6, is a point of the stated data where the solution does not vanish: $x_s = 0$ on Figs 3b, 3d and 5, $x_s = 1$ on Figs 4a and 4b, $x_s = \frac{1}{2}$ on Figs 3c, 4c and 4d, and on two axes a point of the data slice where the stated function equals one, moved with the heat equation's data to $t = -1$ in the problem as run. The constraint $\mathcal{T}\big(\mathbb{B}(\pm 1) \otimes I\big)$ is read as two rows, one for each sign.

## I.2 All panels

**Table I.1: The exact solution reproduces the printed scale factor of every ordinary panel, to within one unit of the last printed digit, and of no partial differential or nonlinear panel.** Per panel of Wu et al. at their printed $n$: the printed scale factor $\eta$, the exact solution's $\eta_e = \|b\|^2$ ($b$, its Chebyshev coefficients; Equation (4.1)) and the paper-faithful ground state's $\eta$; then the $L^\infty$ error of the panel as printed (the paper-faithful variant) and of the problem as run, at the printed $n$, and the problem's least error over $n$, on 1001 points ($101 \times 101$ on two axes). $^\dagger$Ground state not trusted (degenerate kernel, or gap ratio below 20); $^*$one unit low in the last printed digit; –, the doubled space (R6), whose kernel is never trusted. Classical computation.

| | | | Scale factor $\eta$ | | | | $L^\infty$ error | | |
|---|---|---|---|---|---|---|---|---|---|
| Fig. | Problem | $n$ | Printed | Ideal $\eta_e$ | Ground state | Agrees | As printed | As run | As run, best ($n$) |
| 3b | R2a | 3 | 1.29 | 1.29009 | 1.28731 | ✓ | $1.2 \times 10^{-3}$ | $1.2 \times 10^{-3}$ | $3.2 \times 10^{-13}$ ($n = 5$) |
| 3c | R2b | 3 | 32.47 | 32.47741 | 32.22636$^\dagger$ | ✓$^*$ | $1.8 \times 10^{-2\dagger}$ | $1.8 \times 10^{-2\dagger}$ | $2.3 \times 10^{-11}$ ($n = 5$) |
| 3d | R2c | 5 | 451.71 | 451.70962 | 451.52312 | ✓ | $2.2 \times 10^{-3}$ | $2.2 \times 10^{-3}$ | $5.6 \times 10^{-12}$ ($n = 6$) |
| 4a | R3a | 3 | 1.75 | 1.74512 | 1.74512 | ✓ | $1.9 \times 10^{-15}$ | $1.9 \times 10^{-15}$ | $1.9 \times 10^{-15}$ ($n = 3$) |
| 4b | R3b | 3 | 1.49 | 1.48779 | 1.48779 | ✓ | $3.0 \times 10^{-15}$ | $3.0 \times 10^{-15}$ | $2.3 \times 10^{-15}$ ($n = 4$) |
| 4c | R3c | 5 | 108.51 | 108.51142 | 109.70433$^\dagger$ | ✓ | $8.7 \times 10^{-1\dagger}$ | $4.0 \times 10^{-14}$ | $4.9 \times 10^{-15}$ ($n = 3$) |
| 4d | R3d | 5 | 135.38 | 135.38360 | 137.46159$^\dagger$ | ✓ | $1.2^\dagger$ | $1.5 \times 10^{-13}$ | $8.0 \times 10^{-15}$ ($n = 3$) |
| 5a | R4a | 3 | 1.35 | 1.34649 | 1.34478 | ✓ | $3.1 \times 10^{-3}$ | $1.8 \times 10^{-6}$ | $3.8 \times 10^{-14}$ ($n = 4$) |
| 5b | R4b | 3 | 14.74 | 14.74797 | 14.72531 | ✓$^*$ | $1.9 \times 10^{-2}$ | $7.8 \times 10^{-3}$ | $8.5 \times 10^{-12}$ ($n = 5$) |
| 5c | R4c | 4 | 0.82 | 0.82116 | 0.82521 | ✓ | $7.2 \times 10^{-3}$ | $4.9 \times 10^{-10}$ | $1.6 \times 10^{-13}$ ($n = 5$) |
| 5d | R4d | 4 | 77.51 | 77.51167 | 76.93737 | ✓ | $6.1 \times 10^{-2}$ | $4.9 \times 10^{-7}$ | $5.1 \times 10^{-12}$ ($n = 5$) |
| 6b | R5a | 3 | 5.21559 | 5.21653 | 5.24407 | × | $3.9 \times 10^{-3}$ | $1.4 \times 10^{-4}$ | $2.1 \times 10^{-13}$ ($n = 4$) |
| 6d | R5b | 4 | 603.863 | 598.86606 | 598.87966 | × | $1.9 \times 10^{-4}$ | $4.7 \times 10^{-5}$ | $4.6 \times 10^{-12}$ ($n = 5$) |
| 6f | R5c | 5 | 216.469 | 239.82712 | 239.82712 | × | $8.5 \times 10^{-8}$ | $7.3 \times 10^{-8}$ | $6.1 \times 10^{-11}$ ($n = 6$) |
| 7b | R6a | 2 | 3.75367 | 3.52344 | 3.52344$^\dagger$ | × | $2.2 \times 10^{-16\dagger}$ | $2.2 \times 10^{-16\dagger}$ | – |
| 7c | R6b | 2 | 0.07984 | 0.04383 | 0.04182$^\dagger$ | × | $6.8 \times 10^{-3\dagger}$ | $6.8 \times 10^{-3\dagger}$ | – |

Only the ordinary panels' scale factors reproduce, and the agreement is the exact solution's; as printed, the panels' own ground states fall short of it. R2b, R3c and R3d have no trusted ground state at the printed $n$. R3c and R3d, multiplied through by $1-x^2$, have solutions with a branch point at each end, which a Chebyshev series approximates only algebraically (Appendix B.3), and Fig. 5's panels hold at the floors of Appendix I.3. As run, every ordinary problem converges spectrally until the standard form's conditioning stops it. Wedin's bound $\epsilon_{\text{mach}}\sigma_{\max}/(\sigma_1 - \sigma_0)$ (Proposition 35) grows by a factor of about 16 per qubit on a second-order equation with constant coefficients, whose $\sigma_{\max}$ grows as $N^4$, so the least error is reached between $n = 3$ and 6 and the error rises beyond it, to $1.4 \times 10^{-6}$ on R2a at $n = 10$. The partial differential and doubled-space panels reproduce no printed scale factor, by the exact solution or by any ground state (Appendix I.4).

## I.3 The Fig. 5 triage

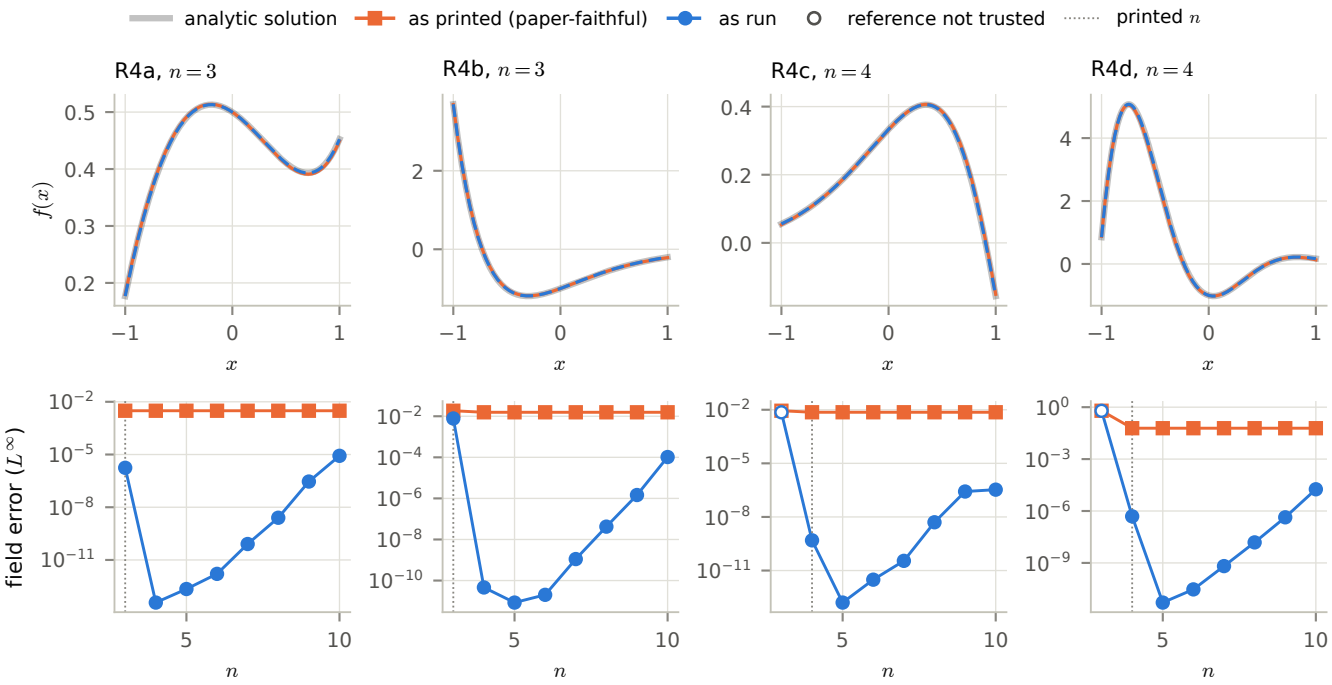



**Figure I.1: As printed, each panel of Wu et al.'s Fig. 5 holds at an error floor that no resolution removes; as run, it converges spectrally.** R4a–R4d. Top: the analytic solution, the paper-faithful ground state (rounded constraint point, Maclaurin source) and the problem as run (exact root, Chebyshev source) at the printed $n$; the three coincide at plot scale, the paper-faithful errors there lying between $3.1 \times 10^{-3}$ (R4a) and $6.1 \times 10^{-2}$ (R4d). Bottom: the $L^\infty$ field error against $n$ for both variants (open where the reference is not trusted), the printed $n$ dotted. The paper-faithful error holds at its floor (Table I.2); as run it falls to between $3.8 \times 10^{-14}$ and $8.5 \times 10^{-12}$ at $n = 4$ or 5, then rises with the standard form's conditioning (Proposition 35). On R4b the printed $n = 3$ is itself under-resolved: as run, the error there is $7.8 \times 10^{-3}$. Classical computation (dense SVD).

**Table I.2: Each printed choice alone sets an error floor; only both departures together converge.** $L^\infty$ error on Wu et al.'s Fig. 5 panels over $n = 4, \ldots, 10$: the paper-faithful variant (rounded constraint point, Maclaurin source), each departure alone (exact root; Chebyshev source coefficients), and both together at their best $n$. –, R4a's polynomial source is already exact, so its rounded point acts alone. Classical computation.

| Fig. | Problem | Faithful | Exact root | Chebyshev source | Both, best |
|---|---|---|---|---|---|
| 5a | R4a | $3.1 \times 10^{-3}$ | – | – | $3.8 \times 10^{-14}$ ($n = 4$) |
| 5b | R4b | $1.6 \times 10^{-2}$ | $4.0 \times 10^{-2}$ | $3.6 \times 10^{-2}$ | $8.5 \times 10^{-12}$ ($n = 5$) |
| 5c | R4c | $7.2 \times 10^{-3}$ | $4.8 \times 10^{-3}$ | $1.0 \times 10^{-2}$ | $1.6 \times 10^{-13}$ ($n = 5$) |
| 5d | R4d | $6.1 \times 10^{-2}$ | $8.1 \times 10^{-2}$ | $1.1 \times 10^{-1}$ | $5.1 \times 10^{-12}$ ($n = 5$) |

R4a's source is a polynomial, represented exactly, so its rounded point acts alone: the printed $f'(-0.2) = 0$ stands for an extremum at $x \approx -0.196$, and the error holds at $3.1 \times 10^{-3}$ for every $n$ from 4 to 10, against $3.8 \times 10^{-14}$ at $n = 4$ at the exact point. On R4b to R4d each printed choice alone sets a floor, and only both departures together converge. The source dominates on R4d, whose order-7 Maclaurin polynomial departs from the source near $x = -1$ by more than the source's own largest value: with the exact root, raising the order to 11 and to 15 lowers the error at $n = 4$ from $8.1 \times 10^{-2}$ to $7.4 \times 10^{-4}$ and $1.6 \times 10^{-5}$, against $4.9 \times 10^{-7}$ with the Chebyshev source.

## I.4 Panels not reproduced, and printed errors

**Printed errors.** Wu et al. print the solution of their Laplace example as $\operatorname{csch}\pi \cosh(\pi x/2) \sinh(\pi(y+1)/2)$ (claim A-05). It is not harmonic, its two second derivatives summing to $\frac{\pi^2}{2} f$, and it violates

$f(\pm 1, y) = 0$, since $\cosh(\pi/2) \neq 0$. The solution, and the reference throughout, is $\cos(\pi x/2)\sinh(\pi(y+1)/2)/\sinh\pi$. The printed scale factor was not computed from the printed form either, which gives $\eta_e = 48.57$ at $n = (3, 3)$. Wu et al.'s heat data, stated at $t = 0$ inside the time domain $[-1, 1]$, pose a backward heat problem on $[-1, 0]$ (Appendix B.2).

**Table I.3: Each of Wu et al.'s invariant sets for their partial differential examples admits a family of solutions, which the truncation suppresses only near the printed resolution.** The constraints Wu et al. impose on Laplace's equation (R5a, their Fig. 6b), the heat equation $f_t = \frac{1}{25} f_{xx}$ (R5b, Fig. 6d) and the wave equation $f_{tt} = 4 f_{xx}$ (R5c, Fig. 6f); the solutions that satisfy the equation and every constraint, $\ell$ a positive integer and $\phi_\ell$ a phase fixed by the constraints in $t$; the printed $n$ per axis; and the least $n$ per axis above the printed one at which the paper-faithful residual's near-kernel is degenerate, with its dimension. Classical computation (dense SVD).

| Problem | Printed invariant set | Solutions satisfying it | Printed $n$ | Degenerate fron |
|---|---|---|---|---|
| R5a | $f(\pm 1, y) = 0,\ f(x, -1) = 0$ | $\sin\frac{\ell\pi(x+1)}{2}\ \sinh\frac{\ell\pi(y+1)}{2}$ | 3 | $(5, 5)$, dimensio |
| R5b | $f(t, \pm 1) = 0,\ \partial_x f(t, \pm\frac{3}{4}) = 0$ | $\sin\frac{\ell\pi(x+1)}{2}\ e^{-(\ell\pi/2)^2 t/25},\ \ell \equiv 4 \pmod 8$ | 4 | $(6, 6)$, dimensio |
| R5c | $f(\frac{1}{8}, x) = f(\frac{7}{8}, x) = 0,\ f(t, \pm 1) = 0$ | $\sin\frac{\ell\pi(x+1)}{2}\ \cos(\ell\pi t + \phi_\ell),\ 4 \mid \ell$ | 5 | $(6, 6)$, dimensio |

**The invariant sets.** None of Wu et al.'s sets for their partial differential examples contains the function that fixes the solution. Laplace's set omits the top side, and the heat and wave sets omit the initial function and place interior constraints at zeros of the known solution instead, $\partial_x f = 0$ at $x = \pm\frac{3}{4}$ and $f = 0$ at $t = \frac{1}{8}$ and $\frac{7}{8}$, which a new problem would not supply. Each set therefore admits the family of Table I.3. At the printed resolution the truncation suppresses all but the solution's member, and the near-kernel is one-dimensional and trusted on all three panels. As $n$ grows the other members become representable and the near-kernel grows. On Laplace's equation the bottom of the spectrum is already near-degenerate at $n = (4, 4)$, where the gap of $H$ relative to its norm is $2.2 \times 10^{-25}$, and at $(5, 5)$ the field error is 1.3; the heat and wave sets fail at $(6, 6)$, with errors 166 and 32. The datum slices of Appendix B.2, which impose the omitted function, leave on Laplace's equation a one-dimensional kernel trusted at every resolution computed from $n = (3, 3)$ to $(6, 6)$, with gap ratio $1.7 \times 10^9$ at the last, and on the heat equation, posed forward from $t = -1$, an error of $5.5 \times 10^{-11}$ at $(6, 6)$.

**Fig. 6's scale factors.** The printed values are 5.21559, 603.863 and 216.469; the exact solutions give 5.21653, 598.86606 and 239.82712, and the ground states at the printed $n$, trusted on all three, do not give the printed values either (Table I.1). An unprinted datum point would explain Laplace's value alone: over a $41 \times 41$ grid of datum points its ground state's $\eta$ spans 5.127 to 5.261, which contains the print, while the heat and wave ground states stay within 598.58–599.14 and 239.826–239.828. Reading $\mathcal{T}\big(\mathbb{B}(\pm 1) \otimes I\big)$ as one summed row degenerates the Laplace kernel and moves the heat and wave values further from the print, to 727.47 and 634.66. A time domain of $[0, 1]$ or $[0, 2]$ in place of $[-1, 1]$ gives 38.4 or 25.5 for heat and 249.6 or 239.8 for wave, and a neighbouring resolution gives 149 or 2395 for heat and 59.8 or 959 for wave. Wu et al.'s own preparation, imaginary time from $|0\rangle$ under $H/\|H\|_F$ stopped at a finite time, does not reach the printed values: on Laplace's equation its $\eta$ falls monotonically to the ground state's, which lies above the print, and $|0\rangle$, the constant function, is orthogonal to the heat and wave solutions, which are odd in $x$. The one explanation left untested is a shot-sampled interferometric readout, whose noise would scatter the value; that readout was simulated with shots only at $n = (2, 2)$ on Laplace's equation (§ 7.6), below every printed resolution. Laplace's value is consistent with an unprinted datum point, and the heat and wave values are unexplained.

**Fig. 7's scale factors.** The doubled space has no unique ground state: at the printed $n = 2$ its kernel has dimension 10 on R6a and 8 on R6b (claim A-10). We read out instead the product state $\hat{b} \otimes \hat{b}$ of least residual, the state Wu et al. describe their construction as selecting. On R6a,

whose solution $1 - x^2/8$ is represented exactly, it gives the exact solution's $\eta_e = 3.52344$, and on R6b 0.04182 against the exact 0.04383; the printed values are 3.75367 and 0.07984. Another datum point does not explain them: R6a's product state gives the same $\eta$ at every datum point tried, and R6b's spans 0.0416–0.0479. Nor does another resolution, since $\eta$ doubles with each qubit, to 7.04688 and 14.09375 on R6a and 0.08754 and 0.17532 on R6b at $n = 3$ and 4. The converged filter's output from $|0\rangle$, the projection of $|0\rangle$ onto the kernel, gives 3.72756 on R6a and 0.157 on R6b. Imaginary time from $|0\rangle$ under $H/\|H\|_F$, stopped at a finite time, passes through both printed values, at $\beta \approx 178$ and $\beta \approx 302$; Wu et al. print no time for Fig. 7, so this is consistent with their values without confirming them. The normalisation of the doubled-space readout (their Eq. 39) would need that readout, which was not built.

## I.5 Wu et al.'s claims

**Table I.4: The claims of Wu et al. [1] that we tested, and their verdicts.** $G$, differentiation matrix; $\mathcal{M}$, multiplication operator; $\mathcal{N}_1$, product-to-sum fold; $\mathbb{B}$, $\mathbb{D}^{(0)}$, zero and collapse constraint rows; $\eta$, scale factor; $\beta$, imaginary time; $\lambda_{\max}$, $\lambda_2$, largest and second-smallest eigenvalues of $H$; $x_s$, datum point.

| | Claim | Verdict |
|---|---|---|
| A-01 | $G$, $\mathcal{M}$, $\mathcal{N}_1$ and $\mathbb{B}$ as printed at $n = 2$ | reproduced |
| A-02 | the printed $\eta$ of Figs 3–5 (11 panels) | reproduced: within one unit of the last printed digit, nine of eleven equal when rounded |
| A-03 | the printed $\eta$ of Fig. 6 | not reproduced |
| A-04 | the printed $\eta$ of Fig. 7 | not reproduced |
| A-05 | the printed Laplace solution | a typographical error (cosh for cos) |
| A-06 | the $\mathbb{B}$ circuit (Fig. 2): $n+2$ qubits, prefactor $\sqrt{2^{n+1}}$ | confirmed, exact |
| A-07 | the $G$ circuit (Fig. 9): $2n+3$ qubits, $O(n^3)$ gates, error decreasing exponentially | partly: the mechanism reproduced; built on $2n+5$ qubits; the error falls as about $N^{-5}$; the prefactor valid but loose (Appendix I.6.3) |
| A-08 | the $\mathbb{D}^{(0)}$ circuit (Fig. 8): prefactor $\sqrt{2^{n+1}}/f(x_s)$ | confirmed, exact |
| A-09 | the linear $H$ is positive definite with distinct eigenvalues | partly: positive semi-definite, an exact discrete solution being an exact kernel; the bottom need not be distinct |
| A-10 | the doubled-space $H$ has at least $2^{2n-1}$ zero eigenvalues | confirmed: 962 on R6a at $n = 5$, against $2^{2n-1} = 512$ |
| A-11 | the time rule $\beta \geq \lambda_{\max}/(\lambda_2(n-1))$ suffices | not met: short of the time for $\varepsilon$ by at least $(n-1)\ln(1/\varepsilon)$ (Equation (I.1)); misses $\varepsilon$ on every trusted record |
| A-12 | the necessary terms scale as $O(\mathrm{poly}(n))$ and are efficient | partly: true of the gates, not of the subnormalisation |
| A-13 | the examples are solved accurately at the printed $n$ | partly: Figs 4a and 4b to rounding; R2b, R3c and R3d not trusted; Fig. 5 held at the floors of its printed points and sources |

Rows A-01, A-02, A-05 and A-13 are settled by the panels above, A-03 and A-04 by Appendix I.4, and A-06 to A-08 by the circuits of Appendix I.6. The derivations behind the others follow.

**Definiteness (A-09).** Wu et al. call $H$ "a real symmetric positive-definite matrix with distinct eigenvalues" and elsewhere a positive semi-definite form whose ground state has zero energy [1]. Only the second holds in general. As a Gram sum, $H = R^\dagger R$ is positive semi-definite, and definite exactly when no nonzero state satisfies every row of $R$. The method aims at the opposite case, a solution the truncation represents exactly, whose coefficients span an exact kernel: on R3a, whose solution $P_4$ is a polynomial, $\sigma_0 = 1.0 \times 10^{-15}$ at the printed $n = 3$, which is rounding. Distinctness at the bottom fails when the constraints under-determine the solution, as Laplace's printed set does above its printed resolution (Table I.3). The thesis relies on neither property; it measures the isolation of the ground state as the gap ratio of a trusted kernel (§ 7.2).

**Uniqueness.** Wu et al. select the solution with invariant constraints and fix its scale with one regular datum. For an ordinary equation of order $k$ whose leading coefficient has no zero on $[-1, 1]$, Proposition 12 makes the selection exact when the constraints have rank $k-1$ on the solution space; R2a's single constraint $f(-1) = 0$ has rank one on the span of $e^{-2x}$ and $xe^{-2x}$. The count concerns the continuous problem. Whether the truncated residual has a trusted kernel at a given $n$ is computed, and at the printed $n$ R2b, R3c and R3d do not (Table I.1). Fig. 5's inhomogeneous panels and the singular R3 (Definition 4) lie outside the count and are judged by their computed kernels. A partial differential equation, whose solution space is infinite-dimensional, has no such count, and its printed sets fail as Table I.3 shows.

**The time rule (A-11).** Wu et al. bound the imaginary time below by Equation (3.6), call the bound approximate, and report that $\beta = 15$ and 8 "are enough" for two Legendre examples [1]; no accuracy enters the rule. Write $\tilde\lambda_{\min} < \tilde\lambda_2 \le \cdots \le \tilde\lambda_{\max}$ for the eigenvalues of $\tilde H = H/\|H\|_F$, so that $\tilde\lambda_{\max} \le 1$, and take $n \ge 2$. After $e^{-\beta\tilde H}$ the first excited component keeps the fraction $e^{-\beta(\tilde\lambda_2 - \tilde\lambda_{\min})}$ of its amplitude relative to the ground component. At the least time the rule allows, the exponent is $(\tilde\lambda_2 - \tilde\lambda_{\min})\,\tilde\lambda_{\max}/\big(\tilde\lambda_2(n-1)\big) \le 1/(n-1)$, so that component keeps at least $e^{-1/(n-1)}$, a fraction that tends to one as $n$ grows. Suppression to $\varepsilon$ needs $\beta_\varepsilon = \ln(1/\varepsilon)/(\tilde\lambda_2 - \tilde\lambda_{\min})$, and

$$\frac{\beta_\varepsilon}{\beta_{\text{rule}}} = \frac{(n-1)\ln(1/\varepsilon)\,\tilde\lambda_2}{(\tilde\lambda_2 - \tilde\lambda_{\min})\,\tilde\lambda_{\max}} \;\ge\; (n-1)\ln\frac{1}{\varepsilon}, \tag{I.1}$$

with $\beta_{\text{rule}}$ the rule's least time. The rule permits longer times without saying how much longer, and at its least time the excited weight left is set by the initial state's overlap with the low excited states, not by $\varepsilon$. In exact emulation (T2) from the geometric product state at $\varepsilon = 10^{-6}$, the rule's state misses $\varepsilon$ on every protocol record whose reference is trusted, all at $n \le 4$, with infidelity from 0.042 (R1, $n = 3$) to 0.59 (R3b, $n = 3$), where the gap's own time reaches $\varepsilon$ on every one (R2a's record is given in Appendix H.5). The rule estimates the scale of the time and certifies no accuracy, which Wu et al. do not claim for it.

**The readout identity.** Wu et al.'s interferometric identity takes the preparation to succeed with certainty. Post-selected with probability $p_G < 1$, the system part on the all-zero ancilla is $\sqrt{p_G}\,\psi_g$, and the identity acquires that factor (Equation (F.6), derived in Appendix F.9). The field, a ratio against the regular datum, is unchanged, but $\eta_e$ computed by the printed identity comes out larger by $1/p_G$ (Equation (7.9)). The protocol was built and verified at small $n$ only, and it is not the thesis's readout (§ 7.6).

**Efficiency (A-12).** Every operator's circuit has a gate count polynomial in $n$, as Wu et al. claim (Table I.5). The claim prices one query. The number of queries grows with the subnormalisation and with the norm of $H$, exponentially in $n$ under any encoding (§ 4.6).

## I.6 Wu et al.'s circuits

§ 3.3 gives the idea of the circuit Wu et al. use to load $G$'s entries as rotation amplitudes [1]. This section rebuilds the circuit and bounds its entrywise error, which Wu et al. describe only as "negligible" and decreasing "exponentially" with $n$; the bound is compared with the error measured on the circuit as built (Table I.5).

### I.6.1 The circuits as built

**The constraint rows (Figs 2 and 8).** $\mathbb{B}(x)$ and $\mathbb{D}^{(0)}(x_s)$ are block-encoded by the inverse feature map $U_\tau^\dagger(x)$ followed by $\hat{S}_{n+1} = 2\,|0\rangle\langle 0| - I$ controlled inside a Hadamard sandwich on one ancilla, a two-term LCU on $n+2$ qubits. The encoded block is $\mathbb{B}(x)/\sqrt{2^{n+1}}$, and $\mathbb{D}^{(0)}(x_s)/\big(\sqrt{2^{n+1}}/f(x_s)\big)$ for the collapse row. The global phase of $-1$ in $\hat{S}_{n+1}$ becomes a relative phase once the circuit is controlled, and cannot be dropped.

**The derivative circuit (Fig. 9).** The registers are, in order, a flag qubit acted on by $R_Y(\omega_c)$ with $\omega_c = 2\arccos\sqrt{1/3}$, an $X$ ancilla, an offset register of $m = n$ qubits between Hadamards, a value ancilla, and the $n$-qubit system. Two conditioned subroutines assign the entries on different bases. In $\hat{U}_R$ the offset $|i\rangle$ selects the $(2i+1)$th superdiagonal and the system $|j\rangle$ the $j$th column; it consists of $n$ singly controlled $R_Y(\theta_k)$ gates on the value ancilla, controlled by the bits of $|j\rangle$, followed by $n$ cyclic shifts $\hat{R}_n$. In $\hat{U}_C$ the offset selects the $(i+1)$th row; it consists of $n$ multi-controlled $R_Y(\varphi_k)$ gates, one shift $\hat{L}_n$ and $2n$ swaps. With $\mu = 2/\|G\|_S$ the angles are $\theta_k = 2\arcsin(-2^{k-1}\mu)$ and $\varphi_k = 2\arcsin[-2^{k-1}\mu(1/\sqrt{2}-1)]$. The circuit uses $m+n+3 = 2n+3$ qubits, its gate count is linear in $n$ before the multi-controlled gates are decomposed and $O(n^3)$ after an $O(n^2)$ decomposition of each, and $G$ is the top-left $2^n \times 2^n$ block of the unitary multiplied by $(2^{n-1}+2^n)\|G\|_S$. These are Wu et al.'s statements; their tests follow.

**The rebuild.** The angles of Fig. 9 can be read from its caption, but the wiring of $\hat{U}_R$ and $\hat{U}_C$ cannot be read from the figure, so the rebuild realises the same mechanism with registers of its own. The offset register is prepared in a uniform superposition over odd offsets only. Every non-zero entry of $G$ has $p-k$ odd, and an even offset, $d = 0$ above all, would load a value onto the diagonal and onto entries that must vanish, which the superposition drawn in Fig. 9 does not exclude. The row index is formed by subtracting the offset from the system register with a borrow, the arithmetic of the structured encoding (Appendix C), in place of the cyclic shifts and swaps. A copy of the borrow masks the offsets that would carry the row below zero, and an all-zero test on the difference flags row 0, on which a second ladder loads the row's correction, the role of the angles $\varphi_k$ in $\hat{U}_C$. The value is loaded by the ladder of Appendix I.6.2 with Wu et al.'s angles. Carry, borrow, its copy, the row-0 flag and the value qubit bring the circuit to $2n+5$ qubits.

**Costing the rebuild.** With the ladder coefficient $\mu = 2/\|G\|_S$ and the weight $2/N$ that the odd-offset superposition gives each entry, the rebuilt circuit block-encodes $G$ at $\alpha = N/\mu = N\|G\|_S/2$.

**Lemma 11** (Wu et al.'s norm of $G$)**.** *For $n \geq 3$, $\|G\|_S = \|G^\top G\|_1 = N(N-1)^2(N+1)/3$.*

*Proof.* $(G^\top G)_{pq} = \sum_k G_{kp}G_{kq}$ is non-zero only for $p \equiv q \pmod 2$, the sum running over the $k < \min(p,q)$ of the other parity with weight $4pq/c_k$. For odd $p \leq q$ these $k$ are $0, 2, \ldots, p-1$, and $\sum 1/c_k = \frac{1}{2} + \frac{p-1}{2} = p/2$; for even $p \leq q$ they are $1, 3, \ldots, p-1$, again $p/2$. So $(G^\top G)_{pq} = 2\min(p,q)^2\max(p,q)$, non-negative and increasing in both indices, and the largest column sum is the last, $q = N-1$, over the odd $p \leq N-1$: $2(N-1)\sum_{i=1}^{N/2}(2i-1)^2 = N(N-1)^2(N+1)/3$. The column sums of $GG^\top$ are, to leading order, $N^4/4$, below this from $n = 3$ on. □

The rebuilt circuit's $\alpha = N\|G\|_S/2$ is therefore a third of the printed prefactor $\frac{3N}{2}\|G\|_S$ of Equation (3.5), so every cost quoted for Wu et al.'s encoding is on the kinder side of the printed

one. With $\|G\|_S \sim N^4/3$ and $\|G\|_2 \sim N^2/(2j_{-3/4,1})$ (Theorem 3), the printed prefactor's looseness is $\frac{3N}{2} \cdot \frac{N^4}{3} \cdot \frac{2j_{-3/4,1}}{N^2} = j_{-3/4,1}N^3$, approached from below.

**The claims tested.** Rows A-06 to A-08 of Table I.4 give the verdict on each claim Wu et al. make for their circuits. The rows of Figs 2 and 8 hold exactly. For Fig. 9 the gate count holds as a bound and the error to leading order in $\mu$ (Proposition 38). Its qubit count is not testable, since the rebuild's own registers make $2n + 5$, which neither confirms nor refutes the printed $2n + 3$. The prefactor is valid but exceeds $\|G\|_2$ by a factor growing as $N^3$ (§ 4.5.1).

### I.6.2 The small-angle ladder and its error

**The mechanism.** A ladder of $n$ singly controlled $R_y$ rotations acts on one value qubit prepared in $|1\rangle$: bit $m$ of the column index controls $R_y(\theta_m)$,

$$\theta_m = 2\arcsin\big(-2^m\mu\big), \qquad m = 0, \ldots, n-1, \tag{I.2}$$

for a small real coefficient $\mu$. Rotations about one axis compose by adding their angles, so for a column index $p$ with bit set $S$ the net rotation is $R_y(\Theta(p))$, $\Theta(p) = \sum_{m\in S}\theta_m$, and the value qubit's $|0\rangle$ amplitude is

$$\mathrm{raw}(p) = -\sin\big(\Theta(p)/2\big) = \sin\Big(\textstyle\sum_{m\in S}\arcsin(2^m\mu)\Big). \tag{I.3}$$

Row 0's entries, $\sqrt{2}p$ in place of $2p$, take a second ladder on the same qubit with per-bit coefficient $\mu(1/\sqrt{2}-1)$, controlled on the row-0 flag. The amplitude equals the linear value $\mu p$ only if $\sin\big(\sum_m \arcsin(x_m)\big) = \sum_m x_m$, which fails whenever more than one bit is set, so the load is approximate.

**Proposition 38** (Entrywise error of the ladder construction)**.** *Let $J_3(p) := \sum_{m\in S}(2^m)^3$ for column index $p$ with bit set $S$. Then*

$$\big|\mathrm{raw}(p) - \mu p\big| \;\le\; \frac{\mu^3}{6}\,p^3 \;+\; O(\mu^5 N^5). \tag{I.4}$$

*Proof.* Expand both non-linearities to leading order in $\mu$, $\arcsin(y) = y + y^3/6 + O(y^5)$ and $\sin(x) = x - x^3/6 + O(x^5)$. Summing the first over the set bits and substituting into the second,

$$\sum_{m\in S}\arcsin(2^m\mu) = \mu p + \tfrac{\mu^3}{6}J_3(p) + O(\mu^5N^5), \tag{I.5}$$

$$\begin{aligned}\sin\Big(\sum_{m\in S}\arcsin(2^m\mu)\Big) &= \sin(\mu p) + \cos(\mu p)\cdot\tfrac{\mu^3}{6}J_3(p) + O(\mu^5N^5)\\ &= \mu p + \tfrac{\mu^3}{6}\big(J_3(p) - p^3\big) + O(\mu^5N^5).\end{aligned} \tag{I.6}$$

Since $p = \sum_{m\in S}2^m$ with every term positive, $p^3$ contains every $(2^m)^3$ among the terms of its expansion, so $0 \le p^3 - J_3(p) \le p^3$, with $J_3(p) = p^3$ only when $p$ is a power of two. The worst case $p = N - 1$ gives Equation (I.4). □

The bound is leading order, valid once the largest single argument, $2^{n-1}\mu$, lies well inside arcsin's domain. At the default $\mu = 2/\|G\|_S$ (Equation (3.5)) that holds from $n = 2$; at $n = 1$, $\mu = 1$ and the bound is void.

**Table I.5: Wu et al.'s encoding of $G$ as rebuilt takes two qubits more than printed, and its entry error stays within the ladder's bound, falling as about $N^{-5}$.** $n$, qubits of the system register, $N = 2^n$; $\|G\|_S = \max(\|GG^\top\|_1, \|G^\top G\|_1)$, Wu et al.'s norm; qubits of the rebuilt circuit, against the printed $2n + 3$; two-qubit gates after transpiling; $\epsilon_G$, largest entry error of the encoded $G$: the leading-order bound of Proposition 38 times the encoding's subnormalisation, and the value measured from the circuit's block column by column, for $n \leq 7$ (at $n = 8$ the 21-qubit extraction was not run). Exact classical computation.

| $n$ | $N$ | $\|G\|_S$ | Qubits (claim $2n+3$) | Gates | Bound on $\epsilon_G$ | Measured $\epsilon_G$ |
|---|---|---|---|---|---|---|
| 2 | 4 | $6.15 \times 10^{1}$ | 9 | 160 | $1.9 \times 10^{-2}$ | $6.4 \times 10^{-3}$ |
| 3 | 8 | $1.18 \times 10^{3}$ | 11 | 246 | $1.3 \times 10^{-3}$ | $2.6 \times 10^{-4}$ |
| 4 | 16 | $2.04 \times 10^{4}$ | 13 | 336 | $8.7 \times 10^{-5}$ | $8.9 \times 10^{-6}$ |
| 5 | 32 | $3.38 \times 10^{5}$ | 15 | 422 | $5.6 \times 10^{-6}$ | $2.9 \times 10^{-7}$ |
| 6 | 64 | $5.50 \times 10^{6}$ | 17 | 508 | $3.5 \times 10^{-7}$ | $9.4 \times 10^{-9}$ |
| 7 | 128 | $8.88 \times 10^{7}$ | 19 | 594 | $2.2 \times 10^{-8}$ | $3.0 \times 10^{-10}$ |
| 8 | 256 | $1.43 \times 10^{9}$ | 21 | 680 | $1.4 \times 10^{-9}$ | – |

### I.6.3 Measured resources and error

## I.7 Wu et al.'s filter and its parameters

Wu et al. realise imaginary time by QSVT on a block encoding of $\tilde{H} = H/\|H\|_F$ [1]. The target $p(x) \approx e^{-\beta x}$ has indefinite parity, so they find one set of phase angles for its even part and one for its odd part, of maximal degrees $d^e$ and $d^o$, run the two opposite-parity QSVT sequences and sum them by an LCU. Each sequence uses the unitary dilation

$$U(\tilde{H}) = \begin{pmatrix} \tilde{H} & \sqrt{I - \tilde{H}\tilde{H}^\dagger} \\ \sqrt{I - \tilde{H}^\dagger\tilde{H}} & -\tilde{H}^\dagger \end{pmatrix}, \tag{I.7}$$

built from PREPARE and SELECT subroutines over the unitary decomposition of $\tilde{H}$ with $m = \lceil \log_2 L \rceil$ ancilla for $L$ unitaries. The time $\beta$ is set by the approximate lower bound of Equation (3.6), and for the two Legendre examples of their supplement $\beta = 15$ and $\beta = 8$ recover the ground state with $(d^e, d^o) = (6, 7)$. Wu et al. also note that GQSP, GQET and GQSVT could improve the circuit.

Wu et al. expect a preparation time of order $\Delta^{-1}\,\mathrm{polylog}(1/\varepsilon)$ (§ 3.3) and report "a sizable gap for the studied problems" [1], without fixing the units of $\Delta$. If $\Delta$ is the gap of $\tilde{H}$, it is the gap of $H$ divided by $\|H\|_F$, and the norm that the expectation appears to omit enters through the normalisation. A gap that is sizable for $\tilde{H}$ at the printed resolutions is therefore no evidence about the cost until the growth of $\|H\|_F \geq \|H\|$ with $N$ is known, which § 4.6 derives for the standard form.

## I.8 The tau-truncation ablation

Wu et al. solve the equation by least squares over all its rows. The Chebyshev tau method instead drops the top $k$ rows of the equation block, $k$ the order, which leaves an exact kernel, $\sigma_0 = 0$, at every $n$. On R2 its accuracy matches that of least squares (Table I.6), so the choice between them does not set the error. The standard form keeps Wu et al.'s least squares, under which $\sigma_0$ measures how far the truncation is from an exact solution; under tau truncation the kernel is exact by construction and the trust test says nothing about it.

## I.9 The imaginary-time filter's circuit simulations

The reproduction runs Wu et al.'s imaginary-time filter in the standard form, by circuit simulation as well as by exact emulation, at $n = 2$ and 3, on the panels as printed and as run. Each circuit simulation is compared with the emulation before normalising, against the bar of $10^{-10}$ that the

**Table I.6: Dropping the top $k$ rows of the equation block (the Chebyshev tau method) gives an exact kernel with accuracy comparable to Wu et al.'s least squares.** $L^\infty$ field error and least singular value $\sigma_0$ of the residual, least squares against tau truncation, on R2a–R2c at $n = 5$–$7$; $k = 2$, the equation's order. Classical computation.

| Problem | $n$ | Least-squares $L^\infty$ | $\sigma_0$ | Tau-truncated $L^\infty$ | $\sigma_0$ |
|---|---|---|---|---|---|
| R2a | 5 | $3.2 \times 10^{-13}$ | $3.0 \times 10^{-11}$ | $1.6 \times 10^{-13}$ | 0 |
| R2a | 6 | $2.2 \times 10^{-12}$ | $4.9 \times 10^{-10}$ | $3.8 \times 10^{-12}$ | 0 |
| R2a | 7 | $2.8 \times 10^{-11}$ | $7.8 \times 10^{-9}$ | $1.0 \times 10^{-10}$ | 0 |
| R2b | 5 | $2.3 \times 10^{-11}$ | $5.3 \times 10^{-11}$ | $3.7 \times 10^{-12}$ | 0 |
| R2b | 6 | $1.8 \times 10^{-10}$ | $8.5 \times 10^{-10}$ | $7.4 \times 10^{-11}$ | 0 |
| R2b | 7 | $9.8 \times 10^{-10}$ | $1.4 \times 10^{-8}$ | $8.3 \times 10^{-10}$ | 0 |
| R2c | 5 | $2.2 \times 10^{-3}$ | $1.6 \times 10^{-2}$ | $1.8 \times 10^{-3}$ | 0 |
| R2c | 6 | $5.6 \times 10^{-12}$ | $5.5 \times 10^{-10}$ | $1.8 \times 10^{-11}$ | 0 |
| R2c | 7 | $2.4 \times 10^{-10}$ | $1.1 \times 10^{-8}$ | $1.9 \times 10^{-10}$ | 0 |

minimax records all meet (§ 8.1). All but 3 of the 46 imaginary-time records meet it too (Table I.7). The misses are among the highest degrees run, of order $10^5$, where the statevector's rounding accumulates over the controlled walks, but the degree alone does not set the difference: on R4a at $n = 3$ a record at a comparable degree passes just under the bar, and on the wave equation R5c records at higher degrees pass by more than an order of magnitude. The difference therefore depends on the problem's spectrum and filter as well as on the degree. The bar is kept at every degree, so these are reported as misses and not re-judged. They concern the circuit simulator's arithmetic, not the filter: the emulation is exact for the circuit as built (§ 7.5), and the minimax filter, whose degrees are smaller at equal suppression, meets the bar on every record.

**Table I.7: Three imaginary-time circuit simulations miss the $10^{-10}$ bar at degrees of order $10^5$; others at comparable or higher degrees pass.** Largest difference between circuit simulation (T1) and exact emulation (T2), before normalising, from a unit input. Below the rule, passing records for comparison: the largest passing difference (R4a) and the highest degree run (R5c). R6b is the doubled space, whose filter output is a projection.

| Problem | $n$ | Qubits | Degree | T1 against T2 |
|---|---|---|---|---|
| R6b | 2 | 17 | 211 721 | $2.6 \times 10^{-10}$ |
| R6b as printed | 2 | 17 | 211 722 | $2.1 \times 10^{-10}$ |
| R4a as printed | 3 | 20 | 81 195 | $1.5 \times 10^{-10}$ |
| R4a (passes) | 3 | 20 | 80 960 | $9.5 \times 10^{-11}$ |
| R5c (passes) | 3 | 19 | 287 359 | $4.3 \times 10^{-12}$ |

# J Software and Reproducibility

The software behind every result is `pihm`, a Python package written for this thesis, in the `pihm` directory of the repository https://github.com/DhyanP0564/Research. Production ran at the commit tagged `results-v1`; the thesis cites the repository at the commit of its final export of numbers, whose source differs from the tag's in documentation alone, and never at its main branch, which may change after submission. The repository holds what the thesis leaves out: the method as built (`docs/02_methods.md`) and the derivations from which the appendices are drawn (`docs/companion/`), the campaign files that define every run (`configs/campaigns/`), the tests, and the reports and notebooks generated from the records. The production's records themselves are not under version control.

**Architecture.** The package is ordered in layers, from the Chebyshev basis and the operators through the assembly of a residual, the circuits, their verification and state preparation to the pipeline

and the plotting, and a unit imports only from the layers below it, which an architecture test enforces. Each concept has one implementation, its modes typed parameters resolved through a dispatch table, so both forms and every problem share the code of every step they have in common.

**Records and provenance.** A problem is frozen, declarative data with a stable hash of what defines it; a constraint at a root of the reference is stored as its recipe and resolved when the problem is built. A run is specified by its problem, form, resolution, stage and the stage's options, and the hash of that specification names its record: the specification, its status (ok, failed, infeasible or skipped) and its provenance, the scalar metrics in strict JSON, the arrays and the log. The provenance holds the git commit, the versions of the package and of every dependency, the host, the Slurm job and a source digest, a hash of every module the run imports. A record made by a source the thesis's configuration does not accept is flagged for rerun, and each campaign's records are collected into one summary table (`tools/run_campaign.py -collect`).

**Campaigns and computational resources.** A campaign is a list of groups, each a grid of problems, resolutions and options for one stage, which expands into run specifications. Execution is idempotent, a run with a completed record being skipped, and each stage rebuilds its own residual, so that no run waits on another's files. Production ran all 25 campaigns, 3155 runs and 494 hours of computation, on Setonix [85]: a campaign of many short runs as one Slurm job, and one of long or large runs as Slurm arrays of one task per run, grouped by allocation. Every request was one node of the work partition with 16 CPUs, between 32 and 230 GB of memory and between 2 and 24 hours. Every circuit simulation first estimated its own memory and time against its allocation, and a run that would have exceeded it was recorded as infeasible with its estimate (Appendix G.5).

**Tests.** Every commit is gated by lint, strict type checking, an Australian-English spelling check, the test suite and a documentation build. The suite's tiers are unit and property tests of each module and of the operator identities over random $n$ and $x$; golden tests against Wu et al.'s printed $n = 2$ matrices and fixtures of the earlier code; the reproduction of Wu et al.'s panels; every problem built end to end at its smallest resolution; documentation tests that every report and notebook is current against the records; the architecture test; and the claims tier, one test per checkable claim, which writes the claims report (`docs/18_claims.md`). Larger-$n$ verifications and the circuit-simulation ceilings are marked slow and run at the project's gates.

**Regenerating the results.** From the cited commit, production is rerun by submitting `configs/productio` on Setonix and collecting each campaign, as `docs/05_production.md` sets out. The report scripts and `tools/notebooks/make_notebooks.py -all` then regenerate the reports and notebooks from the records, the scripts in the thesis's `Figs/scripts/` draw its figures from the same records, and `tools/thesis/export_numbers.py -final` writes every number the thesis quotes into its `0_results/genera` refusing unless each one is current and was made by an accepted source.

**Environment.** The package runs on Python 3.11 with NumPy 2.4.6 [83], SciPy 1.17.1 [84], Qiskit 2.1.2 and Qiskit Aer 0.17.2 [82] pinned exactly, and the full dependency closure locked in `requirements.txt`. The laptop installs that file into a conda environment that supplies only the interpreter, and Setonix into a virtual environment, so the two differ only in the platform's linear-algebra library (Accelerate against OpenBLAS); tolerances at the SVD's noise floor are therefore absolute.

# K Research Proposal

The research proposal of May 2025, reproduced below, set out to extend Wu et al.'s method to complex-valued nonlinear equations through a generalised QSVT, and to add data-informed terms to its Hamiltonian. The order of the work changed once their examples had been reproduced and costed. The number of queries a preparation needs, whose growth with the resolution their Discussion

leaves open, grows as a power of $N$, partly through a loose encoding of the dense differentiation matrix and partly through the norm that matrix gives the Hamiltonian (§ 4). An extension to complex or nonlinear equations would inherit both costs, so the thesis first separates them, then reformulates the residual to lower the second, and only then carries both forms to coupled and nonlinear equations. Table K.1 records each deviation and its reason.

**Table K.1: The proposal's extensions were deferred until the method's cost was settled.** Each aim of the research proposal of May 2025, what the thesis did in its place, and why.

| Proposed | Outcome | Reason |
|---|---|---|
| Extending QSVT to the generalised construction for complex-valued, non-Hermitian operators | GQSP applies every filter, without a parity constraint. Complex-valued linear equations are carried by realification (R7) and Fourier bases on real ones (R9, R10); one natively complex encoding is built, and no complex residual is prepared (§ 6.3). | GQSP already lifts QSVT's restrictions to real coefficients and a definite parity. The filter acts only on the Hermitian $R^\dagger R$, so a complex or non-Hermitian residual needs no generalised transform. The cost had to be settled before any extension. |
| Nonlinear PDEs: Burgers, nonlinear Schrödinger, complex Ginzburg–Landau | Burgers' equation and two-dimensional incompressible Navier–Stokes, by the Carleman lift (§§ 6.5 and 6.7); the Schrödinger and Ginzburg–Landau equations were not attempted. | Wu et al.'s doubled space only projects (§ 6.4), so the lift is the route. Liu et al.'s convergence guarantee assumes a dissipative quadratic system [4]; both equations are cubic, the Schrödinger equation conserves its norm, and the Ginzburg–Landau equation amplifies its longest waves. |
| Data-informed Hamiltonians | Not pursued. An observation would enter the residual as one more constraint row (§ 10.2). | The cost of the physics-informed Hamiltonian came first, and noisy observations replace the exact kernel by a least-squares fit whose gap has not been studied. |
| (not proposed) | The standard form and the reproduction of Wu et al.'s examples; the structured encoding of $G$; the descriptor reformulation; coupled systems in both forms; the verified preparation pipeline. | Each locates or lowers the cost: the reproduction fixes the baseline, the structured encoding separates the encoder's cost from the Hamiltonian's, the descriptor lowers the Hamiltonian's, and one pipeline measures both forms alike. |

# Masters Research Proposal

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May 25, 2025

**Title:** Solving Nonlinear PDEs via Quantum Singular Value Transformation

**Supervisor:** Professor Jingbo Wang

## 1 Introduction

### 1.1 Aims

This thesis aims to develop and evaluate quantum algorithms for solving nonlinear partial differential equations (PDEs) using the Quantum Singular Value Transformation (QSVT) framework. Based on recent work that represents differential equations and their boundary conditions as effective Hamiltonians, this work extends the QSVT to the Generalised Quantum Singular Value Transformation (GQSVT). This extension enables the application of this method to complex-valued differential equations which arise naturally in many areas of physics and engineering. Additionally the project will explore how to incorporate data-informed modifications into these Hamiltonians, with the goal of modelling data-rich regimes.

### 1.2 Significance

Solving complex nonlinear partial differential equations (PDEs) on quantum hardware is a rapidly emerging prospect with broad applications ranging from fluid dynamics and turbulence modelling to quantum field theory. Established algorithms such as the Harrow-Hassidim-Lloyd (HHL) algorithm and variational methods typically rely on discretisation, which inhibits their scalability to stiff, nonlinear, multiscale or dense systems with fine grid sizes. In contrast, the effective Hamiltonian approach uses spectral methods to encode the solution into a Chebyshev polynomial basis, allowing it to surpass performance when fine-grid encoding is required. The quantum singular value transformation (QSVT) provides a flexible and modular approach to quantum linear algebra, allowing operations like matrix inversion under broader conditions than HHL [1]. However, the current method restricts effective Hamiltonians to real values, thereby limiting the class of operators and solutions to purely real outcomes. Moreover, the HHL paper only addresses linear partial differential equations or nonlinear ordinary differential equations

through a process called linearisation [3] and variational algorithms require classical optimisation [7].

This research is significant because it extends the effective Hamiltonian based framework to complex-valued nonlinear PDEs, which are common in frontier areas of various fields. In domains like geophysical inversion, deep turbulence and real-time weather prediction, incorporating data-informed terms into the Hamiltonian provides a promising path towards optimising quantum solutions to problems where complete physical models are unavailable or difficult to create. By enabling hybrid physics and data driven Hamiltonians, this work could expand the applicability of quantum PDE solvers to data-rich, real world problems.

# 2 Background

## 2.1 Problem Statement

Many physical systems are described by partial differential equations (PDEs), but solving them remains computationally costly. In the absence of closed-form solutions, many advanced classical algorithms implement numerical methods which are difficult to scale and lose accuracy in systems with a need for more fine grid or time-step discretisations. This is especially true of nonlinear PDEs which are the primary focus of this research.

Quantum algorithms provide a potentially scalable alternative. Quantum Singular Value Transformation (QSVT) allows quantum circuits to encode real-valued polynomials of definite parity. Recent theoretical developments have shown that it is possible to represent differential equations and their boundary conditions as effective Hamiltonians, which are structured matrices that can be processed using QSVT. However, due to QSVT's restriction to real-valued Hermitian Hamiltonians, the range of solvable PDEs is restricted. Systems with damping, gain/loss or complex boundary interactions can lead to complex-valued or non-Hermitian operators.

While novel frameworks such as the Generalised QSVT (GQSVT) have been proposed to handle complex-valued operators and remove parity constraints, they have not been systematically applied to nonlinear PDEs.

The existing effective Hamiltonian approach is largely physics-driven, assuming known governing models. Yet in many high-dimensional, data-rich domains such as geophysical imaging, real-time weather prediction, or turbulence modelling, the governing equations may be incomplete or unavailable. In these settings, data-informed Hamiltonian construction offers a promising path forward towards more flexible and accurate quantum solvers.

This research addresses the gap in current quantum PDE solving frameworks by developing and applying GQSVT based methods for nonlinear, complex-valued PDEs and exploring hybrid physics and data driven constructions for real world data-intensive applications.

## 2.2 Literature Review

### 2.2.1 Exploration of Problems

**Burgers Equation (Viscous)**

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \nu\frac{\partial^2 u}{\partial x^2} \tag{1}$$

This equation appears in turbulence theory and has a closed form solution in one real spatial dimension with $\nu > 0$. However, complex versions do exist and the equation remains unsolvable analytically outside these circumstances. The nonlinearity term is $u\frac{\partial u}{\partial x}$.

**Nonlinear Schrödinger Equation**

$$i\frac{\partial \psi}{\partial t} + \nabla^2\psi + |\psi|^2\psi = 0 \tag{2}$$

This equation describes the evolution of complex wave functions in nonlinear media. Its nonlinearity term is $|\psi|^2\psi$.

**Complex Ginzburg-Landau Equation**

$$\frac{\partial A}{\partial t} = A + (1 + i\alpha)\nabla^2 A - (1 + i\beta)|A|^2 A \tag{3}$$

This equation models phenomena such as superconductivity and fluid convection patterns. The nonlinearity term is $|A|^2A$.

The current physics-informed Hamiltonian method is able to address Burgers equation in its real case, but the Schrödinger and Ginzburg-Landau equations involve are complex-valued fields. This limitation motivates the development of a more robust framework capable of handling a broader class of nonlinear PDEs.

### 2.2.2 Existing Classical and Quantum Solving Methods

Classical methods for solving PDEs employ mesh based discretisation techniques such as finite difference and finite element methods or spectral methods [11]. These methods approximate the continuous domain onto a discrete grid, converting the PDE into a large system of algebraic equations that can be solved numerically. These methods are well-established and reliable, but for high-resolution or multi-dimensional systems the computational cost increases rapidly. For higher dimensional problems, direct matrix or classical variational methods may be employed [4].

The Harrow-Hassidim-Lloyd (HHL) algorithm was one of the first to demonstrate exponential speedup for solving systems of linear equations, which arise from discretised PDEs [3]. HHL encodes the linear system $A\vec{x} = \vec{b}$ into a quantum state and uses phase estimation to prepare a quantum state proportional to the solution vector $\vec{x}$. However, it is limited to sparse, well-conditioned Hermitian matrices. This makes it impractical for solving PDE problems, particularly nonlinear ones. Importantly, in the 2021 "Grand Unification" paper, it is shown that HHL can be viewed as a special case of QSVT, where

we're looking to find a target polynomial that inverts the singular values of the matrix $A$.

Variational quantum algorithms use a hybrid quantum-classical loop to optimise a parameterised quantum circuit that approximates the solution to a PDE [7]. This approach also discretises the domains into a grid. While more flexible than HHL, these methods require classical optimisation and suffer from challenges such as barren plateaus and scaling issues for larger systems.

### 2.2.3 Quantum Singular Value Transformation

The quantum singular value transformation (QSVT) is a powerful framework that generalises quantum linear algebra operations by enabling quantum circuits to apply polynomial transformations to the singular values of block encoded matrices [1]. QSVT enables the application of any parity-constrained polynomial $f(\Sigma)$ to the singular values $\Sigma$ of a matrix $A$ block encoded within a unitary.

$$U = \begin{bmatrix} A & \cdot \\ \cdot & \cdot \end{bmatrix} \text{ with } \|A\| \leq 1 \tag{4}$$

Assuming we have access to such a unitary $U$ and the ability to apply a sequence of controlled-phase rotations by an angle $\phi$, we can construct a unitary $V_{\vec{\phi}}$, dependent on the values of $\vec{\phi}$ we select for our phase rotations, that looks like the following:

$$V_{\vec{\phi}} = \begin{cases} \tilde{\Pi}_{\phi_1} U \left[ \prod_{k=1}^{(d-1)/2} \Pi_{\phi_{2k}} U^\dagger \tilde{\Pi}_{\phi_{2k+1}} U \right] = \begin{bmatrix} \text{Poly}^{(\text{SV})}(A) & \cdot \\ \cdot & \cdot \end{bmatrix} & \text{for odd Poly}^{(\text{SV})}(A) \\ \left[ \prod_{k=1}^{d/2} \Pi_{\phi_{2k-1}} U^\dagger \tilde{\Pi}_{\phi_{2k}} U \right] = \begin{bmatrix} \text{Poly}^{(\text{SV})}(A) & \cdot \\ \cdot & \cdot \end{bmatrix} & \text{for even Poly}^{(\text{SV})}(A) \end{cases} \tag{5}$$

where the sequences of phases $\vec{\phi} = (\phi_0, \phi_1, \ldots, \phi_d)$ is chosen specifically to encode a desired polynomial $f(x)$ of degree $d$ [1]. $\tilde{\Pi}_\phi$ and $\Pi_\phi$ represent a controlled phase rotation by $\phi$ where $\tilde{\Pi}$ and $\Pi$ are the left and right unitary projectors respectively such that $A := \tilde{\Pi} U \Pi$. By carefully selecting which phase angles to rotate by, we can approximate any real, fixed parity continuous polynomial [6] acting on the singular values of $A$.

It can be shown that QSVT serves as a unifying framework for many quantum algorithms including quantum search, Hamiltonian simulation, factoring, quantum walks and more [8]. A key limitation of QSVT is that it only supports real-valued polynomials applied to matrices. This restricts its applicability to problems where both the matrix and the function being applied are real-valued [1]. For differential equations, this implies only PDEs with real-valued operators and real-valued solutions can be handled.

For physics-informed partial differential equation (PDE) solvers, QSVT is used in ground-state preparation via Hamiltonian simulation. In this case the target polynomial is $p(x) = e^{-tx}$ [11].

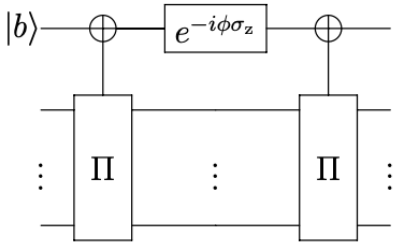



Figure 1: Quantum Circuit Diagram of the implementation of a Projector Controlled Phase Gate, $\Pi_\phi$ [2]

#### 2.2.4 Physics-Informed Effective Hamiltonians

This algorithm encodes a differential equation in a latent Chebyshev basis and finds its solution by preparing the ground state of a specifically constructed Hamiltonian [11]. First, consider some differential equation with a differential operator $\mathcal{L}$, solution $u(x)$ and source function $f(x)$:

$$\mathcal{L}[u(x)] = f(x), \qquad x \in [a, b]. \tag{6}$$

We can represent the solution as a state overlap in the Chebyshev basis with a scale factor s, namely

$$u(x) = s\langle T(x)|\psi\rangle \tag{7}$$

Here $|T(x)\rangle$ is known as a "feature map" state, a superposition of computational basis states with Chebyshev coefficients. $|\psi\rangle$ is a parameterised quantum state carrying the unknown amplitudes. Thus, we've effectively represented our solution in a latent Chebyshev basis [11].

To translate differential operators, we use the known differentiation matrices associated with the Chebyshev basis [5]. An example translation is shown below where $\mathbb{G}$ is the generator matrix corresponding to differentiation:

$$\frac{d^2 f(x)}{dx^2} + af(x) = 0 \longrightarrow \mathbb{A} = \mathbb{G}^{\intercal 2} + a\mathbb{I}. \tag{8}$$

Here, $\mathbb{A}$ is the matrix corresponding to the latent space representation of the DE. Crucially, differential operators act simply in the Chebyshev basis, as the following

$$\frac{d}{dx} f(x) = \frac{d}{dx}\langle T(x)|\psi\rangle = \langle T(x)|D^\intercal|\psi\rangle. \tag{9}$$

As a result, all terms in the DE can be rewritten as linear operators acting on $|\psi\rangle$.

To implement constraints, we use $\mathbb{B}(x)$, a data transformation operator dependent on $x$. For example, for specific constraints such as $f(x_0) = 0$ or $f'(x_1) = 0$, we can use $\mathbb{B}(x_0)$ or $\mathbb{B}(x_1)\mathbb{G}^\intercal$ [11]. From these, we can create our effective Hamiltonian:

$$\mathbb{H} = \mathbb{A}^\intercal \mathbb{A} + \mathbb{B}^\intercal \mathbb{B}. \tag{10}$$

We may add additional boundary constraints as required, which would effectively be more $\mathbb{B}$ matrices. Therefore, by construction, solving the original DE is equivalent to solving the eigen-problem:

$$\mathbb{H}_{\text{eff}}|\psi\rangle = 0, \tag{11}$$

and the desired solution corresponds to the ground state of the effective Hamiltonian. H is a real symmetric positive-definite matrix with distinct eigenvalues and mutually orthonormal eigenvectors [11].

We then block encode the effective Hamiltonian, and use a ground state preparation routine, QSVT, to amplify the amplitude of the ground state. Specifically, we use quantum imaginary time evolution, constructing the polynomial approximation of $e^{-t\mathbb{H}}$. This exponentially suppresses the excited eigen-components, leaving a higher probability of measuring the ground state. Importantly, the algorithm directly prepares the lowest energy solution without classical optimisation [11].

Finally, to measure the quantum state, we use a Chebyshev transformation and Hadamard test to measure the state we have prepared. This controlled overlap test effectively measures the overlap $\langle T(x)|\psi\rangle$. In practice this is done via interferometric measurement [11].

This same framework is extendable to multi-variable and nonlinear DEs. For a function of two variables $f(x, y)$, we use a tensor product Chebyshev basis with two registers, preparing a joint feature state $|T(x)\rangle \otimes |T(y)\rangle$. Take Laplace's equation for example:

$$\frac{d^2 f}{dx^2} + \frac{d^2 f}{dy^2} = 0 \longrightarrow \mathbb{A} = \mathbb{G}_x^{\intercal 2} \otimes \mathbb{G}_y^{\intercal 2}. \tag{12}$$

For nonlinear terms, we add extra registers. For example, a term like $f'(x)^2$ leads to a product of two copies of $|\psi\rangle$ in latent space. We create a new, higher dimensional latent Chebyshev space where $|\Psi\rangle \equiv |\psi\rangle \otimes |\psi\rangle$, so that the nonlinearity term becomes linear in the larger space. This naturally handles all nonlinearity. Inhomogeneous terms are handled via Maclaurin series expansion. For all these cases, we then construct the effective Hamiltonian as usual and solve [11]. This approach allows us to encode the solution over the entire domain, rather than using grid discretisation. Therefore, it is possible to evaluate the solution function at any arbitrary input values. However, as discussed above, QSVT can only handle real values. This restricts the class of differential equations we're able to solve.

**Example Solutions:**

Below are results from the application of this algorithm on example differential equations [11]. The paper did not attempt to solve nonlinear PDEs, but there are examples of both linear PDEs and nonlinear DEs. $f(x)$ is the known solution and $f_Q^*(x)$ is the quantum solution.

1. Heat Equation

$$\frac{\partial f(t,x)}{\partial t} - k\frac{\partial^2 f(t,x)}{\partial x^2} = 0, k = \frac{1}{25}, \mathcal{BC} = \{f(0,x) = \sin(2\pi x), f(t,\pm 1) = 0\} \tag{13}$$




Figure 2: Solution for the Heat Equation Example. [11]

2. Nonlinear Example 1

$$4\frac{d^2 f(x)}{dx^2} + 2\left(\frac{df(x)}{dx}\right)^2 + f(x) = 0, \mathcal{BC} = \{f(0) = 1, f'(0) = 0\} \tag{14}$$

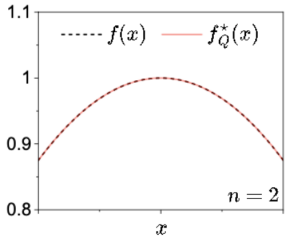



Figure 3: Solution of Nonlinear Example 1 [11]

3. Nonlinear Example 2

$$\frac{d^2 f(x)}{fx^2} - 2f(x)^2 + x = 0, \mathcal{BC} = \{f(-1) = -0.1, f(1) = 0.1\} \tag{15}$$

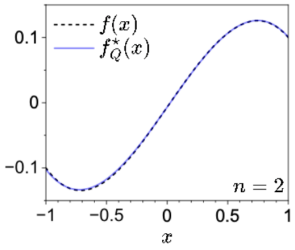



Figure 4: Solution of Nonlinear Example 2 [11]

The model performs well for all examples shown. The errors are almost negligible and decrease as the number of qubits used (shown as $n$ on the figures) increase. The experiments were conducted on classical simulations of quantum hardware and are therefore currently computationally costly, however based on these results it is projected this algorithm will become useful in the early fault-tolerant era [11].

### 2.2.5 Generalised Quantum Singular Value Transformation

The generalised QSVT (GQSVT) framework extends the standard QSVT algorithm by lifting the restrictions on coefficient phases and polynomial parity [9]. The standard QSVT restricts the target polynomial $p(x)$ to have real coefficients and definite parity. However, the generalised version allows complex polynomials and indefinite parity. These generalisations are inherited from developments in generalised quantum signal processing (GQSP) which is itself an extension of the quantum signal processing framework that QSVT was built on [10]. Eliminating the restrictions on real polynomials and Hamiltonians imposed by QSVT will allow us to solve complex-valued differential equations. GQSVT also promises a simplification in complexity for the crucial and time-consuming step of calculating phase factors. The task was of complexity $\mathcal{O}(d^2)$ for QSVT, GQSVT promises a complexity of $\mathcal{O}(d)$ [9]. This should speed up calculations even for problems which are currently possible via the existing method.

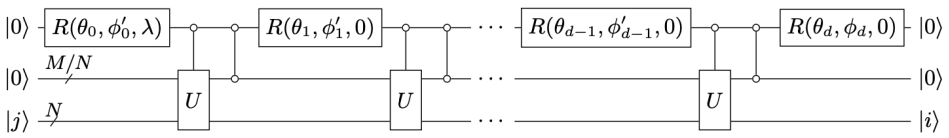



Figure 5: Example circuit for the generalised quantum singular value transformation for a Hermitian matrix $A$, block encoded as $U$. $R$ is a rotation matrix analogous to $\Pi$ for QSVT [9].

# 3 Research Plan

## 3.1 Method

This research will be primarily theoretical and computational. The goal is to design, simulate and evaluate novel quantum algorithms for solving nonlinear, complex-valued partial differential equations (PDEs). Development will be conducted using using *Mathematica* for symbolic manipulation and proof-of-concept derivations. Simulations will be created using *Python* based simulators such as *Qiskit* and *Pennylane* and results will be compared to classical solving algorithms. Efforts wil be split into two phases. First, extending the current QSVT algorithm to GQSVT and second, integrating data-informed terms into effective Hamiltonians to enable solving PDEs in data-rich regimes. This research represents a new research direction within QUISA.

**Proposed Plan:**

1. Reproduce the Current Physics-Informed QSVT Method:
   (a) Implement the standard QSVT framework in *Python*.
   (b) Choose an appropriate problem.
   (c) Implement a physics-informed Hamiltonian PDE solver in *Python*.
   (d) Compare quantum solution against classical benchmarks
2. Extend to GQSVT:
   (a) Create the GQSVT algorithm in *Python*.
   (b) Choose an appropriate problem.
   (c) Implement a new physics-informed Hamiltonian PDE solver in *Python*.
   (d) Compare quantum solution against classical benchmarks
3. Experiment with more data-informed terms in the Hamiltonian

**Timeline:**

Semester 1 2025: Research Proposal.

Semester 2 2025: Recreation of QSVT Physics-informed method. Begin extension to GQSVT.

Semester 1 2026: Completion of a working GQSVT enhanced algorithm, including simulation and evaluation in comparison to classical algorithms.

Semester 2 2026: Experimentation with data-informed terms.

## 3.2 Preliminary Work

In semester 1 2025, work consisted of a thorough literature review, achieving an in-depth understanding of QSVT and the physics-informed Hamiltonian approach for PDEs. This was pivotal in developing a thesis that could both build on the newest developments in the field and be achievable within the time-frame of a Master's course. A working *Python* implementation of QSVT has been created and work in underway on implementing the current physics-informed Hamiltonian approach using QSVT.

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